

Fig. 19-13 shows the perspective view of a hut having its front face in the picture plane. The front face is seen in its true size and shape, while the back parallel face is of the same shape but reduced in size. As the lines AF , BG , CH , DJ and EK are perpendicular to the picture plane, their perspectives $A'F'$, $B'G'$ etc. converge to the centre of vision c' on HL . Note that vertical lines AE , CD etc. remain vertical in perspective. Similarly, horizontal lines ED and KJ , and sloping lines AB , BC , FG and GH (which are all parallel to the picture plane) remain respectively horizontal and sloping in perspective.

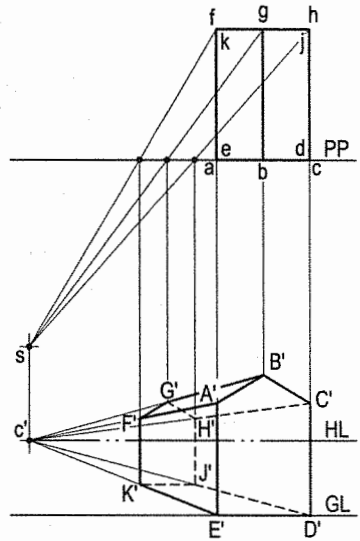


FIG. 19-13



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 47 for the angular or two point perspective.

(2) Angular perspective or two point perspective: When an object has its two faces inclined to the picture plane, its perspective is called *angular perspective* also called *two point perspective* as the edges of the object converge to two vanishing points.

Problem 19-4. (fig. 19-14): A rectangular block, 30 mm $\times$ 20 mm $\times$ 15 mm, is lying on the ground plane on one of its largest faces. A vertical edge is in the picture plane and the longer face containing that edge makes an angle of 30° with the picture plane.

The station point is 50 mm in front of the picture plane, 30 mm above the ground plane and lies in a central plane which passes through the centre of the block. Draw the perspective view of the block.

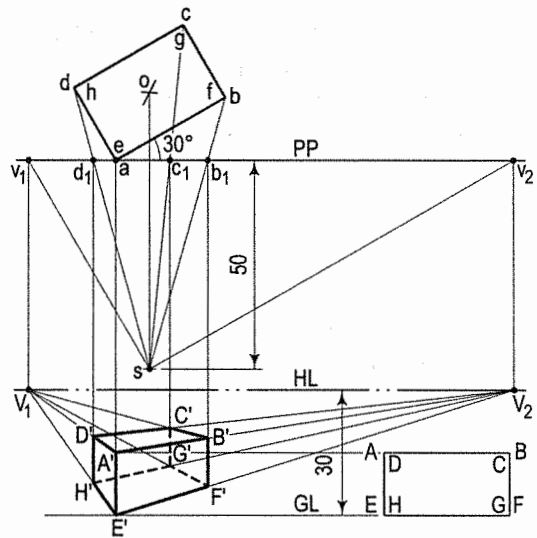


FIG. 19-14

- Draw the top view $abcd$ with a in PP and the longer edge ab inclined at 30° to PP . Mark its centre o . Mark s , the top view of the station point, on a vertical line through o and 50 mm below PP .
- Draw lines joining s with corners b , c and d , and intersecting PP at points b_1 , c_1 and d_1 .
- Draw the ground line GL at any distance below PP and the horizon line HL , 30 mm above GL .

- (iv) Through s , draw lines parallel to ad and ab cutting PP at points v_1 and v_2 respectively. Project v_1 to V_1 and v_2 to V_2 on HL. V_1 and V_2 are the vanishing points. Perspectives of edges AD , EH , BC and FG will converge to V_1 and those of edges AB , CD , EF and GH will converge to V_2 . Perspectives of vertical edges AE , BF , CG and DH will remain vertical.
- (v) As AE is in the picture plane, its perspective will be equal to the true length and the end E will lie on GL. Therefore, through a , draw a vertical line to a point E' on GL and on it, mark $A'E' = AE$.
(This length may be measured directly or may be projected from the front view as shown.)
- (vi) Draw lines joining A' and E' with V_1 and V_2 . Through b_1 , draw a vertical line to intersect $A'V_2$ at B' and $E'V_2$ at F' . Similarly, draw a vertical line through d_1 and obtain points D' and H' .
- (vii) Draw lines joining B' and F' with V_1 and D' and H' with V_2 , intersecting at points C' and G' respectively. They must lie on the vertical line through c_1 . Note that lines meeting at G' are all hidden and therefore, shown dotted.

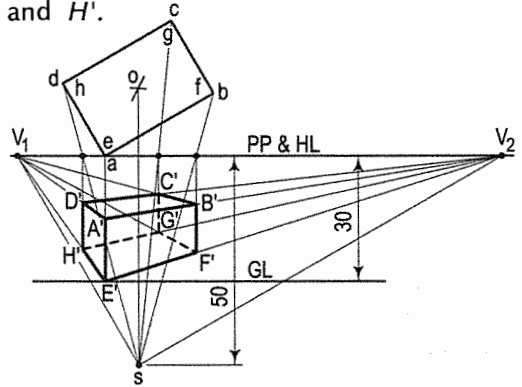


FIG. 19-15

In fig. 19-15, the ground line GL has been so drawn that HL coincides with PP. Hence, V_1 and V_2 coincide with v_1 and v_2 respectively on PP. The perspective view is obtained in the same manner as described above.



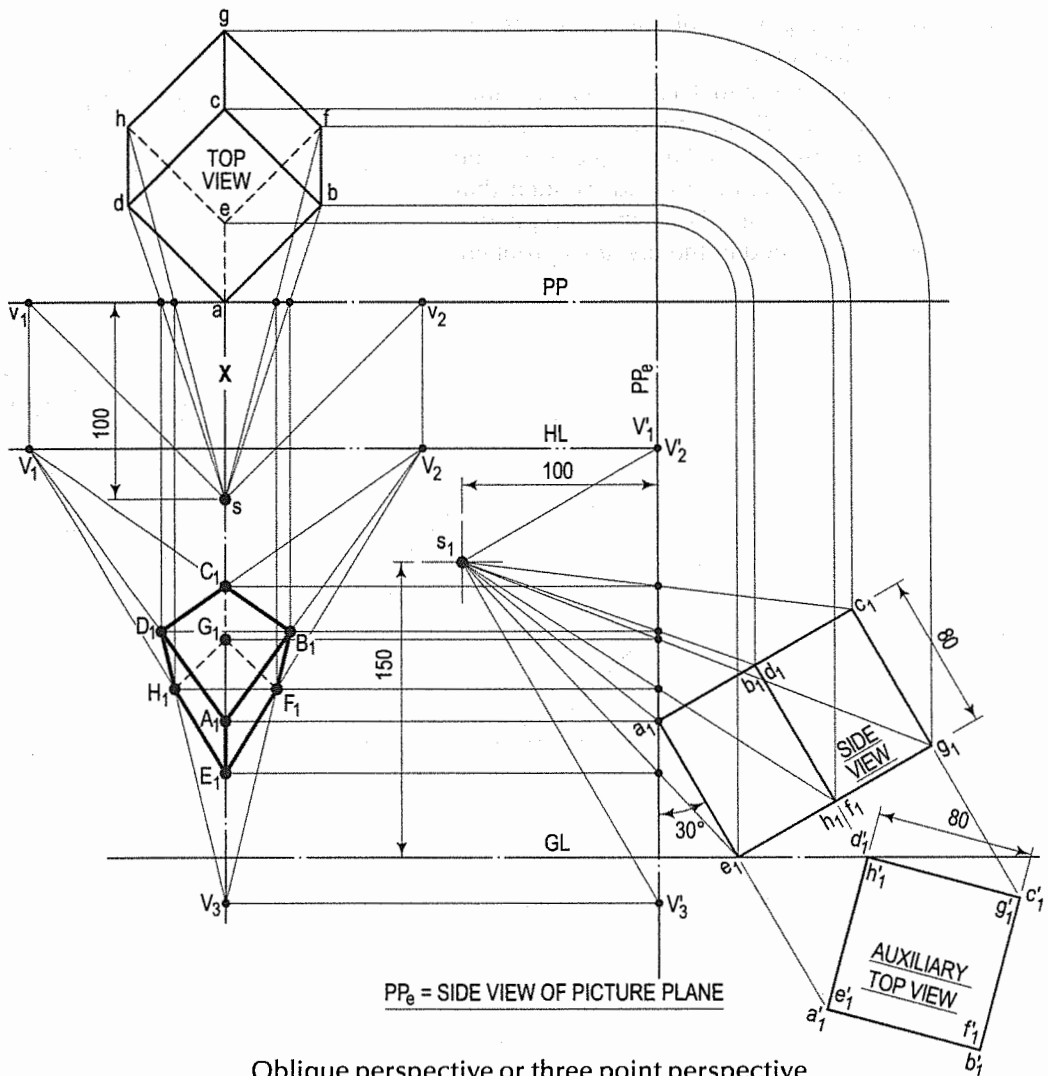
This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 48 for the oblique or three point perspective.

(3) Oblique perspective or three point perspective: When an object has its three faces inclined to the picture plane, its perspective is called *oblique perspective* also called *three point perspective* as edges of the object converge to three vanishing points, as shown in fig. 19-16.

Problem 19-5: Draw the perspective view of a cube of 80 mm side having its one corner of the edge on the ground plane and the other corner of the edge resting on the picture plane such that the edge is inclined at 30° to the picture plane. The other two edges of the corner are equally inclined with the picture plane. The station point is 100 mm in front of the picture plane, 150 mm above the ground plane and lies in a central plane which passes through the centre of the cube.

Refer fig. 19-16.

- (i) Draw projection plane lines PP and PP_e , perpendicular to each other.
- (ii) Draw side view on PP_e from auxiliary top view.
- (iii) Draw top view on PP by taking projection from side view.
- (iv) Mark the station point s for F.V. and s_1 for S.V. at a distance of 100 from PP and PP_e respectively and 150 from G.L.
- (v) Obtain the three vanishing points V_1 , V_2 and V_3 and complete the perspective view as shown.



19-9. DISTANCE POINTS

Vanishing points for all horizontal lines inclined at 45° to the picture plane are given special name of *distance points* on account of their definite positions.

They are equidistant from the centre of vision, the distance of each from the centre of vision being equal to the distance of the station point from the picture plane.

In fig. 19-17, $p_1c' = p_2c' = sc$. Thus, perspectives of all horizontal lines inclined at 45° to the picture plane converge to distance points on the horizon line.

Problem 19-6. (fig. 19-17): Draw the perspective view of a cube of 25 mm edge, lying on a face on the ground plane, with an edge in the picture plane and all vertical faces equally inclined to the picture plane. The station point is 50 mm in front of the picture plane, 35 mm above the ground plane and lies in a central plane which is 10 mm to the left of the centre of the cube.

- (i) Draw the top view of the cube and of the station point.
- (ii) Draw lines GL and HL. Obtain distance points P_1 and P_2 in the same manner as the vanishing points. Or, project s to c' on HL and mark points P_1 and P_2 such that $P_1c' = P_2c' = sc$. Draw the perspective view as described in the previous problem.

Problem 19-7: Fig. 19-18 shows the two point perspective of model of steps with the base on ground plane and vertical edge in picture plane and inclined at 45° to picture plane. The station point is 2 m in front of vertical edge and 1 m from ground plane.

Observe that $v_1-a = v_2-a = s-a = 2.0$.

See fig. 19-18.

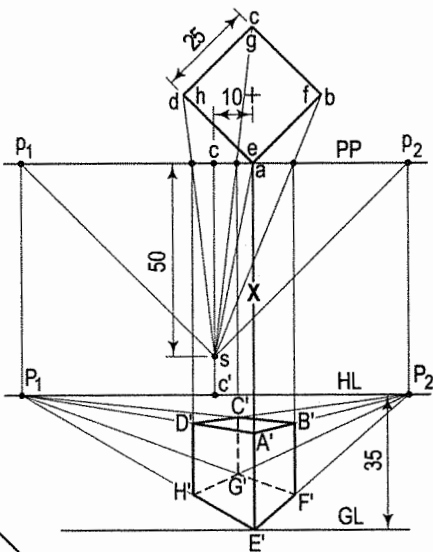


FIG. 19-17

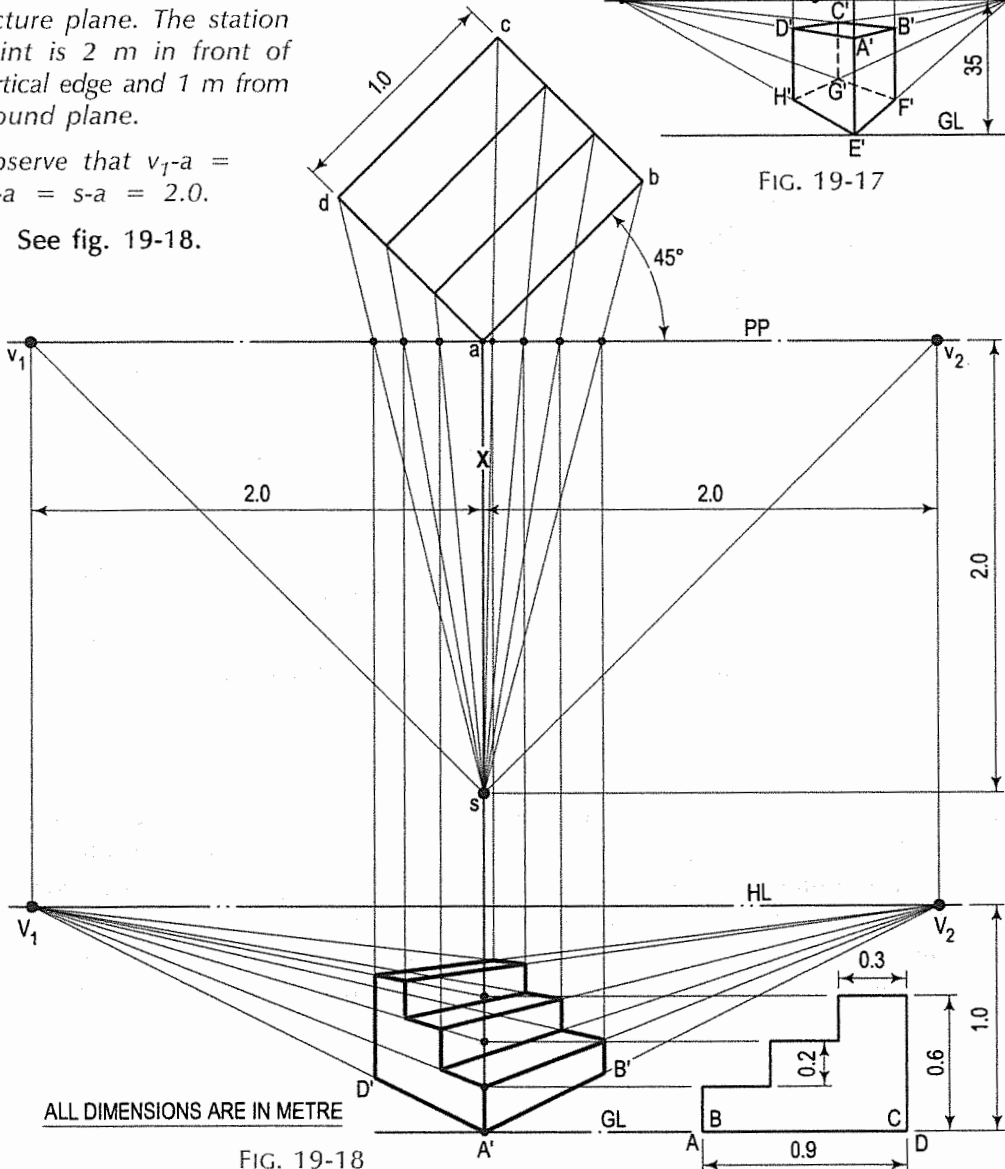


FIG. 19-18

19-10. MEASURING LINE OR LINE OF HEIGHTS

We have seen that when a line is in the picture plane, it is seen in its true length in perspective. When a line is behind the picture plane, it is foreshortened in its perspective view.

In fig. 19-19, ab is the top view of a rectangle $ABCD$ whose surface is vertical and inclined to the picture plane, the edge DC is on the ground plane and the edge AD is in the picture plane.

In its perspective view $A'B'C'D'$, $A'D'$ is equal to the true length AD , while $B'C'$ is shorter. $A'D'$ lies on a vertical line drawn through a . The length $B'C'$ is derived from $A'D'$. Let us now consider the rectangle $EBCF$ within the rectangle $ABCD$. $E'B'C'F'$ is its perspective view. $E'F'$ is shorter than EF or AD . Its length has been derived from $A'D'$.

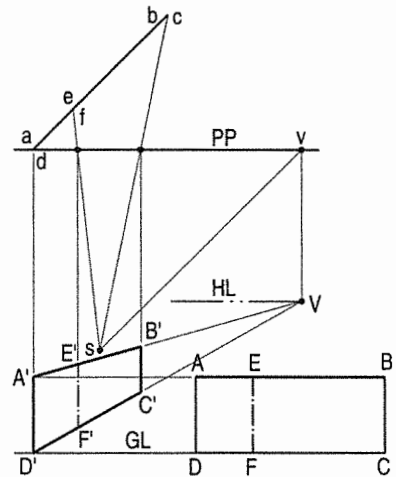


FIG. 19-19

Thus, aD' is the measuring line or the line of heights for points E , F , B and C . It is obtained by producing be to intersect PP at the point a ; through a , a perpendicular is dropped to meet GL at a point D' . In other words, if we imagine that the rectangle $EBCF$ is moved along the line eb to meet the picture plane, its edge EF will strike it at the line $A'D'$ showing its true length.

Thus, the measuring line or the line of heights is the trace or the line of intersection with the picture plane, of the vertical plane containing the point or points whose heights are to be determined. Heights of points lying in different vertical planes can be measured from their respective lines of heights. Heights on this line may be measured directly with a scale or may be projected from the front view.

Problem 19-8. (fig. 19-20): Determine the line of heights for points lying in the line ac which is the top view of a regular hexagon $ABCDEF$ (the front view of which is shown on GL) and then, draw its perspective view from the given position of the station point.

As ac is behind the picture plane, all the sides of the hexagon will be foreshortened.

- (i) To determine the measuring line, produce ca to meet PP at a point h ; through h , draw a vertical to meet GL at a point H . Then hH is the measuring line for heights of all points in the hexagon.
- (ii) To draw the perspective view, determine the vanishing point V . Project horizontally, points A , B etc. from the front view to points a' , b' etc. on hH . Draw lines joining these points with V . Draw lines joining s with a , b and c . Through the points of intersection of these lines with PP , draw verticals to meet the corresponding lines converging to V at points A' , B' ,..... F' . Join these points in correct sequence. The resulting figure is the perspective view of the hexagon.

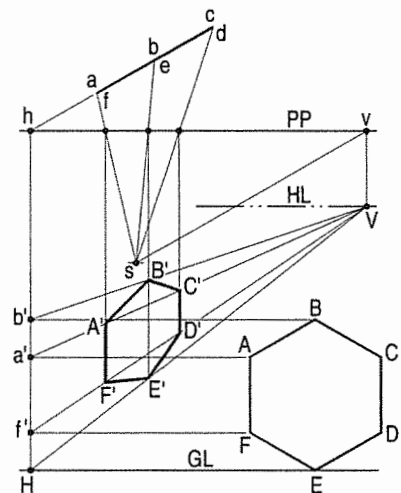


FIG. 19-20

19-11. PERSPECTIVES OF CIRCLES AND SOLIDS



A circle will appear as a circle in its perspective view when it is parallel to the picture plane. In other positions its perspective will be an ellipse (except when it lies in the central plane).

To obtain points for the ellipse, the circle is enclosed or *crated* in a square. The mid-points of the sides of the square are the four points for the ellipse. Points of intersection of the diagonals with the circle are the other four points. Lines are drawn through these points, parallel to the sides of the square. Perspective of the square along with the parallel lines is then drawn, locating the eight points for drawing the ellipse.

Problem 19-9. (fig. 19-21): *Draw the perspective view of a circle of 50 mm diameter, lying on the ground plane and touching the picture plane. The station point is 80 mm in front of the picture plane and 60 mm above the ground plane. The central plane passes through the centre of the circle.*

- (i) With o as centre, draw the circle touching PP. Enclose it in a square $abcd$ with ab in PP. Draw the horizontal and vertical diameters and the diagonals of the square. Mark points 1 to 8 as shown. Through the points on the diagonals, draw lines ef and gh parallel to ad .
- (ii) Draw lines GL and HL, and mark s , the top view of the station point. Project s to c' on HL. c' is the centre of vision to which the perspectives of lines perpendicular to PP will converge.
- (iii) Draw the perspective view of the square. Draw the diagonals, intersecting each other at O' . Through O' , draw horizontal and vertical lines, intersecting the sides at points $1'$, $2'$, $3'$ and $4'$. Draw perspectives of lines ef and gh , cutting the diagonals at points $5'$, $6'$, $7'$ and $8'$. Draw the ellipse through points $1'$, $5'$, $2'$ $8'$. The ellipse is the required perspective view of the circle.
- (iv) It may be noted that the centre of the ellipse does not coincide with O' , the centre of the perspective of the square.

Perspectives of concentric circles are not concentric ellipses as can be seen from fig. 19-22. The circles are on the ground plane. The station point is away to the left of the centre of the circles.

Fig. 19-23 shows perspectives of circles inscribed in the faces of a cube resting on the ground plane with an edge in the picture plane and all vertical faces equally inclined to the picture plane. The central plane passes through the centre of the cube. The ground line GL has been so placed that HL coincides with PP. Hence, distance points P_1 and P_2 also lie on PP.

Curve of any shape can similarly be drawn in perspective by enclosing it in a rectangle and then drawing horizontal and vertical lines through a number of points on the curve. Fig. 19-24 shows the perspective view of a moulding, the front view of which is shown on the right-hand side. aH is the line of heights.

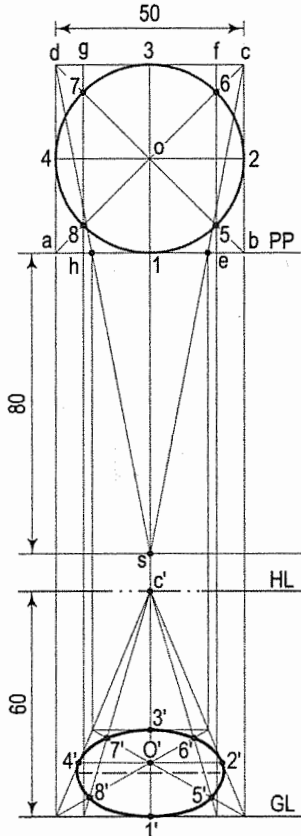


FIG. 19-21

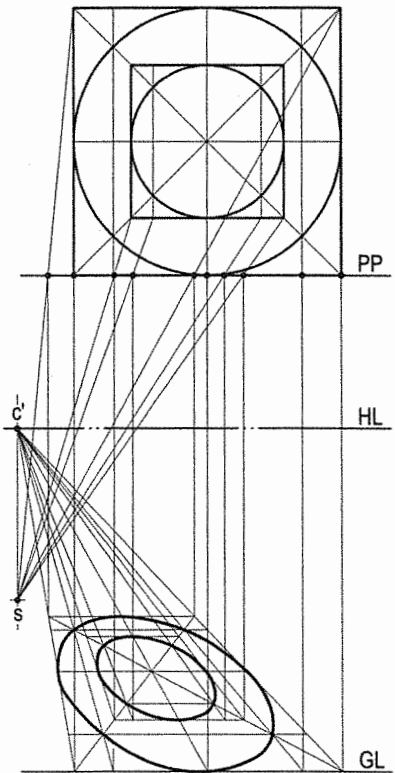


FIG. 19-22

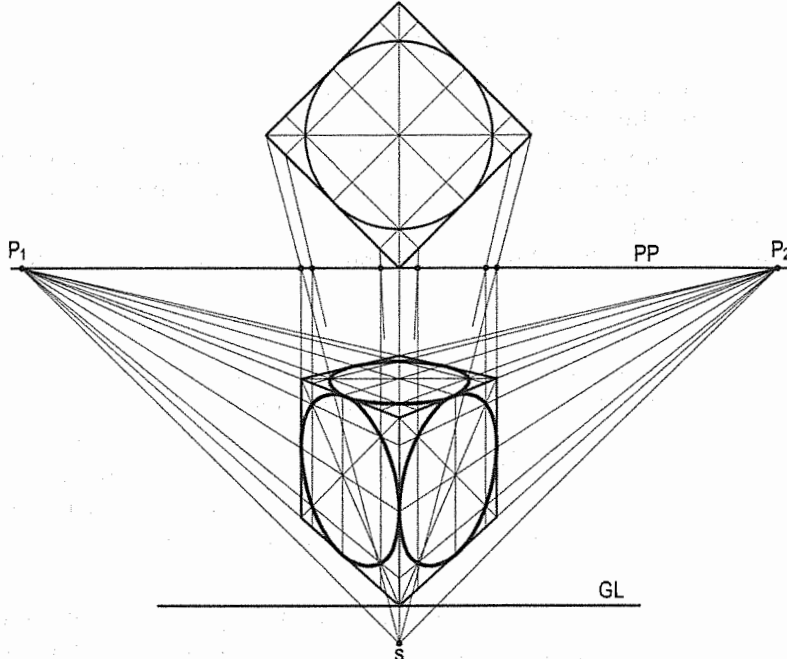


FIG. 19-23

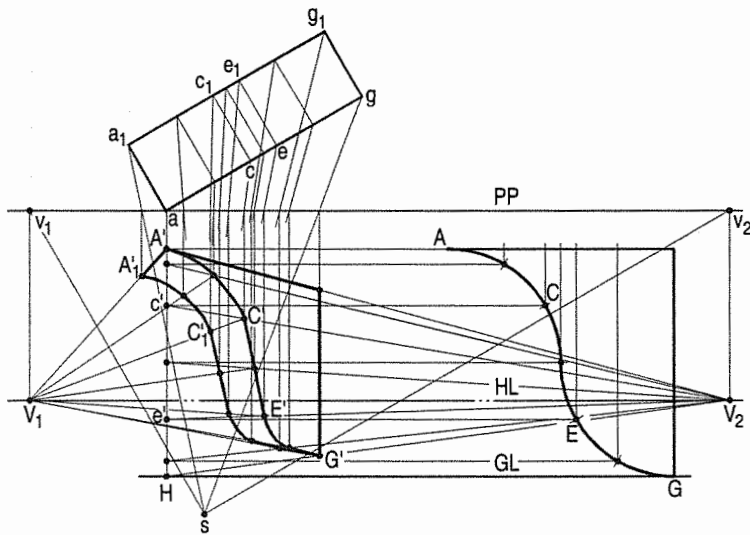


FIG. 19-24

19-12. TYPICAL PROBLEMS OF PERSPECTIVE PROJECTION

Problems 19-10 to 19-15: Draw perspective views of a straight line AB , 30 mm long, in the given positions. The station point is 40 mm in front of the picture plane (PP), 30 mm above the ground plane (GP) and lies in a central plane (CP) in the given positions.

Problem 19-10. (fig. 19-25): AB in GP and inclined at θ to PP; A in PP; CP, 10 mm to the right of A .

(1) **Visual-ray method – by means of the top view and the front view** [fig. 19-25(i)]:

- (i) Draw the top view ab with a in PP.
- (ii) Draw the front view $a'b'$ in GL. Mark s , the top view and s' , the front view of the station point.
- (iii) Join s with a and b , and s' with a' and b' . As A is in PP and on GP, its perspective A' will coincide with a' . B' will lie on $s'b'$ on a vertical through b_1 .
- (iv) Join A' with B' . Then $A'B'$ is the perspective view of AB .

(2) **Visual-ray method – by means of the top view and the side view** [fig. 19-25(ii)]:

- (i) Draw the top view ab and the side view $a''b''$. Mark s , the top view and s'' the side view of the station point.
- (ii) Join s with a and b , and s'' with a'' and b'' . As A is in PP and on GP, its perspective A' will lie on GL on the vertical through a . B' will lie at the point of intersection of the vertical through b_1 and the horizontal through b_2 . $A'B'$ is the required perspective view.

(3) **Vanishing-point method** [fig. 19-25(iii)]:

- (i) Draw the top view ab and mark s , the top view of the station point.
- (ii) Draw sv parallel to ab . Through v , draw a vertical and mark V , the vanishing point on HL. A' will lie on GL on a vertical through a . Join A' with V .

- (iii) Draw a vertical through b_1 and obtain B' on $A'V$. $A'B'$ is the required perspective view.

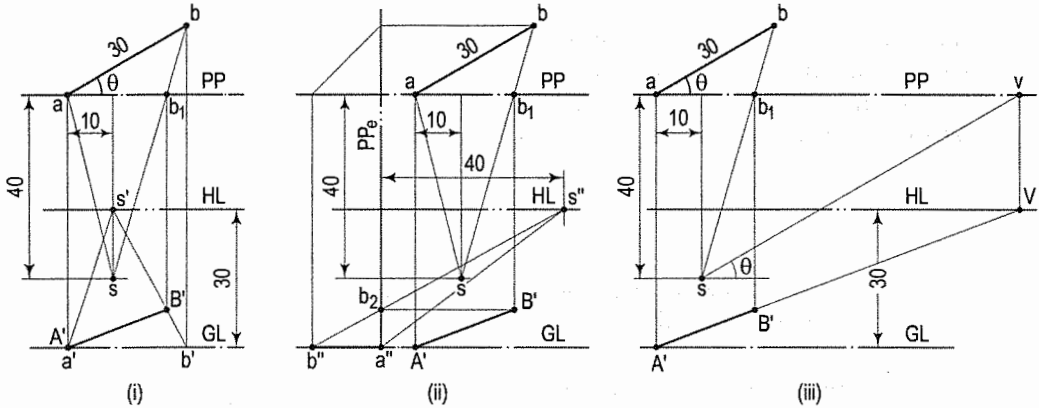


FIG. 19-25

Problem 19-11. (fig. 19-26): AB parallel to GP and 5 mm above it; inclined at 30° to PP ; A , 10 mm behind PP ; C , 10 mm to the right of A .

- Draw the top view ab , mark s and obtain the vanishing point V .
- Join s with a and b . As A is behind PP , the line of heights will be necessary.
- Produce ba to h on PP . Draw the line of heights hH . Mark a point a' on hH , 5 mm above GL . Join a' with V . Obtain points A' and B' on the line $a'V$ as shown $A'B'$ is the required perspective view.

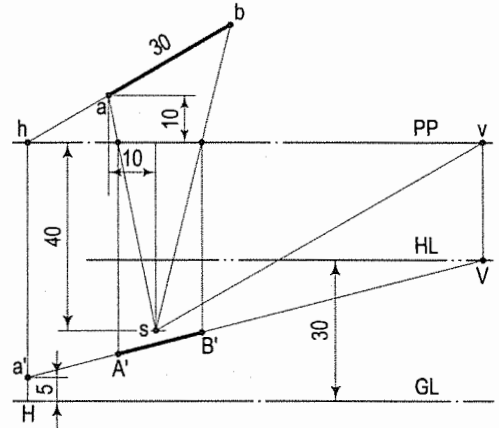


FIG. 19-26

Problem 19-12. (fig. 19-27): AB , perpendicular to PP ; A , 10 mm behind PP and 5 mm above GP , C , 20 mm to the right of AB .

- Draw the top view ab , and the front view $a'b'$. Mark s and s' .
- Join s with a and b , and s' with a' .
- Obtain the perspective $A'B'$ as shown. This is the visual-ray method.

The same figure can be interpreted as the solution by the vanishing-point method. As the line is perpendicular to PP , its vanishing point will be at the centre of vision c' , i.e. at s' on HL . The vertical through a is the line of heights. a' , marked 5 mm above GL , shows the height of A and B both. The perspective $A'B'$ lies on $a's'$, i.e. on $a'c'$.

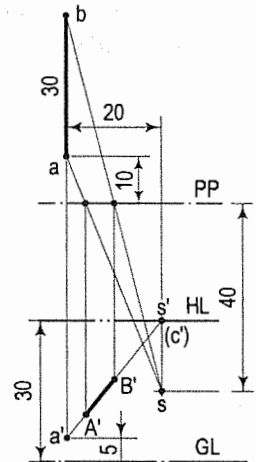


FIG. 19-27

Problem 19-13. (fig. 19-28): AB , parallel to both PP and GP ; 5 mm above GP and 15 mm behind PP ; CP , 15 mm to the left of A .

As the line is parallel to PP , it will have no vanishing point. The perspective view $A'B'$ is drawn by means of the top view and the front view (visual-ray method). It is parallel to $a'b'$. (See note at the end of problem 19-15.)

Problem 19-14. (fig. 19-29): AB , parallel to and 15 mm behind PP ; inclined at 30° to GP ; A , 5 mm above GP ; CP , 15 mm to the left of A .

As AB is parallel to PP , it will have no vanishing point. The perspective $A'B'$ is obtained by means of the top view and the front view (visual-ray method). It is parallel to the front view $a'b'$. (See note at the end of problem 19-15.)

Problem 19-15. (fig. 19-30): AB , perpendicular to GP and 10 mm behind PP ; A , 5 mm above GP ; CP , 20 mm to the right of AB .

As AB is parallel to PP , it will have no vanishing point. a is the top view and $a'b'$ is the front view. The perspective $A'B'$ is obtained by visual-ray method. It is perpendicular to GL .

Note: Problems 19-13, 19-14 and 19-15 may also be interpreted as solved by vanishing-point method. The ends A and B of the line may each be assumed to be the ends of two separate lines perpendicular to the PP and which in perspective view would vanish at the centre of vision c' , i.e. at s' .

These problems may also be termed as cases of parallel perspective. Horizontal, inclined and vertical lines are each parallel to the PP . In their perspective views they remain respectively horizontal, inclined and vertical.

Problem 19-16. (fig. 19-31): A rectangle $ABCD$, 30 mm $\times$ 20 mm, has its surface parallel to and 10 mm above the ground plane. Its shorter edge AD is inclined at 60° to the picture plane which passes through the rectangle so that the corner A is 10 mm in front of it. The station point is 70 mm in front of the picture plane, 40 mm above the ground plane and lies in a central plane which passes through A . Draw the perspective view of the rectangle.

- Draw the line PP and the top view $abcd$ in the given positions. Mark s and join it with the corners a , b , c and d .
- Obtain the vanishing points V_1 and V_2 . Through h (the point at which PP cuts ad), draw the line of heights hH .
- Draw the front view of the rectangle. It is the line AB , parallel to and 10 mm above GL . Project the line to a point H' on hH .

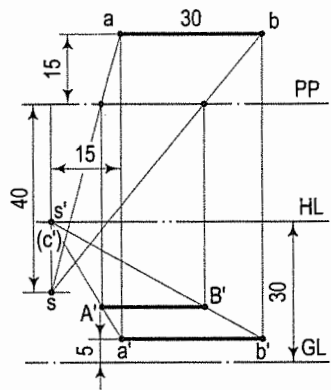


FIG. 19-28

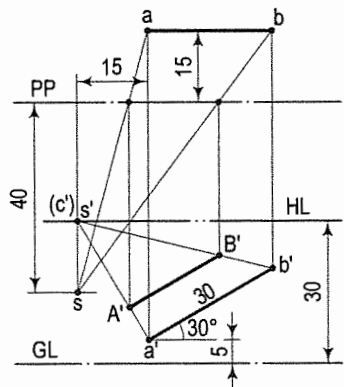


FIG. 19-29

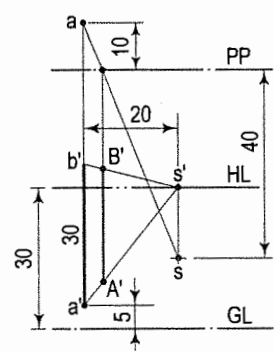


FIG. 19-30

- (iv) Draw a line through H' and V_1 and on it, obtain points A' and D' as shown.
- (v) Draw lines joining V_2 with A' and D' , and on them, obtain points B' and C' .
- (vi) Join B' with C' . Then $A'B'C'D'$ is the perspective view of the rectangle. Note that $B'C'$, if produced, will pass through V_1 .

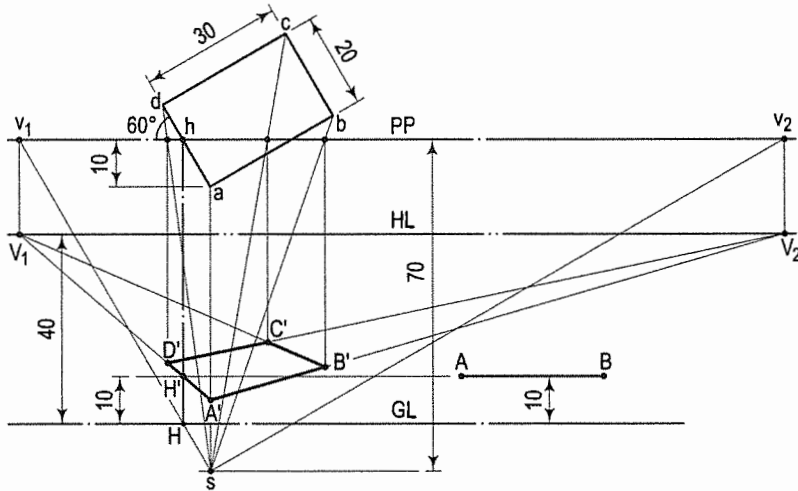


FIG. 19-31

It may also be noted that H' can be marked directly 10 mm above H . Hence, it is not absolutely essential to draw the front view in this particular case. (A line of heights may also be drawn through the point of intersection of ab with PP . In that case, H' will be joined with V_2 and the point B' obtained on $H'V_2$.)

Problem 19-17. (fig. 19-32): Draw the perspective view of a circle of 40 mm diameter, having its surface vertical and inclined at 45° to the picture plane. The centre of the circle is 25 mm above the ground plane. The positions of the station point and the horizon level are as shown in the figure.

- (i) Draw the top view of the circle. Obtain the vanishing point V .
- (ii) Draw the front view of the circle (on the right-hand side) with its centre 25 mm above GL and mark eight points on it. Mark these points on the top view also.
- (iii) Draw the line of heights hH and on it, project horizontally, the eight points from the front view. Join s with the points in the top view, and join V with the points on hH .
- (iv) Draw verticals through the points of intersection of $s1$, $s5$, etc. with PP , and obtain points $1'$, $5'$ etc.

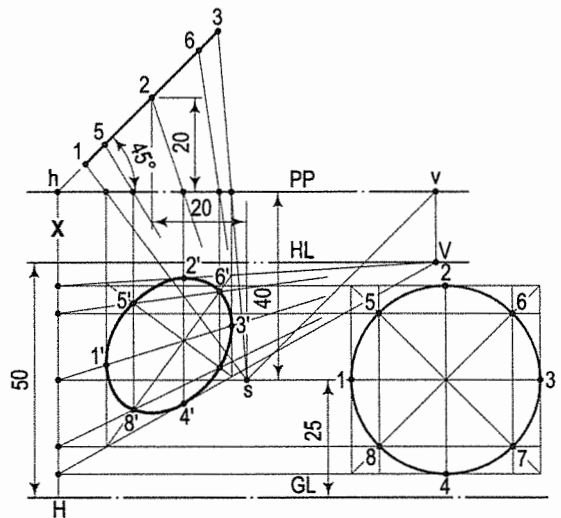


FIG. 19-32

- (v) Draw the ellipse through these points. It is the perspective view of the circle. Note that points 5', 6', 7' and 8' lie on the diagonals of the perspective view of the square in which the circle is enclosed.

Problem 19-18. (fig. 19-33): Draw the perspective view of a pentagonal prism, lying on the ground plane on one of its rectangular faces, the axis being inclined at 30° to the picture plane, and a corner of the base touching the picture plane. The station point is 65 mm in front of the picture plane, and lies in a central plane which bisects the axis. The horizon is at the level of the top edge of the prism.

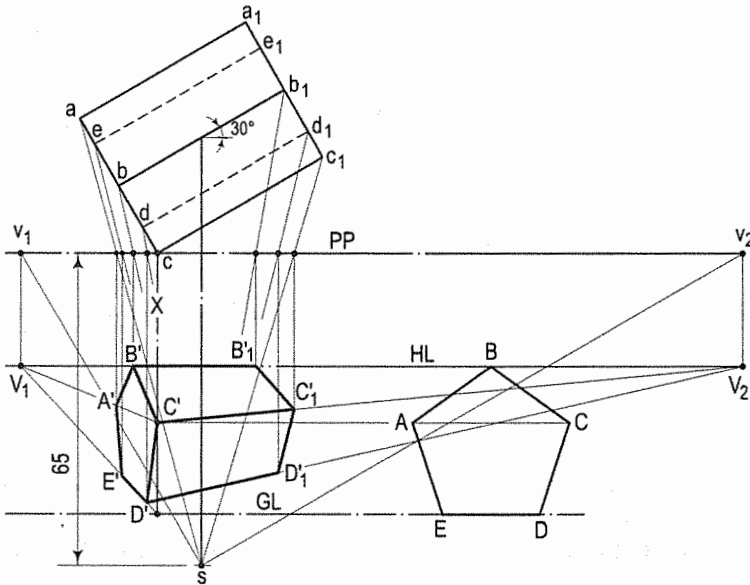


FIG. 19-33

- (i) Draw the top view and the front view of the prism. Mark s and obtain the two vanishing points V_1 and V_2 . Through c , draw the line X which is the line of heights for points $A, B \dots E$. Join s with all the points in the top view.
- (ii) Project horizontally, all the corners in the front view to points on the line X and join these points with V_1 . Obtain points $A', B' \dots E'$ on these lines. Draw lines joining B', C' and D' with V_2 and obtain points B'_1, C'_1 and D'_1 on these lines. Complete the perspective view as shown. The vanishing point V_2 is not shown in the figure. Hidden edges of the prism are also not shown in the perspective view.

Problem 19-19. (fig. 19-34): Draw the perspective view of the frustum of a hexagonal pyramid (top 15 mm side, base 25 mm side and axis 40 mm long), the top view of which is given in the figure, along with the positions of the station point and the horizon.

- (i) Enclose the two hexagons in two rectangles as shown in the figure. (They can be enclosed in different ways also.)
- (ii) Determine the vanishing points V_1 and V_2 . Draw the perspective of the larger rectangle. Note that the corner a is projected back on PP .
- (iii) Draw hH the line of heights for points p and q . On this line, mark the length of the axis viz. $h'H$. Join h' with V_2 and obtain the perspective of the line pq and the whole (small) rectangle.

Problem 19-20. (fig. 19-35): *Letter P.*

Perspectives of semi-circles are drawn by enclosing the outer semi-circle in a rectangle and then marking points at which the diagonals cut the semi-circles.

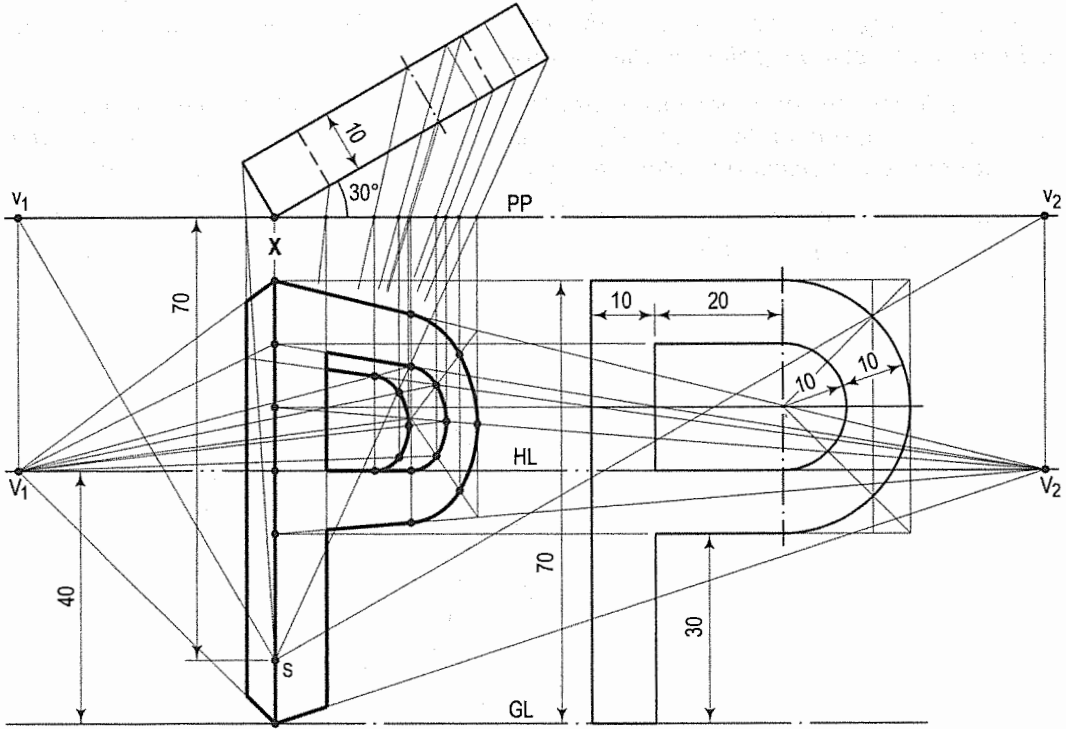


FIG. 19-35

Problem 19-21. (fig. 19-36): *Guide-block.*

This is a problem on parallel perspective. The front face is in PP and hence, it is seen in its true shape and size. It is above the ground plane and therefore above GL. Circles appear as circles. Lines perpendicular to PP converge to the centre of vision c' .

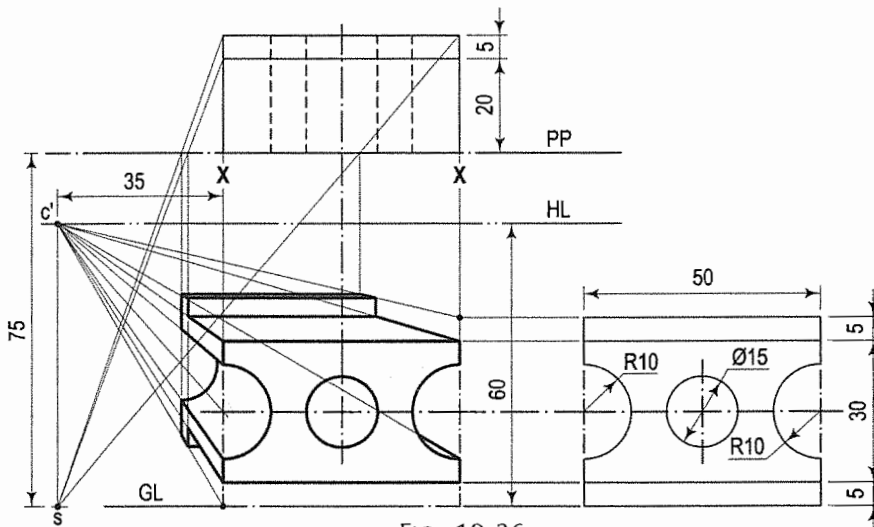


FIG. 19-36

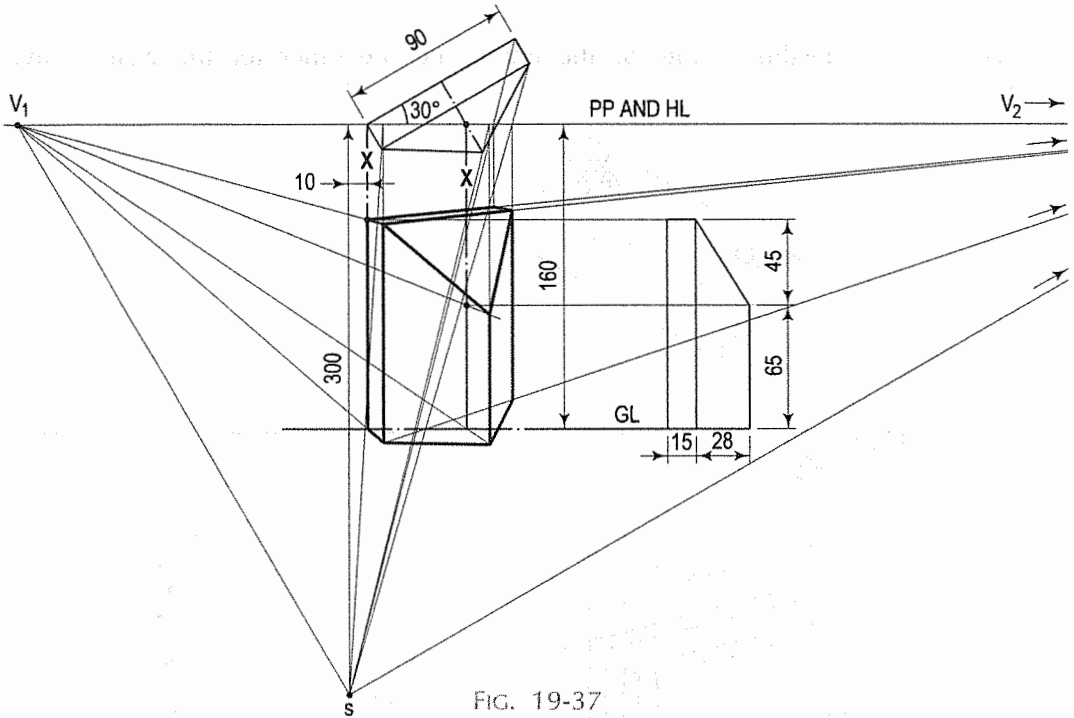


FIG. 19-37

Problem 19-22. (fig. 19-37): *Mile-stone.*

PP passes through the stone.
Hence, the perspective of the front part will appear bigger. A separate line of heights is needed for the front vertical edge.

The station point and vanishing points are not shown in the figure. HL is shown coinciding with PP.

Problem 19-23. (fig. 19-38):
A square prism with a square pyramid placed centrally on its top.

This is also a case of parallel perspective, but some portions of the solids are in front of PP. The front corners are projected back on PP. Hence, these portions appear bigger in size.

Two lines of heights have been used for the prism (though even one is sufficient). Corners of the base of the pyramid are at the mid-points of the sides of the top of the prism. A separate line of heights is needed for the apex.

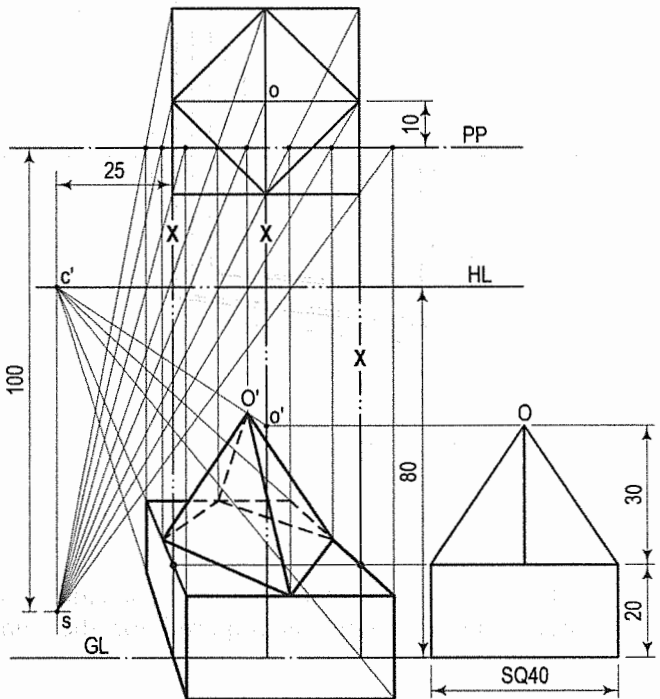


FIG. 19-38

Problem 19-24. (fig. 19-39): *Window-frame placed in a wall.*

Two lines of heights – one for the frame and the other for the wall – have been used.

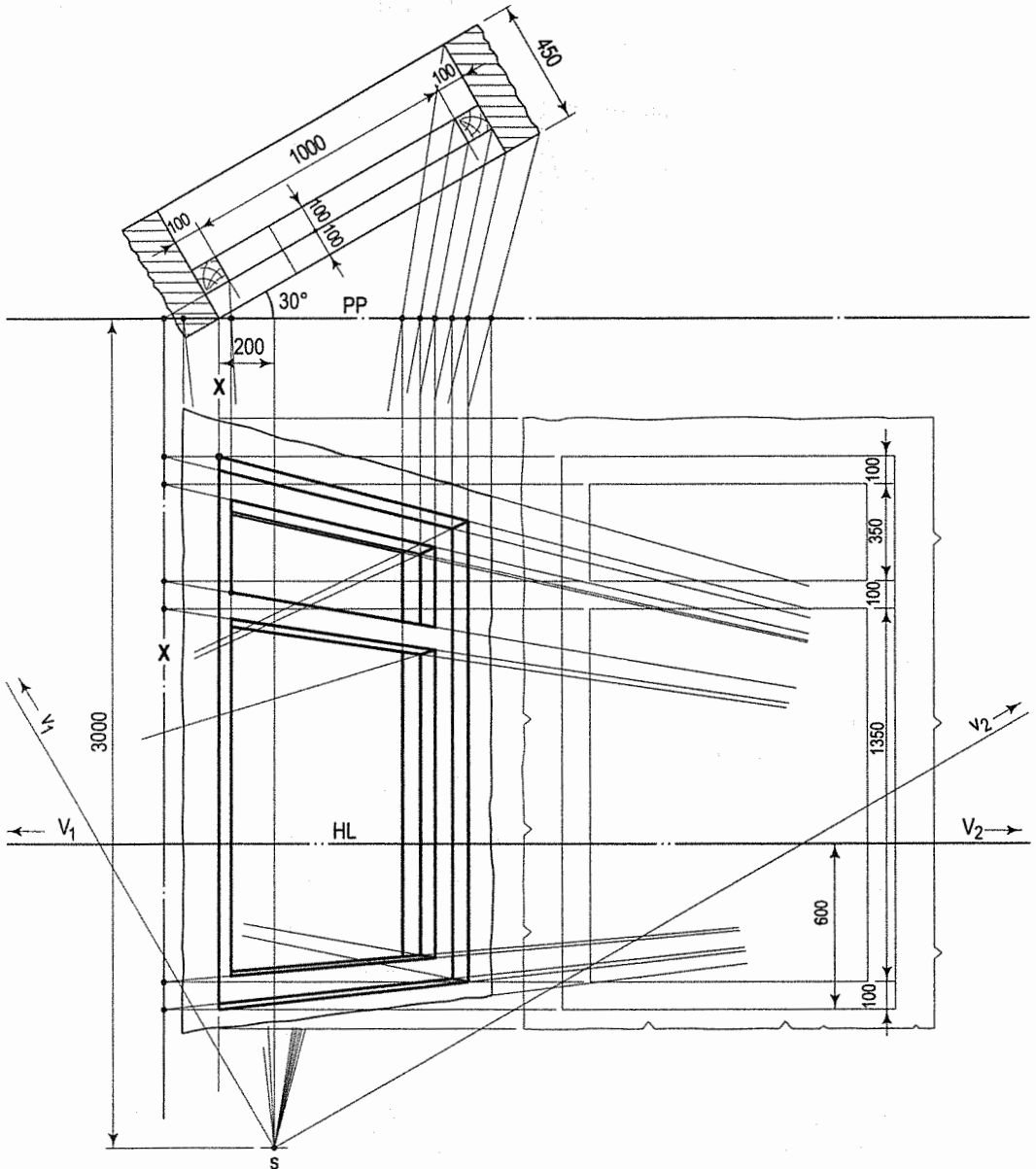


FIG. 19-39

Problem 19-25. (fig. 19-40): *Casting.*

GL has been so placed that HL coincides with PP. It may be taken lower. A separate line of heights is required for the central rib.

Problem 19-26. (fig. 19-41): *Ogee arch in a wall.*

A number of points are marked on the arch and the curves are drawn through the perspectives of these points.

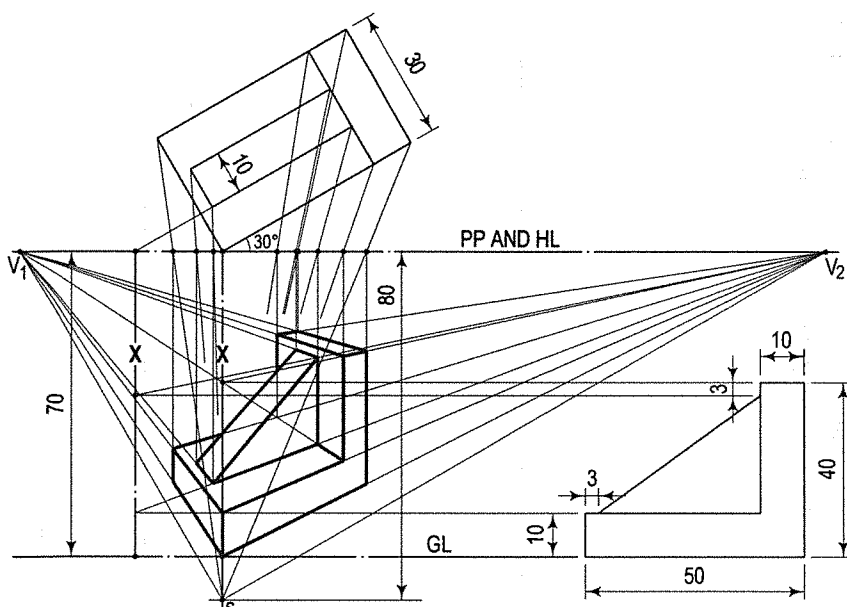


FIG. 19-40

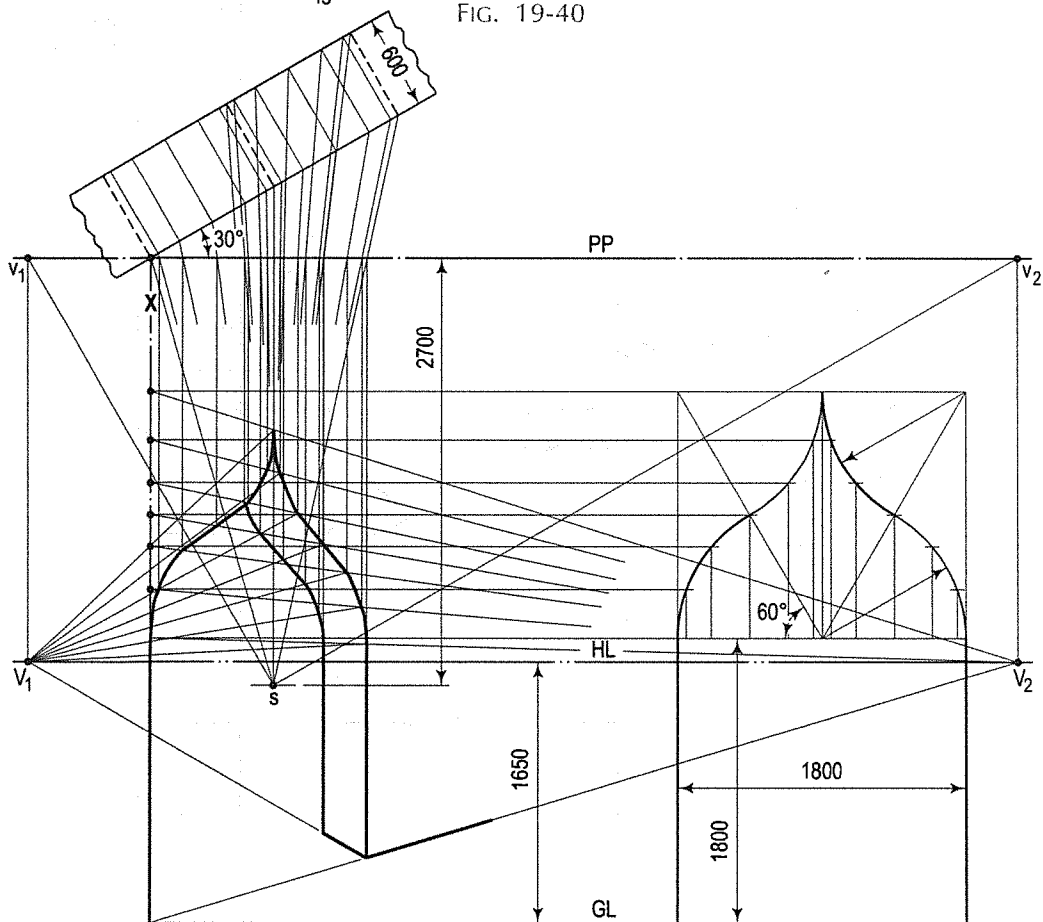


FIG. 19-41

Problem 19-27. (fig. 19-42): *H* door and a side window. All dimensions in metres.

Note that a corner of the roof is 120°. Four lines of heights are needed.

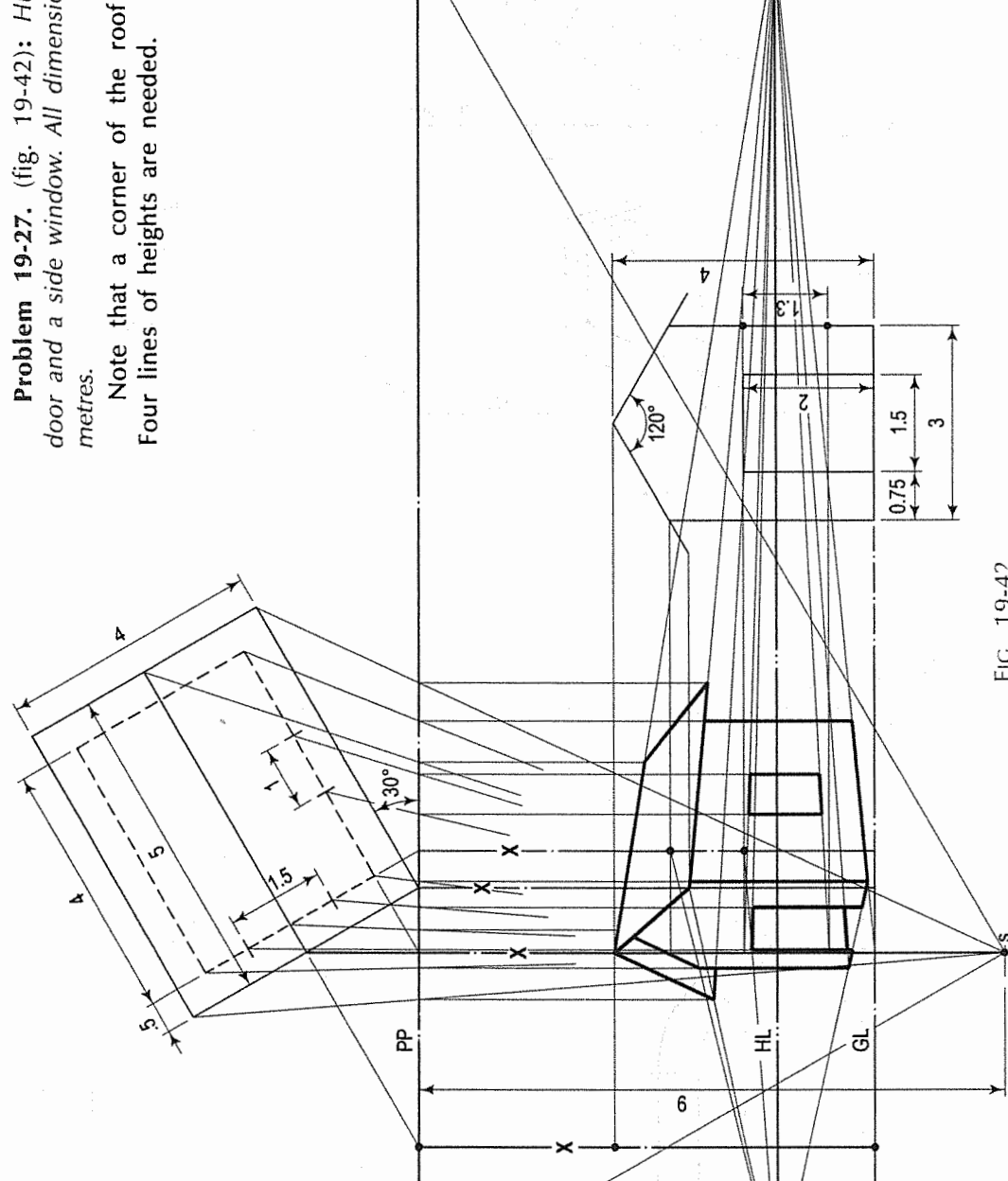


FIG. 19-42

Problem 19-28. (fig. 19-43): *Sign-post.*

Four lines of heights have been used. Vanishing points not shown in the figure.

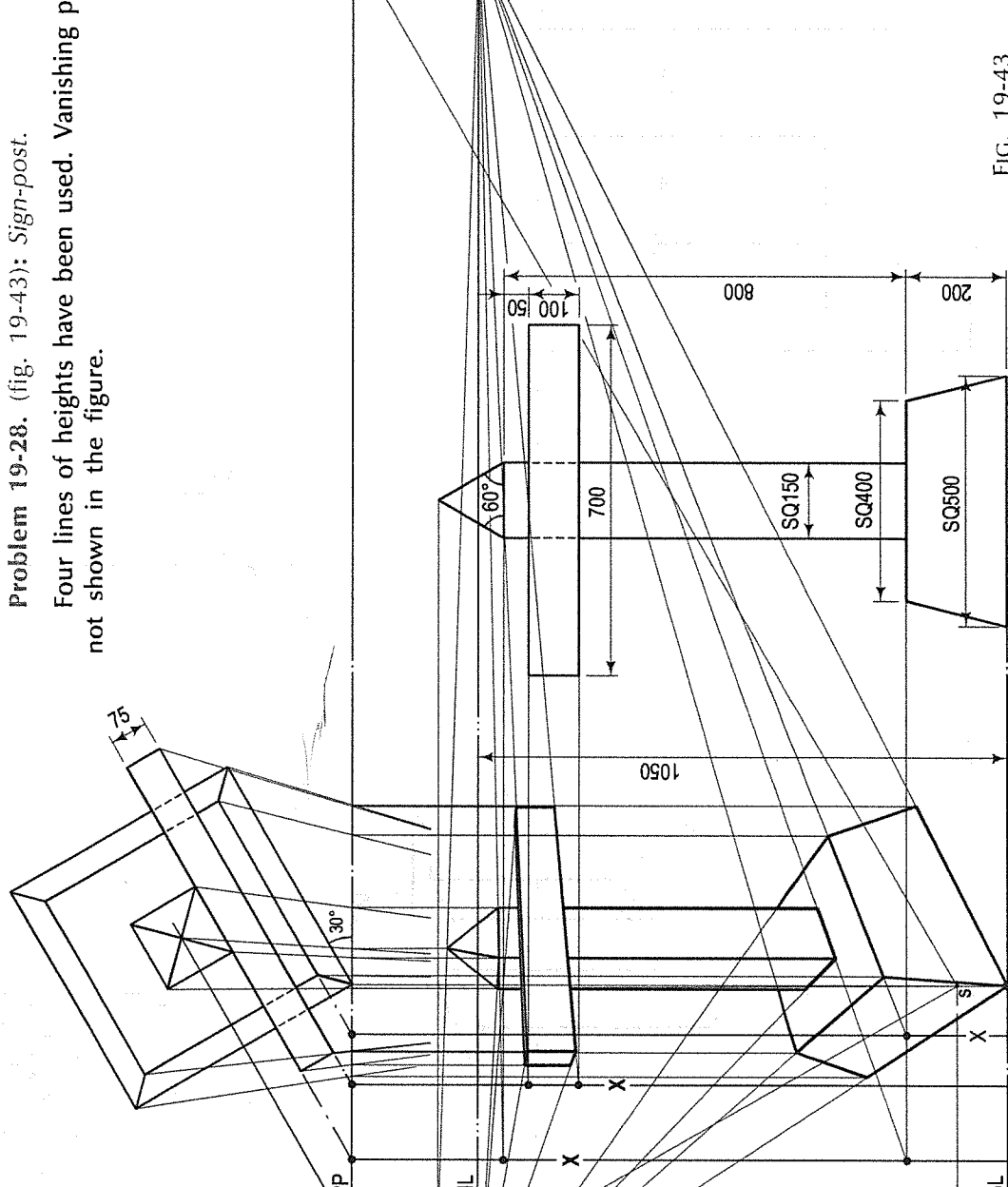


FIG. 19-43

Problem 19-29. (fig. 19-44): *Model of steps.*

This is also a problem on parallel perspective, but the front face is behind PP. Hence, that face will be smaller in size.

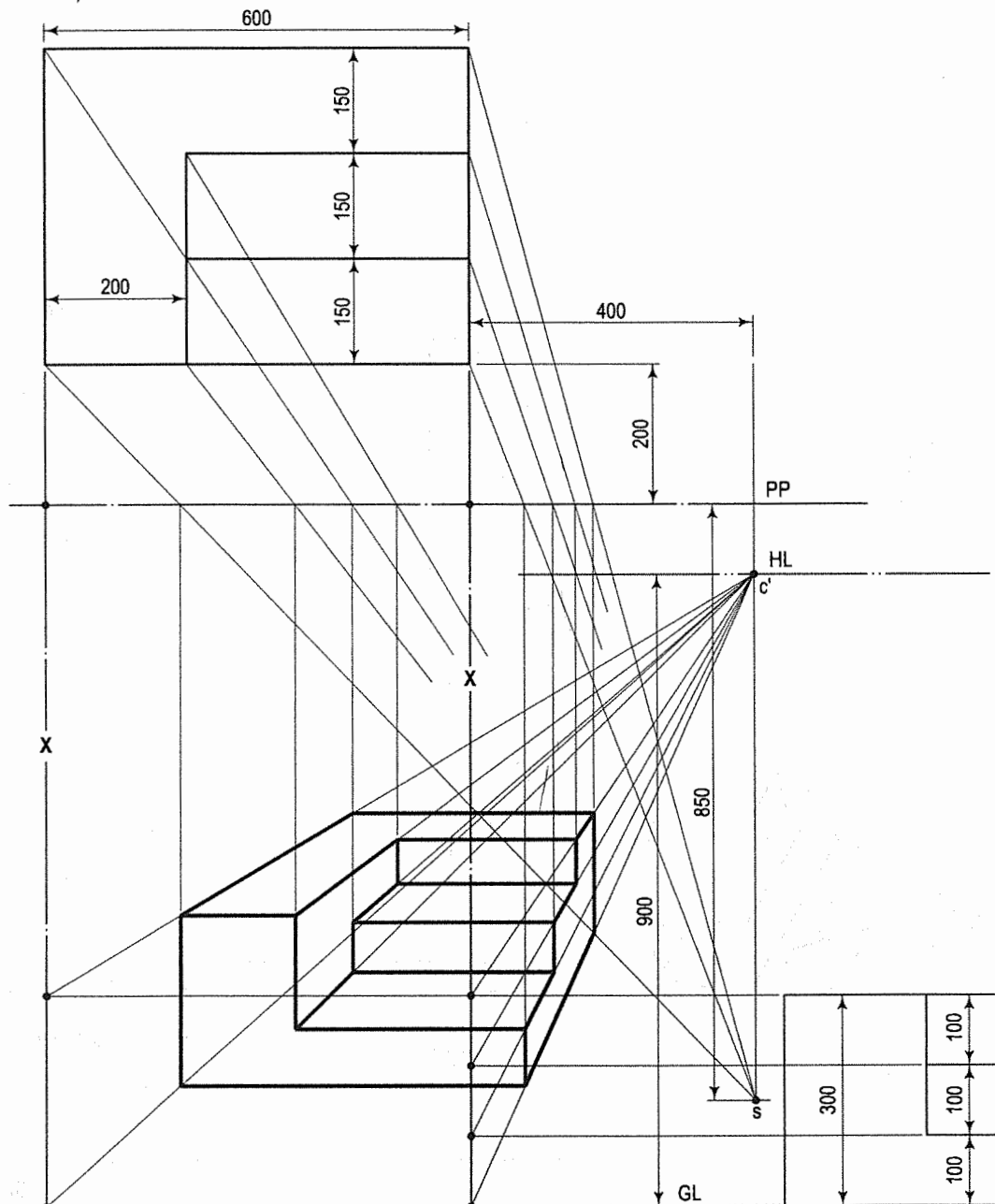


FIG. 19-44

Problem 19-30. (fig. 19-45): *Lamp-post.*

The back edges (of the lamp casing) which are visible through the glass are shown thinner.

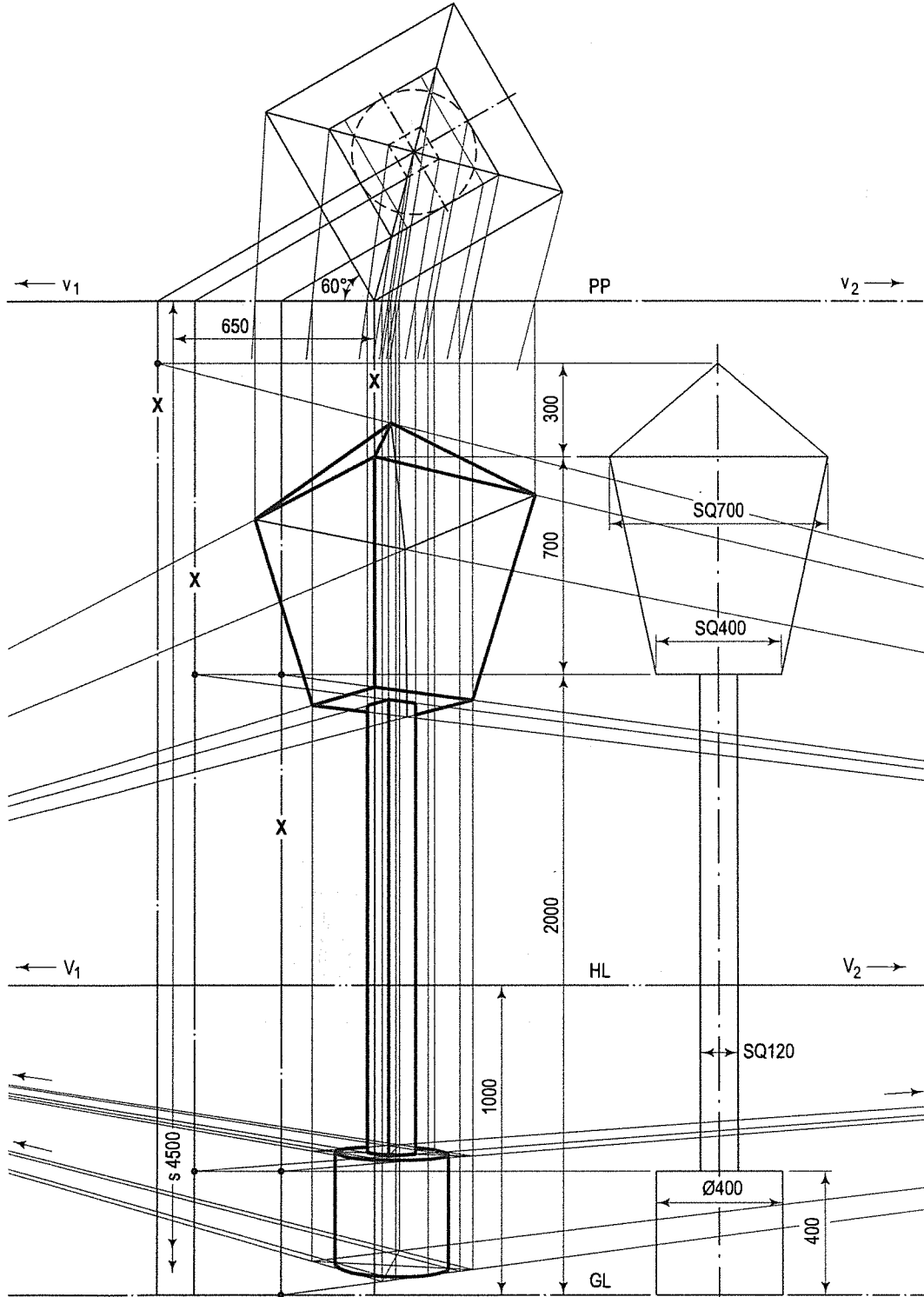


FIG. 19-45

Problem 19-31. (fig. 19-46): Steps with guards and
Two lines of heights are used for convenience.

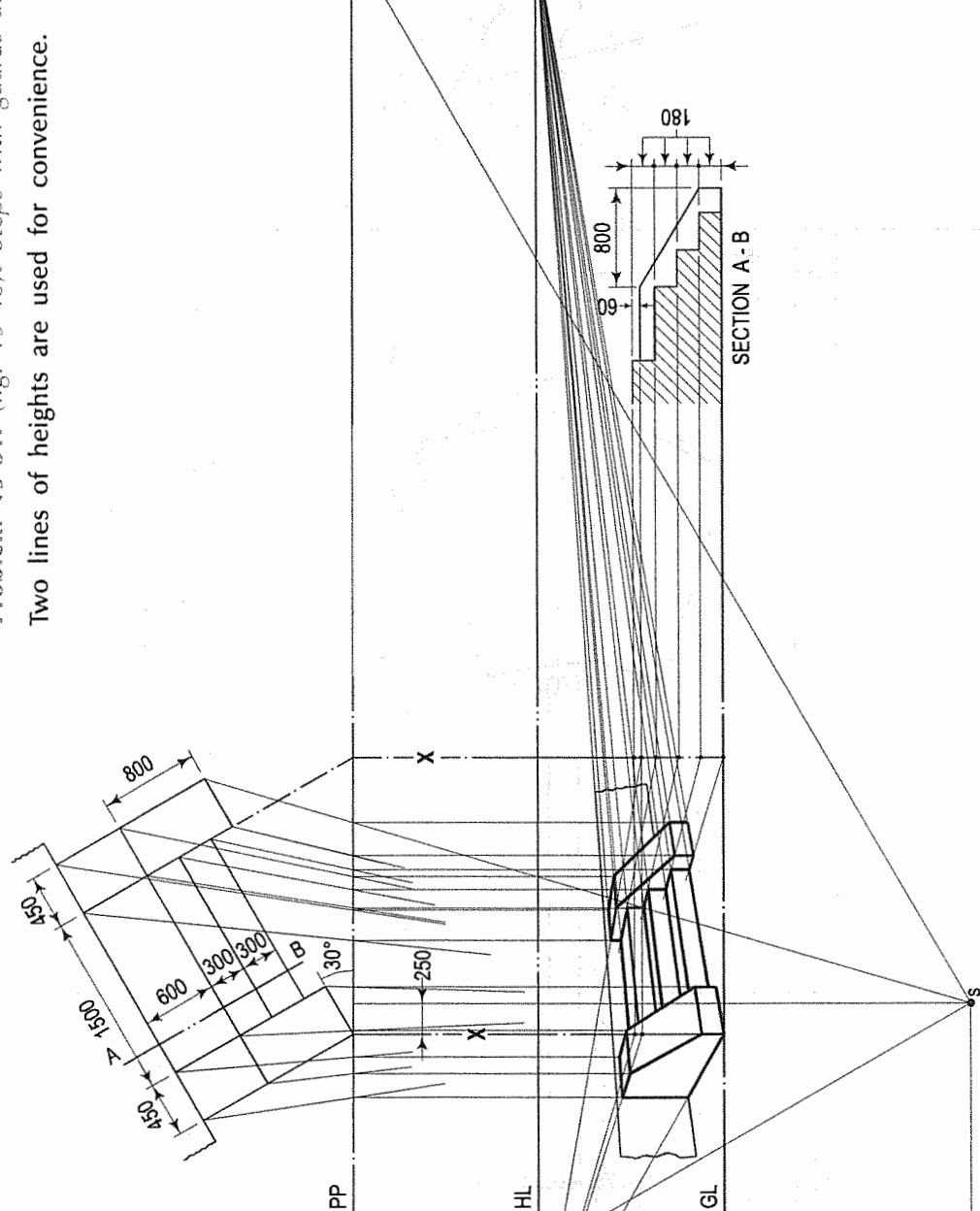


FIG. 19-46

EXERCISES 19

1. A man stands at a distance of 5 m from a flight of four stone steps having a width of 2 m, tread 0.3 m and rise 0.2 m. The flight makes an angle of 45° with the picture plane and touches the same at a distance of 2 m to the right of the centre of vision. Draw the perspective view of the flight.
2. Draw the perspective view of a square pyramid of base 100 mm side and height of the apex 120 mm. The nearest edge of the base is parallel to and 30 mm behind the picture plane. The station point is situated at a distance of 300 mm from the picture plane, 60 mm above the ground plane and 200 mm to the right of the apex.
3. Draw the perspective view of the model of steps shown in fig. 17-57. The position of the steps relative to the picture plane is shown in fig. 19-47. The station point is 200 mm from the picture plane. Take horizon level to be 100 mm above the ground level.
4. Draw the perspective view of the memorial shown in fig. 19-48. The horizon level is 8 m above the ground plane, while the observer is stationed at a distance of 12 m from the picture plane. (All dimensions are in metres.)

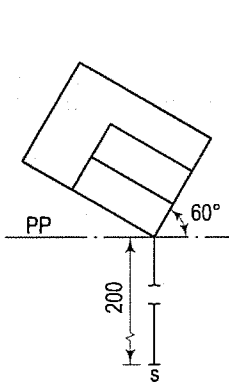


FIG. 19-47

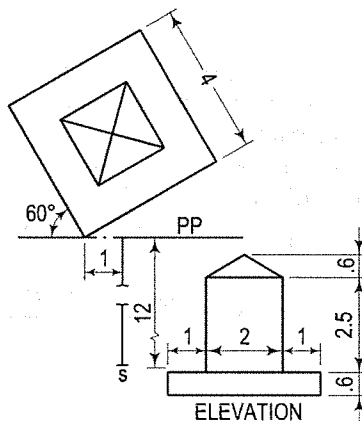


FIG. 19-48

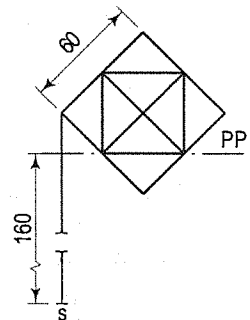


FIG. 19-49

5. Fig. 19-49 shows the top view of a square prism of 40 mm thickness with a square pyramid of 60 mm long axis, placed centrally on it. The station point is 160 mm from the picture plane and 100 mm above the ground plane. Draw the perspective view of the solids.
6. Draw the perspective view of the box shown in fig. 17-61. The top view of the box with the lid 90° open, along with the picture plane and the station point is shown in fig. 19-50. Assume horizon level to be 200 mm above the ground level.
7. Draw the perspective view of a semi-circular arched opening in a wall having the length of 4 m, thickness 0.5 m and height 3.5 m. The opening is 1.5 m wide and the springing points are at a height of 2 m. The wall makes an angle of 45° with the picture plane. Select a suitable position of the spectator.

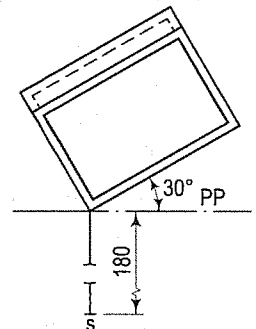


FIG. 19-50

8. Draw the perspective view of the wooden cabinet shown in fig. 19-51. The thickness of wood is 20 mm. The cabinet is placed with one edge touching the picture plane and 1000 mm to the left of the centre of vision. The station point is 2500 mm away from the picture plane and 1500 mm above the ground plane.

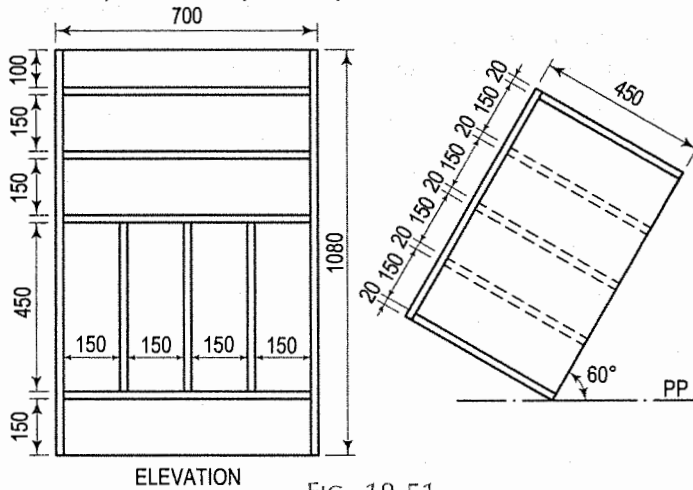


FIG. 19-51

9. Draw the perspective view of the pedestal for a statue (neglecting the sphere) shown in fig. 17-127(5). The edges of the base, as shown in fig. 19-52, are equally inclined to the picture plane. Assume the horizon plane to be 150 mm above the ground plane.

10. The casting shown in fig. 17-127(2) is placed behind the picture plane as shown in fig. 19-53. Draw its perspective view, assuming the horizon plane to be 80 mm above the ground plane.

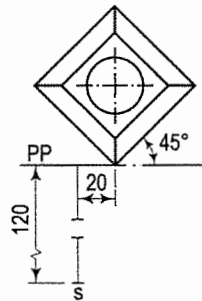


FIG. 19-52

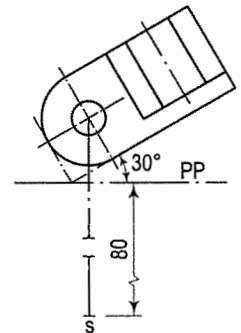


FIG. 19-53

11. Draw the perspective view of the model of a memorial shown in fig. 19-54. The station point is 200 mm in front of the picture plane and 160 mm above the ground plane.

12. Draw the perspective view of a vertical steel cup-board 1 m $\times$ 2 m $\times$ 0.35 m deep, having four shelves. One shutter is open. The front of the cup-board makes an angle of 30° with the picture plane, with a vertical edge touching it. Assume suitable position of the spectator at a height of 4 m above the ground.

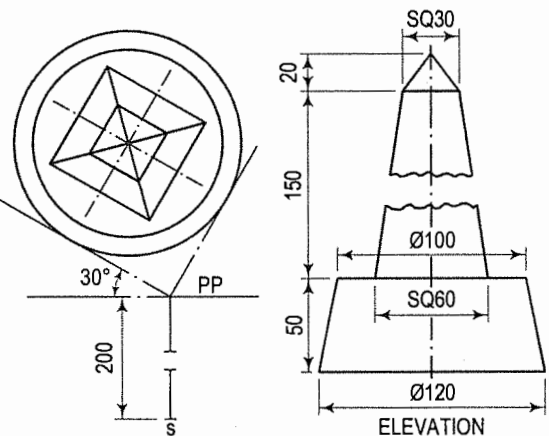


FIG. 19-54

Chapter 20



ORTHOGRAPHIC READING AND CONVERSION OF VIEWS

20-1. INTRODUCTION



Orthographic reading is the ability to visualize the shape of an object from its drawing in orthographic views. Every engineer or technician connected with the work of construction should possess this ability. Without it, it would be difficult for him to execute, independently, any work according to a given drawing.

An engineering drawing is not read aloud. It is read mentally. The whole drawing cannot be read or interpreted at a glance. It should be read systematically and patiently. The easiest way to learn reading such a drawing is to learn how to prepare one.

However, it is not impossible to know how to read it without learning how to draw. In either case, a sound knowledge of the principles of orthographic projection is quite essential for reading the drawing without hesitation.

We studied in chapter 8 that *in orthographic projection, any one view shows only two dimensions of a three dimensional object. Hence, it is impossible to visualize the shape of the object from a single view. The second view shows the third dimension. Thus, at least two views are necessary to determine its shape.* Sometimes, a third view is also necessary to completely visualize an object.

20-2. READING OF ORTHOGRAPHIC VIEWS (BLUE-PRINT READING)



Every object may be imagined as consisting of a number of components having forms of simple solids such as prisms, cylinders, cones, etc. with some additions or subtractions or both. The additions may be in the form of projections, while the subtractions may be in the form of holes, grooves, cavities, etc. It is not possible to determine from only one view whether there is an addition or subtraction. The other view or views must be referred to.

For example in fig. 20-1(i), the meaning of lines AB and CD can be determined only after referring to the other view. They might represent a projection as shown in figs. 20-1(ii) and (iii) or a cavity as shown in figs. 20-1(iv) and (v) or a cavity made by projections as shown in fig. 20-1(vi).

Note that one *front-view* represents six different types of objects whose *top-view* are different as shown in fig. 20-1.

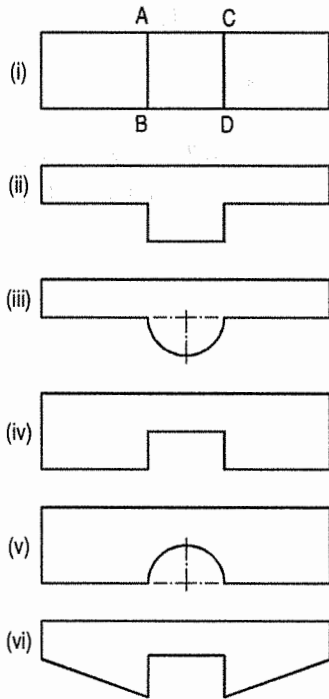


FIG. 20-1

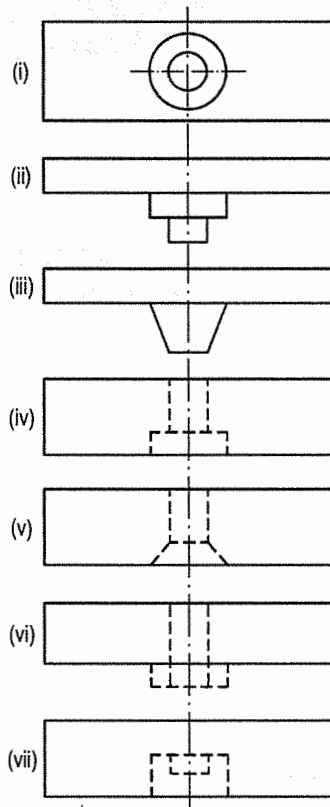


FIG. 20-2

Similarly, the two circles in fig. 20-2(i) may either represent additions [figs. 20-2(ii) and (iii)] or holes [figs. 20-2(iv) and (v)] or both [figs. 20-2(vi) and (vii)]. Thus, every point and every line in an orthographic view has a meaning. A point may represent a corner or an edge. A line may represent an edge or a surface. The meaning of each point or line should be interpreted by systematically referring back and forth from one view to the other. Simultaneously, the shape of the object as a whole should be visualized.

Practice in reading a drawing can be had by either projecting one or more views from the given views, or by converting one or both the given views into sectional views.

20-3. MISSING LINES AND MISSING VIEWS

Most of the students in Engineering Drawing course find it difficult to visualize an object from two orthographic views. To overcome this difficulty, assume the object as basic geometrical solids such as prisms, cylinders, pyramids, cones and so on. It is impossible to determine whether the shape is addition or subtraction by observing only one view. Therefore, the each orthographic view of an object must be referred systematically back and forth. Sound knowledge of projections of points, lines, planes and solids is required in solving the problems of missing lines

or missing views. In orthographic views, if a line or lines are not drawn in order to test the understanding of students, such lines are known as *missing lines*. Similarly, if amongst the three views, only two views are given then third view will be known as *missing view*.

Following procedure can be adopted in order to identify missing lines or missing views.

- (i) From the given orthographic views, try to visualize an object and prepare a pictorial view.
- (ii) From pictorial view, prepare orthographic views and compare with given views. Read carefully each line in each view and find out corresponding projection from another view.

Following conclusion from the projections of lines and planes will be useful for identification of lines and planes in the given views.

- (a) When a line is perpendicular to a plane, its projection on that plane is point, while its projection on the other plane is a line equal to its true length.
 - (b) When a line is inclined to both the planes (H.P. and V.P.), its projections are shorter than the true length in both the plane and inclinations to xy line are greater than true inclinations.
 - (c) When a line is parallel to a plane, its projection on that plane will show its true length and true inclination with the other plane.
 - (d) When a plane is perpendicular to a reference (principal planes H.P. and V.P.) plane, its projection on that plane is a straight line.
 - (e) When a plane is parallel to a reference plane, its projection on that plane shows its true shape and size.
 - (f) When a plane is perpendicular to one of the reference plane and inclined to the other, its inclination is shown by the angle which its projection on the plane to which it is perpendicular, makes with xy line. Its projection on the plane to which it is inclined, is smaller than the plane it self.
- (iii) Remember that each line in one view, represents a plane in another view depending upon the position of the plane with reference to the plane of projection. When a line is observed in any two views, it will be plane in third view.

Study carefully following problems of missing lines or missing views.

Problem 20-1. In fig. 20-3(i) two views are given. Draw its missing view and show missing lines.

See fig 20-3(ii).

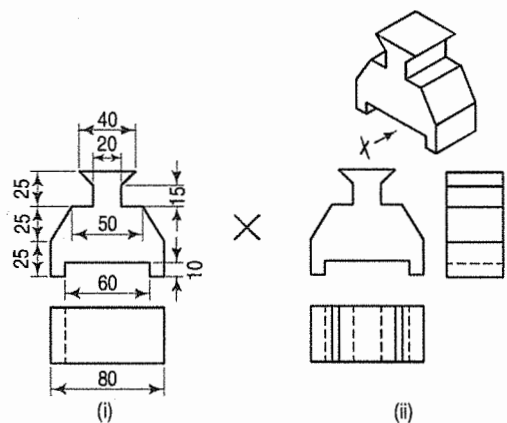


FIG. 20-3

Problem 20-2. In fig. 20-4(i) two views are given. Draw its pictorial view and missing view.

See fig 20-4(ii).

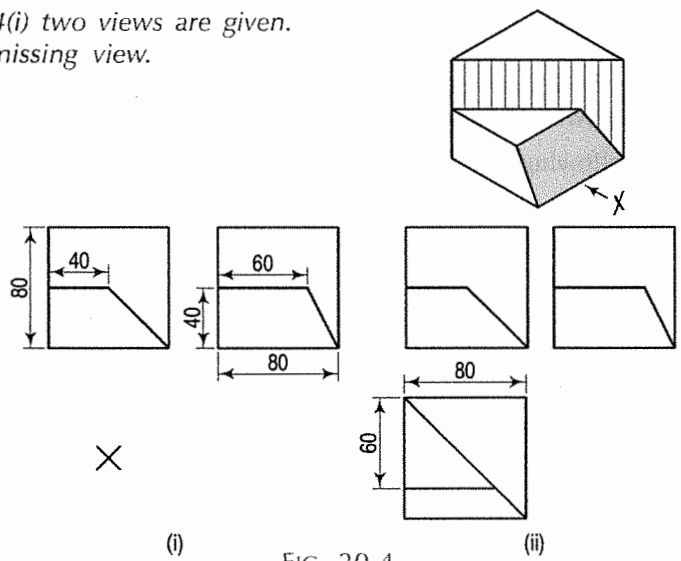


FIG. 20-4

Problem 20-3. In fig. 20-5(i) two views are given. Draw its missing view and show missing lines along with its pictorial view.

See fig 20-5(ii)

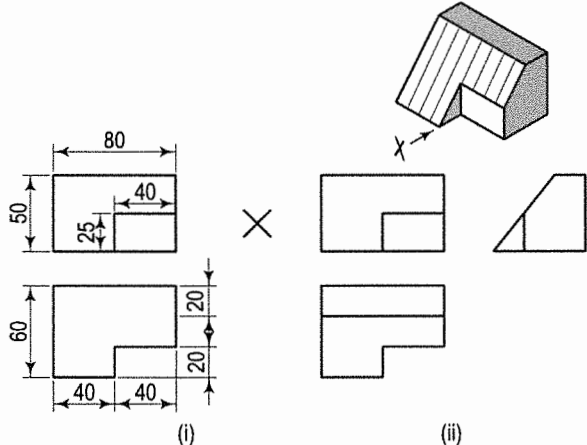


FIG. 20-5

Problem 20-4. In fig. 20-6(i) two views are given. Draw its pictorial view and missing view.

See fig 20-6(ii).

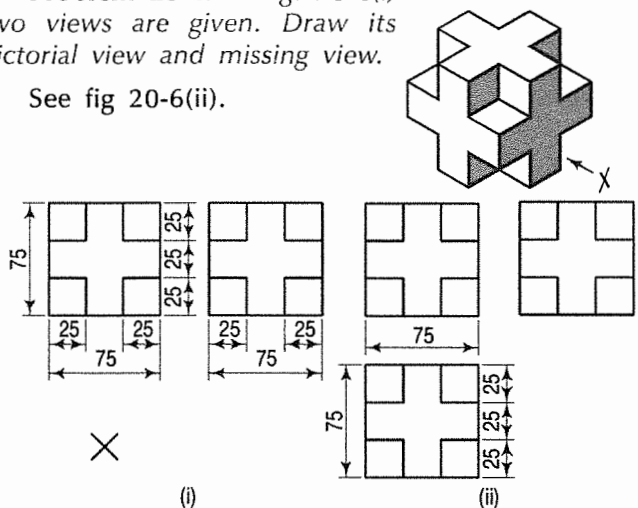


FIG. 20-6

20-4. IDENTIFICATION OF PLANES

Problem 20-5. Two views (front view and top view) are given in fig. 20-7(i). Draw third view (side view) and identify each plane.

See fig. 20-7(ii) along with pictorial view. Read each line and the plane in different views.

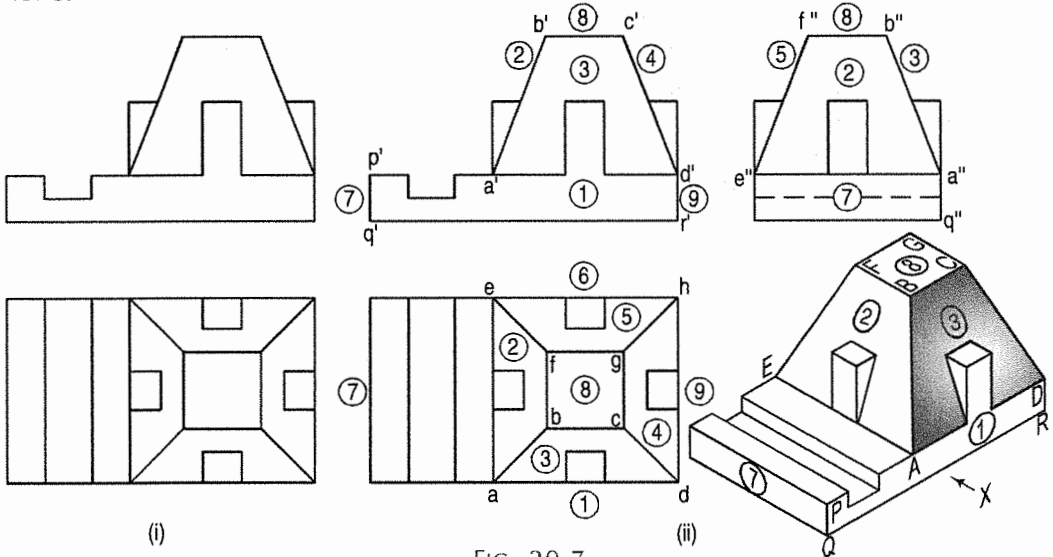


FIG. 20-7

Problem 20-6. Two views of a casting are shown in fig. 20-8. Using both first angle projection method and third angle projection method, draw the given views and add following views

- the side view looking from the right and
- a sectional side view, section A-A.

Step 1: The shape of the casting may be visualized by imagining it to be broken up in three components B, C and D.

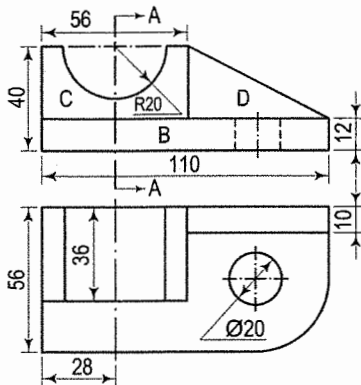


FIG. 20-8

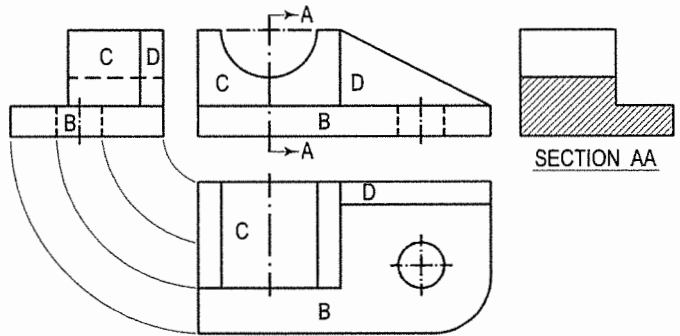


FIG. 20-9

Step 2: After referring to both the views, it can be visualized that the part B is in the form of a rectangular block with one corner rounded, and having a circular hole.

The component C has a semi-circular cavity at its top, while the part D is a wedge-shaped piece of uniform thickness. The side view of each part may be imagined separately and then projected from the two views as shown in fig. 20-9.

Step 3: The sectional side view is obtained by imagining the block to be cut in two parts through the line A-A. The portion that is cut is shown in section by means of section lines. Note that the curved surface of the cavity behind the cutting-plane line is also shown as a rectangle in the sectional view. Dashed lines for hidden features may not be shown in the sectional view.

Fig. 20-10 shows the four views of the casting drawn according to the third-angle projection method. There is no change in shape or size of the views. Only their positions are changed.

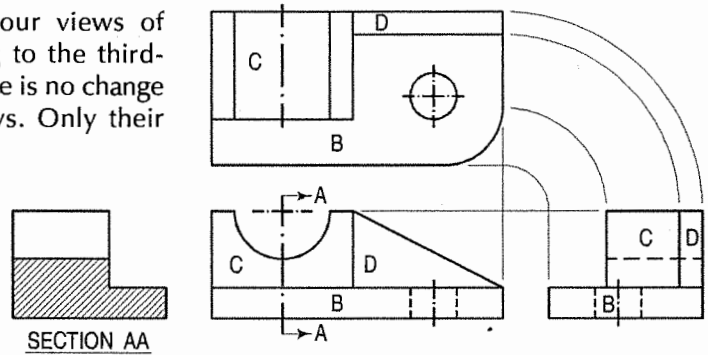


FIG. 20-10

Problem 20-7. Two views of a shaft support are given in fig. 20-11. Draw, scale full size,

- (i) Sectional front view on A-A.
- (ii) Sectional top view on B-B.
- (iii) Side view from the left.

The object is composed of two cylinders and two ribs. Follow step 1, step 2 and step 3 of problem 20-6.

See fig. 20-12. Note that, although the cutting-plane line A-A passes through the rib, it is not shown cut in the front view according to the convention.

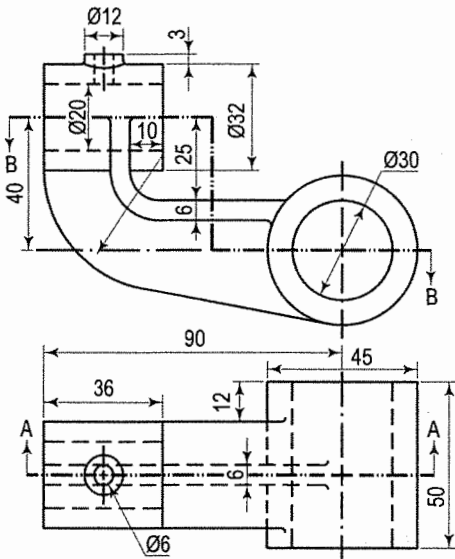


FIG. 20-11

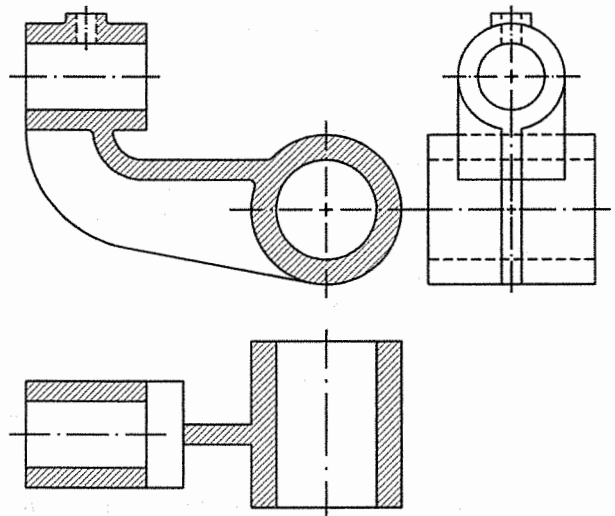


FIG. 20-12

20-5. CONVERSION OF PICTORIAL VIEWS INTO ORTHOGRAPHIC VIEWS



Having grasped the principles of orthographic and isometric projection, we proceed to deal with the application of the same to conversion of pictorial views of objects, for preparing the orthographic views.

Conversion of a pictorial view into orthographic views requires sound knowledge of the principles of pictorial projection and some imagination. A pictorial view may have been drawn according to the principles of isometric or oblique projection. In either case, it shows the object as it appears to the eye from one direction only. It does not show the real shapes of its surfaces or the contour. Hidden parts and constructional details are also not clearly shown. All these have to be imagined.

For converting a pictorial view of an object into orthographic views, the direction from which the object is to be viewed for its front view is generally indicated by means of an arrow. When this is not done, the arrow may be assumed to be parallel to a sloping axis. Other views are obtained by looking in directions parallel to each of the other two axes and placed in correct relationship with the front view.

When looking at the object in the direction of any one of the three axes, only two of the three overall dimensions (viz. length, height and depth or thickness) will be visible. Dimensions which are parallel to the direction of vision will not be seen. *Lines which are parallel to the direction of vision will be seen as points, while surfaces which are parallel to it will be seen as lines.*

While studying a pictorial view, it should be remembered that, *unless otherwise specified:*

- (i) A hidden part of a symmetrical object should be assumed to be similar to the corresponding visible part. For example, in fig. 20-49, the back rib is assumed to be similar as the front rib.
- (ii) All holes, grooves etc. should be assumed to be drilled or cut right through. Refer fig. 20-18 and fig. 20-19.
- (iii) Suitable radii should be assumed for small curves of fillets etc.

An object in its pictorial view may sometimes be shown with a portion cut and removed, to clarify some internal constructional details. While preparing its orthographic views, such object should be assumed to be whole, and the views should then be drawn as required.

20-6. ORTHOGRAPHIC PROJECTION



- (i) It is comparatively easy to prepare a drawing from an actual object. The object is carefully examined and then placed in a suitable position for the front view. The front view of the object is selected in such a way that the maximum details of the object are visible so that the minimum dotted lines are required.
- (ii) All the necessary views are then sketched freehand in a sketch book or on a pad.

- (iii) Measurements of all its details and overall sizes are taken and inserted in the views, along with important notes and instructions.
- (iv) Finally, a scale-drawing is prepared from these sketches.

A pictorial view of a rectangular plate is given in fig. 20-13(i). Its front view when seen in the direction of the arrow X, side view from the left, i.e. in the direction of the arrow Y and the top view in the direction of arrow Z, are shown in fig. 20-13(ii).

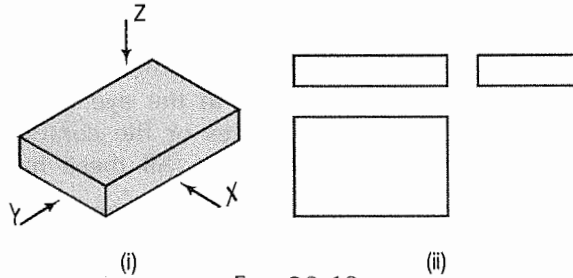


FIG. 20-13

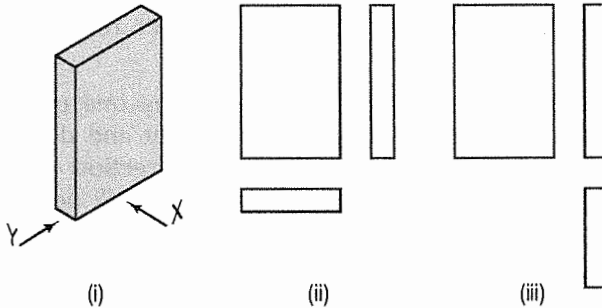


FIG. 20-14

The same plate is shown in fig. 20-14(i) with its longer edges vertical. Its front view looking in the direction of the arrow X, side view from the left and the top view are shown in fig. 20-14(ii). These three views are similar in shape and size to the views shown in fig. 20-13(ii). Only their positions and conditions have changed. Even when the front view is drawn looking in the direction of the arrow Y [fig. 20-14(iii)], the three views remain similar in shape and size.

The same plate is shown cut in various shapes in fig. 20-15. The front view and the side view in each case will be the same as in fig. 20-13(ii). The changed shapes will be seen in the top view of each plate. If these plates are kept in the position shown in fig. 20-14(i), the front view in each case (looking in direction of the arrow X) will show a different shape, while the side view and the top view will be rectangles.

A plate cut in three different ways is shown in fig. 20-16. The front view in each case is the same, viz. a rectangle with a vertical line ab for the edge AB . The side view from the right in each case will be a rectangle with a vertical line cd for the edge CD . The top view in each case shows the shape of the cut.

In the front view of the plate having different shapes of grooves (fig. 20-17), two vertical lines are drawn for the edges of rectangular as well as semi-circular grooves.

In case of the triangular groove, three vertical lines are required. Although edges *AB* and *CD* are cut, they are seen as continuous lines *ab* and *cd*. In the top view, shapes of the grooves are seen. The grooves are not visible in the side view and hence, they are shown by a hidden line.

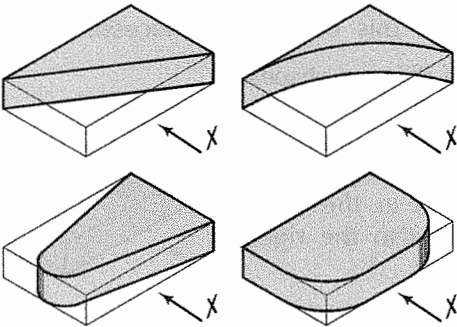


FIG. 20-15

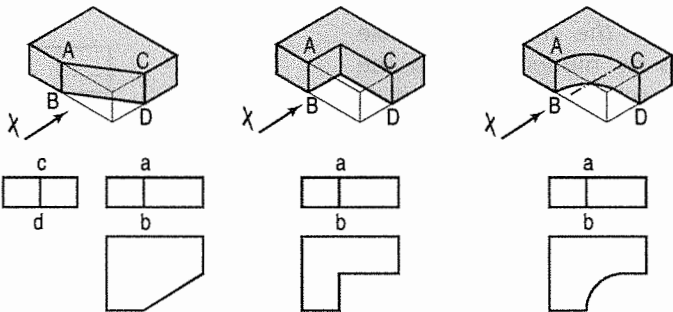


FIG. 20-16

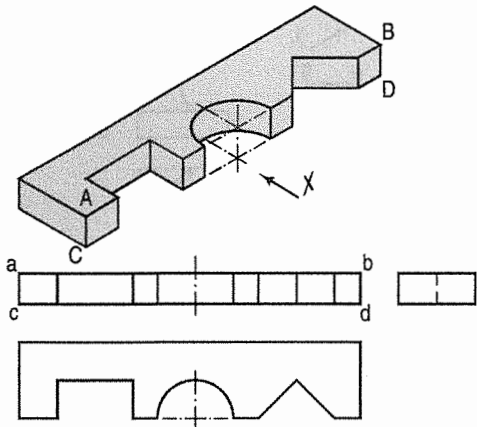


FIG. 20-17

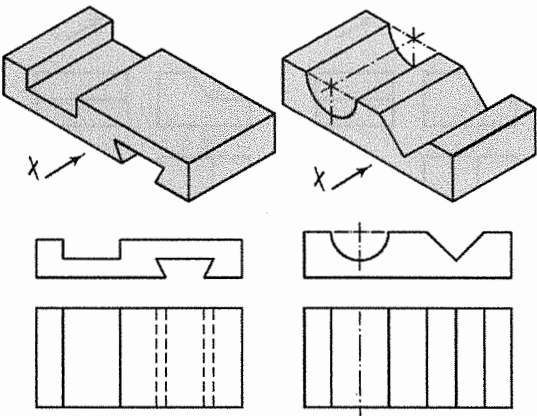


FIG. 20-18

Two plates having grooves in upper and lower surfaces are shown in fig. 20-18. The shapes of the grooves are seen in the front views. In top views, two lines for

the edges of rectangular as well as semi-circular grooves are drawn. For the triangular grooves, three lines are required. The tapered groove in the bottom surface is assumed to be cut throughout the width of the plate. It is not visible from above and hence, its four edges are shown by four hidden lines in the top view. In their side views, each groove should be shown by a horizontal hidden line. Therefore the side view in this case is not necessary.

A plate having holes of different shapes is shown in fig. 20-19. These shapes are seen in the top view. Holes are assumed to be cut or drilled right through. They are shown by hidden lines in the front and the side views. Here also the right side view is not necessary.

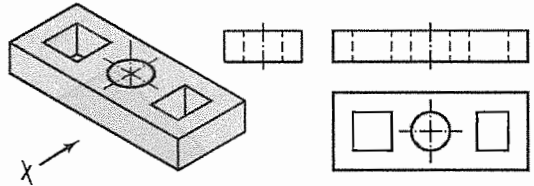


FIG. 20-19

The semi-circular groove in the block shown in fig. 20-20 does not extend upto the opposite surface. *Therefore, although the edge AB is broken, ab is seen as a continuous line in the front view.* The groove is shown by hidden lines in the side view. In the top view it is drawn as a rectangle.

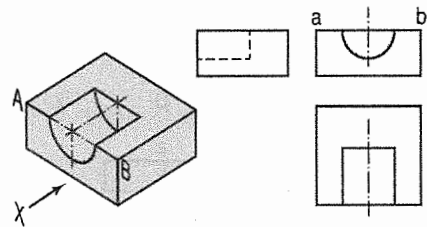


FIG. 20-20

Fig. 20-21 shows a block in the shape of steps, along with its three views. In fig. 20-22, its face ABCD is inclined. Hence, in the top view, two lines *ab* and *cd* are drawn for that face.

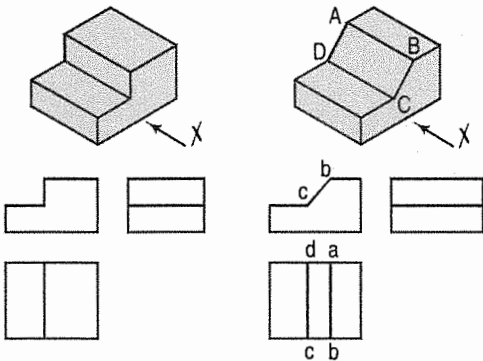


FIG. 20-21

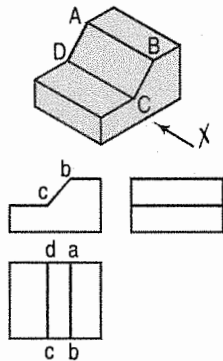


FIG. 20-22

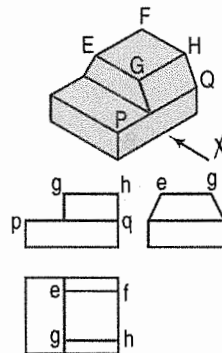


FIG. 20-23

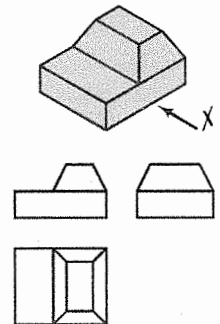


FIG. 20-24

The front and back faces of the upper step are shown inclined in fig. 20-23. Their inclinations are shown in the side view. In the front view, the line *pq* for the edge *PQ* is drawn. In the top view, lines *ef* and *gh* for edges *EF* and *GH* are shown.

In fig. 20-24, all the four side-faces of the upper step are inclined. Their inclinations are seen in the front view and in the side view. In the top view, two rectangles are seen for the upper step. Lines (for the four sloping edges) joining the corners of the rectangles should also be shown.

A casting having a hollow cylinder supported by a vertical rib is shown in fig. 20-25. The width of the rib is equal to the diameter of the cylinder. Hence, in the front view, vertical lines for the rib are tangential to the circle for the cylinder.

In the side view, the line showing the thickness of the rib vanishes just at the centre line. The rib is not visible when seen from top. Hence, it is shown by a hidden line in the top view. Note that the third view (top view) is a 'necessary' view to show the shape of rectangular slot.

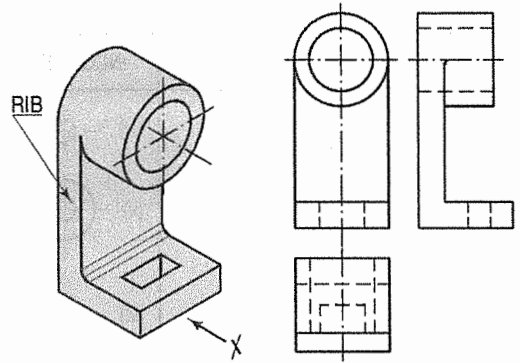


FIG. 20-25

In the bearing block shown in fig. 20-26, the line for the rib is inclined and tangent to the semi-circle in the front view. Vanishing points for the lines for the rib in the side view and the top view are obtained by projecting the tangent-point from the front view.

From the above discussion, the following important points are to be noted:

- (i) An object can be observed from six sides such as
 - (a) front side
 - (b) back side
 - (c) left-hand side
 - (d) right-hand side
 - (e) top side and
 - (f) bottom side.

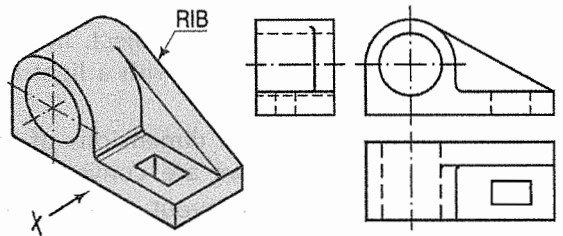


FIG. 20-26

It is not necessary to draw all the possible six views to describe completely the shape of the object. In practice, only those views which are *necessary to describe the shape of the object should be drawn*.

The view should be selected in such a way that minimum dotted lines are necessary to show internal details. The object which has both the right-hand side and left-hand side symmetrical shape requires only two views.

One of the views must be such that it completely describes the shape of the object. Refer fig. 20-16, fig. 20-32, fig. 20-47 and fig. 20-48.

- (ii) Three views are necessary for the object which is not symmetrical. Refer fig. 20-22 and fig. 20-24.
- (iii) The view should not be drawn out of place. That is the side view and top view are not aligned with the front view. This should be avoided.
- (iv) Dotted lines must be drawn correctly as shown in table 20-1. Incorrect points are indicated by circle.

TABLE 20-1

INCORRECT	CORRECT	INCORRECT	CORRECT	INCORRECT	CORRECT
(i)		(iv)		(vii)	
(ii)		(v)		(viii)	
(iii)		(vi)		(ix)	

- (v) (a) When a visible line coincides with either a dotted line or a centre line, *the visible line is shown* and a centre line is extended beyond the outlines of the view.
- (b) When a section-plane line coincides with a centre line, the *centre line is shown* and the section-plane line is drawn outside the out-lines of the object at the ends of centre line by thick dashes.
- (c) When a dotted line coincides with the centre line, *the dotted line should be shown*.

20-7. PROCEDURE FOR PREPARING A SCALE-DRAWING



A scale-drawing must always be prepared from freehand sketches initially prepared from a pictorial view or a real object. In the initial stages of a drawing, always use a soft pencil viz. HB, and work with a light hand, so that lines are thin, faint and easy to erase, if necessary.

1. Determine overall dimensions of the required views. Select a suitable scale so that the views are conveniently accommodated in the drawing sheet.
2. Prepare the sheet layout as described in illustrative problem 20-8. Draw rectangles for the views, keeping sufficient space between them and from the borders of the sheet.
3. Draw centre lines in all the views. When a cylindrical part or a hole is seen as a rectangle, draw only one centre line for its axis. When it is seen as a circle, draw two centre lines intersecting each other at right angles at its centre.
4. Draw details simultaneously in all the views in the following order:
 - (i) Circles and arcs of circles.
 - (ii) Straight lines for the general shape of the subject.
 - (iii) Straight lines, small curves etc. for minor details.
5. After the views have been completed in all the details, erase all unnecessary lines completely. Make the outlines so faint that only their impressions exist.
6. Fair the views with 2H or 3H pencil, making the outlines uniform and intensely black (but not too thick). For doing this, adopt the same working order as stated in step 4 above.
7. Dimension the views completely. Keep all centre lines.
8. Draw section lines in the view or views which are shown in section.
9. Fill up the title block and furnish all other necessary particulars.
10. Check the drawing carefully and see that it is complete in all respects.

20-8. ILLUSTRATIVE PROBLEMS



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 49 for the following problem.

Problem 20-8 [fig. 20-27(i)]. A pictorial view of bearing block is shown in fig. 20-27(i). Draw the front view, left-hand side view and top view according to the First-Angle projection method.

The procedure of preparing the orthographic views is illustrated in the following steps:

Step 1 [fig. 20-27(ii)]:

- (i) Take a half imperial size drawing paper of 560 mm × 380 mm.
- (ii) Draw the border lines taking $A = 30$ mm and $B = 10$ mm as shown in fig. 20-27(ii).

Now clear space in the drawing paper is 520 mm × 360 mm.

- (iii) The scale of drawing is decided from the size of object and number of views required to draw. Let L , W and H be the length, width and height of the object. The spacing between two views can be calculated as under:

Views	Horizontal distance (E)	Vertical distance (F)
(a) For four views (front view, top view and two side views):	$E = \frac{520 - (L + 2W)}{4}$ mm	$F = \frac{360 - (H + W)}{3}$ mm
(b) For three views (front view, top view and side view):	$E = \frac{520 - (L + W)}{3}$ mm	$F = \frac{360 - (H + W)}{3}$ mm

If these distances E and F are less than 20 mm, adopt the suitable standard scale. In the above illustrated problem E and F are 95 mm and 75 mm respectively. Select suitable scale approximately and draw required blocks depending upon the number of views (three in this problem).

Step 2 [fig. 20-27(iii)]:

- (i) Mark centres for circles.
- (ii) Draw centre lines passing through these centres and extend in other views also.

Step 3 [fig. 20-27(iv)]:

- (i) Draw circle of required diameter in the front view as well as in the top view.
- (ii) Draw the other lines in the front view and project them in the side view as well as in the top view.
- (iii) Erase construction lines. Refer fig. 20-27(iv).

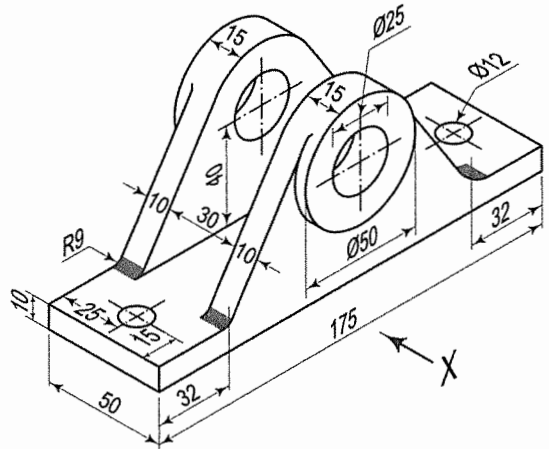


FIG. 20-27(i)

- (ii) Draw extension lines.
- (iii) Draw dimension lines. Insert dimensions. Observe that the height of numerals should not be more than 3 mm. It should be written freehand. Aligned method of writing dimensions is recommended. Follow IS:11669 (1986) for dimensioning.
- (iv) Complete title block by specifying unit of dimensions, symbol for method of projection and scale adopted.
- (v) Fair boundary lines by 2H pencil.

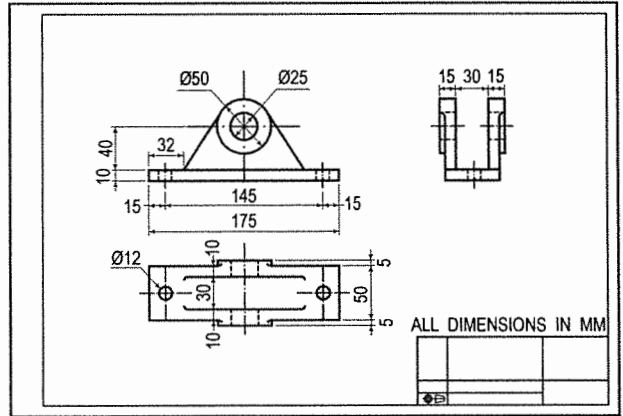


FIG. 20-27(v)

Problem 20-9. Draw the front view and a side view of the wedge-shaped piece shown in pictorial view in fig. 20-28(i).

The edge AB is parallel to the isometric axis oz and hence, it will be seen as a horizontal line in the front view [fig. 20-28(ii)]. The edge CD will be seen as an inclined line.

Problem 20-10. Draw the front view and a side view of the wedge-shaped piece shown pictorially in fig. 20-29(i).

It can be seen from the shape of the circular hole that the pictorial view is drawn according to *oblique projection*. Hence, the front view [fig. 20-29(ii)] will be similar to the front face shown in the pictorial view.

It is interesting to note that the two pictorial views [figs. 20-28(i) and 20-29(i)] look very much similar. But from their orthographic views we find that in fig. 20-29, the piece is in an inverted (upside-down) position.

Problem 20-11. Draw the front view and a side view of the wedge-shaped piece shown pictorially in fig. 20-30(i).

See fig. 20-30(ii). Note that the bottom edge is horizontal.

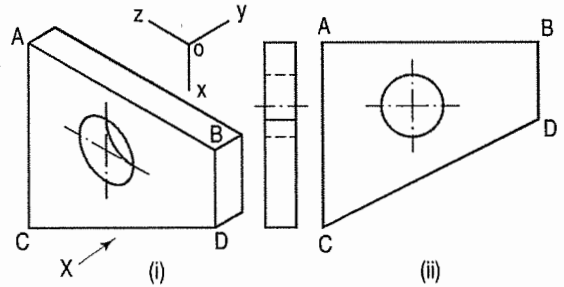


FIG. 20-28

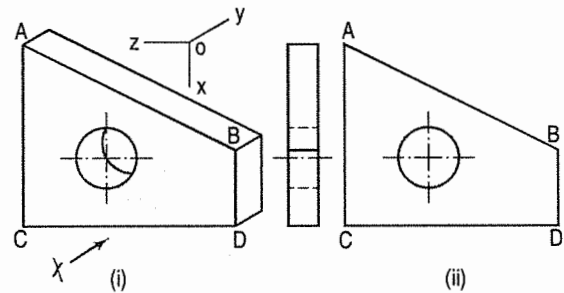


FIG. 20-29

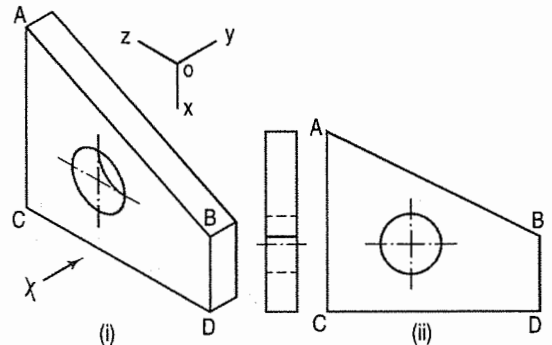


FIG. 20-30

Note: The three axes are shown along with the pictorial views, only for the purpose of explanation of the above problems No. 20-9, 20-10 and 20-11.

Problem 20-12. Draw the following views of the object shown pictorially in fig. 20-31(i).
(i) Front view. (ii) Top view. (iii) Side view from the right.

See fig. 20-31(ii).

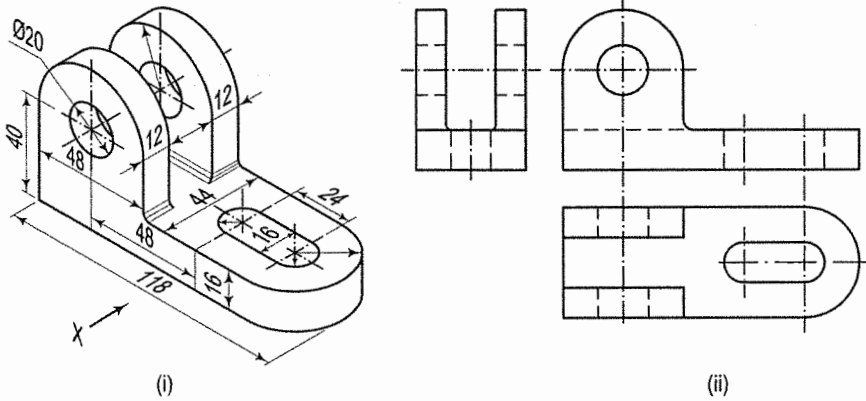


FIG. 20-31

Problem 20-13. Draw the following views of the block shown pictorially in fig. 20-32(i). Use third-angle projection method. (i) Front view. (ii) Top view. (iii) Both side views.

See fig. 20-32(ii).

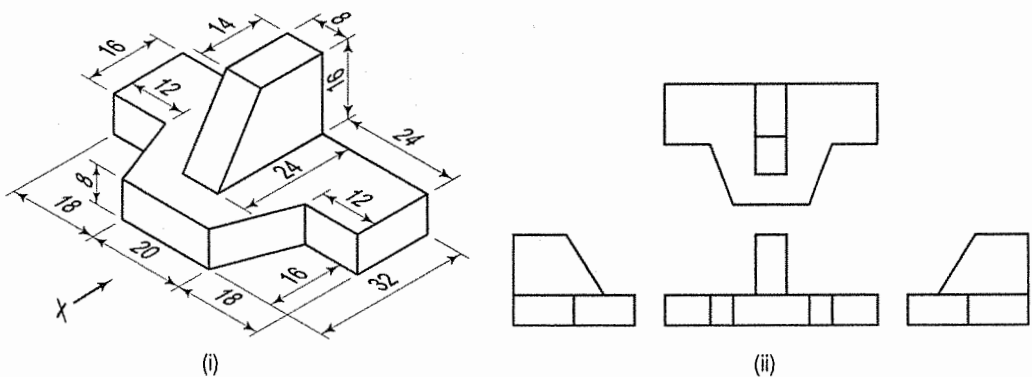


FIG. 20-32

EXERCISES 20

Pictorial views of objects are shown in fig. 20-33 to fig. 20-70. Draw, scale full size, views of each object as stated below. The front view in each case, should be drawn as seen in the direction of the arrow X. Unless otherwise specified, use first-angle projection method. Insert all dimensions in the views.

Solve each exercise independently and then, compare your answer with the given solution.

1. Fig. 20-33: (i) Front view. (ii) Side view. (iii) Top view.

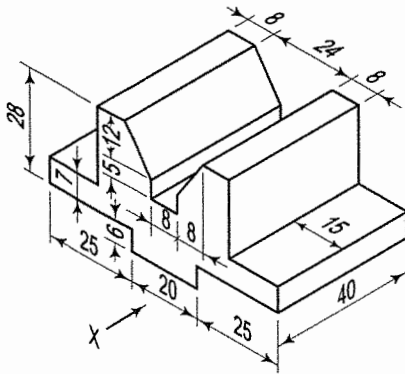


FIG. 20-33

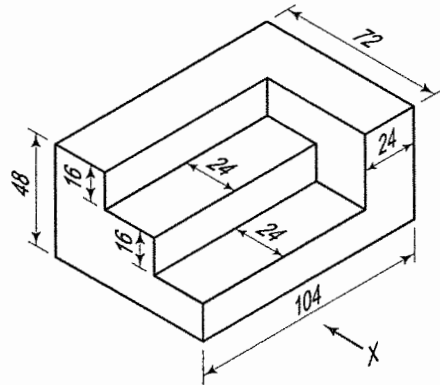


FIG. 20-34

2. Fig. 20-34: (i) Front view. (ii) Side view from the left. (iii) Top view. Use third-angle projection method.

3. Fig. 20-35: (i) Front view. (ii) Side view. (iii) Top view.

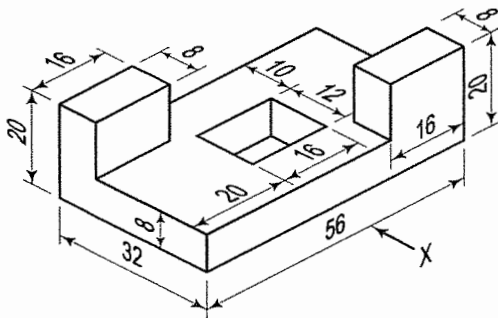


FIG. 20-35

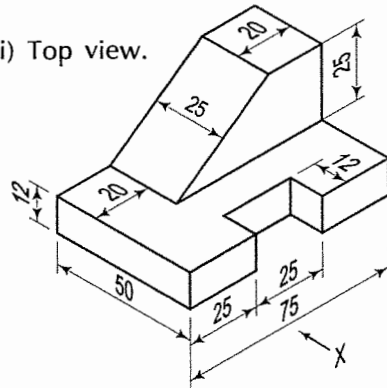


FIG. 20-36

4. Fig. 20-36: (i) Front view. (ii) Side view from the left. (iii) Top view.

5. Fig. 20-37: (i) Front view. (ii) Both side views. (iii) Top view. Use third-angle projection method.

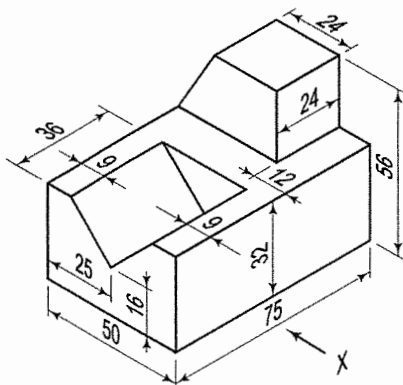


FIG. 20-37

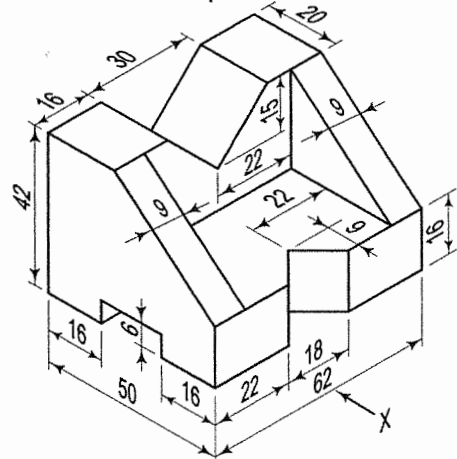


FIG. 20-38

6. Fig. 20-38: (i) Front view. (ii) Side view. (iii) Top view.

7. Fig. 20-39: (i) Front view. (ii) Side view from the left. (iii) Top view.

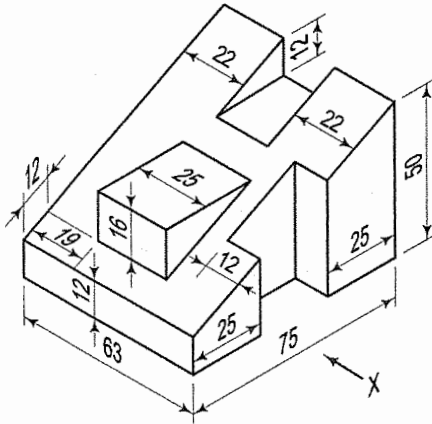


FIG. 20-39

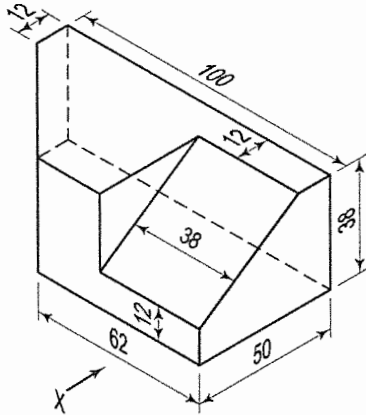


FIG. 20-41

10. Fig. 20-42: (i) Front view. (ii) Both side views. (iii) Top view.
 11. Fig. 20-43: (i) Front view. (ii) Both side views. (iii) Top view. Use third-angle projection method.

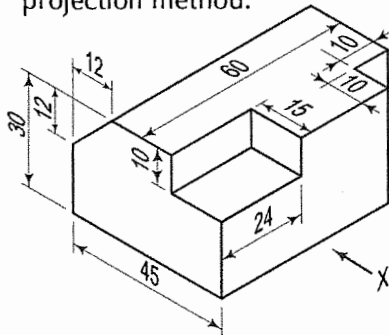


FIG. 20-43

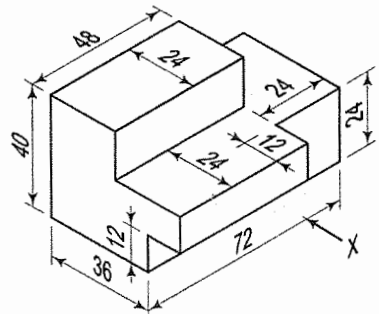


FIG. 20-40

8. Fig. 20-40: (i) Front view. (ii) Both side views. (iii) Top view.
 9. Fig. 20-41: (i) Front view. (ii) Both side views. (iii) Top view. Use third-angle projection method.

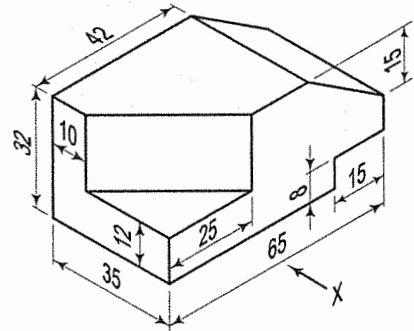


FIG. 20-42

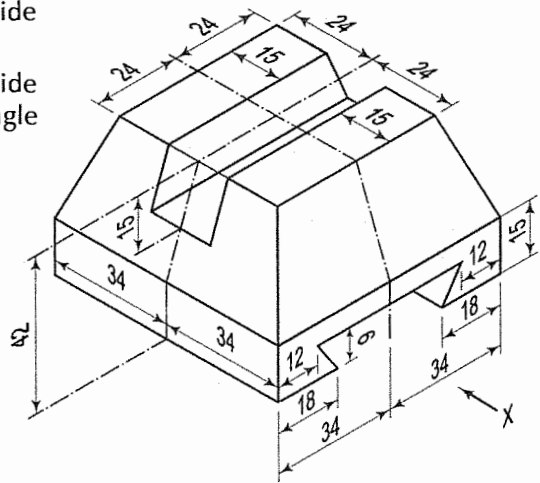


FIG. 20-44

12. Fig. 20-44: (i) Front view. (ii) Side view from left. (iii) Top view.

13. Fig. 20-45:

- (i) Front view.
- (ii) Side view.
- (iii) Top view.

14. Fig. 20-46:

All the six views.
Use third-angle projection method.

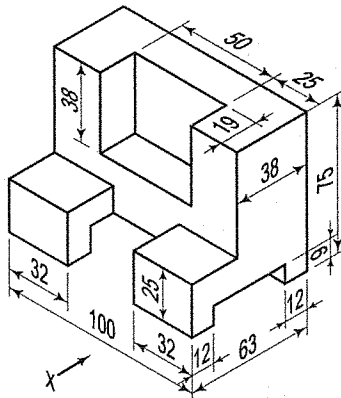


FIG. 20-45

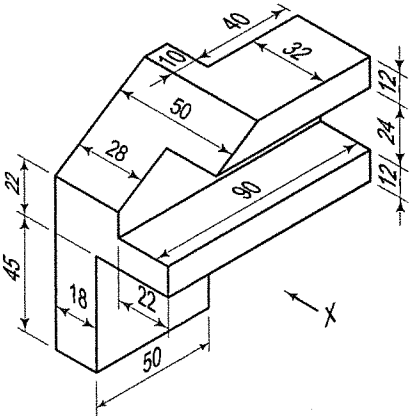


FIG. 20-46

15. Fig. 20-47: (i) Front view. (ii) Side view from the right. (iii) Top view.

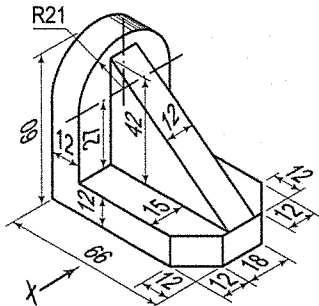


FIG. 20-47

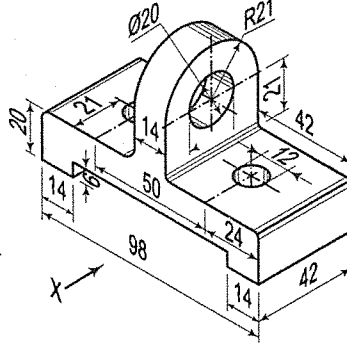


FIG. 20-48

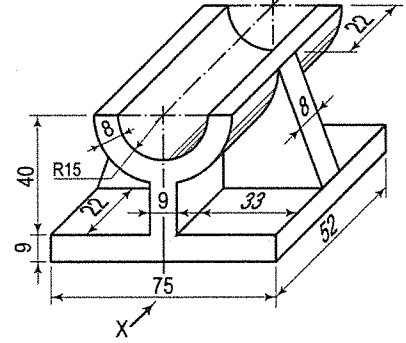


FIG. 20-49

16. Fig. 20-48: (i) Front view. (ii) Side view. (iii) Top view.

17. Fig. 20-49: (i) Front view. (ii) Side view. (iii) Top view.

Use third-angle projection method.

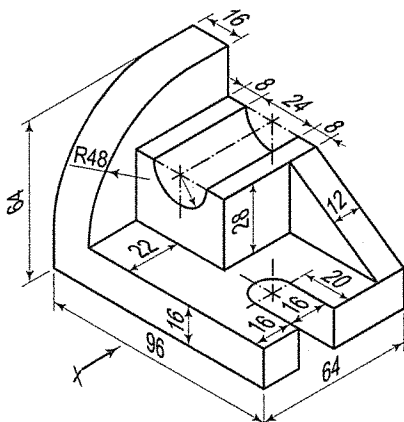


FIG. 20-50

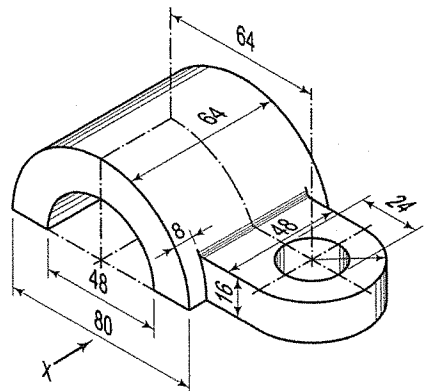


FIG. 20-51

18. and 19. Fig. 20-50 and fig. 20-51:

- (i) Front view. (ii) Side view from the right. (iii) Top view.

20. Fig. 20-52:

- (i) Front view.
- (ii) Both side views.
- (iii) Top view.

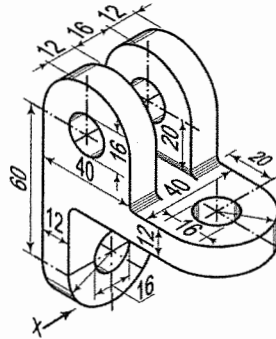


FIG. 20-52

21. Fig. 20-53:

- (i) Front view.
- (ii) Both side views.
- (iii) Top view.

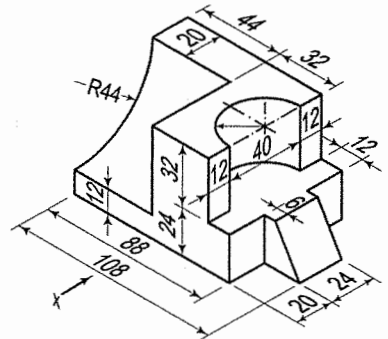


FIG. 20-53

22. Fig. 20-54: (i) Front view. (ii) Both side views. (iii) Top view.

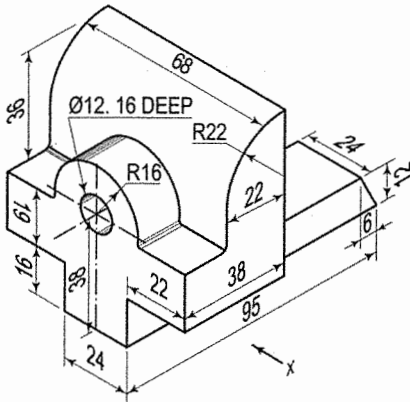


FIG. 20-54

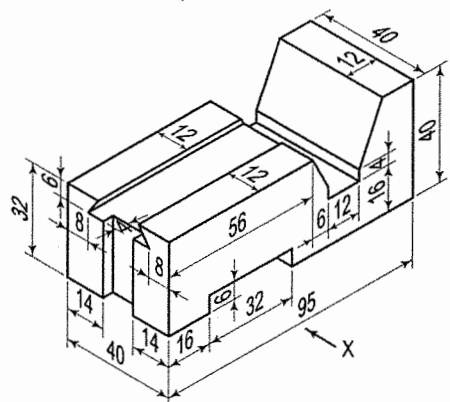


FIG. 20-55

23. Fig. 20-55: (i) Front view. (ii) Side view from the left. (iii) Top view.
Use third-angle projection method.

24. Fig. 20-56: All the six views.

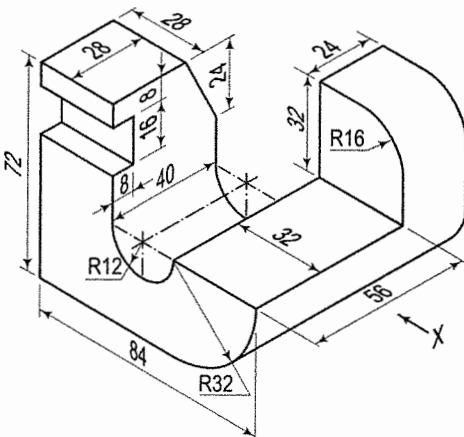


FIG. 20-56

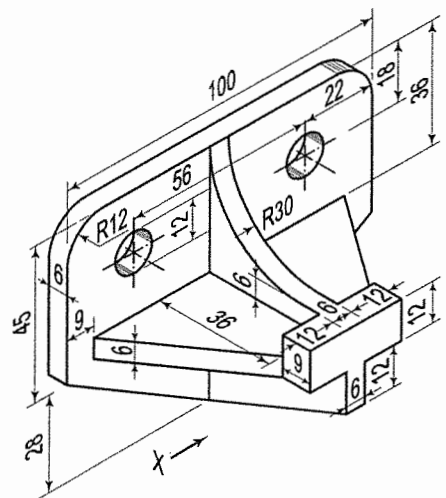


FIG. 20-57

25. Fig. 20-57: (i) Front view. (ii) Side view from the right. (iii) Top view.

26. Fig. 20-58 (i) Front view. (ii) Side view from the left. (iii) Top view.

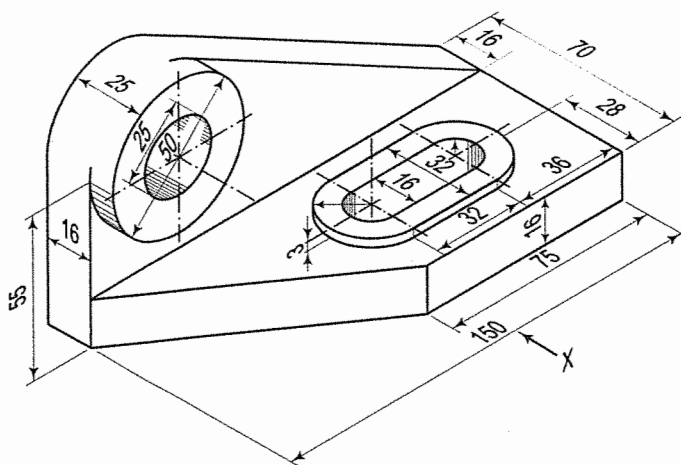


FIG. 20-58

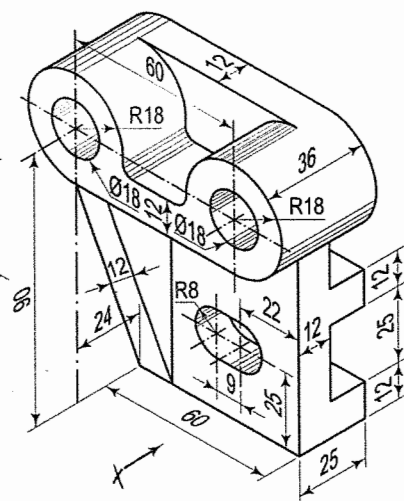


FIG. 20-59

27. Fig. 20-59:

- (i) Front view.
- (ii) Side view from the right.
- (iii) Top view.

Use third-angle projection method.

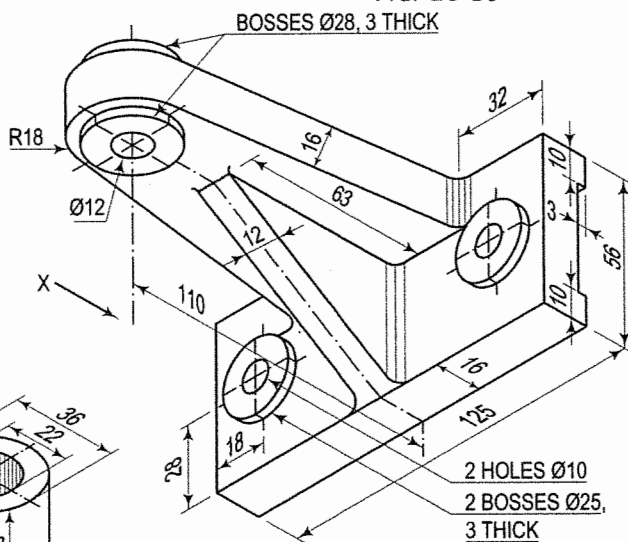


FIG. 20-60

28. Fig. 20-60:

- (i) Front view.
- (ii) Side view.
- (iii) Top view.

29. Fig. 20-61:

- (i) Front view.
- (ii) Side view from the right.
- (iii) Top view.

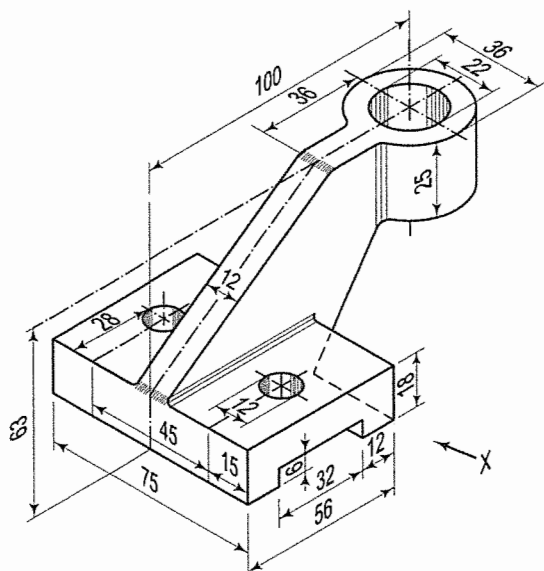


FIG. 20-61

30. Fig. 20-62: (i) Front view. (ii) Side view from the right. (iii) Top view. Use third-angle projection method.

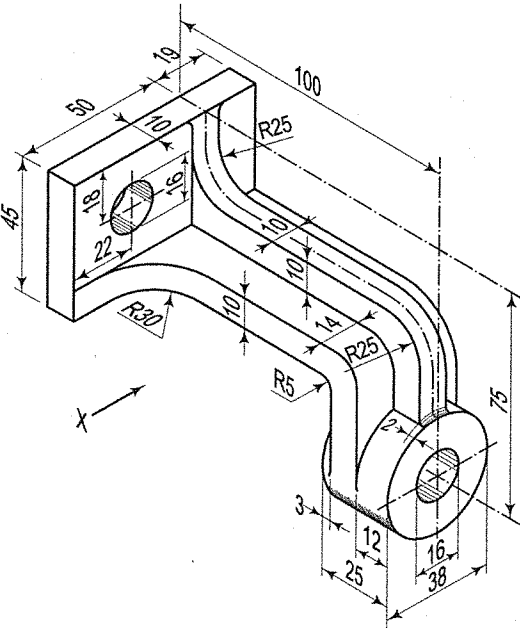


FIG. 20-62

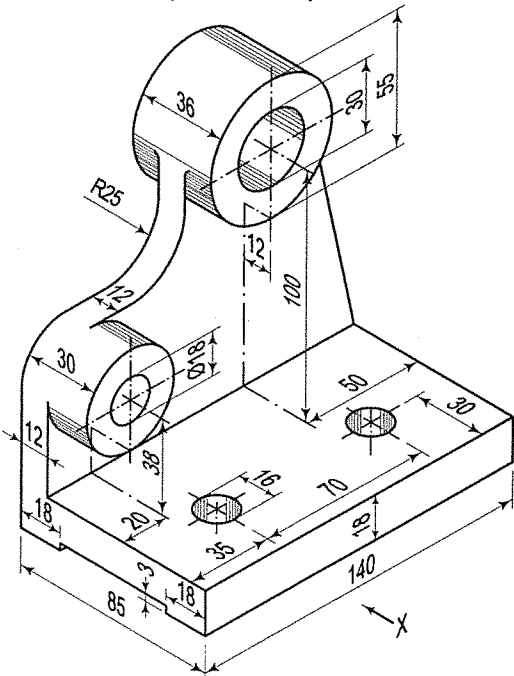


FIG. 20-63

31. Fig. 20-63: (i) Front view. (ii) Side view from the left. (iii) Top view.
32. Fig. 20-64: (i) Front view. (ii) Left-hand side view. (iii) Top view. Use third-angle projection method.

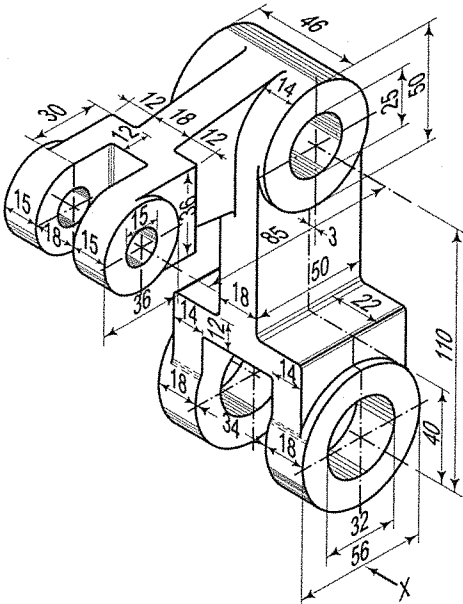


FIG. 20-64

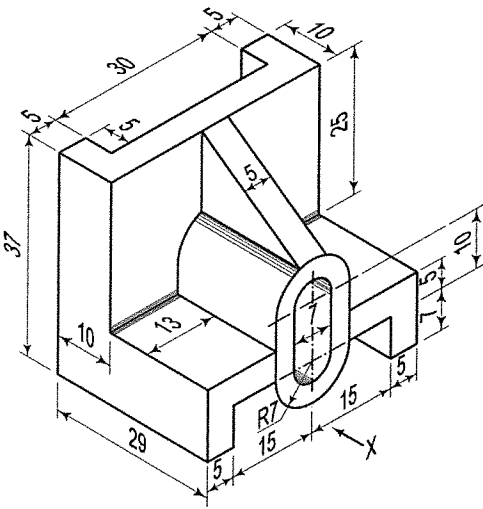


FIG. 20-65

33. Fig. 20-65: (i) Front view. (ii) Left-hand side view. (iii) Top view.

34. Fig. 20-66: (i) Front view. (ii) Right-hand side view. (iii) Top view.

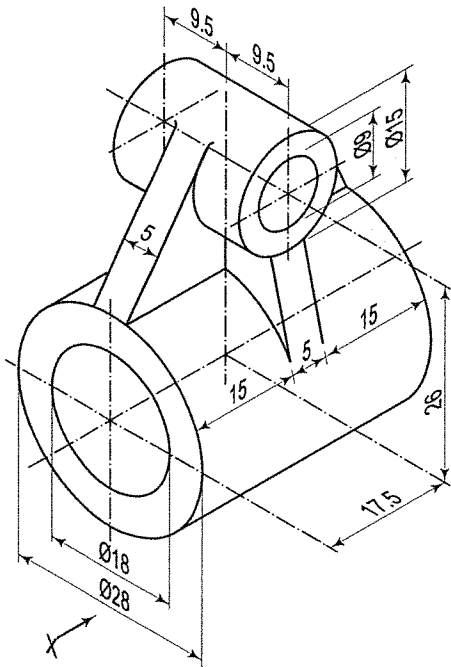


FIG. 20-66

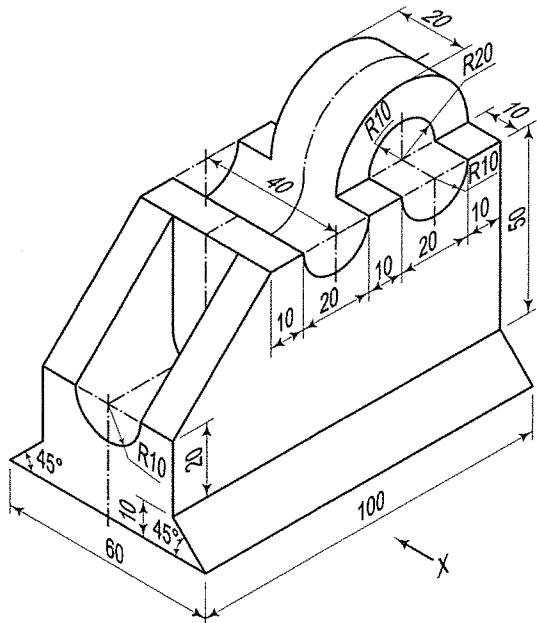


FIG. 20-67

35. Fig. 20-67: (i) Front view. (ii) Left-hand side view. (iii) Top view.

36. Fig. 20-68: (i) Front view. (ii) Side view. (iii) Top view.

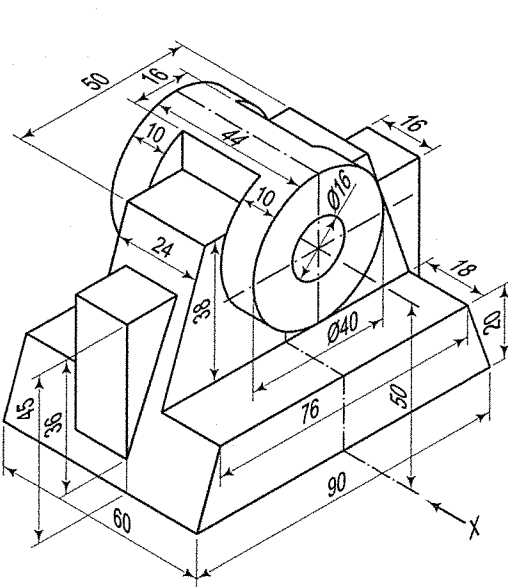


FIG. 20-68

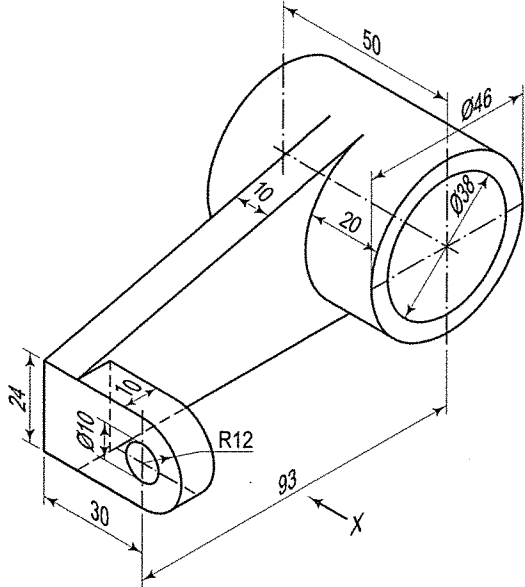


FIG. 20-69

37. Fig. 20-69: (i) Front view. (ii) Left-hand side view. (iii) Top view.

38. Fig. 20-70: (i) Front view. (ii) Left-hand side view. (iii) Top view.

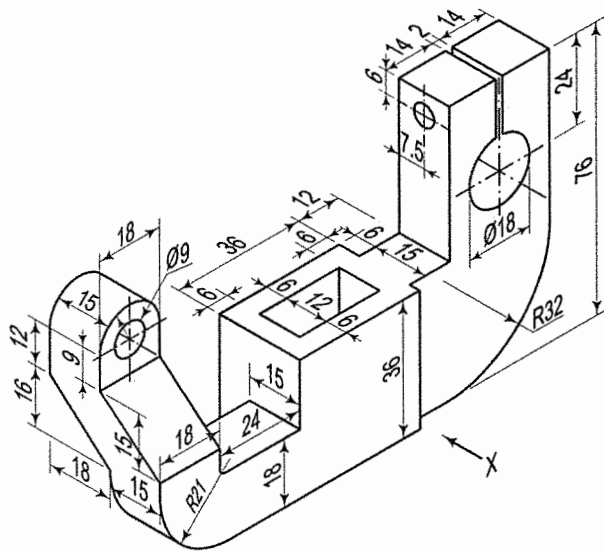
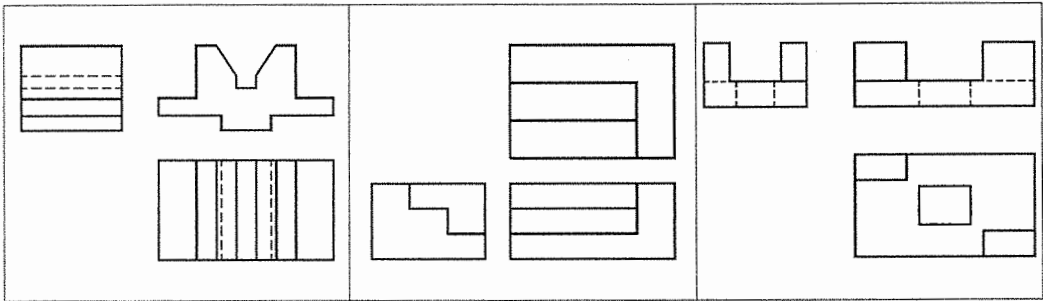


FIG. 20-70

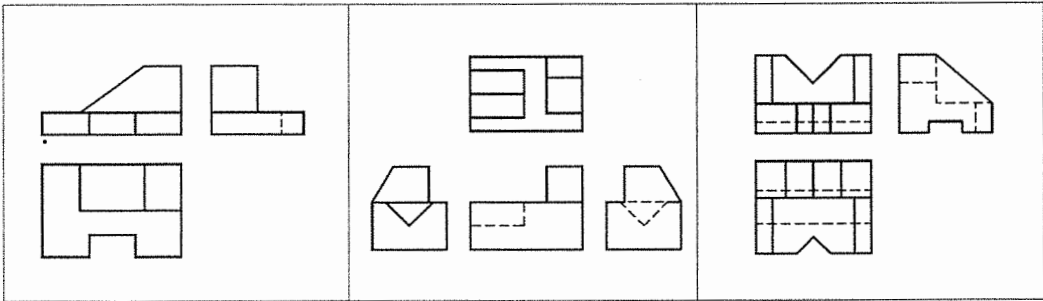
SOLUTIONS TO EXERCISES 20



(Ex. 1. FIG. 20-33)
FIG. 20-71

(Ex. 2. FIG. 20-34)
FIG. 20-72

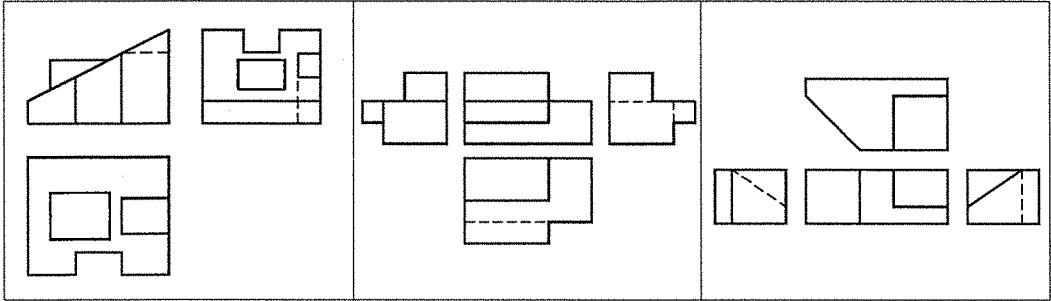
(Ex. 3. FIG. 20-35)
FIG. 20-73



(Ex. 4. FIG. 20-36)
FIG. 20-74

(Ex. 5. FIG. 20-37)
FIG. 20-75

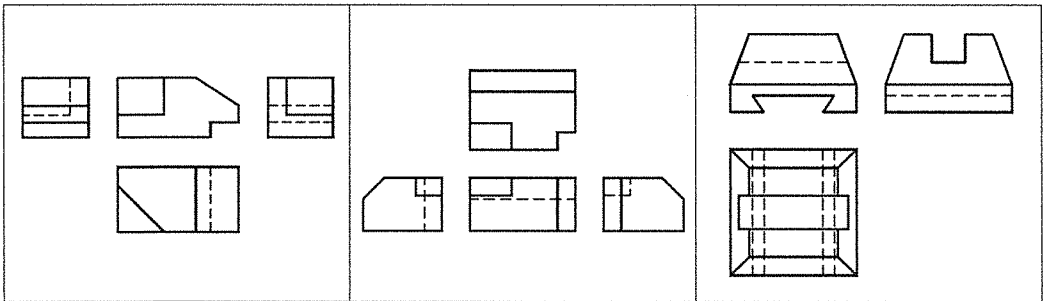
(Ex. 6. FIG. 20-38)
FIG. 20-76



(Ex. 7. FIG. 20-39)
FIG. 20-77

(Ex. 8. FIG. 20-40)
FIG. 20-78

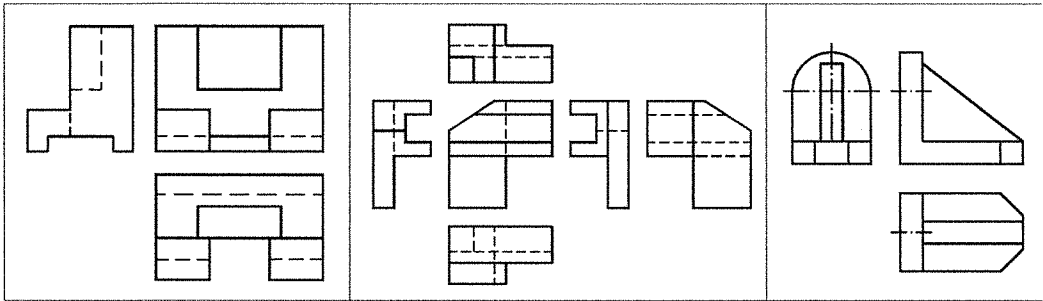
(Ex. 9. FIG. 20-41)
FIG. 20-79



(Ex. 10. FIG. 20-42)
FIG. 20-80

(Ex. 11. FIG. 20-43)
FIG. 20-81

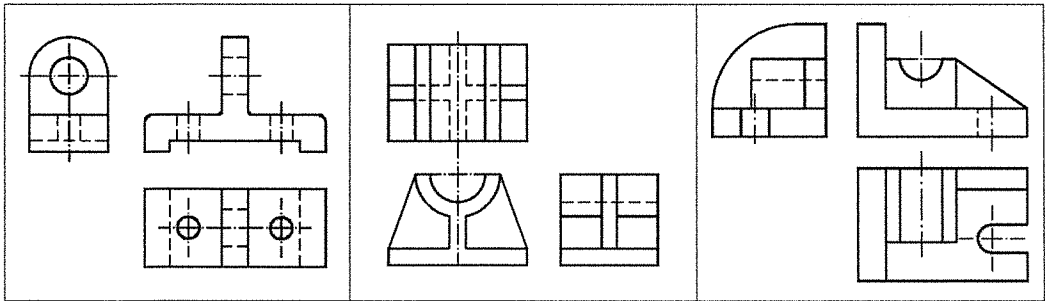
(Ex. 12. FIG. 20-44)
FIG. 20-82



(Ex. 13. FIG. 20-45)
FIG. 20-83

(Ex. 14. FIG. 20-46)
FIG. 20-84

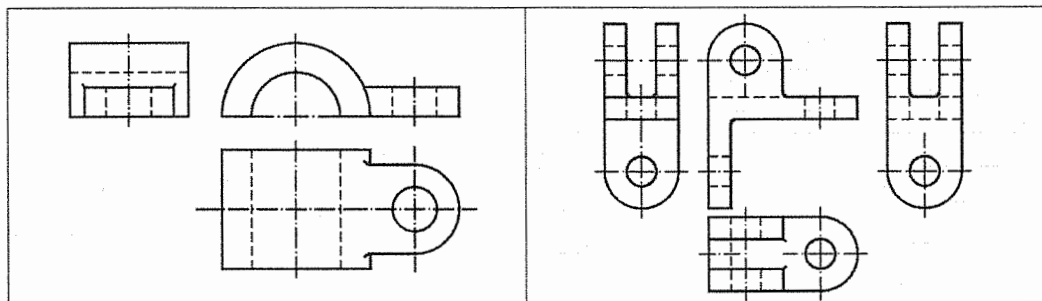
(Ex. 15. FIG. 20-47)
FIG. 20-85



(Ex. 16. FIG. 20-48)
FIG. 20-86

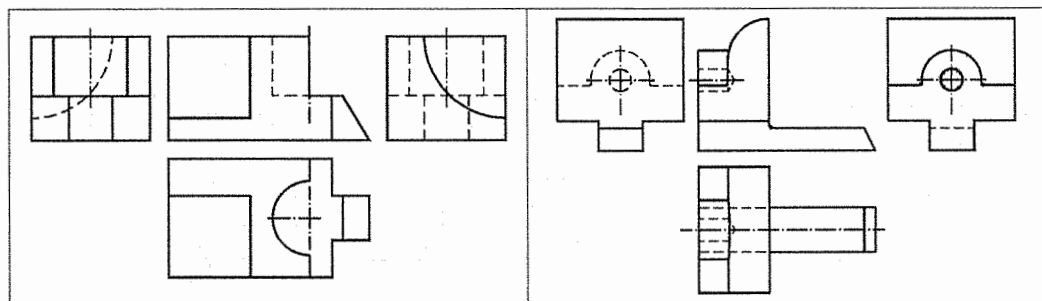
(Ex. 17. FIG. 20-49)
FIG. 20-87

(Ex. 18. FIG. 20-50)
FIG. 20-88



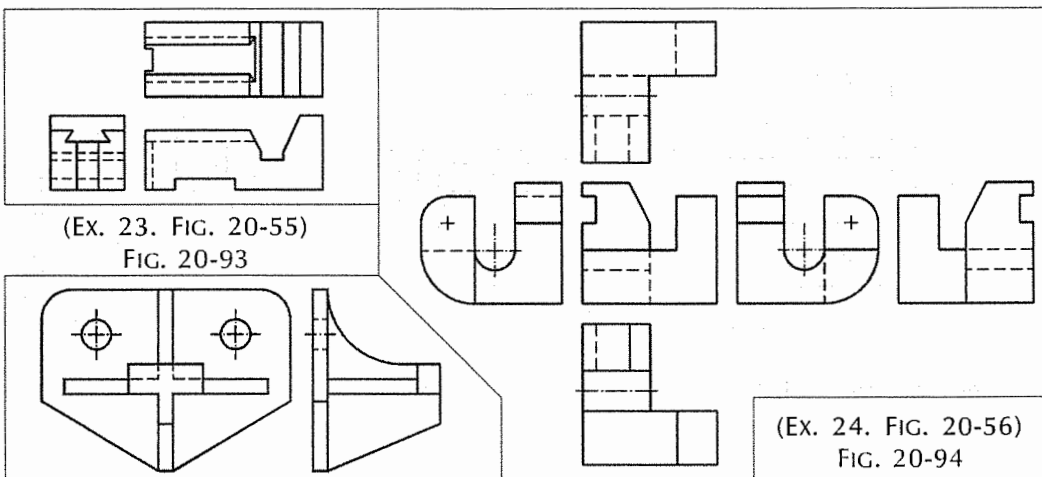
(Ex. 19. FIG. 20-51)
FIG. 20-89

(Ex. 20. FIG. 20-52)
FIG. 20-90



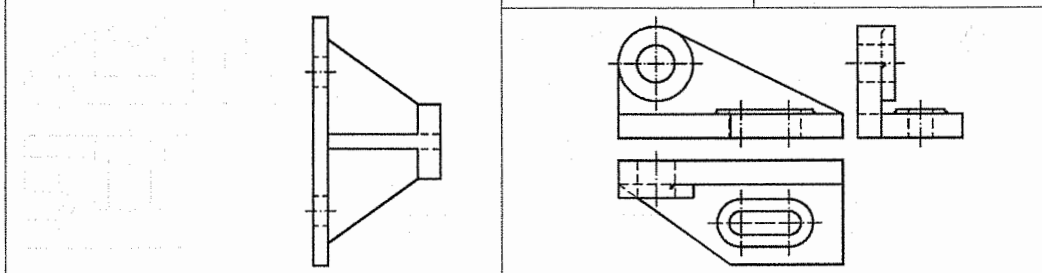
(Ex. 21. FIG. 20-53)
FIG. 20-91

(Ex. 22. FIG. 20-54)
FIG. 20-92



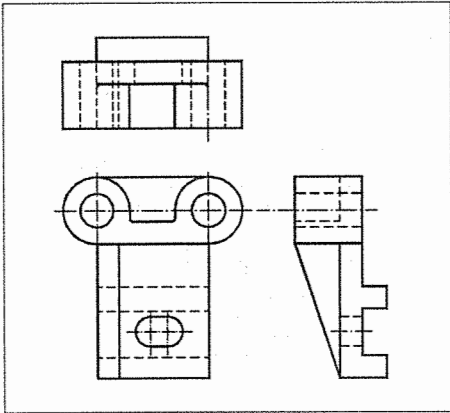
(Ex. 23. FIG. 20-55)
FIG. 20-93

(Ex. 24. FIG. 20-56)
FIG. 20-94

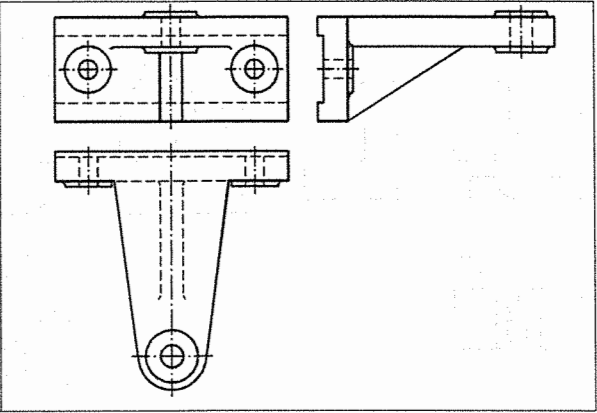


(Ex. 25. FIG. 20-57)
FIG. 20-95

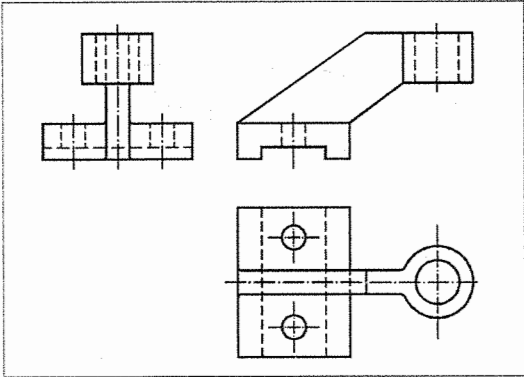
(Ex. 26. FIG. 20-58)
FIG. 20-96



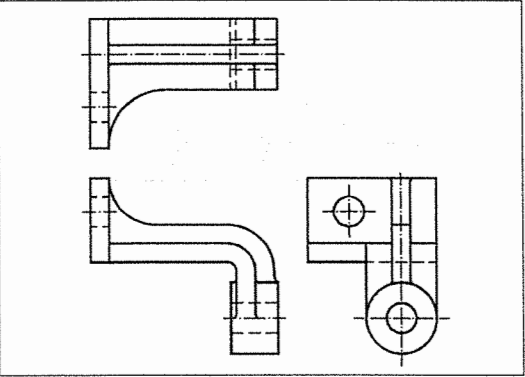
(Ex. 27. FIG. 20-59)
FIG. 20-97



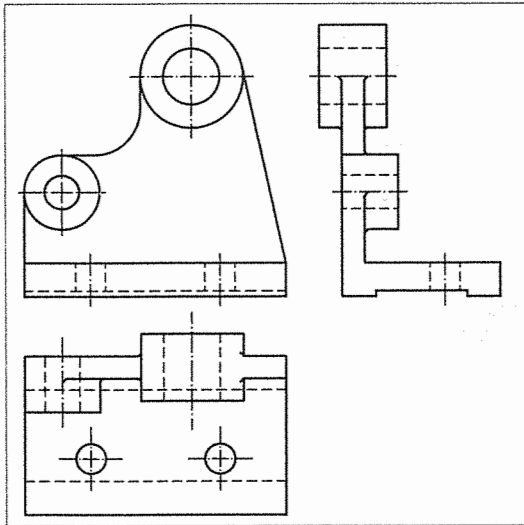
(Ex. 28. FIG. 20-60)
FIG. 20-98



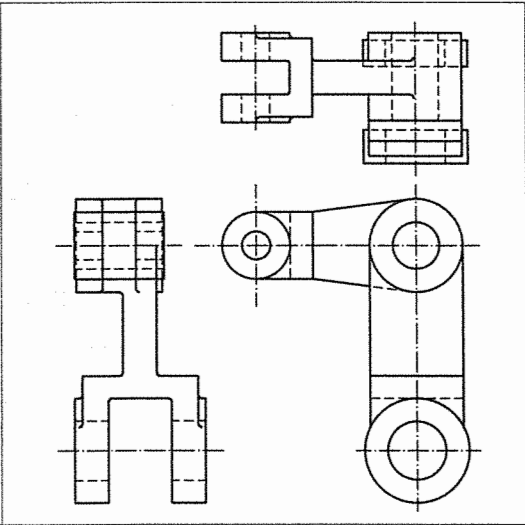
(Ex. 29. FIG. 20-61)
FIG. 20-99



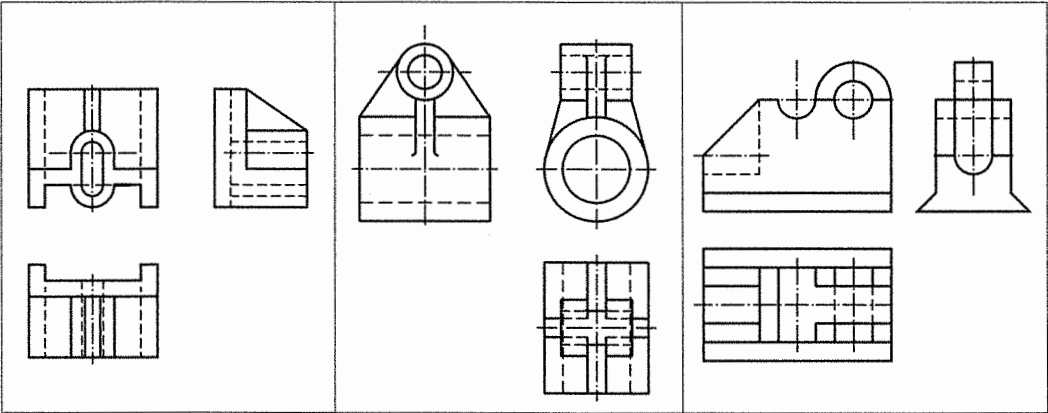
(Ex. 30. FIG. 20-62)
FIG. 20-100



(Ex. 31. FIG. 20-63)
FIG. 20-101



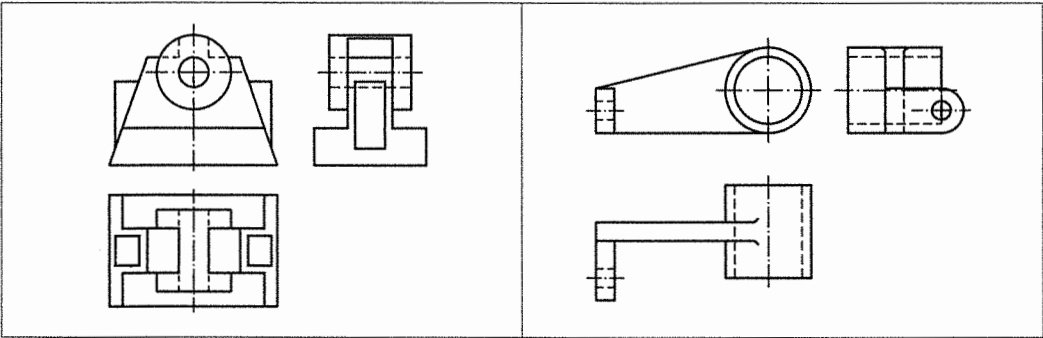
(Ex. 32. FIG. 20-64)
FIG. 20-102



(Ex. 33. FIG. 20-65)
FIG. 20-103

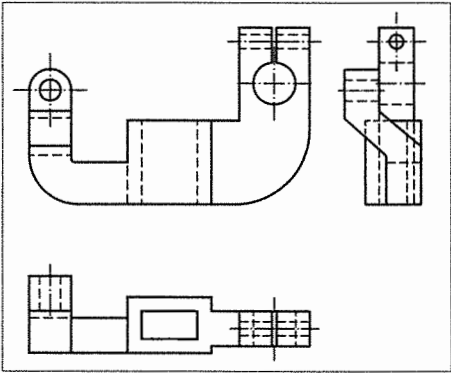
(Ex. 34. FIG. 20-66)
FIG. 20-104

(Ex. 35. FIG. 20-67)
FIG. 20-105



(Ex. 36. FIG. 20-68)
FIG. 20-106

(Ex. 37. FIG. 20-69)
FIG. 20-107



(Ex. 38. FIG. 20-70)
FIG. 20-108

Chapter 21



CENTRES OF GRAVITY AND MOMENTS OF INERTIA OF AREAS

21-0. INTRODUCTION



Centres of gravity and moments of inertia of areas of plane surfaces are frequently used in engineering practice. Therefore, we shall study in this chapter, about these two parameters as shown below:

- (1) Centre of gravity of symmetrical and unsymmetrical areas
- (2) Moment of inertia of plane surfaces.

21-1. CENTRE OF GRAVITY



A body consists of numerous particles on which the pull of the earth, i.e. the forces of gravity act. The resultant of these forces acts through a point. This point is called the *centre of gravity* of the body. It can also be defined as a point about which the weights of all the particles balance. It may not necessarily lie within the body.

In case of an area, the figure is assumed to be a lamina of negligible thickness so that its centre of gravity will be practically on the surface. As the area has no weight this point is also called the *centroid*.

21-1-1. CENTRES OF GRAVITY OF SYMMETRICAL AREAS



When an area is symmetrical about both its axes, the centre of gravity will be at the point of intersection of these axes. Thus, the centre of gravity of a square, rectangle, parallelogram or rhombus will be at the point of intersection of its diagonals. The centre of gravity of a circle will be at the point of intersection of its any two diameters, i.e. at its centre.

When an area is symmetrical about one axis, the centre of gravity will lie on that axis.

The centre of gravity of a triangle lies at the point of intersection of its medians (fig. 21-1). The centre of gravity of a quadrilateral is determined by dividing it into triangles and as explained in problem 21-1.

Problem 21-1. Find the centre of gravity of the quadrilateral ABCD shown in fig. 21-2.

- (i) Draw the diagonal BD and locate the centres of gravity G_1 and G_2 of triangles BCD and ABD respectively. Similarly, draw the diagonal AC and determine the centres of gravity G_3 and G_4 of triangles ABC and ADC respectively.

- (ii) Draw lines joining G_1 with G_2 , and G_3 with G_4 . G , the point of intersection of lines G_1G_2 and G_3G_4 is the centre of gravity of the quadrilateral $ABCD$.

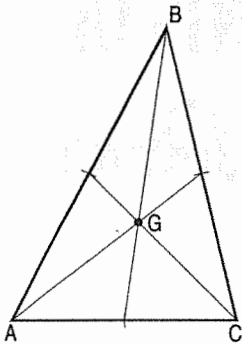


FIG. 21-1

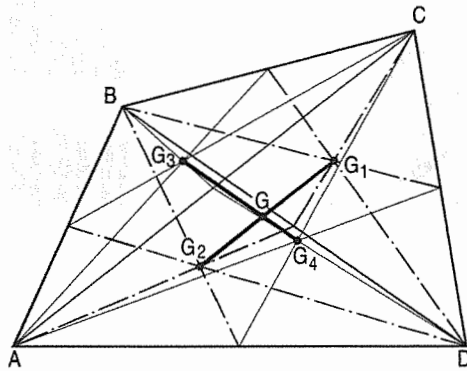


FIG. 21-2

A simple method of finding the centre of gravity of a trapezoid is illustrated in problem 21-2. It can also be located as explained in problem 21-3.

Problem 21-2. Determine the position of the centre of gravity of the trapezoid $ABCD$ shown in fig. 21-3.

- (i) Draw a line joining the mid-points P and Q of the parallel sides AB and DC respectively.
- (ii) Produce AB to a point E so that $BE = DC$. Similarly, produce CD to a point F so that $FD = AB$.
- (iii) Draw the line EF intersecting the line PQ at a point G . Then G is the required centre of gravity.

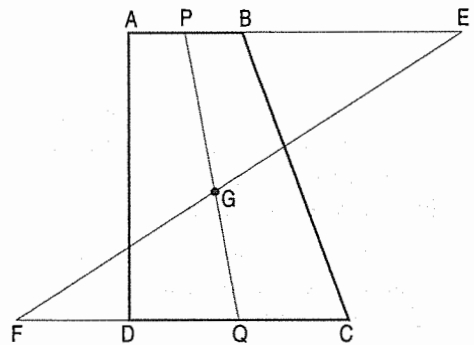


FIG. 21-3

21-1-2. CENTRES OF GRAVITY OF UNSYMMETRICAL AREAS

For locating the position of the centre of gravity of an area of unsymmetrical shape, it is first divided into a number of smaller areas of symmetrical shapes such as triangles, squares, rectangles, trapezoids etc. so that the centre of gravity of each such area is easily determined. Each small area is then treated as a force proportional to the area and acting at the centre of gravity of the area.

A resultant of these forces acting in one direction in the plane of the area is first determined. It will pass through the centre of gravity of the whole area. Another resultant of the same forces acting in the same plane but in another direction is then found out. This will also pass through the centre of gravity of the whole area. The point of intersection of these two resultants will be the centre of gravity of the whole area. The position of the resultant of parallel forces acting in a plane is graphically determined by means of *polar diagram* and *funicular polygon* or *link polygon* as explained in the following illustrative problems.

21-1-3. ILLUSTRATIVE PROBLEMS ON CENTRE OF GRAVITY



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 50 for the following problem.

Problem 21-3. Find graphically the centre of gravity of the I-section shown in fig. 21-4.

As the section is symmetrical about its vertical axis yy' , the centre of gravity must lie on that axis.

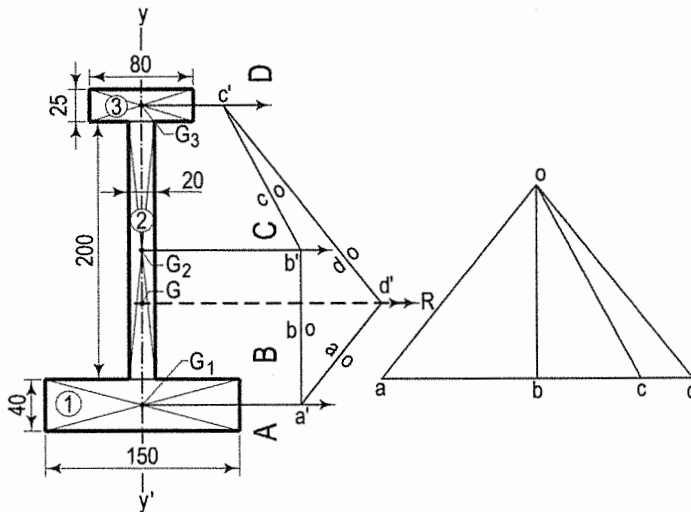


FIG. 21-4

- (i) Draw the given section to a convenient scale. Divide it into three rectangles viz. 1, 2 and 3 and locate their centres of gravity G_1 , G_2 and G_3 respectively at the points of intersection of their respective diagonals. Through these points, draw parallel lines of action of forces representing the area of each rectangle. These lines may be drawn in any direction except vertical, as the centre of gravity lies in the vertical axis. Name these forces as AB , BC and CD respectively according to Bow's Notation. This figure is called the *space diagram*.
- (ii) Calculate the areas of the rectangles 1, 2 and 3. They will be 6000, 4000 and 2000 sq. mm respectively.
- (iii) Draw a line parallel to the lines of action of forces and on it mark a point a . Select any convenient scale, say 1 cm = 1000 sq. mm and cut-off lengths $ab = 6$ cm, $bc = 4$ cm and $cd = 2$ cm in the direction of the forces. The line ad is called the *load line* or *force diagram*.
- (iv) Mark a point o at any convenient position either above or below the line ad and draw lines ao , bo , co and do . The point o is called the *pole* and the whole figure is called the *polar diagram*.
- (v) Mark any point a' on the line of action of the force AB and draw a line parallel to ao . Through the same point, draw a line parallel to bo to intersect the line of action of the force BC at a point b' . Through b' , draw a line parallel to co to intersect

the line of action of the force CD at a point c' . Through c' , draw a line parallel to do to intersect the line drawn parallel to ao at a point d' . Through d' , draw a line parallel to the lines of action to intersect the axis yy' at a point G . This line is the line of action of the resultant R of the forces and G is the centre of gravity of the section. Figure $a'b'c'd'$ is called the *funicular polygon* or *link polygon*.

Problem 21-4. Find graphically the centre of gravity of the channel section shown in fig. 21-5.

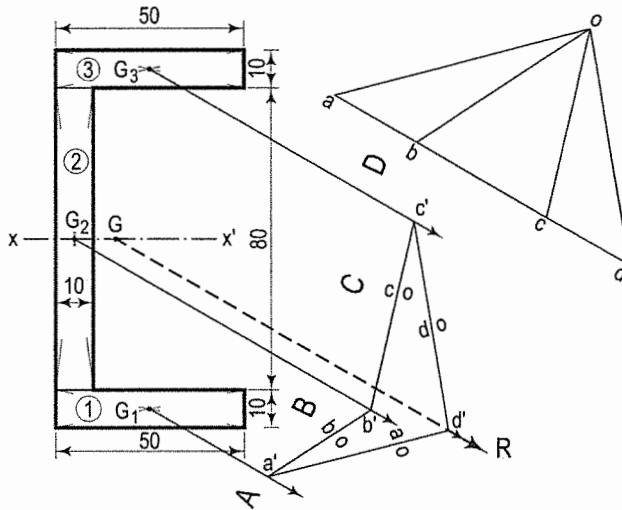


FIG. 21-5

As the section is symmetrical about its horizontal axis xx' , the centre of gravity must lie on this axis.

- (i) Draw the section to any convenient scale. Divide it into three rectangles 1, 2 and 3 and locate their centres of gravity G_1 , G_2 and G_3 respectively. Through these points, draw (in any direction except horizontal) parallel lines of action of forces representing the area of each rectangle.
- (ii) Draw the polar diagram and funicular polygon as explained in problem 21-3 and obtain the point G where the line of action of the resultant R of the forces intersects the axis xx' . G is the centre of gravity of the section.

Note that the line of action of the resultant is drawn through d the point of intersection between the lines drawn through points on the first and the last lines of action and parallel to ao and do respectively.

Problem 21-5. Determine graphically the position of the centre of gravity of the Z-section shown in fig. 21-6.

As the section is not symmetrical about any axis, it will be necessary to obtain two resultants.

- (i) Draw the section to a convenient scale and divide it into three rectangles 1, 2 and 3. (The division can also be done in a different way.)

- (ii) Mark the centres of gravity G_1 , G_2 and G_3 of these rectangles. Through these centres, draw lines of action of the forces in any one direction, say vertical. Calculate the areas of the three rectangles and draw the polar diagram and funicular polygon as explained in problem 21-3 and obtain the position of the resultant R .

Similarly, through points G_1 , G_2 and G_3 , draw horizontal or inclined parallel lines of action of forces and determine the position of the resultant R_1 . The point of intersection G , between the lines of action of the resultants R and R_1 is the required centre of gravity of the section.

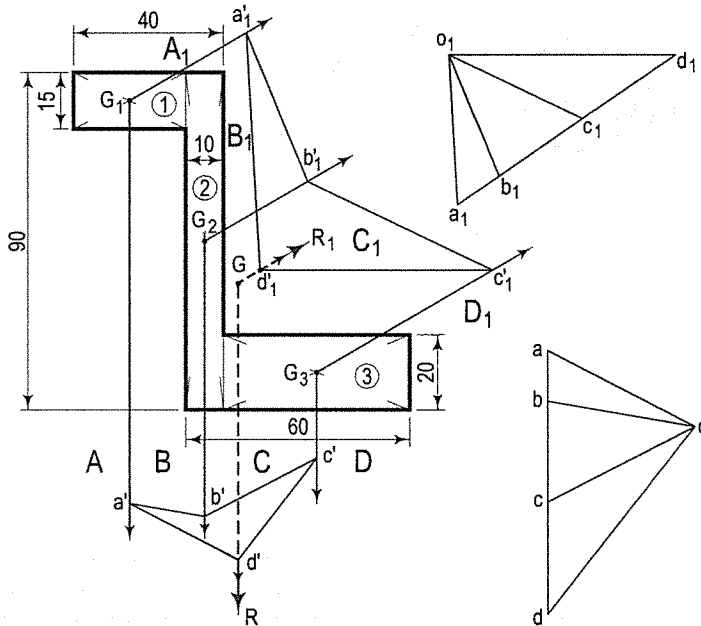


FIG. 21-6

Problem 21-6. Find graphically the centre of gravity of the plate PQRS of negligible thickness having a circular hole in it as shown in fig. 21-7.

- (i) Draw the plate to a convenient scale. Complete the rectangle PTRS.
- (ii) Find the centres of gravity of the circle, the rectangle PTRS and the triangle QTR and name them as G_1 , G_2 and G_3 respectively. Through these points, draw vertical lines of action of forces representing the areas.

As the areas of the circle and the triangle are removed from the area of the rectangle, the direction of action (i.e. the sense) of the forces representing them, viz. AB and CD will be opposite (as shown by the arrows) to that of the force BC representing the area of the rectangle. Calculate the areas of the circle, the rectangle and the triangle. They will be 707, 6000 and 1200 sq. mm respectively.

- (iii) Draw a line parallel to the lines of action. Select a convenient scale, say 1 cm = 1000 sq. mm, and on this line mark a length $ab = 0.7$ cm in

direction of the force AB , i.e. upwards. Cut-off a length $bc = 6$ cm in direction of the force BC , i.e. downwards. Again mark a length $cd = 1.2$ cm in direction of the force CD , i.e. upwards. Mark a suitable point o and draw lines ao , bo , co and do , thus completing the polar diagram.

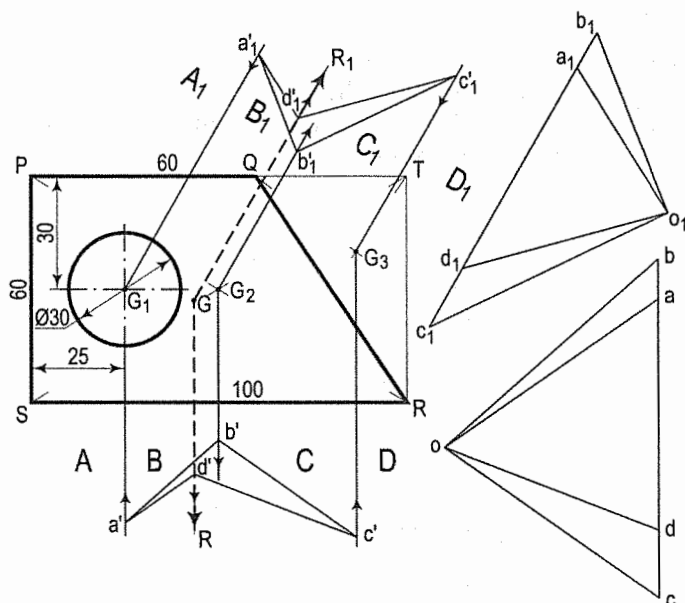


FIG. 21-7

- (iv) Mark any point a' on the line of action of the force AB and through it, draw a line parallel to ao . Through the same point, draw another line $a'b'$ parallel to bo and intersecting the line of action of the force BC at a point b' . Similarly through b' , draw a line $b'c'$ parallel to co and intersecting the line of action of the force CD at a point c' . Through c' , draw a line parallel to do and intersecting the line drawn parallel to ao at a point d' . Through d' , draw a line parallel to the lines of action. This line is the line of action of the resultant R .
- (v) Similarly, through point G_1 , G_2 and G_3 , draw inclined (or horizontal) parallel lines of action and determine the position of the line of action of the resultant R_1 as explained above. Note that this line passes through d'_1 , the point of intersection between the lines drawn through points on the first and the last lines of action and parallel to a_1o_1 and d_1o_1 respectively.

G , the point of intersection between the lines of action of the resultants is the required centre of gravity of the plate.

Note: Instead of subtracting the triangular part QTR from the plate $PTRS$, the plate $PQRS$ may be divided into two parts, viz. a square and a triangle.

Problem 21-7. Find graphically the centre of gravity of the thin circular plate having holes in it as shown in fig. 21-8.

Adopt the same method as explained in problem 21-6. The point G is the required centre of gravity.

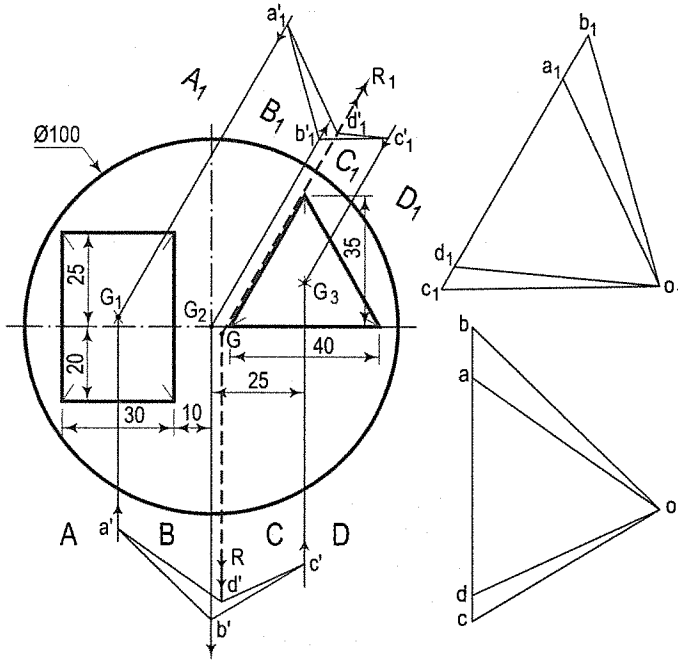


FIG. 21-8

Problem 21-8. Determine graphically the centre of gravity of the quarter-circle *PQR* shown in fig. 21-9.

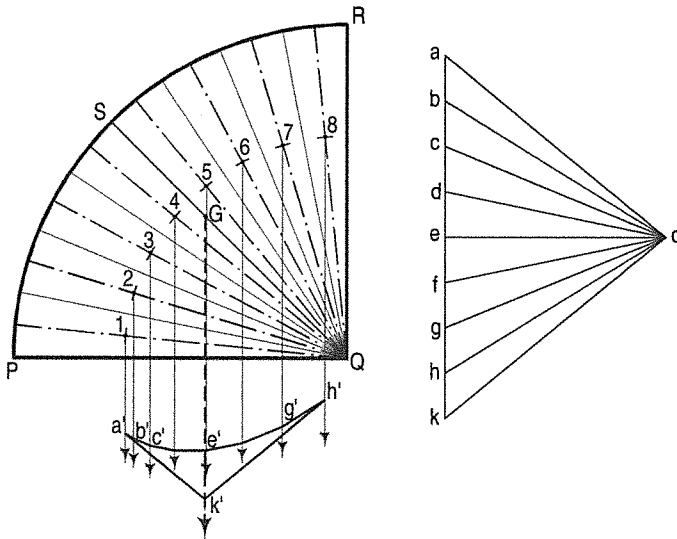


FIG. 21-9

The centre of gravity must lie on the bisector of the quarter-circle, viz. the line *SQ*.

- (i) Divide the quarter-circle into a number of equal sectors, say 8. Greater the number, each sector will tend to become an isosceles triangle.

- (ii) Assuming the sectors to be isosceles triangles, mark the centre of gravity (the point 1) of any one sector, say the end sector. It will lie on the altitude, at a distance of $\frac{2}{3}$ the altitude from the apex Q.
- (iii) With Q as centre and Q1 as radius, draw an arc to cut the other altitudes at the centre of gravity of each sector. Name these points as 2, 3,...8. Through points 1, 2,...8, draw vertical lines of action of forces.
- (iv) Draw the polar diagram taking ab , bc etc. of equal lengths, as all the sectors are of the same area.
- (v) Complete the funicular polygon and draw the line of action of the resultant R through the point k' . G , the point of intersection between this line and the line SQ is the required centre of gravity.

Problem 21-9. Find graphically the centre of gravity of the segment of the circle shown in fig. 21-10.

- (i) Divide the chord PQ into a number of equal parts and through the division-points, draw lines perpendicular to PQ , thus dividing the segment into a number of smaller parts. The centre of gravity of the segment must lie on the line ST which bisects the segment.
- (ii) Draw the mid-ordinates of each part and find the mid-point of each mid-ordinate. These points will approximately be the centres of gravity of the respective parts of the segment.

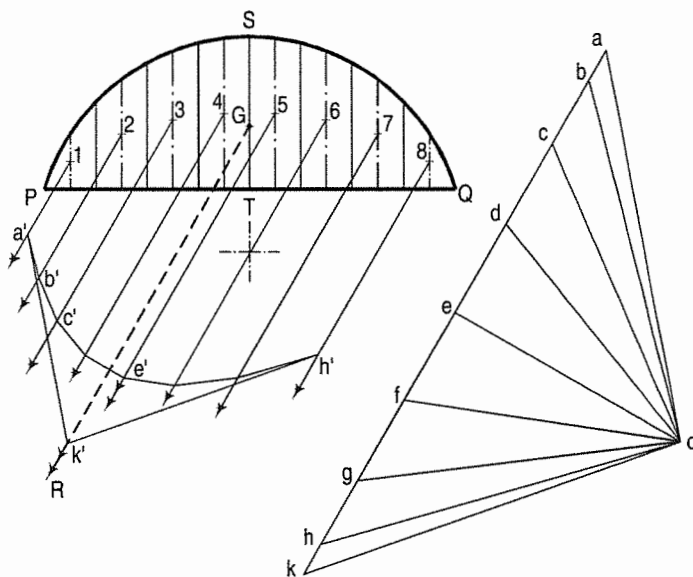


FIG. 21-10

- (iii) Through these points, draw inclined parallel lines of action of forces representing the area of each section. It is rather difficult to accurately determine the area of each part. But it would be approximately correct if the lengths ab , bc etc. in the polar diagram are taken in a fixed proportion, say 1:2, of the corresponding lengths of the mid-ordinates.
- (iv) Draw the polar diagram and the funicular polygon.

- (v) Through the point k' , draw the line of action of the resultant R . G , the point of intersection between this line and the line ST is the required centre of gravity.

Note: This is an approximate and quick method. More accurate method would be to (i) consider the end parts as triangles and the remaining parts as trapezoids, (ii) determine their centres of gravity, (iii) calculate their areas and then (iv) draw the polar diagram and funicular polygon.

Problem 21-10. Find graphically the centre of gravity of the section of a retaining wall shown in fig. 21-11.

Adopt the same method as explained in problem 21-5. The point G is the required centre of gravity. The section has been divided into two triangles and one rectangle. It can also be divided into two trapezoids and one square by drawing horizontal lines.

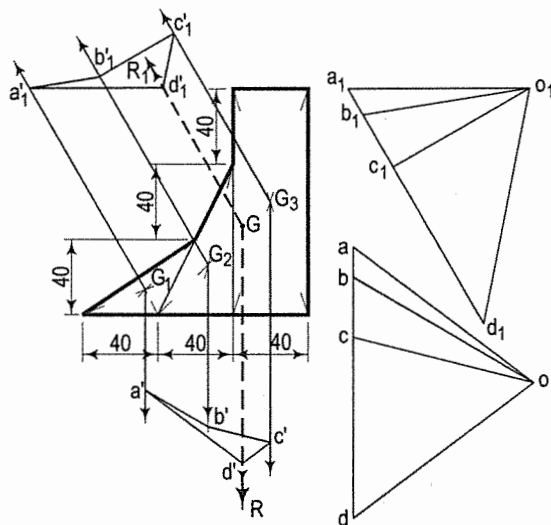


FIG. 21-11

21-2. MOMENTS OF INERTIA OF AREAS

Moment of inertia of an area of a plane figure about an axis in the plane of the area is a frequent requirement in engineering practice. The graphical method of determining the moment of inertia of areas gives fairly accurate results speedily. It is more useful in case of areas of irregular shapes.

(1) Definition: The moment of inertia of an area about an axis in the plane of the area is the sum of the products of the areas of its elements and the squares of the perpendicular distances of the centres of gravity (centroids) of these elements from the axis.

It is obtained by multiplying the whole area by the mean value of the squares of perpendicular distances of the centroids of its elements from the axis. An axis passing through the centroid of the whole area is called centroidal axis.

(2) Unit: When the area is calculated in centimetre units, i.e. in square centimetres (cm^2) and the distances are also measured in centimetres, i.e. the squares of the distances will be square centimetres (cm^2) the unit of moment of inertia (area $\times$ distance²) will be in centimetres raised to the fourth power, i.e. cm^4 .

In S.I. Units, the area is calculated in square metres (m^2) and the distances are also measured in metres. Therefore, the unit of moment of inertia ($\text{area} \times \text{distance}^2$) will be in metres raised to the fourth power, i.e. m^4 .

(3) **Graphical method:** The moment of inertia is graphically determined by Culmann's method with the help of space diagrams, force diagrams, polar diagrams and funicular polygons as explained in the following illustrative problems.

21-3. ILLUSTRATIVE PROBLEMS ON MOMENTS OF INERTIA



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 51 for the following problem.

Problem 21-11. Find graphically the moment of inertia of the rectangular section 10 cm wide $\times$ 20 cm deep shown in fig. 21-12, about the axis yy' passing through its centre of gravity.

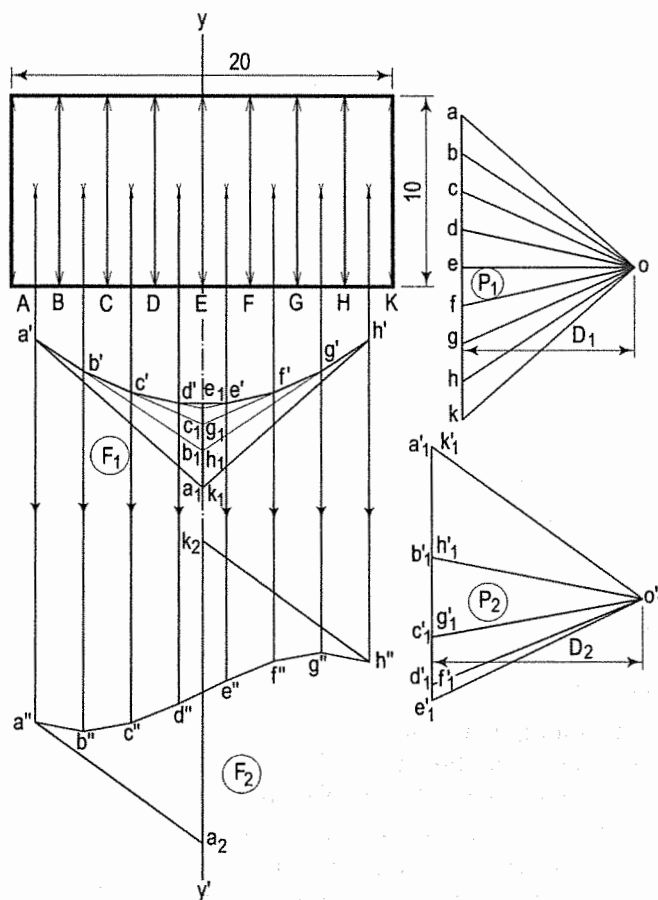


FIG. 21-12

- (i) Draw the given rectangle to a suitable scale, say $1 \text{ cm} = 2 \text{ cm}$ and draw the axis yy' passing through its centre of gravity.
- (ii) Divide the rectangle into a number of equal strips, say 8, parallel to the axis.
- (iii) Draw lines of action of forces representing the area of each strip, parallel to the axis and passing through the respective centres of gravity of the strips. This is the space diagram.

Calculate the area of each strip. Select a convenient scale, say $1 \text{ cm} = 20 \text{ sq. cm}$ and draw the load line ak as explained in problem 21-3. Lengths ab, bc etc. will each be equal to 1.25 cm each. Mark the pole o at any convenient polar distance D_1 , say 5 cm , from the line ak and complete the polar diagram P_1 by joining o with points $a, b, \dots, k$.

- (iv) Draw the funicular polygon F_1 as explained in problem 21-3. Produce all the sides of the polygon to intersect the axis at points $a_1, b_1, \dots, k_1$. Note that in this problem points a_1, b_1, c_1 and d_1 coincide with points k_1, h_1, g_1 and f_1 respectively.
- (v) Measure the intercepts $a_1 b_1, b_1 c_1$ etc. Assuming these intercepts to be forces and selecting a convenient scale, say $1 \text{ cm} = 0.333 \text{ cm}$, draw the second load line. While drawing this line it should be noted that the intercepts for the strips on one side of the axis should be drawn in one direction, while those for the strips on the other side of the axis should be drawn in the opposite direction.

Thus the intercepts from a'_1 to e'_1 coincide with those from k'_1 to e'_1 . Note that according to the selected scale, length $a'_1 e'_1$ is three times the length $a_1 e_1$. Select a pole o_1 at a suitable distance D_2 , say 6 cm , and complete the polar diagram P_2 .

- (vi) Draw the corresponding funicular polygon F_2 . The intercept $a_2 k_2$ of this polygon on the axis yy' is the moment of inertia of the rectangle about the axis yy' to the *moment scale* which is determined as shown below: If,
 - (i) the scale of the space diagram is $1 \text{ cm} = n \text{ cm}$
 - (ii) the area scale for the first load line is $1 \text{ cm} = n_1 \text{ cm}^2$
 - (iii) the intercepts from the funicular polygon F_1 are represented on the second load line to the scale $1 \text{ cm} = n_2 \text{ cm}$
 - (iv) the polar distances in polar diagrams P_1 and P_2 are $D_1 \text{ cm}$ and $D_2 \text{ cm}$ respectively,

$$\begin{aligned}
 \text{then the moment scale} &= n^2 \times n_1 \times n_2 \times D_1 \times D_2 \\
 &= 2^2 \times 20 \times 0.333 \times 5 \times 6 \\
 &= 800 \text{ cm}^4.
 \end{aligned}$$

The length of the intercept $a_2 k_2$ is 8.4 cm .

$$\begin{aligned}
 \therefore \text{The moment of inertia} &= \text{length of the intercept } a_2 k_2 \text{ in cm} \\
 &\quad \times \text{the moment scale} \\
 &= 8.4 \times 800 = 6720 \text{ cm}^4.
 \end{aligned}$$

Problem 21-12. Find graphically the moment of inertia of the isosceles triangle, base 9 cm long and altitude 9 cm, shown in fig. 21-13, about the vertical centroidal axis yy' .

Adopt the same method as explained in problem 21-11. The areas of the strips are unequal. The scales and distances selected are as under:

- (i) The scale for the space diagram is 1 cm = 1 cm.
- (ii) The scale for the first load line is 1 cm = 4 sq. cm.
- (iii) The scale for the second load line is 1 cm = 0.2 cm.
- (iv) The polar distances D_1 and D_2 are each 6 cm.

$$\begin{aligned}\text{The moment scale} &= 1^2 \times 4 \times 0.2 \times 6 \times 6 \\ &= 28.8 \text{ cm}^4.\end{aligned}$$

$$\text{The length of the intercept } a_2g_2 = 6.3 \text{ cm.}$$

$$\begin{aligned}\therefore \text{The moment of inertia} &= 6.3 \times 28.8 \\ &= 181.4 \text{ cm}^4.\end{aligned}$$

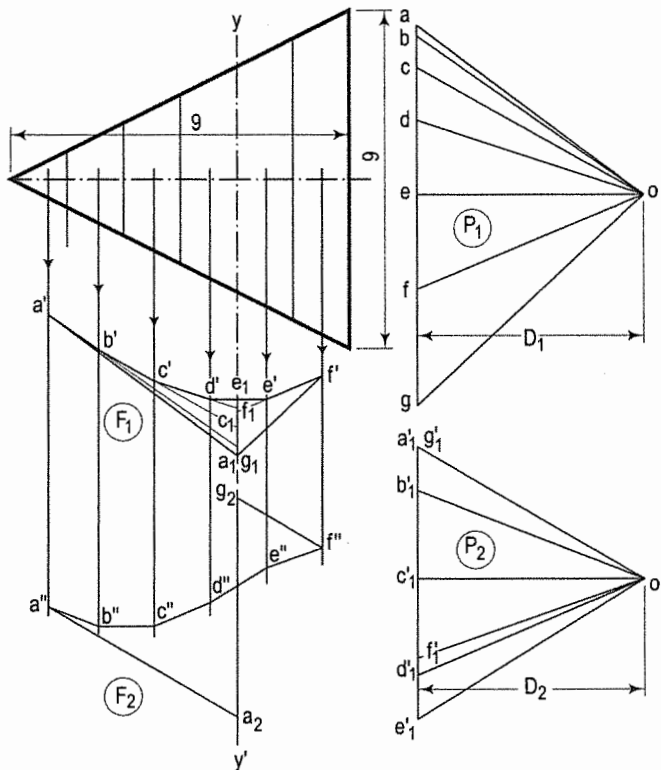


FIG. 21-13

Problem 21-13. Determine graphically the moment of inertia of the semi-circle of 6 cm radius shown in fig. 21-14, about the vertical axis yy' passing through its centre of gravity.

See fig. 21-14. Adopt the same method as explained in problem 21-11 and obtain the intercept $a_2 g_2$. The scales and distances selected are as under:

- (i) The scale for the space diagram is 1 cm = 1 cm.
- (ii) The scale for the first load line is 1 cm = 5 sq. cm.
- (iii) The scale for the second load line is 1 cm = 0.2 cm.
- (iv) The polar distances D_1 and D_2 are each 6 cm.

$$\begin{aligned}\text{The moment scale} &= 1^2 \times 5 \times 0.2 \times 6 \times 6 \\ &= 36 \text{ cm}^4.\end{aligned}$$

The length of the intercept $a_2 g_2$ is equal to 3.9 cm.

$$\begin{aligned}\therefore \text{The moment of inertia} &= 3.9 \times 36 \\ &= 140.4 \text{ cm}^4.\end{aligned}$$

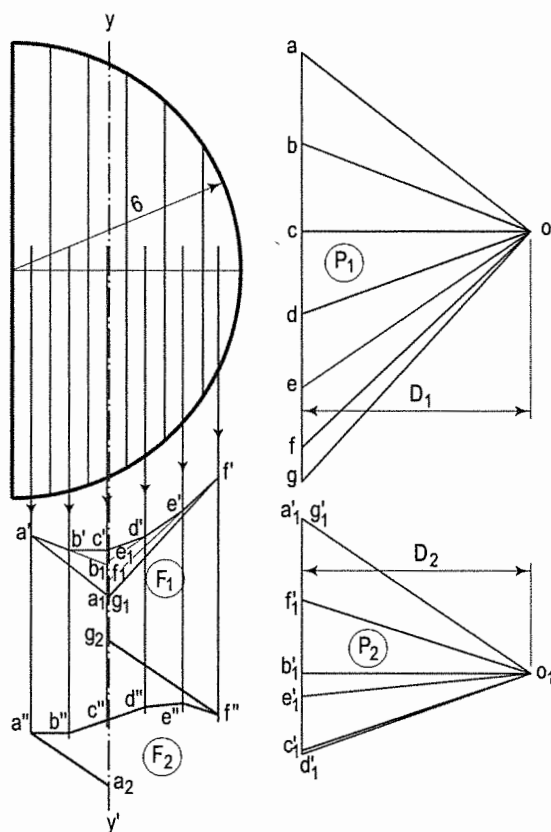


FIG. 21-14

Problem 21-14. Determine graphically the moment of inertia of the T-section shown in fig. 21-15, about the axis yy' situated at a distance of 15 cm from the centre of gravity of the section.

- (i) Draw the polar diagram P_1 and the funicular polygon F_1 and locate the position of the centre of gravity G .

- (ii) Draw the axis yy' at a distance of 15 cm from G .
- (iii) Obtain the intercepts a_1b_1 , b_1c_1 etc. by producing the lines $a'g'$, $a'b'$ etc. to intersect yy' at points a_1b_1 , etc.
- (iv) Draw the polar diagram P_2 and the funicular polygon F_2 as explained in problem 21-11 and obtain the intercept a_2g_2 on yy' . Measure its length and calculate the moment of inertia as shown below:

The scales and distances selected in the figure are as under:

- (i) The scale for the space diagram is 1 cm = 2 cm.
- (ii) The scale for the first load line is 1 cm = 4 sq. cm.
- (iii) The scale for the second load line is 1 cm = 1 cm.
- (iv) The distances D_1 and D_2 are each 8 cm.

$$\text{The moment scale} = 2^2 \times 4 \times 1 \times 8 \times 8 = 1024 \text{ cm}^4.$$

The length of the intercept a_2g_2 is 9 cm.

$$\therefore \text{The moment of inertia} = 9 \times 1024 = 9216 \text{ cm}^4.$$

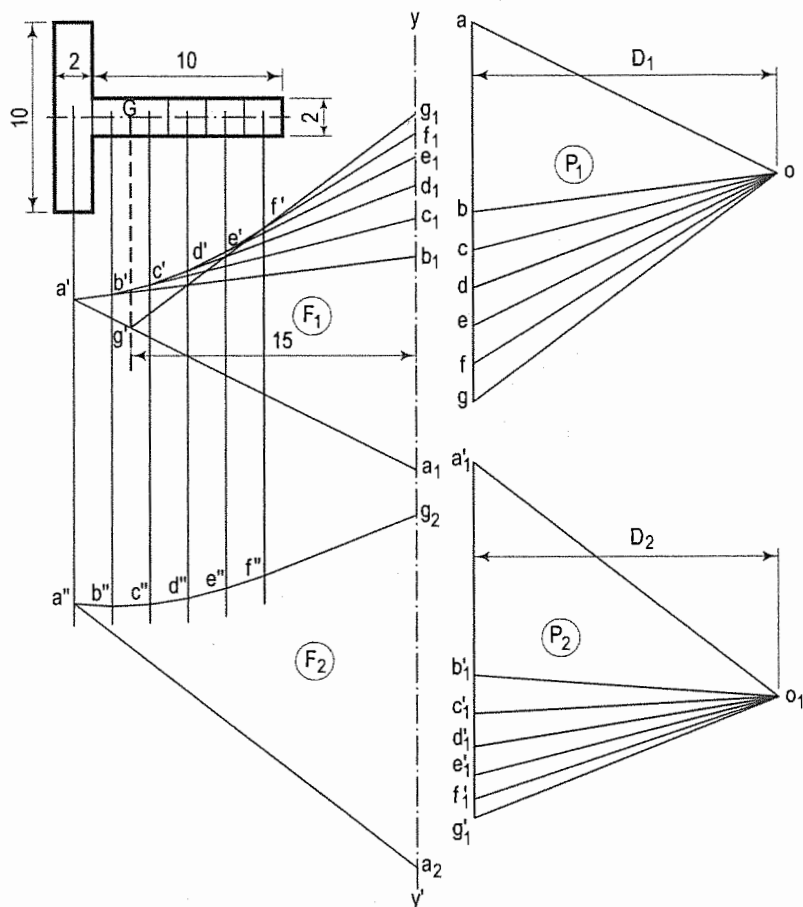


FIG. 21-15

EXERCISES 21

1 to 12. Determine graphically the position of the centre of gravity of each area shown in fig. 21-16 to fig. 21-27. All dimensions are in millimetres.

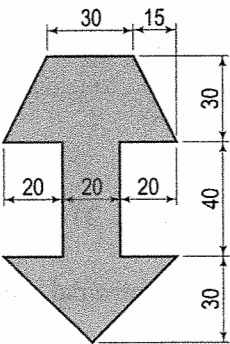


FIG. 21-16

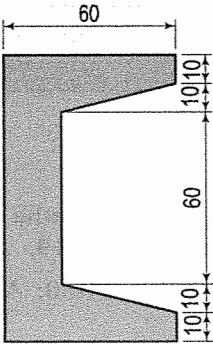


FIG. 21-17

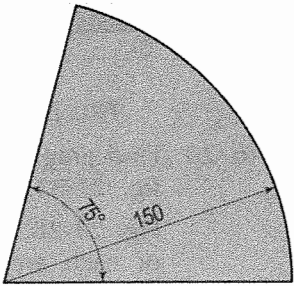


FIG. 21-18

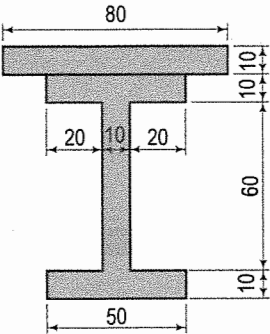


FIG. 21-19

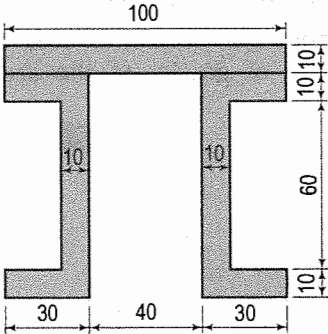


FIG. 21-20

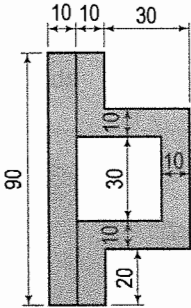


FIG. 21-21

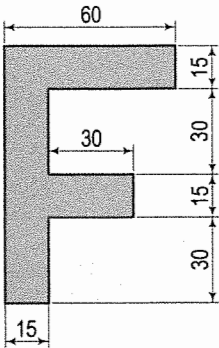


FIG. 21-22

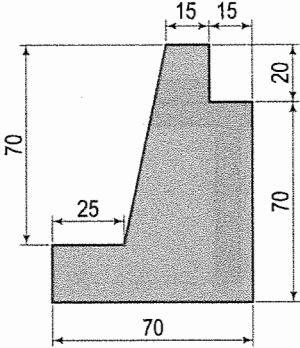


FIG. 21-23

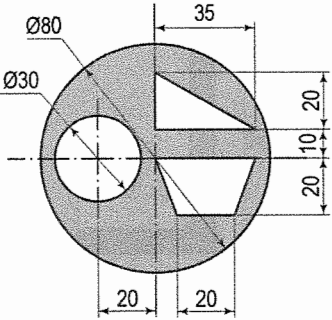


FIG. 21-24

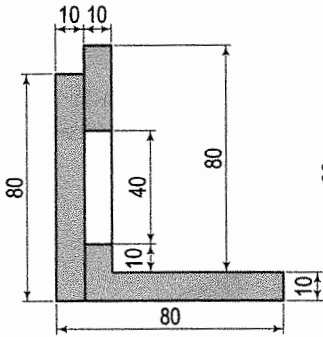


FIG. 21-25

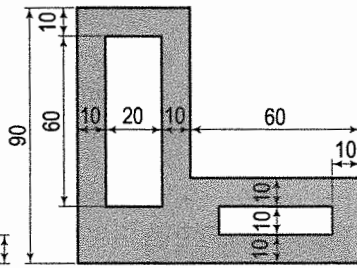


FIG. 21-26

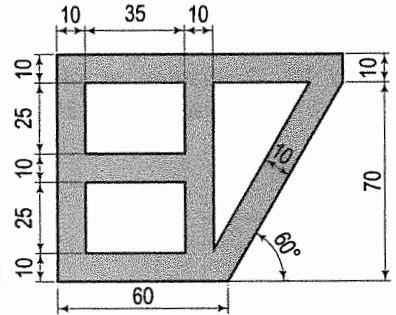


FIG. 21-27

13 to 20. Find graphically the moments of inertia of areas shown in fig. 21-28 to fig. 21-33, fig. 21-17 and fig. 21-19, as stated below:

Fig. 21-28, fig. 21-29 and fig. 21-30: About the vertical centroidal axis of each area.

Fig. 21-31 and fig. 21-32: About the horizontal centroidal axis of each area.

Fig. 21-33: About the base AB .

Fig. 21-17: About the horizontal axis situated at a distance of 100 mm below the centroid of the area.

Fig. 21-19: About the horizontal axis situated at a distance of 60 mm above the centre of gravity of the area.

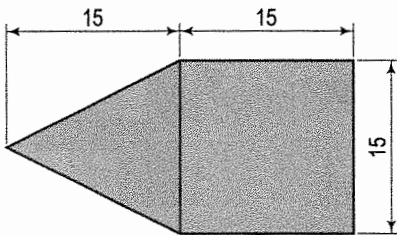


FIG. 21-28

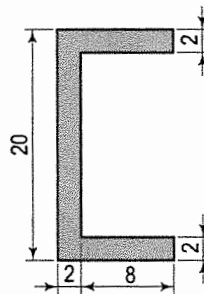


FIG. 21-29

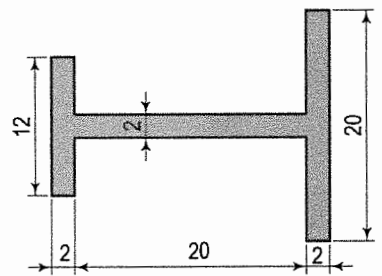


FIG. 21-30

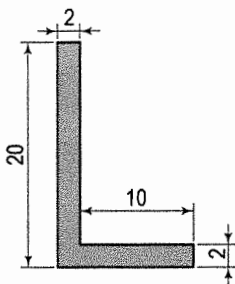


FIG. 21-31

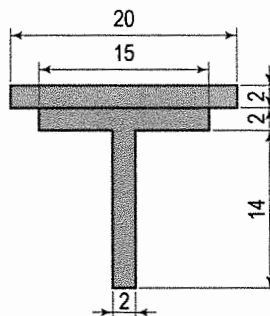


FIG. 21-32

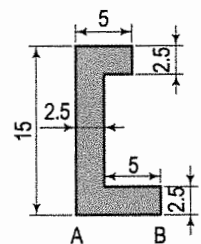


FIG. 21-33

Note: Dimensions in fig. 21-28 to fig. 21-33 are in centimetres.

Chapter

22



NOMOGRAPHY

22-0. INTRODUCTION



The term *nomography* is derived from the Greek words *nomos* (law) and *graphein* (to write) which refers to the graphical representation of mathematical laws or relationships. The law or relationship so represented is always in the form of a mathematical equation.

The *nomographs* which provide quick graphical answers where numerical data are substituted into formulae and *empirical equations* which are the mathematical expressions of data plotted in the form of curves.

A working knowledge of nomographs is useful not only to engineers but also to physicists, doctors and other scientific and professional groups. Applications of nomographs may be found in many phases of industry, such as production, management, sales, research and development. The nomographs are most advantageous when only an approximate answer, say within a few per cent is needed and the same equation is repeatedly required for several sets of values of the variables.

In this chapter, we shall learn about nomographs as shown below:

- (1) Types of nomographs
- (2) Definitions of various terms
- (3) Principle of construction of nomographs of three variables
- (4) Layout of nomographs
- (5) Z-type nomographs.

22-1. TYPES OF NOMOGRAPHS



Nomographs are combination of more than two straight or curved scales aligned in a definite way so that when a straight line drawn across the scales intersects them at values satisfying the represented equation. This straight line is known as *index line*. The nomographs are classified as shown below:

- (i) Parallel scales nomographs
- (ii) Z-type nomographs
- (iii) Circular nomographs
- (iv) Combination of nomographs.

The discussion of last two types of nomographs is beyond the scope of this book.

22-2. DEFINITIONS OF VARIOUS TERMS



The various terms used in the construction of nomographs are defined as under:

- (i) **Constant:** A constant is any quantity that always has the same value. For example

$$z = 3x + 4y, z = ax^2 + by^2 \text{ where } 3, 4, a \text{ and } b \text{ are constants.}$$

- (ii) **Variable:** A variable is any quantity that always has the different values. For example

$$z = 3x + 4y, z = ax^2 + by^2 \text{ where } x, y \text{ and } z \text{ are variables.}$$

- (iii) **Scales:** A graphical representation of physical quantity on a straight or a curve line of a selected length. The scales are further classified as follows:

- (a) *Uniform scales or linear scales or natural scales* are scales on which the spaces between division marks are constant.
- (b) *Logarithmic scales or non-linear scales* are scales in which the spaces between division marks are not uniform but vary according to the logarithms of the numbers that are represented on the scales.

- (iv) **Function:** This is a mathematical equation which expresses the relationship of a group of variables and constants. For example $y = x^2 + \frac{3}{x^3}$. The variable x and constant term 3 are related with y . In short form, we can write as $y = f(x)$.

- (v) **Functional modulus:** This is defined as the length of the scale for a unit value of the functional variable. Mathematically, we can write as

$$m = \frac{L}{f(x_{\max}) - f(x_{\min})} \dots \dots \dots (22-1)$$

where m is the functional modulus

$$L = \text{scale length.}$$

$x_{\max}$ and $x_{\min}$ represent the values of the variable x corresponding to the maximum and minimum values of the function respectively.

In equation (22-1), a convenient modulus can be chosen to simplify the construction of the scale and the length can be determined, which is a preferred procedure. Sometimes, it is difficult to select a functional modulus that will result in a scale length long enough or short enough to fit the paper size to be used. [Based on A4 (210 mm × 297 mm) size of paper.] In such cases, a suitable scale length can be assumed.

- (vi) **Scale modulus (M):** This is defined as the product of functional modulus and common constant coefficient of a function variable.

i.e. Scale modulus $M = \text{Constant coefficient of function} \times \text{Functional modulus}$

For example $z = 2x + 2y = 2(x + y)$

$\therefore M = 2 \times m.$

22-3. PRINCIPLE OF CONSTRUCTION OF NOMOGRAPHS OF THREE VARIABLES



In chapter 4, we have discussed how equation involving two variables can be represented in a graphical form by a chart of two scales side by side, one for each variable. Such a chart of two scales is also known as comparative scales or conversion scales. (Refer to chapter 4.)

For equations containing three or more variables, a graphical solution must have three or more scales, one for each variable. These scales are properly matched and spaced. Such charts are known as *Nomographs* or *Alignment charts*. The construction of nomographs for the equation with three variables, involving the sum, difference or multiplication of two variables, is based on the following geometrical derivation.

Consider the general equation of three variables

(i) $f(z) = f(x) \pm f(y)$ [22-2(a)]

(ii) $f(z) = f(x) \times f(y)$ [22-2(b)]

Equation of type (ii) is required to transform into type (i) by taking logarithms of the functions.

i.e. $f(z) = f(x) \times f(y)$

$\therefore \log f(z) = \log f(x) + \log f(y).$

The ranges of values are given for the variables x and y .

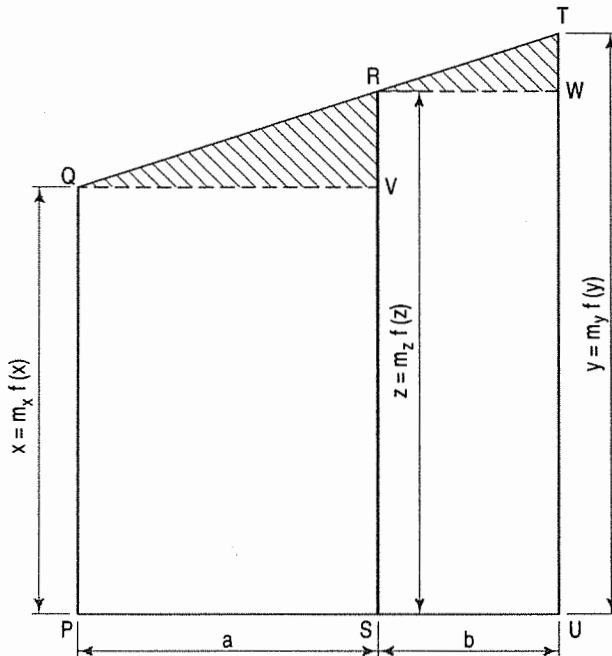


FIG. 22-1

In fig. 22-1, PQ , SR and UT represent three parallel scales. The origins are aligned by the horizontal line PU and an index line QT crosses the three scales in any direction. Lines QV and RW are drawn parallel to the horizontal line (base line) PU .

From similar triangles QVR and RWT

$$\frac{VR}{QV} = \frac{WT}{RW}$$

$$\text{i.e.} \quad \frac{SR - SV}{a} = \frac{UT - UW}{b}$$

$$\text{i.e.} \quad \frac{SR - PQ}{a} = \frac{UT - SR}{b} \quad (\text{as } SV = PQ \text{ and } UW = SR)$$

Substituting values of PQ , SR and UT from fig. 22-1

$$\frac{m_z f(z) - m_x f(x)}{a} = \frac{m_y f(y) - m_z f(z)}{b}$$

where m_x , m_y and m_z are functional moduli.

$$\therefore b m_z f(z) - b m_x f(x) = a m_y f(y) - a m_z f(z)$$

$$\therefore b m_z f(z) + a m_z f(z) = a m_y f(y) + b m_x f(x)$$

Divide both sides by b

$$\therefore m_z f(z) + \frac{a}{b} m_z f(z) = \frac{a}{b} m_y f(y) + m_x f(x)$$

$$\therefore m_z \left(1 + \frac{a}{b}\right) f(z) = m_x f(x) + \frac{a}{b} m_y f(y) \quad \dots\dots\dots (22-3)$$

But $f(x) + f(y) = f(z)$ only if the coefficients of the three terms in eq. (22-3) are equal.

$$\text{i.e.} \quad m_z \left(1 + \frac{a}{b}\right) = m_x = \frac{a}{b} m_y \quad \dots\dots\dots (22-4)$$

From equation (22-4), we have

$$m_x = \frac{a}{b} m_y \quad \text{i.e.} \quad \frac{a}{b} = \frac{m_x}{m_y} \quad \dots\dots\dots (22-4a)$$

$$\text{and} \quad m_x = \left(1 + \frac{a}{b}\right) m_z \quad \dots\dots\dots (22-4b)$$

Substituting eq. (22-4a) in eq. (22-4b), we have

$$m_x = \left(1 + \frac{m_x}{m_y}\right) m_z \quad \text{i.e.} \quad m_x = \left(\frac{m_x + m_y}{m_y}\right) \cdot m_z$$

$$\therefore m_z = \frac{m_x \times m_y}{(m_x + m_y)} \quad \dots\dots\dots (22-5)$$

Again from equation (22-4) we have

$$m_z \left(1 + \frac{a}{b}\right) = \frac{a}{b} m_y \quad \dots\dots\dots (22-5a)$$

Replacing value of m_z from eq. 22-5 in eq. 22-5a, we have

$$\begin{aligned} \frac{m_x \times m_y}{(m_x + m_y)} \left(1 + \frac{a}{b}\right) &= \frac{a}{b} m_y \\ \therefore a &= \frac{m_x (a + b)}{(m_x + m_y)} \quad \dots\dots\dots (22-6) \end{aligned}$$

Equation (22-5) is used to determine the modulus of the centre-scale. Equation (22-6) represents the relationship for spacing scales PQ and UT after selecting moduli for these scales.

22-4. METHOD OF CONSTRUCTING PARALLEL SCALE NOMOGRAPHS

For illustrations, following three types of equations are considered here:

- (1) Addition of two functional values,
say $z = x + y$, where x and y vary from 0 to 10.
- (2) Subtraction of two functional values,
say $z = x - y$, where both x and y variables vary from 0 to 10.
- (3) Multiplication of two functional values,
say $z = x \times y$, where both x and y variables vary from 0 to 10.

The problems of type-1 and type-2 are in the required standard form for three parallel scales. In such cases linear or natural scale can be used.

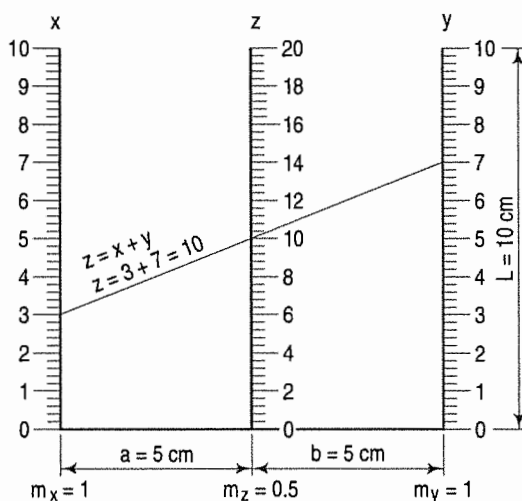


FIG. 22-2 (Refer table 22-1)

While the equation $z = x \times y$ of problem 22-3 is not in the required form of parallel scales. Therefore, it is necessary to transform in the required form by taking logarithms of the functions. That is $z = x \times y$, is transformed to $\log z = \log x + \log y$. Instead of, the natural scale, the logarithmic scale is used. The steps of construction of nomographs for above three types of problems are described in table 22-1, table 22-2, table 22-3 and corresponding fig. 22-2, fig. 22-3, fig. 22-4 respectively.

In plotting parallel scales nomographs following important points must be noted.

- (i) When positive values of variable are laid out in one direction (upward), negative values must be laid out in the opposite direction.
- (ii) Note that a function with a negative sign has an inverted scale and the origins are in alignment.

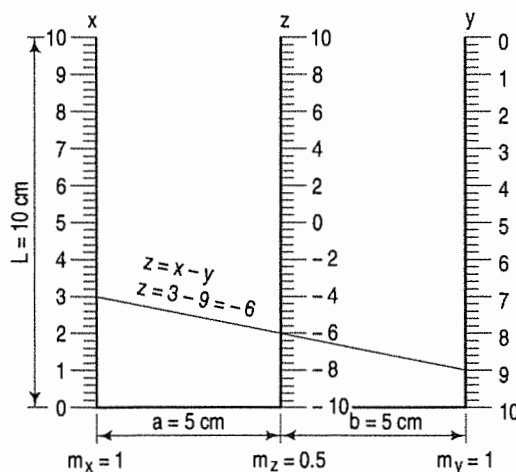


FIG. 22-3 (Refer table 22-2)

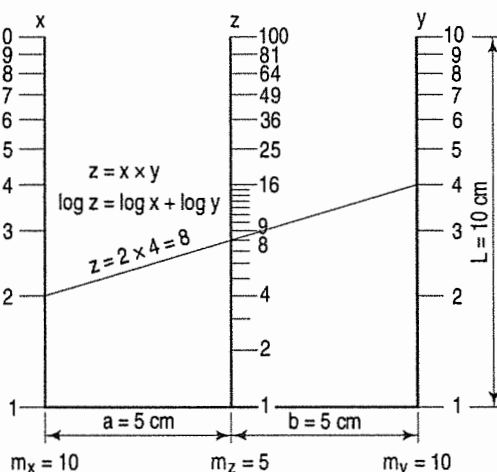


FIG. 22-4 (Refer table 22-3)

TABLE 22-1 (Fig. 22-2)

Step	Description	Computation																																				
(i)	Write down values of variables x and y	<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>y</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr></table>	x	0	1	2	3	4	5	6	7	8	9	10	y	0	1	2	3	4	5	6	7	8	9	10												
x	0	1	2	3	4	5	6	7	8	9	10																											
y	0	1	2	3	4	5	6	7	8	9	10																											
(ii)	Calculate the values of function z by substituting values of step (i)	$z = x + y$ <table><tr><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td><td>14</td><td>16</td><td>18</td><td>20</td></tr></table>	0	2	4	6	8	10	12	14	16	18	20																									
0	2	4	6	8	10	12	14	16	18	20																												
(iii)	To construct nomograph, select a suitable length of scale to fit on A4 size paper Also compute m_x , m_y and m_z	For x-scale, we select the length of scale, L as 10 cm. Using equation (22-1), $m_x = \frac{L}{f(x_{\max}) - f(x_{\min})}$ $= \frac{10}{10 - 0} = 1.$ The scale modulus M_x can be determined by multiplying m_x by constant coefficient of function x. $M_x = 1 \times m_x = 1 \times 1 = 1$. Similarly, we can determine the functional modulus m_y and scale modulus for the variable y, by selecting same scale length (10 cm). $\therefore m_y = 1, M_y = 1.$ Functional modulus m_z for the middle z-scale for variable z is determined by equation (22-5). $m_z = \frac{m_x \times m_y}{(m_x + m_y)} = \frac{1 \times 1}{1 + 1} = \frac{1}{2}$ $\therefore \text{The scale length for variable } z$ $= m_z [f(z_{\max}) - f(z_{\min})]$ $= \frac{1}{2} [20 - 0] = 10 \text{ cm.}$																																				
(iv)	Compute the horizontal distance of x -scale and y -scale from the middle z -scale. Draw three vertical parallel lines of 5 cm apart from each other, denoting x -scale, z -scale and y -scale.	Let a and b be the distances of x-scale and y-scale from z-scale. Using equation (22-6), $a = \frac{m_x(a+b)}{(m_x + m_y)},$ assume $(a+b) = 10$ cm on the basis of available width of the paper (21 cm). $\therefore a = \frac{1 \times 10}{1 + 1} = 5 \text{ cm}$ $\therefore b = 5 \text{ cm.}$																																				
(v)	Multiply functional moduli m_x , m_y and m_z by the functional values. These are measuring lengths for the graduations on the scale for the values of function.	$m_x = 1, m_y = 1$ and $m_z = 0.5$ <table><tr><td>$m_x \cdot x$</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>$m_y \cdot y$</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>$m_z \cdot z$</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr></table>	$m_x \cdot x$	0	1	2	3	4	5	6	7	8	9	10	$m_y \cdot y$	0	1	2	3	4	5	6	7	8	9	10	$m_z \cdot z$	0	1	2	3	4	5	6	7	8	9	10
$m_x \cdot x$	0	1	2	3	4	5	6	7	8	9	10																											
$m_y \cdot y$	0	1	2	3	4	5	6	7	8	9	10																											
$m_z \cdot z$	0	1	2	3	4	5	6	7	8	9	10																											
(vi)	Lay-off the distances obtained in step (v) on the x -scale, y -scale and z -scale.																																					
(vii)	Denote the values of variables (not function) at the corresponding points determined.																																					
(viii)	See fig. 22-2.																																					

TABLE 22-3 (Fig. 22-6)

Step	Description	Computation																																	
(i)	Write down values of variables x and y . Compute values of variable z .	<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>y</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>$z = x.y$</td><td>1</td><td>4</td><td>9</td><td>16</td><td>25</td><td>36</td><td>49</td><td>64</td><td>81</td><td>100</td></tr></table>	x	1	2	3	4	5	6	7	8	9	10	y	1	2	3	4	5	6	7	8	9	10	$z = x.y$	1	4	9	16	25	36	49	64	81	100
x	1	2	3	4	5	6	7	8	9	10																									
y	1	2	3	4	5	6	7	8	9	10																									
$z = x.y$	1	4	9	16	25	36	49	64	81	100																									
(ii)	Now convert eq. $z = x.y$ in the standard form by taking logarithms on both the sides i.e., $\log z = \log x + \log y$ and compute $\log x$, $\log y$ and $\log z$ of variables x , y and z	<table><tr><td>$\log x$</td><td>0</td><td>0.301</td><td>0.477</td><td>0.602</td><td>0.698</td><td>0.778</td><td>0.845</td><td>0.903</td><td>0.954</td><td>1</td></tr><tr><td>$\log y$</td><td>0</td><td>0.301</td><td>0.477</td><td>0.602</td><td>0.698</td><td>0.778</td><td>0.845</td><td>0.903</td><td>0.954</td><td>1</td></tr><tr><td>$\log z$</td><td>0</td><td>0.602</td><td>0.954</td><td>1.204</td><td>1.396</td><td>1.556</td><td>1.690</td><td>1.806</td><td>1.908</td><td>2</td></tr></table>	$\log x$	0	0.301	0.477	0.602	0.698	0.778	0.845	0.903	0.954	1	$\log y$	0	0.301	0.477	0.602	0.698	0.778	0.845	0.903	0.954	1	$\log z$	0	0.602	0.954	1.204	1.396	1.556	1.690	1.806	1.908	2
$\log x$	0	0.301	0.477	0.602	0.698	0.778	0.845	0.903	0.954	1																									
$\log y$	0	0.301	0.477	0.602	0.698	0.778	0.845	0.903	0.954	1																									
$\log z$	0	0.602	0.954	1.204	1.396	1.556	1.690	1.806	1.908	2																									
(iii)	Select suitable length for the x -scale, y -scale and z -scale. Determine the functional moduli m_x , m_y and m_z .	For x-scale and y-scale, we select the length of the scale as 10 cm. Hence the functional modulus $m_x = \frac{l}{f(x_{\max}) - f(x_{\min})} = \frac{10}{\log 10 - \log 1} = \frac{10}{1 - 0} = 10.$ Similarly, we can have the functional modulus m_y for the y-scale $m_y = \frac{l}{f(y_{\max}) - f(y_{\min})} = \frac{10}{\log 10 - \log 1} = \frac{10}{1 - 0} = 10.$ The functional modulus for the z-scale is determined by $m_z = \frac{m_x \times m_y}{(m_x + m_y)} = \frac{10 \times 10}{10 + 10} = \frac{100}{20} = 5.$																																	
(iv)	Same as table 22-3.	$x = \frac{m_x (a + b)}{(m_x + m_y)}, \text{ assume } (a + b) = 10$ $\therefore x = \frac{1 \times 10}{1 + 1} = 5 \text{ cm.} \quad \therefore b = 5 \text{ cm.}$ The length of z-scale is calculated as follows: $l = m_z [f(z_{\max}) - f(z_{\min})]$ $= 5 [\log 100 - \log 1] = 5 [2 - 0] = 10 \text{ cm.}$																																	
(v)	Draw three vertical parallel lines of length of 10 cm and 5 cm apart, representing x -scale, z -scale and y -scale. Multiply functional moduli m_x , m_y and m_z by the functional values. These are measuring length for the graduations on the scale for the values of function.	$m_x = 10, m_y = 10$ and $m_z = 5$ <table><tr><td>$m_x \log x$ cm</td><td>0</td><td>3.01</td><td>4.77</td><td>6.02</td><td>6.98</td><td>7.78</td><td>8.45</td><td>9.03</td><td>9.54</td><td>10</td></tr><tr><td>$m_y \log y$ cm</td><td>0</td><td>3.01</td><td>4.77</td><td>6.02</td><td>6.98</td><td>7.78</td><td>8.45</td><td>9.03</td><td>9.54</td><td>10</td></tr><tr><td>$m_z \log z$ cm</td><td>0</td><td>3.01</td><td>4.77</td><td>6.02</td><td>6.98</td><td>7.78</td><td>8.45</td><td>9.03</td><td>9.54</td><td>10</td></tr></table>	$m_x \log x$ cm	0	3.01	4.77	6.02	6.98	7.78	8.45	9.03	9.54	10	$m_y \log y$ cm	0	3.01	4.77	6.02	6.98	7.78	8.45	9.03	9.54	10	$m_z \log z$ cm	0	3.01	4.77	6.02	6.98	7.78	8.45	9.03	9.54	10
$m_x \log x$ cm	0	3.01	4.77	6.02	6.98	7.78	8.45	9.03	9.54	10																									
$m_y \log y$ cm	0	3.01	4.77	6.02	6.98	7.78	8.45	9.03	9.54	10																									
$m_z \log z$ cm	0	3.01	4.77	6.02	6.98	7.78	8.45	9.03	9.54	10																									
(vi)	Lay-off the distances obtained in the step (v) on the x -scale, y -scale and z -scale.																																		
(vii)	Denote the values of variables x , y and z (1 to 10, 1 to 10 and 1 to 100) at the corresponding points determined in the step (vi)																																		
(viii)	Refer to fig. 22-6.																																		

22-5. LAYOUT OF NOMOGRAPHS



Following illustrations of different types of nomographs reveal the application of principle of construction and layout of the nomographs.

Problem 22-1. Construct the nomograph to read the velocity of free-falling body for which the equation of motion is

$$V_f = V_o + gt$$

where

V_f = velocity of free-falling body

V_o = initial velocity (0 to 10 m/sec)

g = acceleration due to gravity (9.81 m/sec²)

t = time taken to attain V_f (0 to 10 sec).

- (i) Equation $V_f = V_o + gt$ can be expressed in the standard form

$$f(V_f) = f(V_o) + gf(t).$$

- (ii) Choose convenient lengths for parallel scales for both the variables V_o and t , say 20 cm.
- (iii) Determine the functional moduli and scale moduli, m_x , m_y , M_x and M_y for V_o and t scales. Use equation (22-1).

$$\begin{aligned} \text{(a) } m_x &= \frac{L}{f(V_o)_{\max} - f(V_o)_{\min}} \\ &= \frac{20}{[10 - 0]} = 2 \end{aligned}$$

$$\begin{aligned} M_x &= 1 \times m_x \\ &= 1 \times 2 = 2. \end{aligned}$$

$$\begin{aligned} \text{(b) } m_y &= \frac{L}{g[f(t)_{\max} - f(t)_{\min}]} \\ &= \frac{20}{9.81[10 - 0]} \\ &= 0.2038. \end{aligned}$$

$$\begin{aligned} M_y &= 9.81 \times m_y \\ &= 9.81 \times 0.2038 \simeq 2 \end{aligned}$$

- (c) Equation (22-5) can be used to determine the functional modulus m_z for the variable V_f .

$$m_z = \frac{m_x \times m_y}{(m_x + m_y)} = \frac{2 \times 0.2038}{2 + 0.2038} = 0.185.$$

$$M_z = 1 \times m_z = 1 \times 0.185 = 0.185.$$

- (d) The maximum range of V_f is determined by substituting maximum values of variables V_o and t in the equation

$$V_f = V_o + gt = 10 + (9.81 \times 10) = 108.1 \text{ m/sec.}$$

- (iv) Distance of V_o -scale and t -scale from the centre scale (V_f -scale) is determined by equation (22-6). Assume $(a + b) = 10.81$ cm.

$$\begin{aligned}\therefore a &= \frac{m_x (a + b)}{(m_x + m_y)} \\ &= \frac{2 \times 10.81}{(2 + 0.2038)} = 9.81.\end{aligned}$$

$$\therefore a = 9.81 \text{ cm and } b = 1 \text{ cm;}$$

- (v) Prepare table 22-4 as shown below:

TABLE 22-4

Equation	Variable	Function	Variable range	Functional modulus m	Scale modulus M	Scale length cm	Distance from centre scale cm	Plotting equations
$V_f = V_o + gt$	V_o	$f(V_o)$	0 to 10 m/sec	2	2	20	$a = 9.81$	$x = m_x f(V_o)$
	t	$g.f(t)$	0 to 10 sec	0.2038	2	20	$b = 1$	$y = m_y \cdot gf(t)$
	V_f	$f(V_f)$	0 to 108.1 m/sec	0.185	0.185	20		$z = m_z f(V_f)$

- (vi) Calculate plotting length for each variable taking suitable values within given ranges of the variables as shown in table 22-5:

TABLE 22-5

1 V_o m/sec	2 Plotting equation $x = 2f(V_o)$ cm	3 t sec	4 Plotting equation $y = 0.2038 \times 9.81 \times f(t)$ cm	5 V_f m/sec	6 Plotting equation $z = 0.185f(V_f)$ cm
0	0	0	0	0	0
1	2	1	2	10	1.85
2	4	2	4	20	3.70
3	6	3	6	30	5.55
4	8	4	8	40	7.40
5	10	5	10	50	9.25
6	12	6	12	60	11.10
7	14	7	14	70	12.95
8	16	8	16	80	14.80
9	18	9	18	90	16.65
10	20	10	20	108.1	20

- (vii) Draw three parallel lines of 20 cm representing V_o -scale, V_f -scale and t -scale separated at a distance of $a = 9.81$ cm and $b = 1$ cm from V_f -scale as shown in fig. 22-5.
- (viii) Mark divisions on V_o -scale, t -scale and V_f -scale, using columns 2, 4 and 6 of table 22-5.
- (ix) Complete the nomogram as shown in fig. 22-5. Note that a base line is the horizontal line passing through origins.

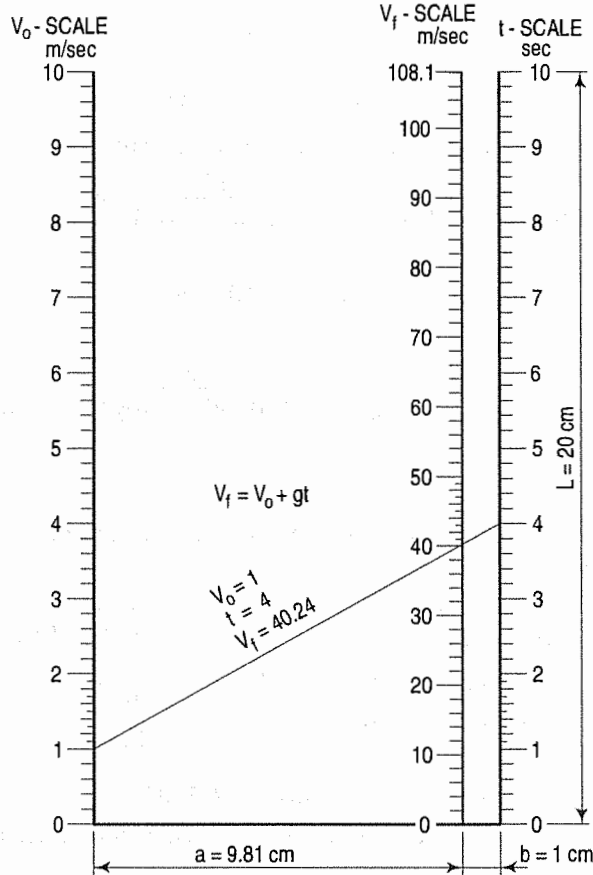


FIG. 22-5

Problem 22-2. Prepare a nomograph for finding moment of inertia of rectangular cross-section for which the equation is

$$I = \frac{bd^3}{12} \dots\dots\dots (i)$$

where I = moment of inertia of rectangular cross-section, cm^4

b = breadth of the cross-section which varies from 1 cm to 30 cm

d = width of the cross-section which varies from 1 cm to 30 cm.

Above equation (i) contains the product of two functions rather than the sum of two functions. In this form, it does not fit the parallel scales nomographs. However, above equation can be transformed into the required form by taking logarithms.

$$\log I + \log 12 = \log b + 3 \log d \dots\dots\dots (ii)$$

Equation (ii) is a standard form of three variable functions

i.e. $f(z) = f(x) + f(y)$

where $f(x) = f(\log b)$, $f(y) = 3 f(\log d)$

and $f(z) = f(\log I + \log 12)$

- (i) Choose scale length say 15 cm, for variables b and d .
- (ii) The minimum and maximum range of variable I can be calculated by substituting minimum and maximum values of b and d in equation (i).

$$I_{(\min)} = \frac{1}{12} bd^3 = \frac{1}{12} (1) (1)^3 = 0.0833 \text{ cm}^4.$$

$$I_{(\max)} = \frac{1}{12} bd^3 = \frac{1}{12} (30) (30)^3 = 67500 \text{ cm}^4.$$

(iii) Determine the functional moduli and scale moduli for b and d .

$$(a) \quad m_x = \frac{L}{[f(b)_{\max} - f(b)_{\min}]} = \frac{15}{\log 30 - \log 1} = \frac{15}{[1.4771 - 0]} = 10.155$$

$$M_x = 1 \times m_x = 1 \times 10.155 = 10.155.$$

(b) Similarly,

$$m_y = \frac{L}{[f(d)_{\max} - f(d)_{\min}]} = \frac{15}{3 \log 30 - 3 \log 1} = \frac{15}{4.4313 - 0} = 3.385$$

$$M_y = 3 \times m_y = 3 \times 3.385 = 10.155.$$

(c) The functional modulus for the functional variable $f(I)$ is determined as under:

$$m_z = \frac{m_x \times m_y}{(m_x + m_y)} = \frac{(10.155 \times 3.385)}{(10.155 + 3.385)} = 2.5387$$

OR

$$m_z = \frac{L}{\log f(I)_{\max} - \log f(I)_{\min}} = \frac{15}{\log 67500 - \log 0.0833} = 2.5387$$

$$M_z = 1 \times m_z = 1 \times 2.5387 = 2.5387.$$

(iv) To compute the horizontal distances of b -scale and d -scale from the center I -scale.

Let a be the distance between b -scale and I -scale

Let b be the distance between d -scale and I -scale

and assume $(a + b) = 15 \text{ cm}$

$$\therefore a = \frac{m_x (a + b)}{(m_x + m_y)} \dots \dots \dots \text{(using equation 22-6)}$$

$$= \frac{10.155 \times 15}{10.155 + 3.385} = 11.25 \text{ cm.}$$

$$\therefore a = 11.25 \text{ cm, } b = 3.75 \text{ cm.}$$

(v) Prepare table 22-6 as shown below.

TABLE 22-6

Equation	Variable	Function	Variable range	Functional modulus m	Scale modulus M	Scale length cm	Distance from centre scale cm	Plotting equations
$I = \frac{1}{12} bd^3$	b	$\text{Log } b$	1 to 30 cm	10.155	10.155	15	$a = 11.25 \text{ cm}$	$x = m_x \log b$
	d	$3 \text{Log } d$	1 to 30 cm	3.385	10.155	15	$b = 3.75 \text{ cm}$	$y = m_y 3 \text{Log } d$
	I	$(\text{Log } I + \text{Log } 12)$	0.0833 to 67500 cm ⁴	2.5387	2.5387	15		$z = m_z (\text{Log } I + \text{Log } 12)$

- (vi) Calculate plotting length for each variable taking suitable values within given range, as shown in table 22-7.
- (vii) Draw three parallel lines of 15 cm representing b -scale, I -scale and d -scale separated at a distance of $a = 11.25$ and $b = 3.75$ cm from I -scale as shown in fig. 22-6.
- (viii) Draw a horizontal base line through the initial marks 1, 0.0833 and 1 of b -scale, I -scale and d -scale respectively as shown in fig. 22-6.
- (ix) Mark division on b -scale, I -scale and d -scale from initial marks by taking distances from columns 3, 5 and 7 of table 22-7, then write values as given in columns 2, 4 and 6 of table 22-7 against the each division mark respectively. Note that the scale generated is logarithmic scale in which the spaces between division marks are not uniform.
- (x) Using above steps complete the nomograph as shown in fig. 22-6.

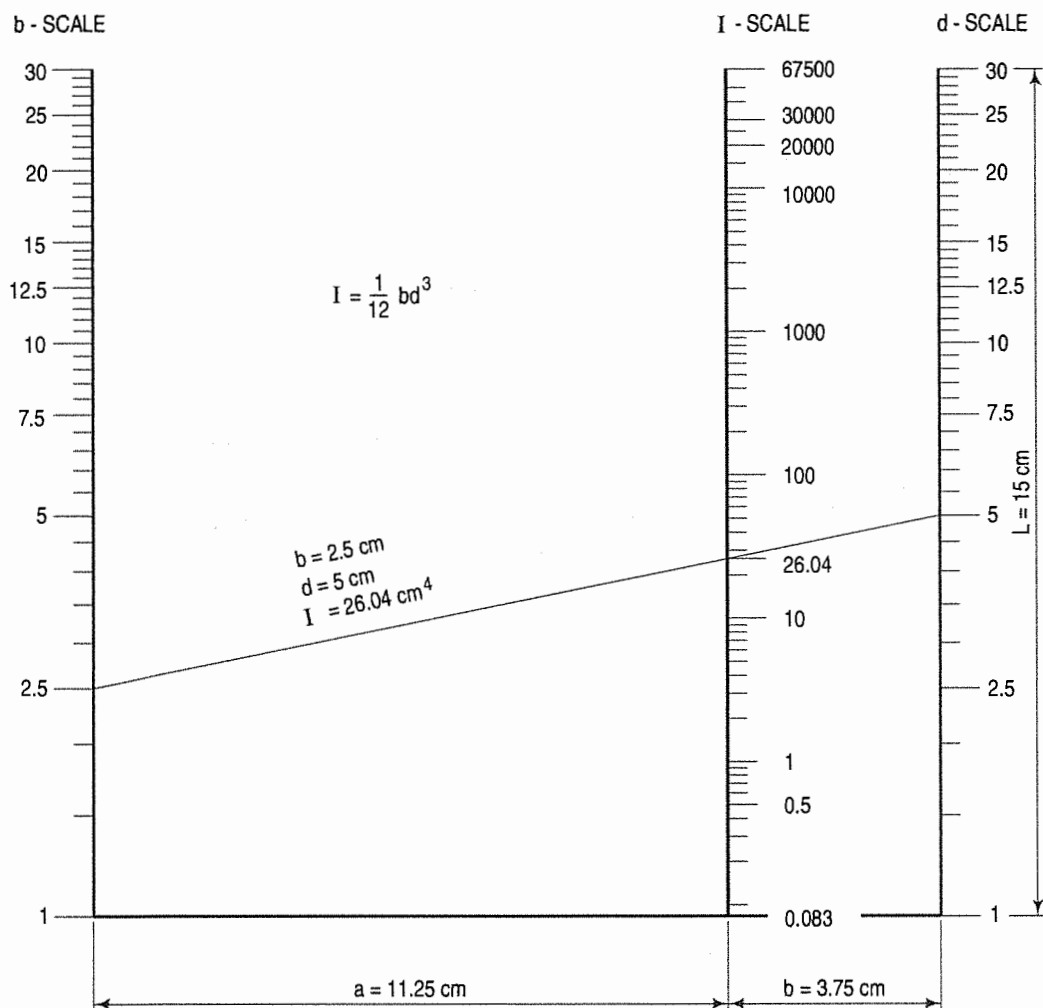


FIG. 22-6

TABLE 22-7

1	2	3	4	5	6	7
SR. NO.	b cm	Plotting equation $x = m_x \times \log b$ cm	d cm	Plotting equation $y = m_y \times 3 \log b$ cm	I cm ⁴	Plotting equation $z = m_z (\log I + \log 12)$ cm
1	1.0	0.00	1.0	0.00	0.083	0.00
2	2.5	4.04	2.5	4.04	10	5.28
3	5.0	7.10	5.0	7.10	100	7.82
4	7.5	8.89	7.5	8.89	1000	10.36
5	10.0	10.15	10.0	10.15	10000	12.89
6	12.5	11.14	12.5	11.14	20000	13.66
7	15.0	11.94	15.0	11.94	30000	14.11
8	20.0	13.21	20.0	13.21	40000	14.42
9	25.0	14.20	25.0	14.20	50000	14.67
10	30.0	15.00	30.0	15.00	67500	15.00

22-6. Z-TYPE NOMOGRAPHS

Certain nomographs for three variables can be plotted on non-parallel scales. For the equation of the type

$f(z) = \frac{f(x)}{f(y)}$ or $f(x) = f(y) \times f(z)$, in which Z is the dependent variable, a nomograph in the shape of Z can be prepared by using non-logarithmic scales.

The equation $f(x) = f(y) \times f(z)$ can be transformed into a form similar to equation (22-2) by taking logarithms. A method of constructing nomograph for such equations is illustrated as follows.

Problem 22-3. Construct a nomograph to read a section modulus of beam for which the equation is

$$Z = \frac{M}{\sigma_b}$$

where M = Bending moment (0 to 2×10^6 Nm)

σ_b = Maximum fibre stress (0 to 1.8×10^6 N/m²)

Z = Section modulus in cubic metre.

Remember that in the Z-shape nomograph, the scales for the variables must begin at zero, otherwise, a different and more complicated relationship would have been produced. The function is expressed as

$$f(z) = \frac{f(M)}{f(\sigma_b)}.$$

- (i) Select suitable length for variables M and σ_b , say 18 cm. The functional moduli for variables M and σ_b are determined by using the following equations:

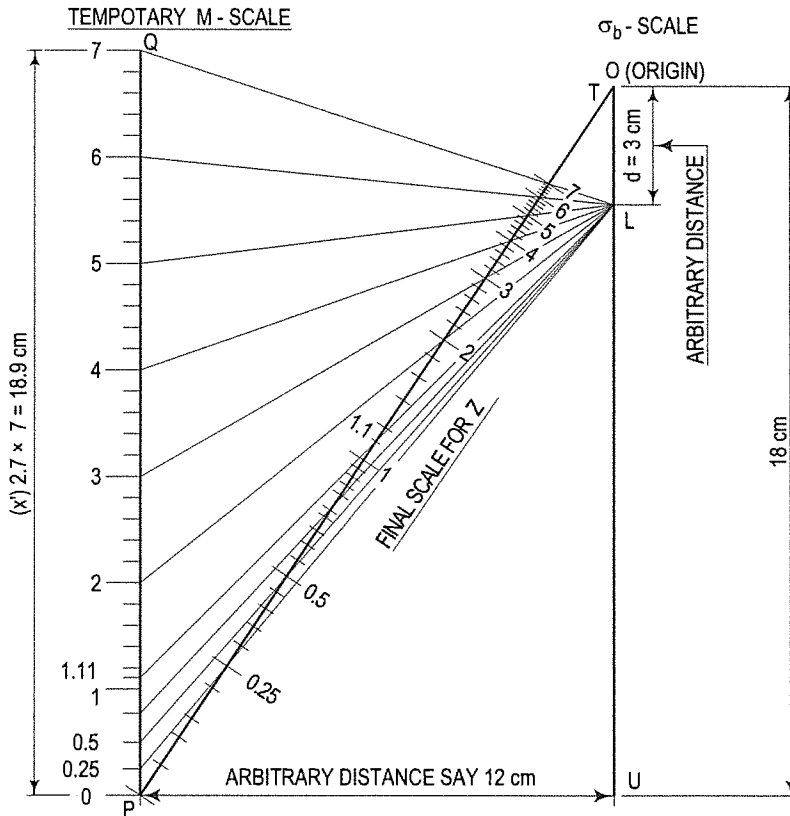
$$(a) \quad m_x = \frac{L}{[f(M)_{\max} - f(M)_{\min}]} = \frac{18}{[2 \times 10^6 - 0]} = 9 \times 10^{-6}$$

$$(b) \quad m_y = \frac{L}{[f(\sigma_b)_{\max} - f(\sigma_b)_{\min}]} = \frac{18}{[18 \times 10^5 - 0]} = 1 \times 10^{-5}$$

$$(c) \quad Z_{\min} = \frac{M}{\sigma_b} = 0.$$

$$Z_{\max} = \frac{M}{\sigma_b} = \frac{2 \times 10^6}{1.8 \times 10^6} = 1.11 \text{ m}^2.$$

- (ii) Before plotting final nomograph, it is necessary to prepare temporary scale to obtain diagonal scale. The method of constructing temporary scale is described as follows.



Construction of the temporary scale

FIG. 22-7

- (a) The plotting equation for temporary M -scale is

$$x' = \frac{m_x}{m_y} \times d \times Z$$

where d = Distance from the origin of σ_b -scale (3 cm selected here)

$$\therefore x' = \frac{9 \times 10^{-6}}{1 \times 10^{-5}} \times 3 \times Z = 2.7Z.$$

- (b) Draw two parallel lines PQ of 18.9 cm long and UT of 18 cm long for temporary M -scale and σ_b -scale with arbitrary distance say 12 cm between them. Join them by diagonal line PT to generate final scale for z as shown in fig. 22-7.
 - (c) Mark divisions on line of temporary M -scale. One unit will measure 2.7 cm. Join the division points to the L -point as shown in fig. 22-7. Intersections of these lines with the diagonal PT gives the required final scale for the variable Z .
- (iii) Prepare table 22-8 and table 22-9 as shown below.
- (iv) Redraw two parallel lines of 18 cm long for M -scale and σ_b -scale, 12 cm apart. Mark divisions on M -scale and σ_b -scale by taking distances from column 3 and 5 of table 22-9.
- (v) Redraw diagonal scale for z from fig. 22-7.
- (vi) Complete the nomograph as shown in fig. 22-8.

TABLE 22-8

Equation	Variable	Function	Variable range	Function modulus m	Scale modulus M	Scale Length cm
$Z = M/\sigma_b$	M	$f(M)$	0 to 2.0×10^6	9×10^{-6}	9×10^{-6}	18
	σ_b	$f(\sigma_b)$	0 to 1.8×10^6	1×10^{-6}	1×10^{-5}	18
	Z	$f(Z)$	0 to 1.11			

TABLE 22-9

1	2	3	4	5	6
SR. NO.	M (N/mm ²)	Plotting equation $x = m_x \cdot M$ cm	σ_b (N/mm ²)	Plotting equation $y = m_y \cdot \sigma_b$ cm	$Z = M/\sigma_b$
1	0	0	0	0	0
2	0.25×10^6	2.25	0.225×10^6	2.25	1.11
3	0.50×10^6	4.50	0.450×10^6	4.50	1.11
4	0.75×10^6	6.75	0.675×10^6	6.75	1.11
5	1.00×10^6	9.00	0.900×10^6	9.00	1.11
6	1.25×10^6	11.25	1.125×10^6	11.25	1.11
7	1.50×10^6	13.50	1.350×10^6	13.50	1.11
8	1.75×10^6	15.75	1.575×10^6	15.75	1.11
9	2.00×10^6	18.00	1.800×10^6	18.00	1.11

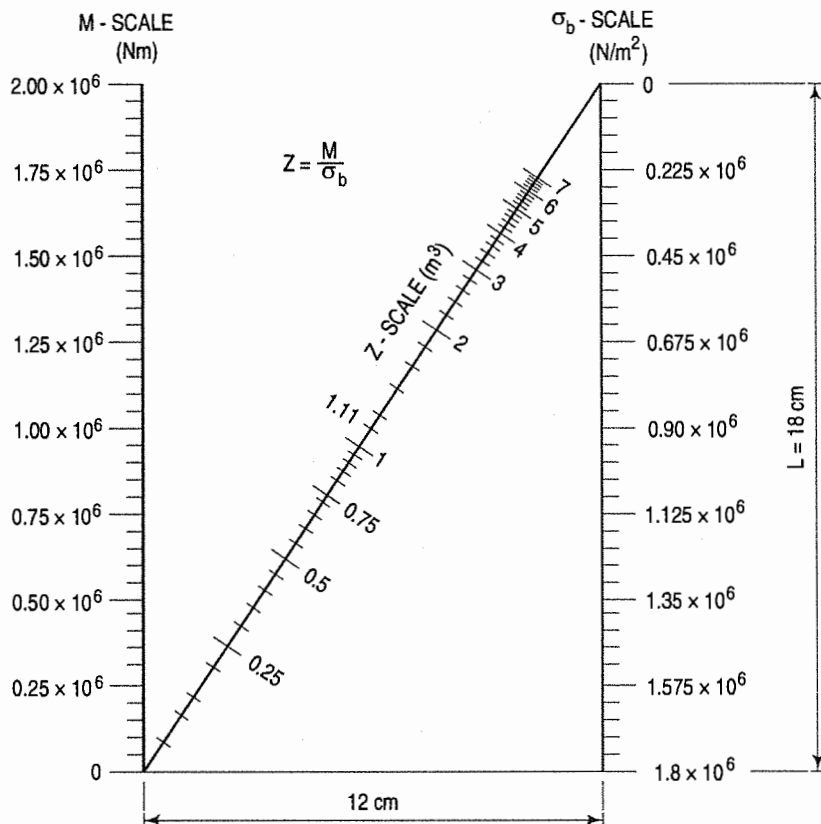


FIG. 22-8

EXERCISES 22

- Construct a nomograph to read the velocity of free-falling body for which the equation is

$$V_f^2 = 2gs$$

where

V_f = final velocity, m/sec

V_o = initial velocity (3 to 15 m/sec)

G = gravitational constant = 9.81 m/sec²

S = displacement of body, 2 m to 16 m.

- Prepare parallel scales nomographs to read rectangular moment of inertia for circular hollow cylinder

$$I = 0.049 [D^4 - d^4]$$

where

I = Moment of inertia, cm⁴

D = Outside diameter (1.5 to 15 cm)

d = Inside diameter (1 cm to 14 cm).

3. Prepare parallel scales nomograph to determine voltage through transmission line

$$V = I R$$

where

V = Potential (volts)

I = Electric current (0.1 to 10 amperes)

R = Resistance, 1 to 100 ohms.

4. A railway wagon runs at a speed of v m/sec. Its kinetic energy can be expressed as

$$\text{K.E.} = \frac{1}{2} M \cdot v^2$$

where

M = Mass of wagon

v = Speed of wagon.

Prepare alignment chart to read kinetic energy when mass of the wagon and the speed vary from 1000 kg to 50000 kg and 6 m/sec to 12 m/sec respectively.

5. Construct alignment chart (Z-type) to read coefficient of friction for which the equation is

$$\mu = \frac{F}{N}$$

where

F = Frictional force, N

μ = Coefficient of friction (0.15 to 0.6)

N = Normal force (25 N to 750 N).

6. Kinetic energy of rotating disk is given by

$$\text{K.E.} = \frac{1}{2} I \cdot \omega^2$$

where

I = mass moment of inertia (kg-m^2)

ω = angular velocity, rad/sec.

I and ω vary from 10 to 80 kg-m^2 and 5 rad/sec to 30 rad/sec respectively. Construct alignment chart.

7. Frequency of compound pendulum is given by expression

$$f = \frac{1}{2\pi} \sqrt{\frac{gh}{k^2 + h^2}}$$

where

f = frequency cycle/second

g = gravitational constant = 9.81 m/sec^2

h = distance of the centre of gravity from the suspension point (0.1 m to 0.5 m)

k = radius of gyration (0.5 m to 1.2 m)

Prepare a nomograph to read frequency.

Chapter

23



SCREW THREADS

23-0. INTRODUCTION



If a cylindrical rod is rotated at a constant speed, and simultaneously, if a pointed tool, just touching the rod is moved parallel to the axis of the rod at a constant speed, the cut made by the tool on the rod will be continuous and of a helical form. A screw thread is formed by cutting a helical groove on a cylindrical surface. *The threaded rod is called a screw.* It engages in a corresponding threaded hole inside a nut or a machine part. The screws are used for joining two parts temporary. Therefore such a joint is known as the detachable joint or temporary joint.

Threads are generally cut on a machine called a *lathe*. On a small-size screw, thread is often cut by means of a tool called a *die*. A small-size hole is threaded by means of a tool called a *tap*. Such a hole is called a *tapped hole*.

23-1. DEFINITIONS



Various parts of a screw thread are shown in fig. 23-1 and defined below.

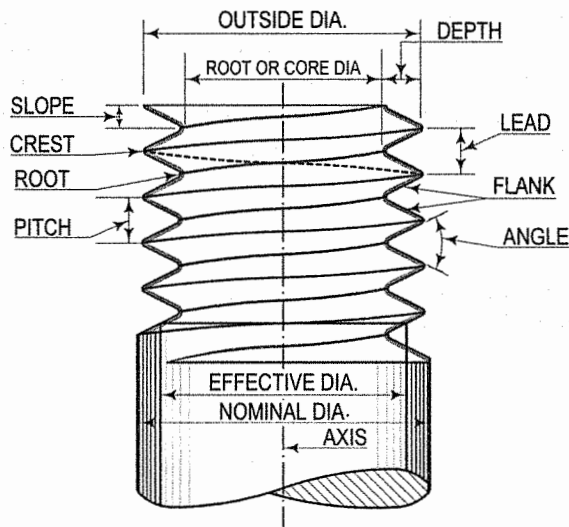
- (1) **Crest:** The crest is the outer-most part of a thread.
- (2) **Root:** The root is the inner-most portion of a thread.
- (3) **Flank:** The surface between the crest and the root is called the flank of the thread.
- (4) **Angle:** It is the angle between the flanks, measured on an axial plane.
- (5) **Depth:** The depth is the distance between the crest and the root, measured at right angles to the axis. It is equal to half the difference between the outside diameter and the core diameter.
- (6) **Nominal diameter:** It is the diameter of the cylindrical piece on which the thread is cut. The screw is specified by this diameter.
- (7) **Outside or major diameter:** It is the diameter at the crest of the thread measured at right angles to the axis of the screw.
- (8) **Core or minor diameter:** It is the diameter at the core or root of the thread. It is the smallest diameter of the screw and is equal to the outside diameter minus twice the depth of the thread.

(9) **Effective diameter:** It is equal to the length of the line, perpendicular to and passing through the axis, and measured between the points where it cuts the flanks of the thread.

(10) **Pitch:** It is the distance measured parallel to the axis, between a point on one thread form and a corresponding point on the adjacent thread form, i.e. from crest to crest or root to root. It may also be described as the reciprocal of the number of thread forms per unit length, i.e. $p = \frac{1}{n}$.

(11) **Lead:** It is the distance measured parallel to the axis from a point on a thread to a corresponding point on the same thread after one complete revolution. It can also be described as the distance moved by a nut in the axial direction in one complete revolution. The lead is equal to the pitch in case of single-start threads. See table 23-5.

(12) **Slope:** The slope of a thread is equal to half the lead.



Screw thread

FIG. 23-1

23-2. FORMS OF SCREW THREADS

The two main forms of screw threads are

- (i) triangular or V thread [fig. 23-2(i)] and
- (ii) square thread [fig. 23-2(ii)].

Other forms are either modified forms of a square thread or a combination of the two forms. A thread, cut on the surface of a rod, is called an *external thread*, while that cut in a hole, is called an *internal thread*.

Threads should be cut so accurately that nuts having a particular form and size of thread, freely engage with screws having the same form and size of thread. The thread-cutting tool should, therefore, be shaped in an exact manner according to the dimensions of the required thread form.

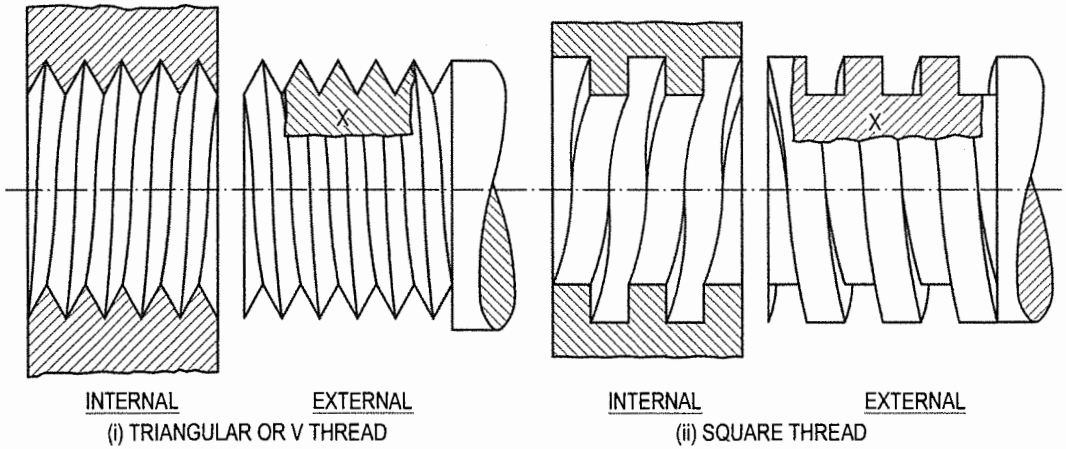
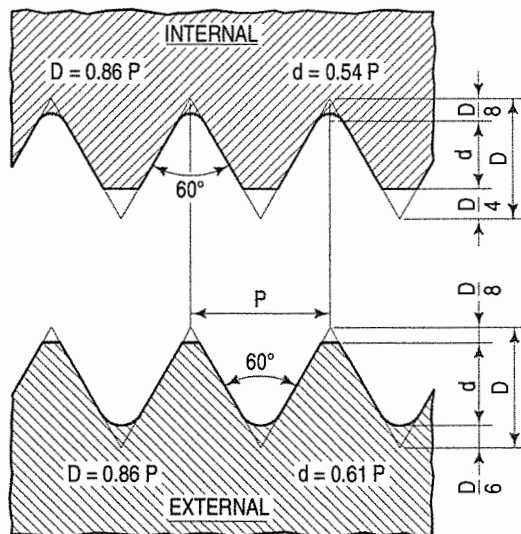


FIG. 23-2

The figures accompanying the description of the various forms of screw threads are shown in section and on an enlarged scale, e.g. fig. 23-4 and fig. 23-7 show, on an enlarged scale, the sectional portions (marked X in fig. 23-2) of the V thread and the square thread respectively.

23-2-1. TRIANGULAR OR V THREADS

(1) **Unified thread** (fig. 23-3): In 1947, the International Organization for Standardization (I.S.O.) of which India, U.S.A., United Kingdom, Canada and a number of other countries are members, came into being. It decided to adopt the Unified screw thread profile as the I.S.O. basic profile. It also decided to recognize two separate I.S.O. series based on inch and metric systems of measurement, with this common basic profile for threads.



Unified thread

FIG. 23-3

In this form of thread, the external thread (on a screw) varies slightly in shape from the internal thread (inside a nut) as can be seen from fig. 23-3. The angle of the thread is 60°. Roots of both — internal and external threads are rounded, while the crests are cut parallel to the axis of the screw. The root of the internal thread is rounded *within* the depth of $\frac{D}{8}$ as shown in the figure. See table 23-1.

TABLE 23-1

Types of thread	Thread angle	Theoretical depth D	Actual depth d
External thread	60°	$0.866 P$	$0.61 P$
Internal thread	60°	$0.866 P$	$0.54 P$

The maximum depth of engagement between the external and the internal threads is $\frac{5}{8} D$.

Table 23-2 gives useful data of Unified screw thread based on inch system:

(2) **Metric thread** (fig. 23-3): The Bureau of Indian Standards has recommended the adoption of the Unified screw thread profile based on metric system as a standard form for use in India, and has designated it as Metric screw thread with I.S.O. profile. In this system, the pitch of the thread (instead of the number of threads per unit length) is fixed.

Metric thread is designated by the letter M followed by the nominal diameter, e.g. M 20, where 20 is the nominal diameter of the screw in millimetres. Refer to IS: 4218:1976 and IS: 11698:1986.

Table 23-3 gives the values of pitch and core diameters for screws of 6 mm to 39 mm diameters:

(3) **Whitworth thread** (fig. 23-4): This form of thread is also known as British Standard Whitworth (B.S.W.) thread and has been adopted as a standard form in the United Kingdom.

The angle is 55°. The theoretical depth $D = 0.96P$, where P is the pitch of the thread. $\frac{1}{6}$ of the theoretical depth is rounded off at the top and at the bottom.

Therefore, the actual depth $d = 0.64P$.

Table 23-4 gives useful data of B.S.W. threads for screws of diameters $\frac{1}{4}$ " to 2":

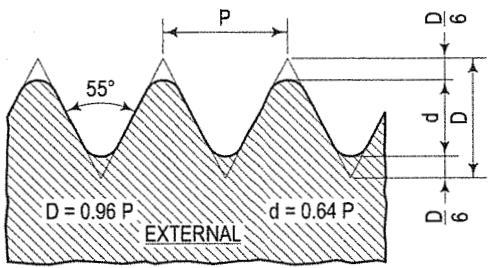
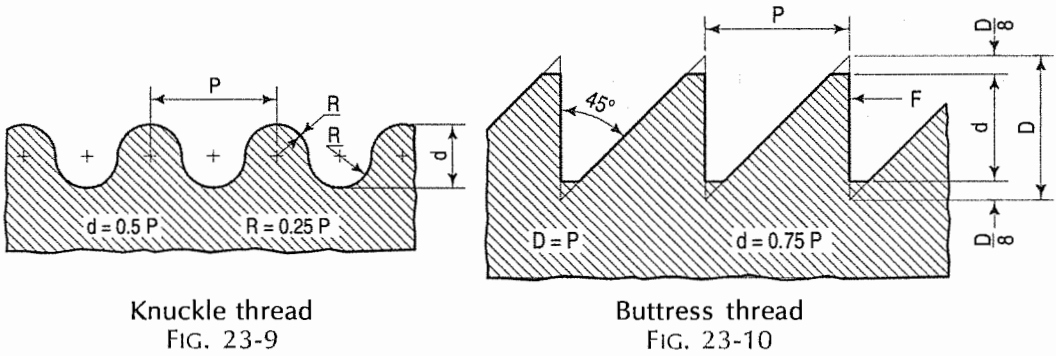


FIG. 23-4

(4) **British Standard Fine (B.S.F.) and British Standard Pipe (B.S.P.) threads:** These have the same Whitworth profile but their pitches are finer and hence, the depths smaller. Thus, they have large effective and core diameters than the B.S.W. threads. *B.S.F. threads are generally used in automobile and aircraft work. B.S.P. threads are used for gas, steam or water pipes. They are specified by the bore of the pipe and not by the outside diameter.* Thus, the outside diameter of a threaded pipe having a bore of 1" nominal diameter is 1.309". Pipes of 1" to 6" diameters have the same number of threads per inch, viz. 11. The Bureau of Indian Standards recommends pipe threads according to IS: 2643:1975.



23-3. CONVENTIONAL REPRESENTATION OF THREADS

SP: 46-2003

The true form of a screw thread is helical and it would take considerable time and labour to draw the same. In actual practice, threads are usually shown by conventional methods.

Method 1:

This conventional method is recommended for use by the Bureau of Indian Standards for Metric screw threads and square threads for which Standards have been published.

(1) External threads: These threads in outside view [fig. 23-11(i)] are shown by means of two continuous thin lines drawn parallel to the axis, thus indicating the minor or root diameter of the threads. The limit of the length of the thread is shown by a continuous thick line drawn perpendicular to the axis and upto the major or outside diameter of the threads. The runout of the thread is shown by lines drawn at an angle of 30° or 45° to the axis. The runout may not be shown if there is no likelihood of any misunderstanding.

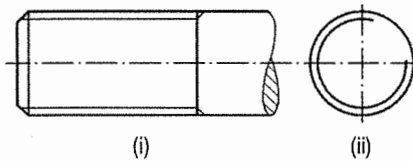


FIG. 23-11

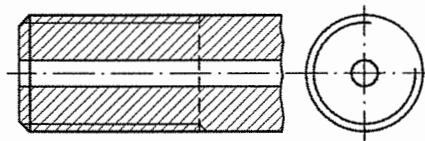


FIG. 23-12

External thread in section (fig. 23-12) is shown in the same manner, except that the limit of the length of the thread is shown by a medium dashed line. The section lines (hatching) are drawn crossing the thin lines.

(2) Internal threads: These threads in outside view [fig. 23-13(i)] are shown by medium dashed lines indicating major and minor diameters.

Internal thread in section [fig. 23-13(ii)] is shown by continuous thick and thin lines indicating respectively the minor and major diameters. The section lines are drawn crossing the thin lines.

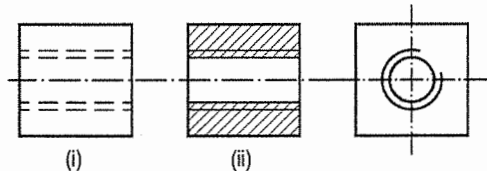


FIG. 23-13

In the side view of the external thread [fig. 23-11 and fig. 23-12] the minor or root diameter is represented by a part of a continuous thin-line circle about three quarters of the circumference; while in the side view of the internal thread (fig. 23-13) the major or outside diameter is shown in the similar manner.

In sectional views where threaded parts are assembled together, externally threaded parts are always shown covering the internally threaded parts as shown in fig. 23-14.

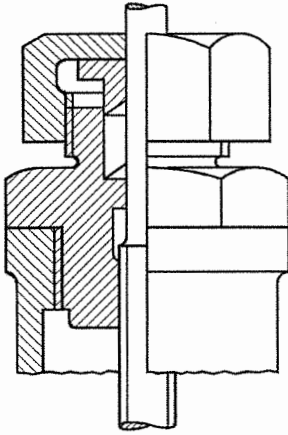


FIG. 23-14

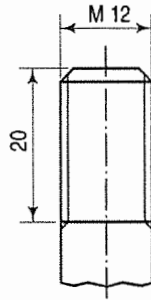


FIG. 23-15

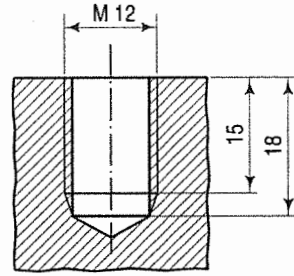


FIG. 23-16

The method of designating and dimensioning Metric screw threads is shown in fig. 23-15 and fig. 23-16. In case of square thread (fig. 23-17), the type, size and pitch are clearly indicated.

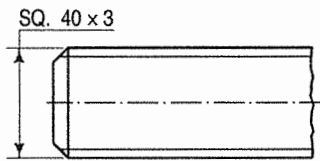


FIG. 23-17

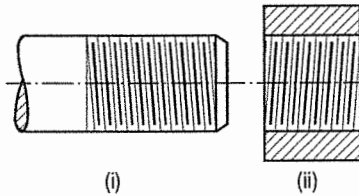


FIG. 23-18

Method II:

(1) **External V thread:** This thread in its outside view is shown by sloping straight lines alternately thin and thick, and spaced one-half the pitch apart [fig. 23-18(i)]. The slope is kept equal to half the lead. The thick lines are drawn shorter than the thin lines by a distance equal to one-half the pitch on each side, thus indicating the roots of the thread.

(2) **Internal V thread:** In its outside view, this thread is shown as described in method I. In its sectional view, it is shown by thick and thin lines [fig. 23-18(ii)], the slope of the lines being opposite to that of the external thread, because the other side is visible.

(3) **External square thread:** This thread is represented by sloping parallel lines spaced one-half the pitch apart as shown in fig. 23-19(i). The depth of the thread is also equal to one-half the pitch. In case of a square thread of large size, the back portion of the thread is also drawn as shown in fig. 23-20.

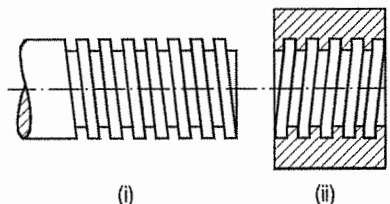


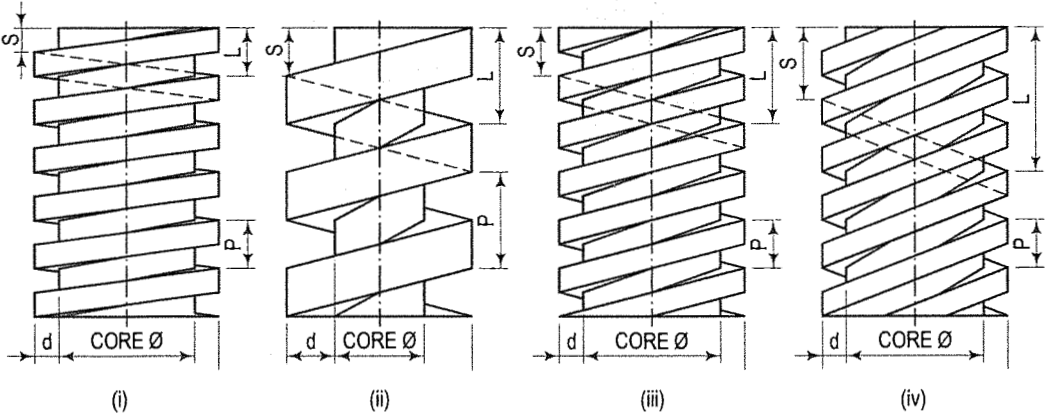
FIG. 23-19

(4) Internal square thread: In its outside view, this thread also is shown as described in method I. In its sectional view [fig. 23-19(ii)] it is shown by parallel lines sloping in opposite direction. The depth of the thread is shown at right angles to the axis.

The present practice is to follow B.I.S. recommendations. Students must show threads in their drawings by method I only.

23-4. MULTIPLE-START THREADS

In a single-start thread, the pitch is equal to the lead [fig. 23-20(i)]. Since the depth of the thread is dependent on the pitch, greater the lead, greater will be the depth of the thread and smaller will be the core diameter. When a nut is required to move a considerably long axial distance in one revolution (i.e. when the lead is large), the core diameter of the screw, in a single-start thread, will be so much reduced as to make the screw too weak [fig. 23-20(ii)]. This is avoided by cutting what are known as multiple-start threads, in which two or more threads having the same pitch as in a single-start thread, but with increased lead, run parallel to one another.



Single-start and multiple-start threads
FIG. 23-20

The pitch being the same, the depth of the thread remains the same as in a single-start thread and the core diameter also remains unaffected. The depth of the thread in each case remains the same, i.e. $D = \frac{1}{2} P$, while the slope S is equal to one-half the corresponding lead L . The relationship between lead and pitch is shown in table 23-5.

TABLE 23-5

No. of start	Pitch, P	Lead, L
Single start	P	$L = P$
Double start	P	$L = 2P$
Triple start	P	$L = 3P$
In general, 'N' start	P	$L = N \cdot P$

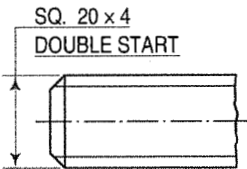


FIG. 23-21

Conventional method of designating and dimensioning multiple-start threads is shown in fig. 23-21.

23-5. RIGHT-HAND AND LEFT-HAND THREADS

If a nut, when turned in clockwise ↻ direction screws on a bolt, the thread is a right-hand thread; but if it screws off the bolt when turned in the same direction, the thread is said to be a left-hand thread. Both these types of threads are shown in fig. 23-22. Note that when the axis of the screw is vertical, the lines slope downwards from right to left in case of the right-hand thread. They slope in the reverse direction, i.e. from left to right downwards when it is a left-hand thread. For indicating a left-hand thread, an abbreviation L.H. is used. Unless otherwise stated, a thread should always be assumed to be a right-hand one.

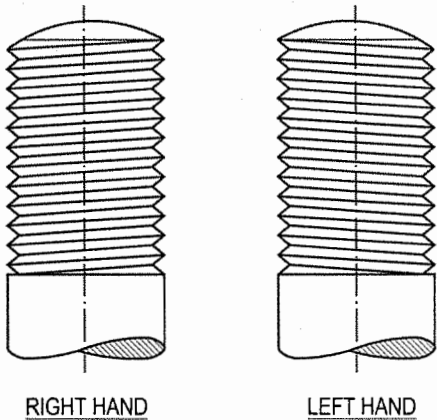


FIG. 23-22

Conventional method of designating and dimensioning a left-hand thread is shown in fig. 23-23.

Practical application of these threads is made in a coupler-nut or turn-buckle shown in fig. 23-24. The length of a tie-bar can be adjusted by this nut. Looking from the right, if the nut is turned in clockwise ↻ direction, the ends of the rods will move closer to each other. They will move further apart when the nut is turned in anti-clockwise ↺ direction.

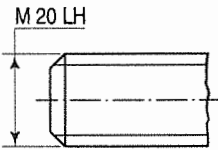
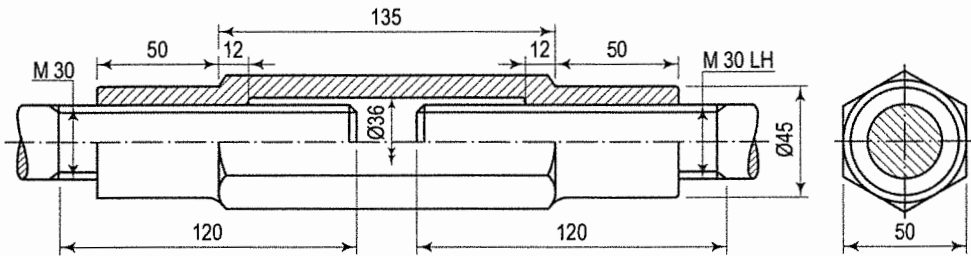


FIG. 23-23



Coupler-nut
FIG. 23-24

**EXERCISES 23**

1. Define the following terms used in connection with a screw thread:
Core diameter; outside diameter; crest; flank; depth; pitch. Show each on a sketch of the threaded end of a screw.
2. Sketch neatly, any three types of profiles of triangular threads. State the angle of the thread in each case and dimension the depth, assuming a pitch of 10 mm.
3. Give dimensioned sketches of the following forms of screw threads:
Acme, Buttress; Unified; Whitworth.
4. Show by means of neat and dimensioned sketches, the difference in profiles of internal and external Unified threads.
5. State the difference between a right-hand and a left-hand thread. Illustrate by means of a sketch, the use of both on the same piece.
6. What is the lead in a screw thread? How does it differ from pitch in a double-start thread? Illustrate your answer by means of a sketch of a square-threaded screw.
7. Show by means of neat sketches, the following threads conventionally:
 - (a) External V thread.
 - (b) External V thread in section.
 - (c) Internal V thread in section.
8. Show by means of sketches, conventional methods of showing the following screw threads:
 - (a) Metric thread, nominal diameter 12 mm.
 - (b) Square thread, nominal diameter 20 mm and pitch 3 mm.
 - (c) Left-hand square thread, nominal diameter 20 mm and pitch 3 mm.
 - (d) Double-start square thread, nominal diameter 40 mm and pitch 3 mm.
9. Prepare a neat sketch of a coupler-nut connecting a tie-bar of 25 mm diameter.
10. Fill-up the blanks in the following sentences with appropriate words selected from the list of words given below:
 - (a) A screw thread is formed by cutting a _____ groove on a cylindrical surface.
 - (b) The thread cut on the surface of a rod is called _____ thread, while that cut on the surface of a hole is called _____ thread.
 - (c) In a small-size hole the thread is cut by a tool called _____.
 - (d) Large-size threads are cut on a machine called _____.
 - (e) The diameter of the screw at the crests is called _____ diameter, while that at the roots is called _____ diameter.
 - (f) The depth of the thread is equal to half the difference between the _____ diameter and the _____ diameter.

- (g) The slope of the thread is equal to one-half its _____.
- (h) In a double-start thread the lead is equal to _____ the pitch.
- (i) The angle between the flanks of Whitworth thread is _____, while that in case of Unified thread is _____.
- (j) The _____ of Unified threads are rounded while its _____ are cut parallel to the axis of the screw.
- (k) The square thread is used for _____ transmission. Its flanks make _____ angles with the axis.
- (l) _____ thread is used in the lead screw of the lathe.
- (m) Knuckle thread is a modified form of _____ thread.
- (n) The combination of square and V threads is made in _____ thread which is used in the screw of _____.

List of words for Exercise 10:

Acme; Bench-vice; Buttress; Conical; Crests; Die; Drill; Effective; External; Flanks; Half; Helical; Internal; Lathe; Lead; Major; Minor; Nominal; Pitch; Power; Roots; Sellers; Slope; Square; Tap; Twice; Unified; V; 29°; 47.5°; 55°; 60°; 90°.

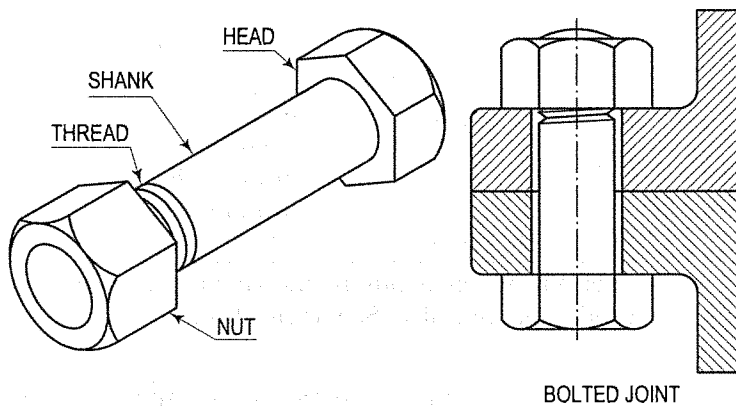
Chapter 24



SCREWED FASTENINGS

24-0. INTRODUCTION

A machine is an assembly of different parts arranged in definite order and is used to transform energy for doing some useful work. The connection between two parts can be either temporary or permanent. The temporary joints are (i) Screwed joints, (ii) Keys, cotter and pin joints, (iii) Pipe joints. As the parts thus connected can be easily separated by screwing off the nut or removing the cotter or the pin, the fastening is said to be temporary. In permanent joints, the connected parts cannot be easily separated. They are riveted and welded joints.



Screw pair
FIG. 24-1

A nut and a screw or a bolt comprise what is known as a screw pair (fig. 24-1). Such a pair is used for fastening together, parts used in engineering construction.

24-1. TYPES OF NUTS

Nuts are generally in the form of hexagonal or square prisms. Besides these, cylindrical and other forms are also used to suit particular requirements.

Between the hexagonal and the square nuts, the former is generally given preference. The spanner used for turning the nut can have better hold on a square nut than on a hexagonal nut, but the angle through which the spanner will have to be turned to get another hold is only 60° in case of a hexagonal shape, while it is 90° in case of a square shape. Hence, it is more convenient to screw-on a hexagonal nut than a square nut. An

octagonal nut would require only $\frac{1}{8}$ turn, but there would be greater tendency for the spanner to slip. Hence, this shape is seldom used. Nuts of forms other than the above two are usually provided with special facilities for screwing them on or off the bolts.

24-1-1. HEXAGONAL NUT (FIG. 24-2)

The upper corners of this nut are rounded-off or chamfered. The chamfering is generally conical. The angle of chamfer is 30° to 45° with the base of the nut. Due to chamfering, an arc is formed on each vertical face, and a circle is formed on the top surface of the nut.

The dimensions of the hexagonal nut cannot be expressed exactly in terms of the nominal diameter of the bolt.

The standard proportions for nuts and bolt-heads may be obtained from the standard tables published by B.I.S. For elementary work, the following approximately standard dimensions may be adopted (fig. 24-2(i)). See table 24-1.

Let D = the nominal diameter of the bolt.

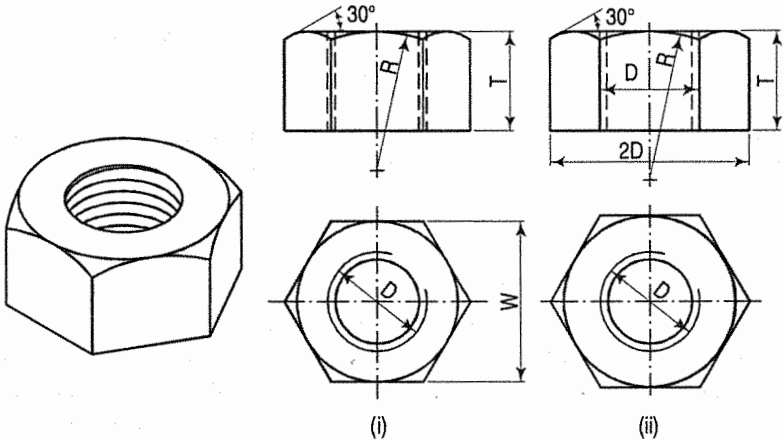
TABLE 24-1
APPROXIMATE STANDARD DIMENSIONS

No.	Description	Dimensions
1.	Height or thickness of the nut, T	$T = D$
2.	Width across flats, W	$W = 1.5D + 3 \text{ mm}$
3.	Angle of chamfer	30°
4.	Radius of chamfer arc, R	$R = 1.4D \text{ (approx.)}$

TABLE 24-2
ROUGH RULE DIMENSIONS

No.	Description	Dimensions
1.	Height or thickness of the nut, T	$T = D$
2.	Distance across diagonally opposite corners	$2D$
3.	Angle of chamfer	30°
4.	Radius of chamfer arc, R	$R = 1.5D \text{ (approx.)}$

Very often, and especially when a nut is shown in one view only, the rough-rule dimensions are used [fig. 24.2(ii)]. See table 24-2.



Hexagonal nut
FIG. 24-2

Problem 24-1. To draw three views of a hexagonal nut for a bolt diameter D , by approximately standard dimensions (table 24-1) (fig. 24-3).

Step 1: Begin with the top view. Draw a circle of diameter W equal to $1.5D + 3$ mm. Circumscribe a regular hexagon about this circle with two sides horizontal. Project the front view of the hexagonal prism, taking T equal to D . Project the circle to points $A-A$ on the upper face. Through points $A-A$, draw lines AB inclined at 30° to the upper face. Draw the line BB cutting the edges of the central face at points $C-C$.

Step 2: Draw the central arc passing through points $C-C$ and touching the line AA , as shown by construction lines. Its radius R will be approximately equal to $1.4D$. Similarly, draw arcs passing through points B and C and touching the line AA . The radius R_1 and the centre for these arcs may be found by trial.

Step 3: Project the side view. Only two faces of the nut will be seen. The distance between the outer edges will be equal to W . Project the line BB and obtain points $E-E$ on the vertical edges. Draw arcs in each face passing through points $E-E$ as shown.

Step 4: Finalise the three views by adding two circles in the top view and dashed lines for the screwed hole in the front and side views. The diameter of the inner circle will be equal to the core diameter. The outer circle is kept thinner and partly cut, approximately 75% according to convention.

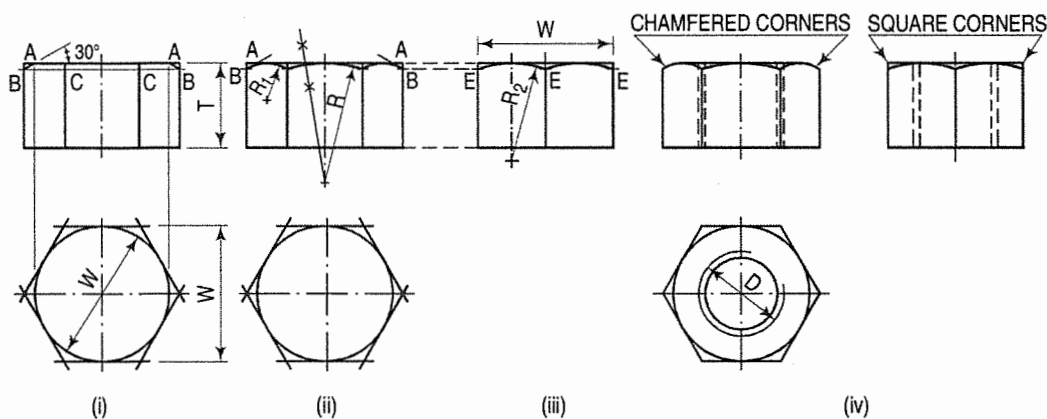


FIG. 24-3

Note carefully that in the front view where three faces of the nut are seen, the upper outer corners should be shown chamfered. When only two faces are visible, as in the side view, the corners must be shown square. The remaining vertical edges are seen terminated at the arcs on the chamfered face.

Alternative method (assuming $R = 1.4D$):

Draw the top view and project the three faces of the prism in the front view. With radius R , draw an arc CC in the central face. Draw a line through points $C-C$, cutting the outer lines at points $B-B$. Draw arcs in each outer face, passing through the points B and C and touching the upper line. Complete the front view by drawing chamfer lines AB at an angle of 30° . Project the side view as already described.

Problem 24-2. To draw three views of a hexagonal nut by rough-rule dimensions (table 24-2) (fig. 24-4).

Step 1: Draw a circle of diameter equal to $2D$ and inscribe a regular hexagon in it, with two sides horizontal. Inscribe the chamfer circle inside the hexagon. Its diameter, and the width W , across the flat sides will be equal to $\sqrt{3} \times D$. Project the front view and draw arcs as described in problem 24-1. The radius R will be approximately equal to $1.5D$.

Step 2: Complete the three views as shown in the figure.

Alternative method (assuming $R = 1.5D$):

As the distance across corners is equal to $2D$, the width of the central face in the front view will be equal to D .

Beginning can therefore be made with the front view taking the distance between the outer lines to be equal to $2D$ and that between the inner lines to be equal to D . Draw the central arc with radius R equal to $1.5D$ and then complete the front view as explained in the alternative method of problem 24-1. Draw the top view and the side view as shown in fig. 24-4(ii).

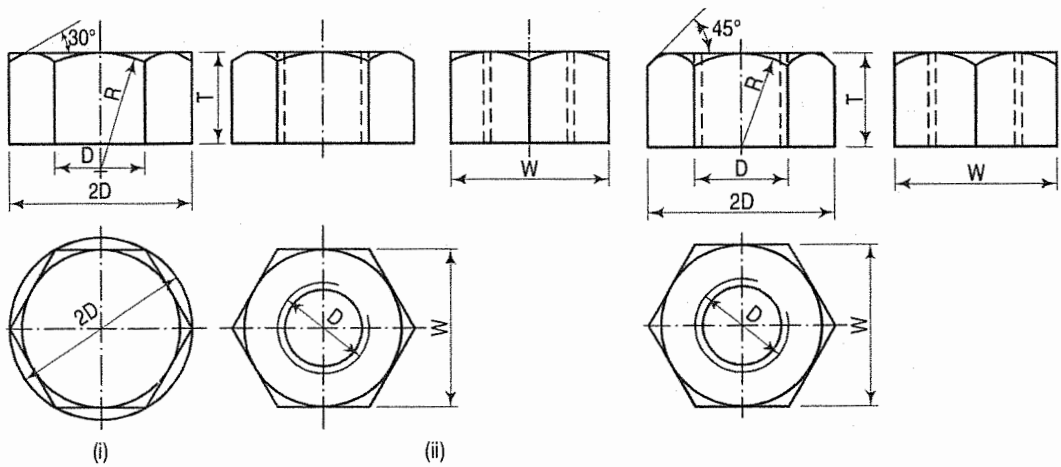


FIG. 24-4

FIG. 24-5

Fig. 24-5 shows three views of the nut having 45° chamfer and drawn according to the alternative method. Note that the radius R for the arc in the central face is equal to D .

24-1-2. SQUARE NUT (FIG. 24-6)

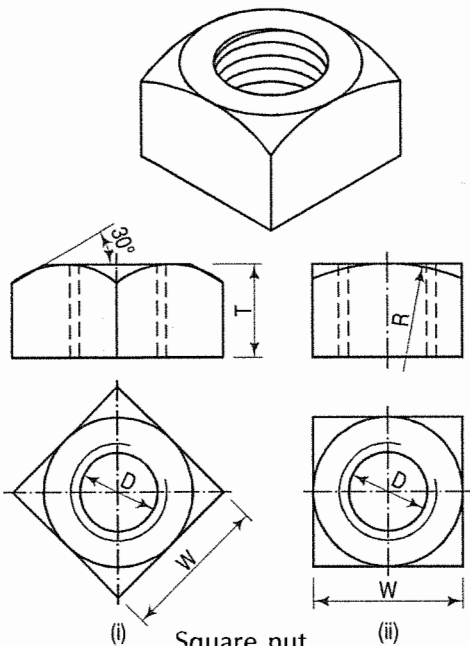
The upper corners of the square nut are also chamfered in the same manner as those of the hexagonal nut. The widths across flats of a square nut and a hexagonal nut, for the same size of bolt, are also equal.

Let D = the nominal diameter of the bolt.

Problem 24-3. To draw the front view and the top view of a square nut. Use table 24-3.

TABLE 24-3
SQUARE NUT DIMENSIONS

No.	Description	Dimensions
1.	Height or thickness of the nut, T	$T = D$
2.	Width across flats, W	$W = 1.5D + 3 \text{ mm}$
3.	Angle of chamfer	30°
4.	Radius of chamfer arc, R	$R = 2D \text{ (approx.)}$



(i) Square nut
FIG. 24-6

(ii)

Step 1: When its two faces are equally seen in the front view [fig. 24-6(i)].

Draw a square in the top view with a side equal to W , i.e. $1.5D + 3$ mm, and all sides equally inclined to the horizontal. Complete the top view by drawing the chamfer circle and circles for the screwed hole.

Project the two faces in the front view and complete the view as explained in case of the hexagonal nut.

Step 2: When only one face of the nut is seen in the front view [fig. 24-6(ii)].

The top view is the same except that two sides of the square are horizontal.

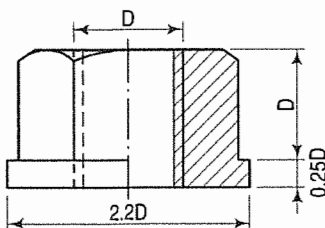
Project the rectangle in the front view and draw the chamfer arc with radius R equal to $2D$.

24-2. TYPES OF NUTS FOR SPECIAL PURPOSE

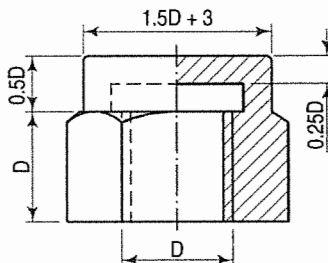
The following different types of nuts are used for special purpose:

(1) **Flanged nut** (fig. 24-7): This is a hexagonal nut with a washer, i.e. a flat circular disc attached to it. It is thus provided with a larger bearing surface. A bolt can be used in a comparatively large-size hole with the help of this nut. It is widely used in automobiles.

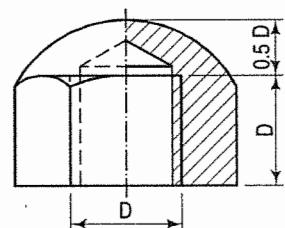
(2) **Cap nut** (fig. 24-8): It is also a hexagonal nut provided with a cylindrical cap at the top to protect the end of the bolt from corrosion. It also prevents leakage through the threads.



Flanged nut
FIG. 24-7



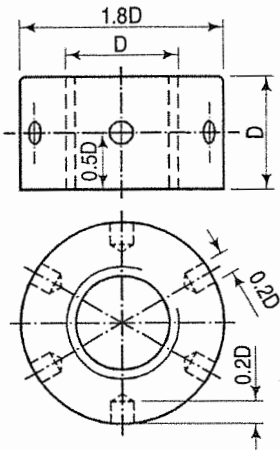
Cap nut
FIG. 24-8



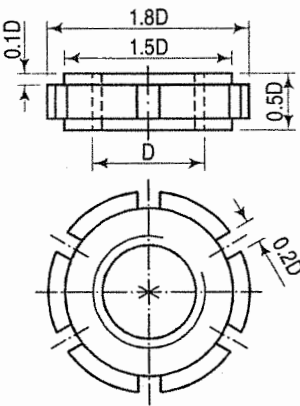
Dome nut
FIG. 24-9

(3) **Dome nut** (fig. 24-9): It is a form of a cap nut having a spherical dome at the top.

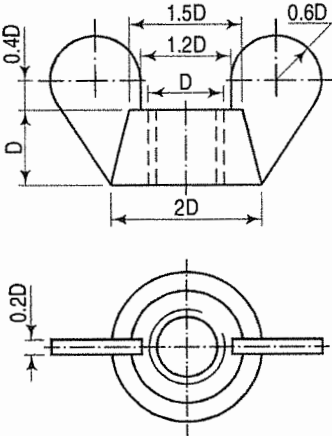
(4) **Cylindrical or capstan nut** (fig. 24-10): This nut is cylindrical in shape. Holes are drilled in the curved surface, for turning it with a tommy bar. Sometimes, holes are drilled in the upper face, for use of a pin-spanner for turning it.



Capstan nut
FIG. 24-10



Ring nut
FIG. 24-11



Wing nut
FIG. 24-12

(5) **Ring nut** (fig. 24-11): It is in the form of a ring provided with slots in the curved surface for a special C-spanner. These nuts are generally used in pairs, one nut acting as a lock-nut for the other.

(6) **Wing nut** (fig. 24-12): This nut can be easily operated by the thumb and a finger and is used where it is required to be adjusted frequently. It is used in a hacksaw.

24-3. WASHERS (FIG. 24-13)

A washer is a cylindrical piece of metal placed below the nut to provide smooth bearing surface for the nut to turn on. It spreads the pressure of the nut over a greater area. It also prevents the nut from cutting into the metal and thus, allows the nut to be screwed-on more tightly. It is sometimes chamfered on the top flat surface.

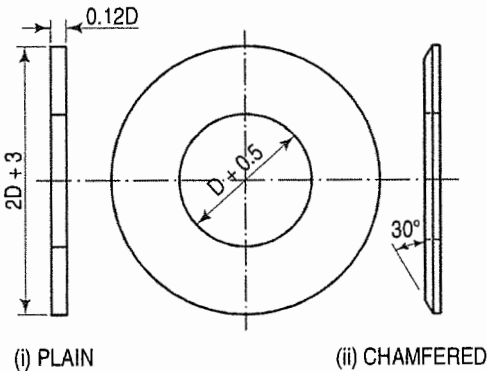
Let D = the nominal diameter of the bolt.

Fig. 24-13(i) shows two views of a plain washer. A chamfered washer is shown in fig. 24-13(ii). See table 24-4.

Washers can also be applied as locking arrangements for nuts. Fig. 24-63 to fig. 24-65 illustrates the *spring washer* in the art. 24-7 of this book.

TABLE 24-4
APPROXIMATE DIMENSIONS OF A WASHER

No.	Description	Dimensions
1.	Diameter of washer	$2D + 3$ mm
2.	Thickness	$0.12D$
3.	Angle of chamfer	30°
4.	Diameter of hole	$D + 0.5$ mm



Washers
FIG. 24-13

24-4. BOLTS (FIG. 24-1)

A bolt comprises of two parts — a shank and a head. The shank is cylindrical and is threaded at the tail end for a sufficient length so as to effectively engage with a nut. The shape of the head depends upon the purpose for which the bolt is required. While considering the length of the bolt, the thickness of the head is not taken into account.

Methods of preventing rotation of a bolt while screwing a nut on or off it:

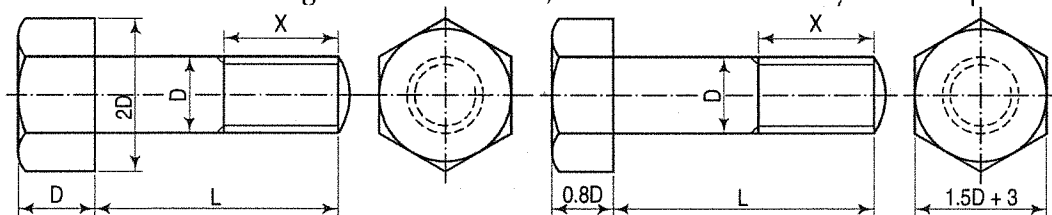
When it is not possible to hold a bolt-head by means of a spanner, the bolt is prevented from rotating by the provision of one of the following, below the bolt-head:

- (i) a square neck (ii) a pin (iii) a snug.

These are shown and described below while dealing with various forms of bolts.

24-5. FORMS OF BOLTS

(1) **Hexagonal-headed bolt** (fig. 24-14 and fig. 24-15): This is the most common form of a bolt. The hexagonal head is chamfered at its upper end. To prevent rotation of the bolt while screwing the nut on or off it, the bolt-head is held by another spanner.



Hexagonal-headed bolt
– rough-rule dimensions
FIG. 24-14

Hexagonal-headed bolt
– approximately standard dimensions
FIG. 24-15

The dimensions of the bolt-head are the same as those of the hexagonal nut, except for the thickness. For elementary work, the thickness is taken as $0.8D$ to D . Fig. 24-14 and fig. 24-15 show two views each of a hexagonal-headed bolt drawn according to *rough-rule dimensions* and *approximately standard dimensions* respectively. Note that the length of the face of the bolt-head is equal to D in fig. 24-14, while it is less than D in fig. 24-15 and X is the length of thread.

Problem 24-4. To draw three views of a hexagonal-headed bolt, 24 mm diameter and 100 mm long, with a hexagonal nut and a washer.

- (i) Assume rough-rule dimensions (fig. 24-16).
- (ii) Determine the dimensions of the nut and the washer as shown below by referring table 24-2, table 24-4 and fig. 24-13.

Dimensions for the nut:

Thickness of the nut, $T \underline{\underline{24}}$ mm

Distance across diagonally
opposite corners = 48 mm

Angle of chamfer = 30°

Radius of chamfer arc, $R = 36$ mm.

Dimensions for washer:

Diameter of the washer $\underline{\underline{2D + 3 = 51}}$ mm

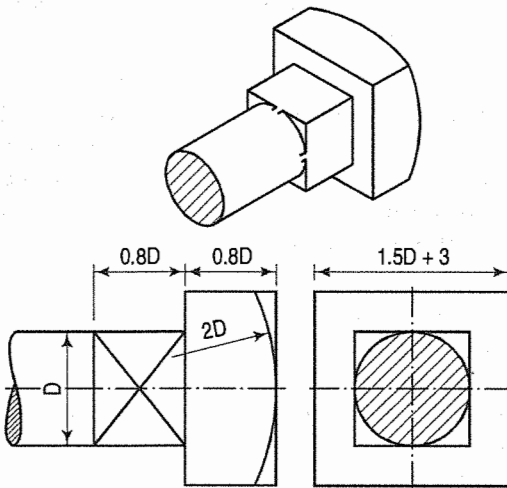
Thickness $\underline{\underline{0.12D = 2.88 \underline{\underline{3}}}}$ mm

Diameter of the hole $\underline{\underline{D + 0.5 = 24.5}}$ mm.

Thickness of bolt-head = $0.8D$ to D

Width across flats = $1.5D + 3$ mm.

When a square-headed bolt is to be used with its head projecting outside, it is provided with a neck of square cross-section (fig. 24-18), which fits into a corresponding square hole in the adjoining part. This prevents rotation of the bolt.



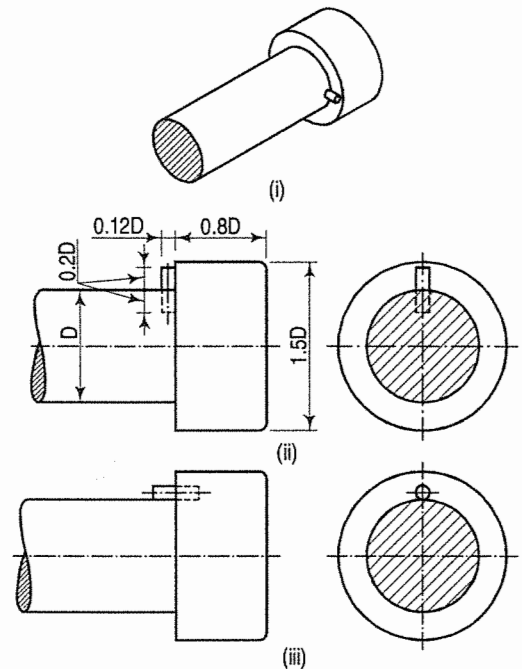
Square-headed bolt (with square neck)

FIG. 24-18

(3) **Cylindrical or cheese-headed bolt** (fig. 24-19): This bolt is used when the space for accommodating the bolt-head is comparatively limited, or where the use of a spanner for holding it is to be avoided. It is commonly used in big ends of connecting rods, eccentrics etc.

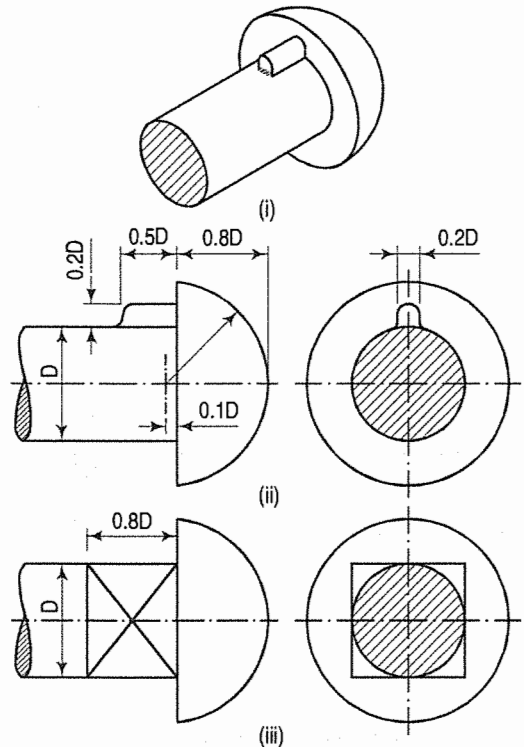
The rotation of the bolt is prevented by means of a pin inserted into the shank just below the head [fig. 24-19(ii)]. The projecting part of this pin fits into a corresponding groove in the adjacent piece. This pin is very often inserted in the bolt-head as shown in fig. 24-19(iii).

(4) **Cup-headed or round-headed bolt** (fig. 24-20): It is provided with a snug forged on the shank just below the head [figs. 24-20(i) and 24-20(ii)]. It fits into a corresponding recess in the adjacent piece to prevent rotation of the bolt. This bolt is often provided with a square neck as shown in fig. 24-20(iii).



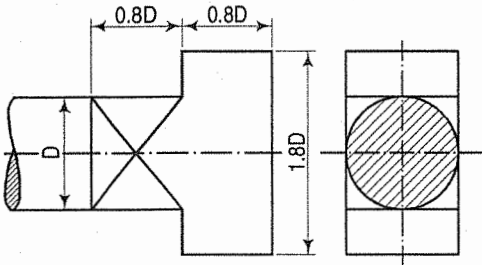
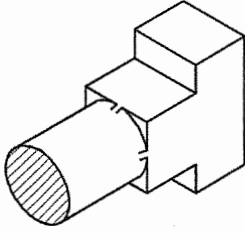
Cylindrical or cheese-headed bolt

FIG. 24-19

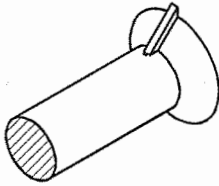


Cup-headed bolt

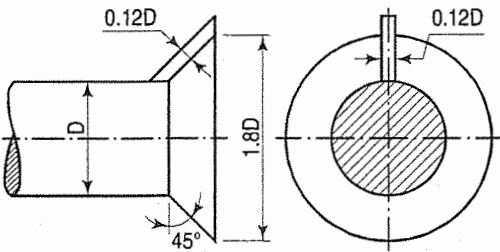
FIG. 24-20



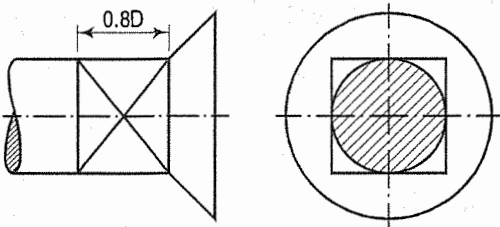
T-headed bolt
FIG. 24-21



(i)



(ii)



Countersunk-headed bolt
FIG. 24-22

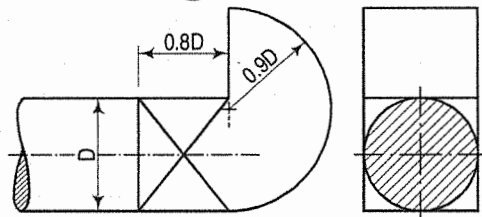
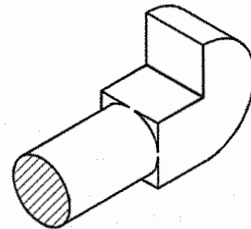
(8) **Headless tapered bolt** (fig. 24-24):

Its shank is tapering and it has no head. It is used mainly in marine shaft couplings.

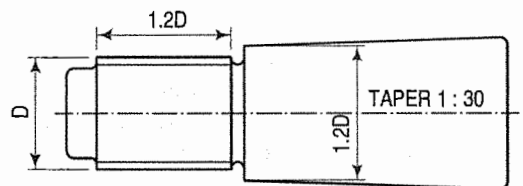
(5) **T-headed bolt** (fig. 24-21): It is used in machine-tool tables in which T-slots are cut to accommodate the T-heads. The neck of this bolt also is usually square in section. The T-headed bolt is often made use of in gland and stuffing box arrangement in boiler mountings such as stop valve, feed-check valve etc. In that case, the square neck becomes unnecessary and hence, it is not provided.

(6) **Countersunk-headed bolt** (fig. 24-22): Where the head of a bolt must not project above the surface of the connected piece, this form of bolt is used. It may be provided with a snug [fig. 24-22(ii)] or a neck of square cross-section [fig. 24-22(iii)].

(7) **Hook bolt** (fig. 24-23): This bolt passes through a hole in one piece only, while the other piece is gripped by the hook-shaped bolt-head. It is used when it is not possible to drill a hole in the piece adjoining the bolt-head. The square neck prevents rotation of the bolt.



Hook bolt
FIG. 24-23

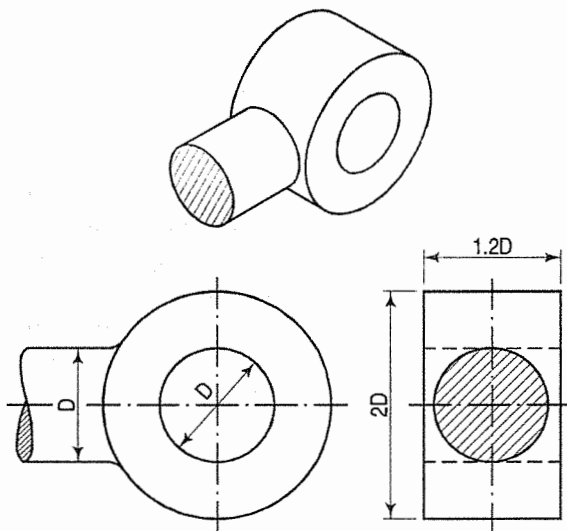


Headless tapered bolt
FIG. 24-24

(9) **Eye-bolt** (fig. 24-25): This bolt has a circular ring of rectangular cross-section as its head, which can be conveniently held to prevent its rotation.

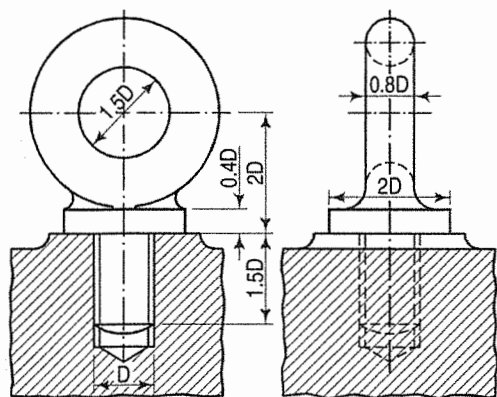
(10) **Lifting eye-bolt** (fig. 24-26): It is used as an appliance for lifting heavy machines. It is screwed inside a threaded hole on the top of the machine, directly above its centre of gravity.

(11) **Tap-bolt or cap-screw** (fig. 24-27): It is a bolt used as a screw, i.e. screwed into a threaded hole in a casting instead of a nut. It is used when it is not possible to accommodate the nut. It passes loosely through a clear hole in the piece A, adjoining the bolt-head, and is screwed into a threaded hole in the casting B.

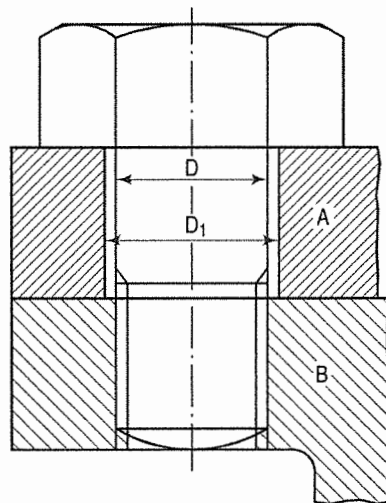


Eye-bolt
FIG. 24-25

Frequent insertion or removal of the tap-bolt is likely to damage the threads in the casting. Owing to this disadvantage, this method of fastening is employed only when parts are not to be disconnected very often. Tap-bolts have various forms of heads, similar to those of set-screws as shown in fig. 24-34 to 24-42. They are used for connecting a cylinder-head with a cylinder of an internal combustion engine.



Lifting eye-bolt
FIG. 24-26

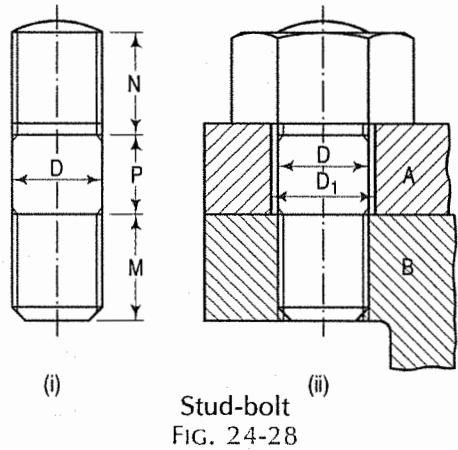


Tap-bolt or cap-screw
FIG. 24-27

(12) **Stud-bolt or stud** [fig. 24-28(i)]: It consists of only a cylindrical shank threaded at both ends. The nut-end N is threaded for a length slightly more than the thickness of a nut or nuts to be used. The other end M, called the metal-end is threaded for a length at least equal to the diameter of the stud. The length

of the plain part P , between the two ends, depends upon the thickness of the piece adjoining the nut. The stud is used in place of a bolt, when there is insufficient space to accommodate the bolt-head or to avoid use of an unnecessarily long bolt. Studs are commonly used to connect cylinder-covers to engine cylinders.

The metal-end M is screwed into the threaded hole in the casting B [fig. 24-28(ii)] by means of a stud-driver which consists of a thick hexagonal nut having a partly threaded hole. The upper piece A has a clear hole (of diameter $D_1 = 1.1D$) through which the stud passes. The two pieces are fastened together by a nut screwed on the nut-end. In this case, it is not necessary to withdraw the stud when disconnecting the two pieces and hence, the threaded hole does not get damaged. The disadvantage of the tap-bolt is thus overcome by using a stud.



When a stud is used for connecting a piece to a very thick block, the hole is drilled in that block and then tapped. Fig. 24-29 shows the drilled hole in section. The diameter d of the drill is equal to the core diameter of the stud. The end of the hole is conical on account of the pointed end of the drill. The depth of the hole is kept at least equal to $1.5D$ (where D is the diameter of the stud). The threaded hole in section is shown in fig. 24-30. In fig. 24-31 the stud is shown in position in the tapped hole, and the two parts are shown connected by means of a nut and a washer.

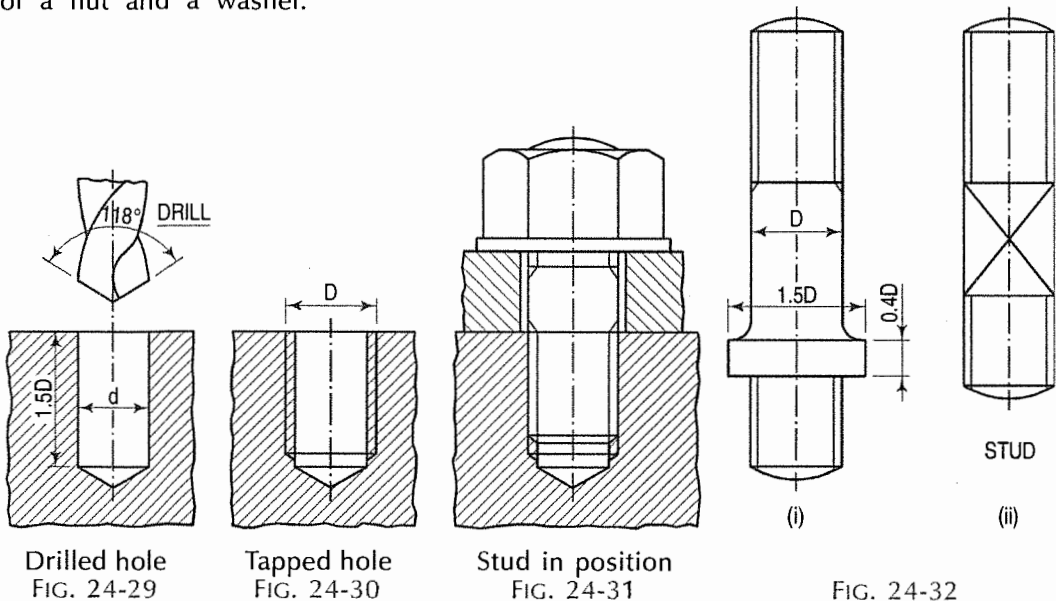


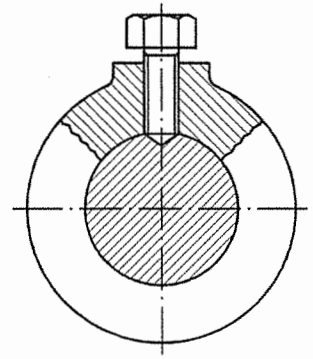
Fig. 24-32(i) shows a stud with a collar. It is called a *collar-stud*. A stud with the middle portion made square in section is shown in fig. 24-32(ii). This facilitates gripping of the stud while screwing or unscrewing it.

24-6. SET-SCREWS

A set-screw is similar to a tap-bolt, but is threaded practically throughout its length.

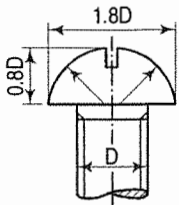
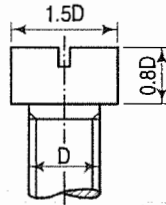
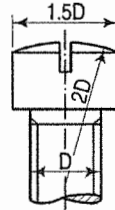
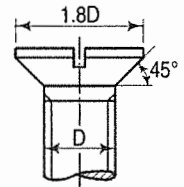
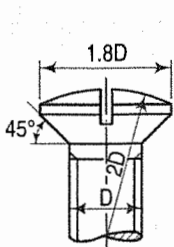
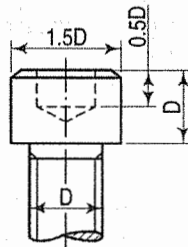
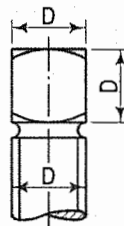
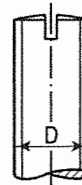
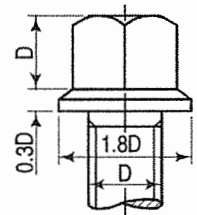
It is used to prevent relative movement between two pieces. It is screwed into a tapped hole (fig. 24-33) in the piece adjoining the screw-head, while its end presses on the other piece, thus preventing relative rotation or sliding. Heads of set-screws except those which can be operated by spanners or wrenches are provided with screw-driver slots.

Set-screws have heads of various forms. Hexagonal and square heads are similar to bolt-heads. The other forms of heads as shown in fig. 24-34 round or cup, fig. 24-35 cylindrical or cheese, fig. 24-36 fillister, fig. 24-37 countersunk, fig. 24-38 rounded countersunk (called instrument screw) and fig. 24-39 socket. The square head is sometimes made of smaller size as shown in fig. 24-40. The grub screw (fig. 24-41) has no head. Fig. 24-42 shows a collar screw with a square head. Width of the slot for the screw driver, in each case, is equal to $0.2D + 0.1 \text{ mm}$, while its depth is equal to $0.25D$ and $0.4D$ in case of flat top and rounded top respectively.

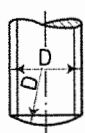


Set-screw

FIG. 24-33

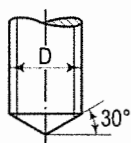
Round or cup
FIG. 24-34Cylindrical or cheese
FIG. 24-35Fillister
FIG. 24-36Countersunk
FIG. 24-37Rounded countersunk
FIG. 24-38Socket
FIG. 24-39Square head
FIG. 24-40Grub screw
FIG. 24-41Collar screw
FIG. 24-42

Ends of set-screws, particularly the grub screws are made in one of the following shapes shown in fig. 24-43 oval, fig. 24-44 conical, fig. 24-45 flat, fig. 24-46 cup, fig. 24-47 half dog and fig. 24-48 full dog.



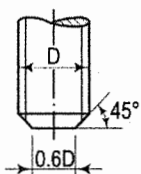
Oval

FIG. 24-43



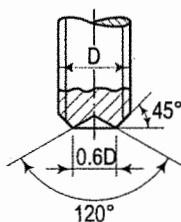
Conical

FIG. 24-44



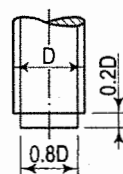
Flat

FIG. 24-45



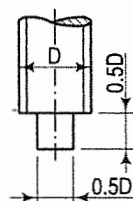
Cup

FIG. 24-46



Half dog

FIG. 24-47



Full dog

FIG. 24-48

24-7. LOCKING ARRANGEMENTS FOR NUTS

Owing to vibrations in moving parts of machines, there is always a tendency for the nuts to get slack and to screw off the bolts slightly. The connected parts might get loose, and lead to serious breakdown. It is, therefore, desirable to secure the nut in some way so as to prevent it from getting loose. Methods employed to do so are called locking arrangements for nuts, a few of which are described below.

(1) **Lock-nut or check-nut:** This nut is used along with an ordinary nut. It is chamfered on both the hexagonal faces. The nut A is first screwed on the bolt as tightly as possible (fig. 24-49).

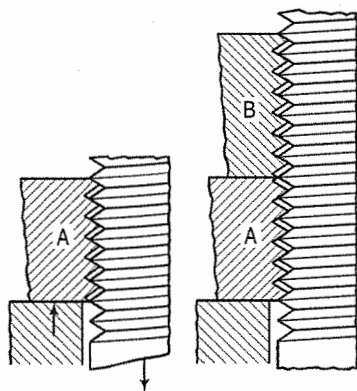


FIG. 24-49

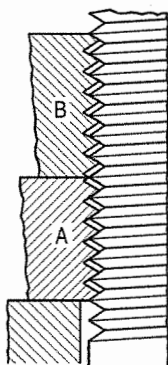


FIG. 24-50

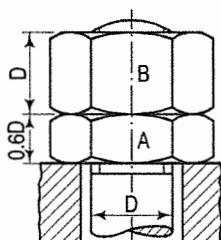


FIG. 24-51

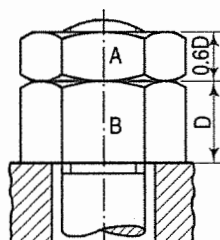


FIG. 24-52

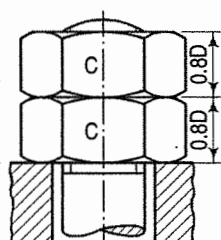


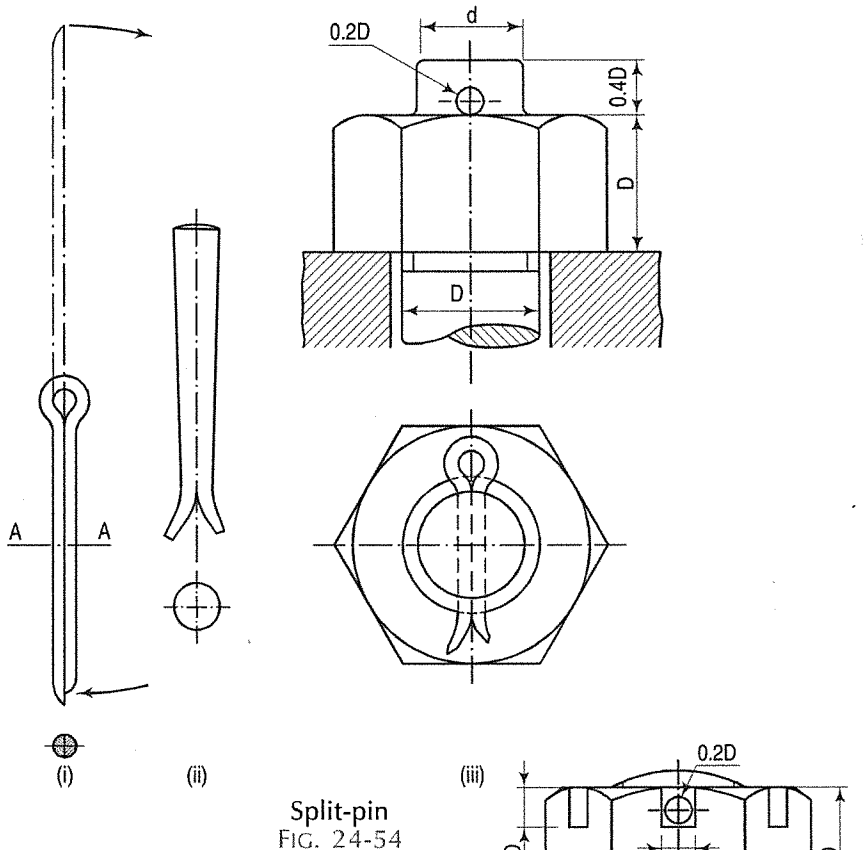
FIG. 24-53

The thread in the nut presses against the bottom side of the thread in the bolt. Another nut B is then screwed on the bolt till it touches the top of the nut A. The nut B is then held by one spanner, while the nut A is turned backwards with another spanner. It will move through a very small angle. The two nuts are thus locked or wedged tightly against each other and against the bolt. This will prevent them from slackening. The action of the threads in the nuts upon the thread on the bolt in this condition is shown in fig. 24-50.

The clearance between the threads has been shown exaggerated. The thread in the nut B presses against the lower side of the thread in the bolt, while that in the nut A presses on the upper side. The nut B, thus, carries practically the entire

axial load on the bolt. It should therefore be of the standard size. The nut *A* which is called a lock-nut, may be thinner. Its thickness is kept equal to 0.6 times the diameter of the bolt (fig. 24-51). To turn the lock-nut backward, a thin spanner would be necessary. As thin spanners are not always readily available, the lock-nut is often placed above the standard nut (fig. 24-52). A compromise is sometimes made by using two nuts of uniform thickness equal to 0.8 times the diameter of the bolt (fig. 24-53).

(2) **Split-pin:** It is made from a steel wire of semi-circular cross-section, bent as shown in fig. 24-54(i). It is inserted in a hole drilled in the bolt so that it bears on the top face of the nut, thus preventing it from turning. The diameter of the hole is kept equal to approximately $0.2D$. The pin is then split open at its tail-end as shown in fig. 24-54(iii). The projecting portion of the bolt is usually turned down to the core diameter. The split-pin is also used in conjunction with special nuts designed for the purpose. A round tapered pin split at its thinner end is also sometimes used [fig. 24-54(ii)].



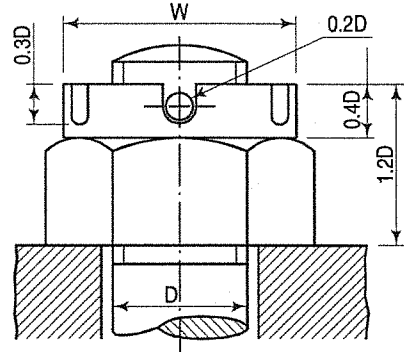
(3) **Slotted nut** (fig. 24-55): It is a hexagonal nut with slots cut in the upper end and through opposite faces. A split-pin is inserted through the slot which comes in line with the hole drilled in the bolt and is then opened out at its end. The strength of the nut is considerably reduced on account of these slots.

(4) **Castle nut** (fig. 24-56): Slots are cut in a cylindrical collar provided on the top of the nut, thus overcoming the disadvantage of the slotted nut. These nuts are widely used in automobile and locomotive engines where they are subjected to sudden shocks and vibrations. It is used in a steam engine.

(5) **Sawn nut or Wiles nut** (fig. 24-57): This is a hexagonal nut with a slot cut half way across it. A cap-screw *A* is passed through a clear hole in the upper part *B* and screwed into a tapped hole in the lower part *C*. On tightening the screw, the friction between the thread in part *B* and that in the bolt increases and prevents slackening of the nut.

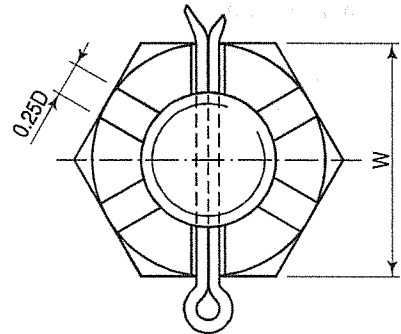
(6) **Simmond's lock-nut** (fig. 24-58): A collar is provided at the upper end of the hexagonal nut and a fibre ring *R* is fitted inside it. The internal diameter of the ring is less than the core diameter of the bolt. The end of the bolt, on reaching the ring, cuts its own thread in the ring when the nut is screwed on. This thread gives a greater grip over the bolt-thread thus preventing the slackening of the nut.

(7) **Penn, ring or grooved nut** (fig. 24-59): This nut has a cylindrical grooved collar at its lower end. This collar fits into a corresponding recess in the adjoining piece. A set-screw is inserted through the nearest face of the piece. The dog-end of the set-screw enters the groove and prevents slackening of the nut. The width *W* of the groove is kept equal to the diameter of the dog-end of the screw. This arrangement is feasible only when the bolt is placed reasonably near a vertical face of the connected piece.



Castle nut

FIG. 24-56



Sawn nut

FIG. 24-57

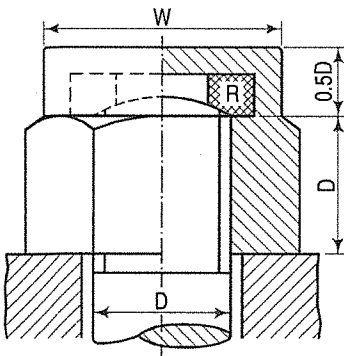


FIG. 24-58

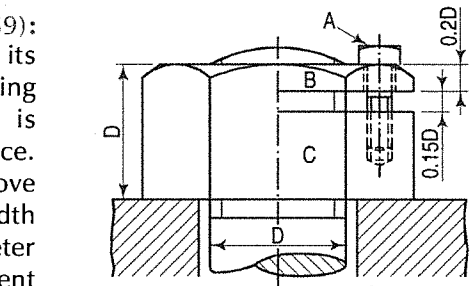


FIG. 24-59

When the bolt is considerably away from the edge, this nut is used in conjunction with a separate cylindrical collar (fig. 24-60). A pin, screwed into the adjoining piece keeps the collar in fixed position.

A small pin (fig. 24-61), screwed in the piece adjoining the nut so that it touches one of the faces of the hexagonal nut, is sufficient to prevent its getting loose. But it becomes an obstruction if the nut is required to be tightened further through a small angle.

(8) **Stop-plate or locking-plate** (fig. 24-62): it is a plate grooved in such a way that it fits hexagonal nut in any position at intervals of $\frac{1}{12}$ th of a revolution, i.e. at 30° intervals. It is fixed around the nut by means of a tap-bolt, thus preventing its rotation.

(9) **Spring-washer:** A single-coiled spring (fig. 24-63) or a double-coiled spring (fig. 24-64) placed under the nut as a washer, offers stiff resistance when compressed by tightening of the nut and keeps the thread in the nut gripped with the thread on the bolt. Fig. 24-65 shows a single-coiled spring washer placed under a nut.

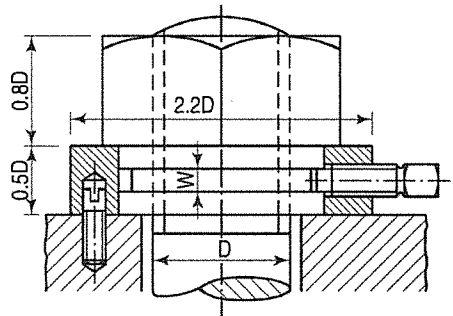


FIG. 24-60

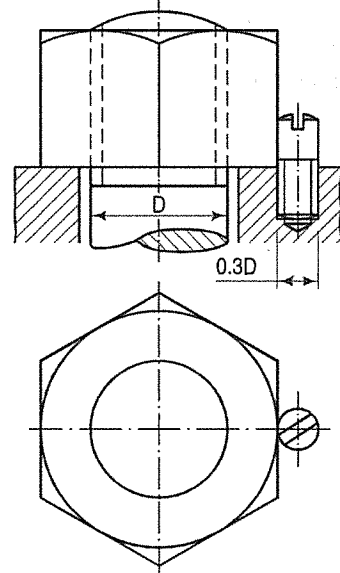
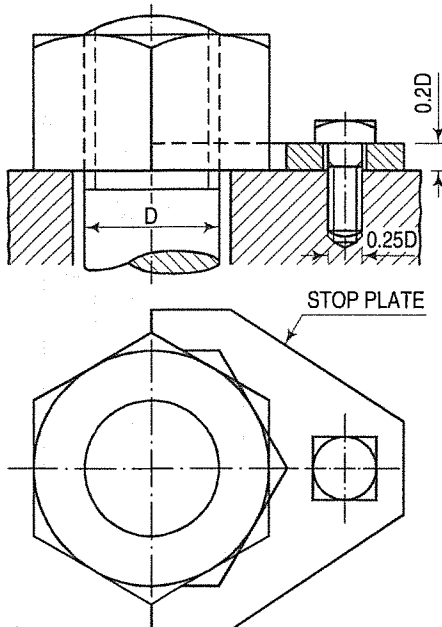
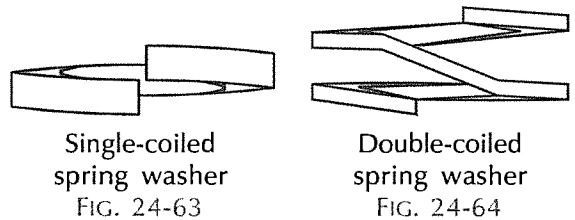


FIG. 24-61

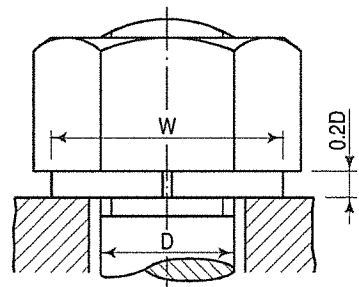


Locking plate
FIG. 24-62



Single-coiled
spring washer
FIG. 24-63

Double-coiled
spring washer
FIG. 24-64

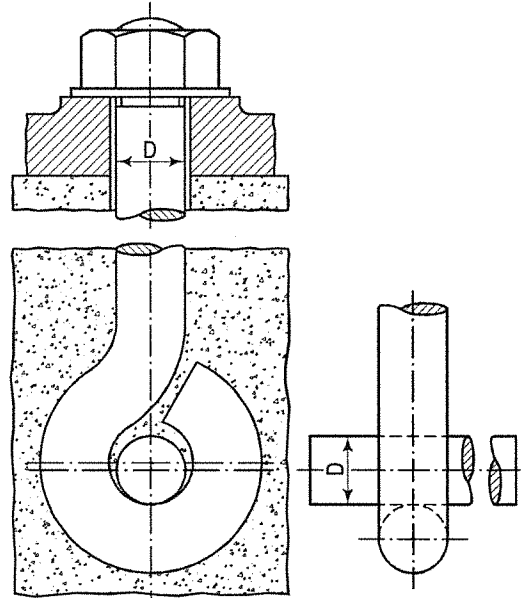


Single-coiled spring washer placed under a nut
FIG. 24-65

24-8. FOUNDATION BOLTS

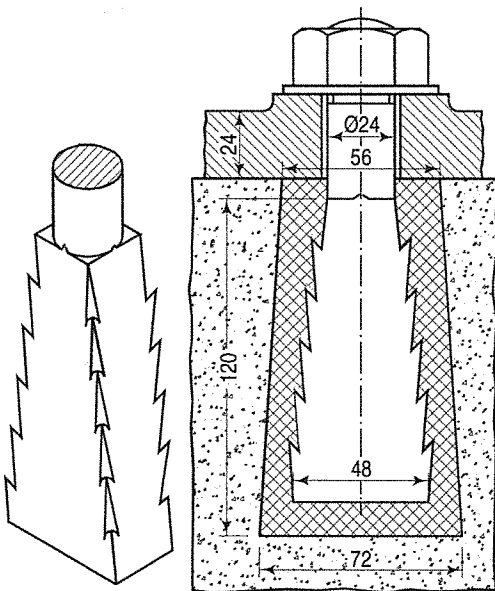
These bolts are used for fixing machines to their foundations.

(1) **Eye or Hoop bolt** (fig. 24-66): Simple forms of these bolts can be quickly forged from a mild-steel or wrought-iron bar. These bolts are suspended in the hole and cement grout is then poured to fill up the space around them. Fig. 24-66 shows two views of an eye-bolt forged from a bar. It has a piece of mild-steel bar passing through the eye and at right angles to it. The stationary engines and lathe machines are fixed on the foundation by these bolts.

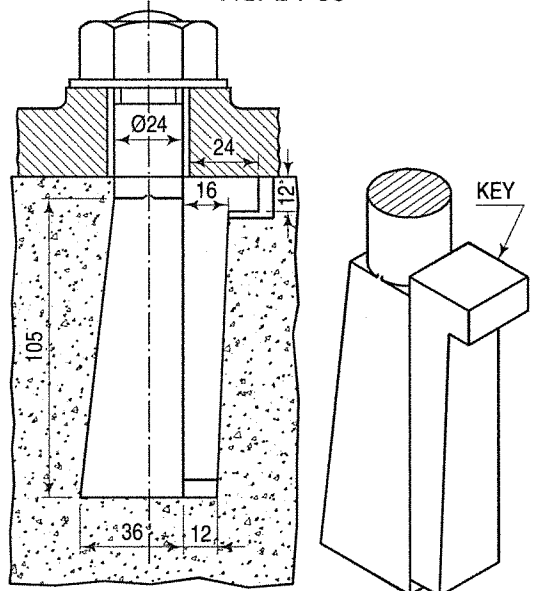


Eye bolt
FIG. 24-66

(2) **Rag bolt** (fig. 24-67): It has its lower part rectangular in cross-section and increasing in width only. Its edges are indented or grooved. The bolt is freely suspended in its position in concrete or stone foundation and the annular space around it is filled with molten lead or sulphur. This bolt cannot be easily dislodged after it has been grouted.



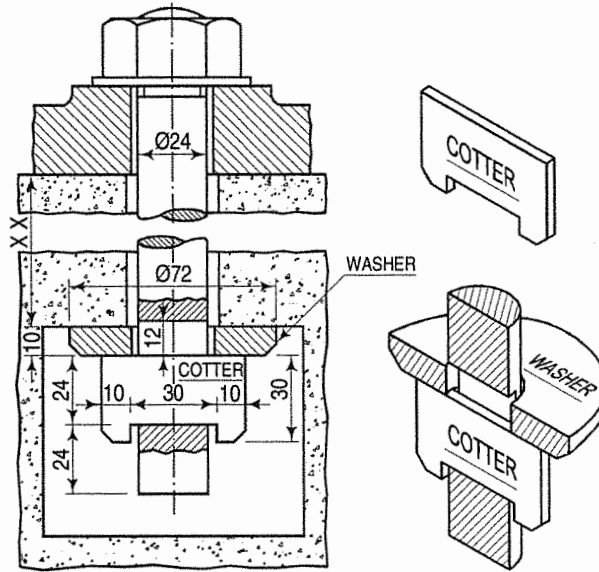
Rag bolt
FIG. 24-67



Lewis bolt
FIG. 24-68

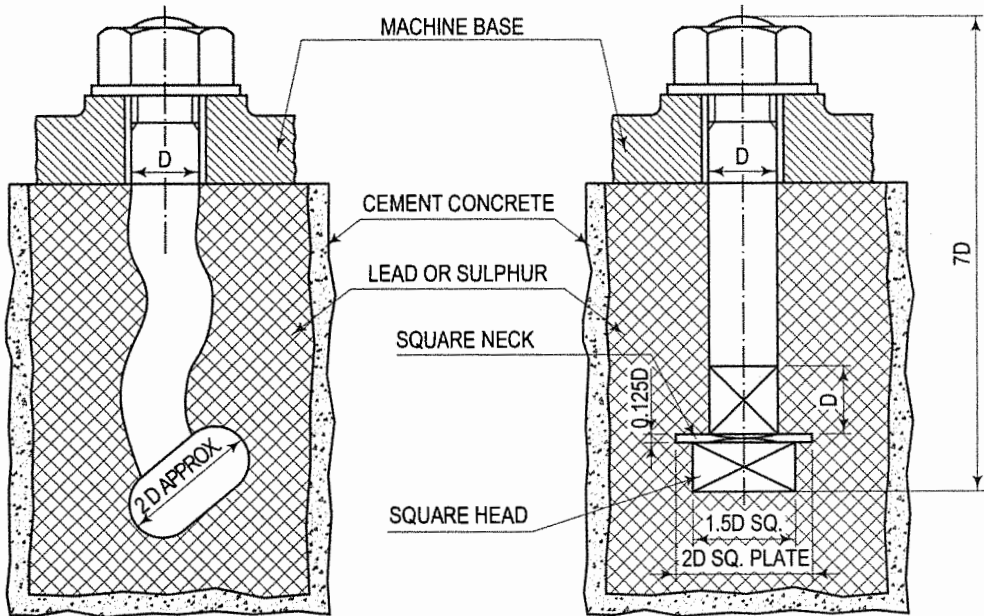
(3) **Lewis bolt** (fig. 24-68): This bolt is fixed in position by inserting a key on the face opposite the tapering face. It can be removed without difficulty by withdrawing the key, which is also tapering.

(4) **Cotter bolt** (fig. 24-69): It is used for fixing heavy machines. It has a rectangular slot to receive a cotter, which is inserted through a hand-hole, previously kept in the foundation for this purpose. A cast-iron washer provides bearing surface for the cotter. Heavy machine-tools are fixed on the foundation by these bolts.



Cotter bolt
FIG. 24-69

(5) **Curved or bent bolt** (fig. 24-70): The shank of this bolt is forged in curved or bent form, set first in lead or sulphur around it and then set in cement concrete, so as not to dislodge. The end is forged approximately twice the shank diameter.



Curved or bent bolt
FIG. 24-70

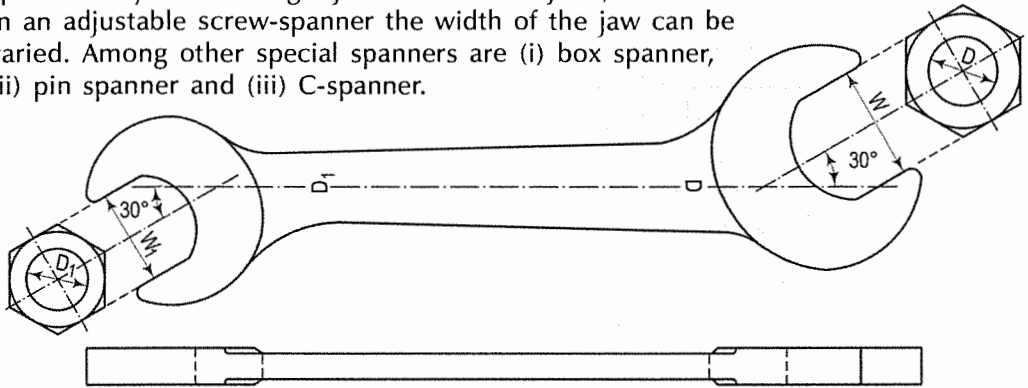
Square-headed bolt
FIG. 24-71

(6) **Squar-headed bolt** (fig. 24-71): As name says, this type of bolt is having simple square head and square neck which is carrying a square plate. Both square head and neck locks the rotation as well as movement of the bolt where as the square plate sets firmly in sulphur which resist the bolt from movement automatically.

24-9. SPANNER

A spanner is made of steel or malleable cast-iron. It is used for turning a nut or for holding a bolt-head while the nut is being screwed on or off the bolt. An ordinary spanner may have a single jaw or have two jaws, one at each end.

In an adjustable screw-spanner the width of the jaw can be varied. Among other special spanners are (i) box spanner, (ii) pin spanner and (iii) C-spanner.



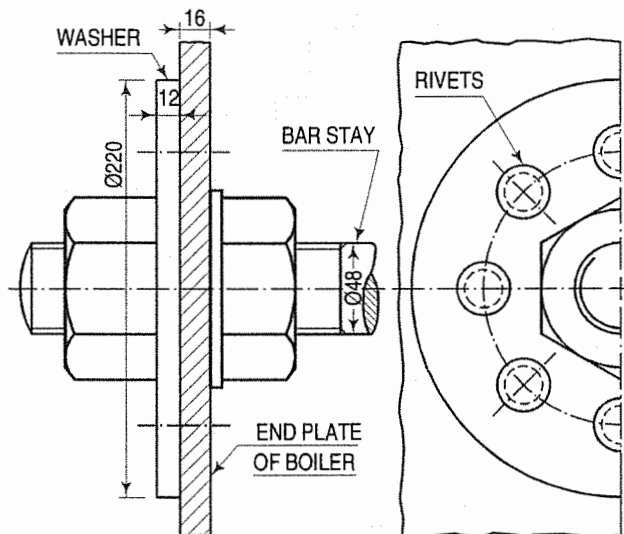
Spanner
FIG. 24-72

Fig. 24-72 shows an ordinary spanner with two jaws. The width of each jaw is slightly more than the width across the flats of the corresponding nuts. A *spanner is specified by the nominal diameter of the bolt* (for the nut of which, it is used). This diameter is usually stamped near the jaw. Jaws are usually made thicker than the body of the spanner.

24-10. LONGITUDINAL OR BAR STAY

Stays are used to prevent flat ends of boiler-shells from bulging out due to high steam pressure. They are generally made of wrought-iron.

Fig. 24-73 shows an end of a bar stay. It is screwed to receive two nuts. The end-plate of the boiler, through which it passes, is strengthened by a thick and large-size washer under the outer nut. This washer is riveted to the end-plate.



Bar stay
FIG. 24-73

24-11. CONVENTIONAL SYMBOLS FOR NUTS AND BOLTS

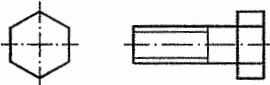
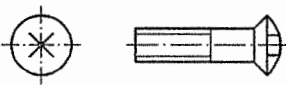
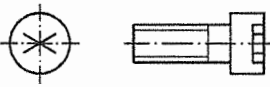
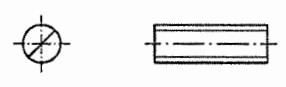
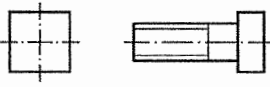
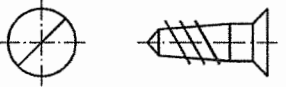
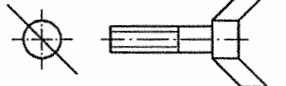
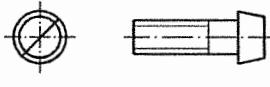
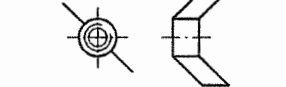
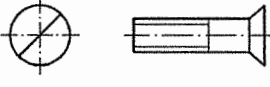
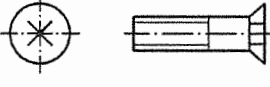
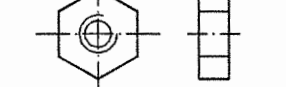
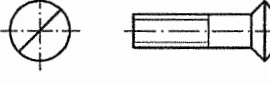
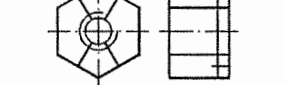
No	Type	Conventional symbol	No	Type	Conventional symbol
1	Hexagonal headed bolt		9	Oval countersunk headed screw cross slot	
2	Hexagonal socket bolt		10	Set screw slot	
3	Square headed bolt		11	Wood and self-tapping screw slot	
4	Cylinder screw cross slot		12	Wing screw	
5	Cylinder screw pan head type slot		13	Wing nut	
6	Countersunk headed screw slot		14	Square nut	
7	Countersunk headed screw cross slot		15	Hexagonal nut	
8	Oval countersunk headed screw slot		16	Crown nut	

FIG. 24-74

EXERCISES 24

1. Draw three views of a hexagonal nut for a 24 mm diameter bolt, according to approximately standard dimensions.
2. Sketch neatly, giving dimensions, three forms of nuts (except the hexagonal and square types) for a 20 mm diameter bolt.
3. Draw three views of a square-headed bolt 20 mm diameter, 120 mm long, with a hexagonal nut, lock-nut, washer and a split-pin, using rough-rule dimensions for the nuts. The length of the threaded part of the bolt is 40 mm.

4. Draw neat and dimensioned sketches of any five forms of bolts showing clearly the method used for preventing rotation in each case.
5. Explain by means of sketches, any three methods of preventing a bolt from rotating while screwing the nut on or off it.
6. Illustrate by means of sketches, difference between a tap-bolt and a stud-bolt.
7. Draw two views of a 24 mm diameter stud, 100 mm long, with a castle nut and a split-pin.
8. Draw the sectional front view and the top view of two 20 mm thick plates fastened together by means of 20 mm diameter stud, a hexagonal nut and a washer. Insert important dimensions.
9. Show by means of neat sketches, sectional views of (a) the drilled hole and (b) the tapped hole for 24 mm dia. stud, in a block 75 mm thick. Take the length of the metal-end of the stud to be 30 mm.
10. Give neat sketches (a) to show the difference between, and (b) to illustrate the uses of, a set-screw and a cap-screw.
11. Show clearly with the help of neat sketches the difference between (i) bolt and nut fastening, (ii) stud-bolt and nut fastening and (iii) tap-bolt fastening.
12. Sketch, giving dimensions, any four forms of set-screws and name them. Show a different type of end in each case and name it. Assume the diameter to be 12 mm and length 50 mm.
13. Show by means of neat sketches, any three methods employed for preventing nuts from getting loose on account of vibrations.
14. Sketch neatly two types of bolts used to secure a machine to its foundation showing clearly how each is fixed.
15. Give by means of neat sketches, an example of each of the following methods of locking a nut:
 - (i) By a split-pin
 - (ii) By a set-screw
 - (iii) By a washer
 - (iv) By a fibre-ring.
16. Sketch neatly, giving important dimensions:
 - (i) Any two forms of studs
 - (ii) Any three methods of locking a nut
 - (iii) Any two forms of foundation bolts.
17. Give neat and dimensioned sketches of:
 - (i) Castle nut
 - (ii) Wing nut
 - (iii) Rag foundation bolt
 - (iv) Fillister-headed screw
 - (v) Eye-bolt
 - (vi) Collar-stud
 - (vii) Curved or bent bolt
 - (viii) Square-headed bolt.
18. A tie bar is to be made in two parts to facilitate adjustment in length during assembly. Sketch the arrangement.
19. Sketch in two views a two-jaw spanner for nuts for 18 mm and 20 mm diameter bolts. The jaws are thicker than the body. Give important dimensions.
20. Show by means of neat sketches how a bar-stay is attached to the flat end of a boiler.

Chapter 25



RIVETED JOINTS AND WELDED JOINTS

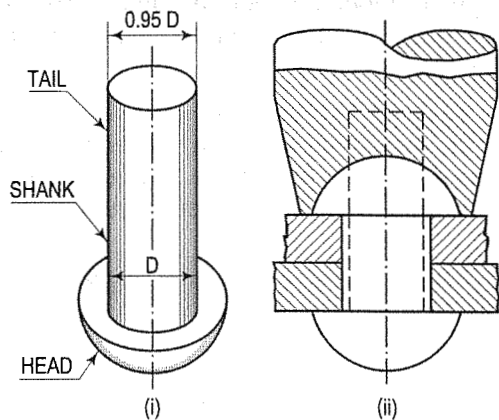
25-1. INTRODUCTION

Rivets are used to fasten permanently two or more plates or pieces of metal. Joints made with rivets are called riveted joints. They are commonly used in ship-building and for the construction of steel buildings, bridges, boilers, tanks etc. Plates joined together by means of a riveted joint cannot be disconnected without chipping-off rivet-heads from one side of the joint.

Rivets are usually made of C-30. In its initial form [fig. 25-1(i)]. A rivet comprises of following:

- (i) a head,
- (ii) cylindrical body of shank and
- (iii) a slightly tapered tail.

A rivet is specified by the diameter of its shank. The length of the tail, out of which another head is formed, is kept about 1.25 times the diameter of the rivet.



Rivet and riveting

FIG. 25-1

25-2. RIVETING

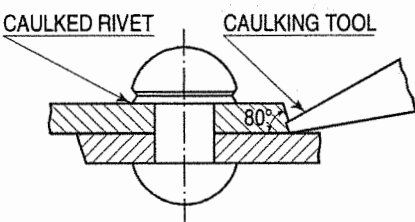
The process of forming another rivet-head, after the rivet is placed in the hole previously drilled or punched through the plates, is called riveting. The diameter of this hole is kept slightly larger (about 1 mm to 1.5 mm) than the diameter of the rivet. Any burr formed at the edges of the hole due to drilling, is removed by a little counter-sinking. The rivet is made red-hot in a charcoal furnace and then inserted within the hole in the plates. The head of the rivet is held fast against the adjoining plate, while the tail is hammered and another rivet-head is forged. The rough surface of the new head is smoothened by the use of a special tool having a cavity of the desired form of the head [fig. 25-1(ii)]. A hot rivet is easier to work on. It also binds the plates more closely together on cooling, on account of contraction of the metal. Riveting is done cold (i.e. without heating the rivet) in case of rivets of small sizes or when they are made of soft ductile metals such as copper, aluminium etc. Machine riveting, which is a common modern practice, is employed when the work is to be

done speedily or on a large scale. The rivet-head is formed under a steady force applied by means of hydraulic or pneumatic pressure. Due to the steady pressure, the hole in the plates is filled up more completely.

25-2-1. CAULKING AND FULLERING

To prevent leakage through the joint, the plates are firmly forced together by caulking or fullering processes. The edges of the plates are hammered and driven-in by a caulking tool (fig. 25-2) or a fullering tool (fig. 25-3). The caulking tool is in the shape of a blunt chisel. The thickness of the fullering tool is about the same as that of the plates. To facilitate these operations the edges of the plates are usually machined to an angle of about 80° before joining them together. This angle is increased to about 85° after the fullering process.

Leakage through the hole is prevented by the caulking operation on the edge of the rivet-head (fig. 25-2). Both these processes are generally performed with the aid of pneumatic power.



Caulking
FIG. 25-2

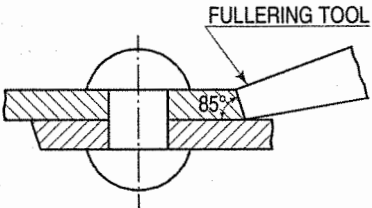
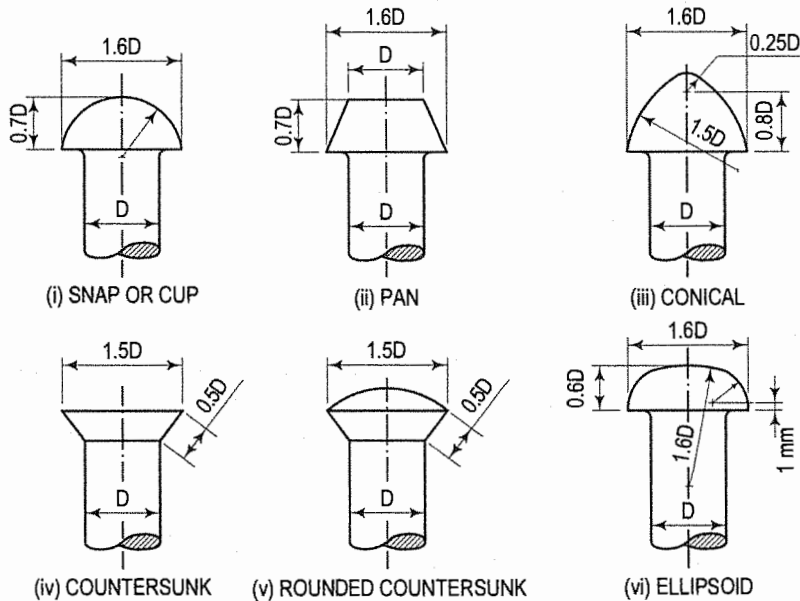


FIG. 25-3

25-3. FORMS AND PROPORTIONS OF RIVET-HEADS



Forms and proportions of rivet-heads

FIG. 25-4

Fig. 25-4 shows some of the various forms of rivet-heads used for general work. The proportions given are for rivets of nominal diameters between 12 mm and 50 mm.

The snap or cup head is the most common form in use. The countersunk head is used when the surface of the plate is required to be free from projecting heads. Conical and ellipsoid heads are generally used in boiler work.

25-4. FAILURE OF RIVETED JOINTS

A riveted joint may fail in any one of the following ways (fig. 25-5):

- (i) Tearing of the plate between the holes if they are very near each other [fig. 25-5(i)].
- (ii) Tearing of the plate between the edges of the plate and the rivet-hole, if the hole is too near the edge [fig. 25-5(ii)].
- (iii) Shearing of the rivet if the diameter of the rivet is smaller than necessary [fig. 25-5(iii)].
- (iv) Crushing of the plate or the rivet [fig. 25-5(iv)].

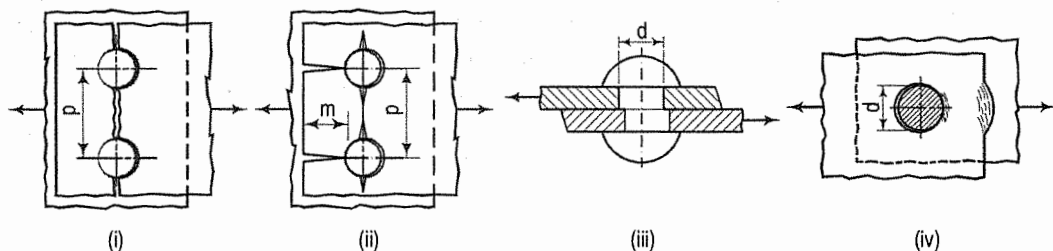


FIG. 25-5

25-5. DIMENSIONS OF A RIVETED JOINT

To prevent failure, the joint should be carefully designed. For elementary work, suitable values of the rivet diameter, positions of holes etc. for a given thickness of the plates, may be obtained by using the following empirical formulae:

- (i) $d = 6 \sqrt{t}$
- (ii) $p = d$
- (iii) $m = d$

where t = thickness of plates in mm

d = diameter of rivets

p = pitch, i.e. the distance between centres of adjoining rivets in the same row (parallel to the edge of the plate)

m = margin, i.e. the distance between an edge of the plate and the nearest rivet-hole.

As $m = d$, the distance of the centre line of the nearest row of rivets from the edge of the plate is equal to $1.5 d$.

Approximate values of d (diameter of rivet) for different values of t (thickness of plates) are given in table 25-1.

TABLE 25-1

Thickness of plate, t mm	8	9	10	11	12	14	16	18	20	22	25
Diameter of rivet, d mm	17	18	19	20	21	22	24	26	27	28	30

25-6. TYPES OF RIVETED JOINTS

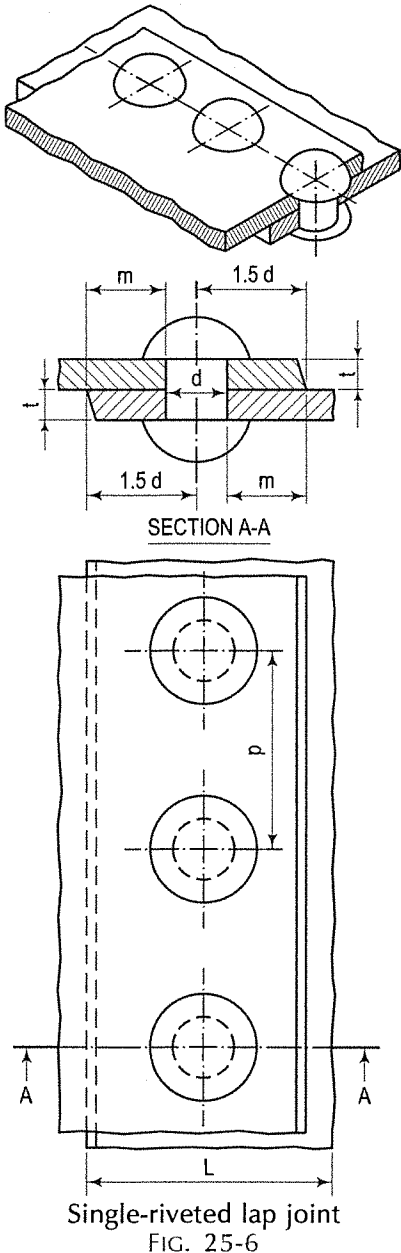


There are two types of riveted joints:

- (1) Lap joint
- (2) Butt joint.

Each of the above riveted joints will now briefly described.

25-6-1. LAP JOINT



In a lap joint, the plates to be connected overlap each other. When the joint is made with only one row of rivets, it is called a *single-riveted lap joint*.

Fig. 25-6 shows the sectional front view and top view of a single-riveted lap joint along with its pictorial view. The width of overlap L is equal to $3d$.

A joint is said to be *double-riveted*, *triple-riveted* etc. according to the number of rows of rivets in it.

When two or more rows of rivets are required, rivets may be arranged in one of the following formations:

- (i) *chain formation* or
- (ii) *zigzag formation*.

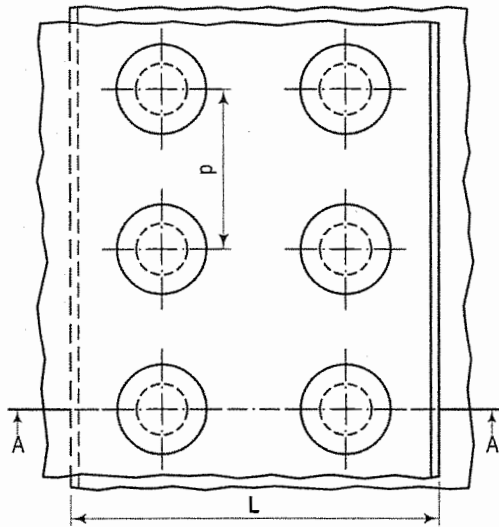
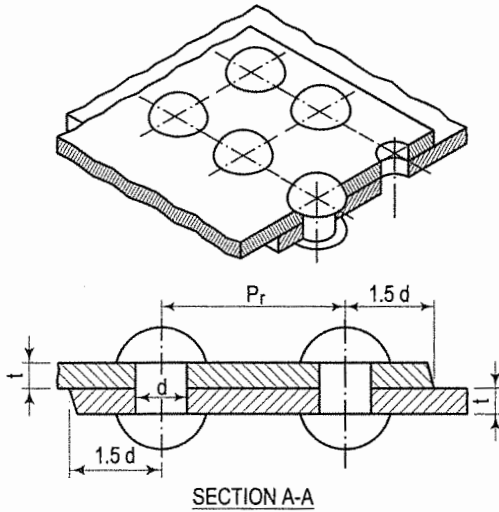
In chain arrangement shown in the double-riveted lap joint in fig. 25-7, rivets in the adjoining rows are placed directly opposite each other.

In zigzag formation they are staggered as shown in fig. 25-8. The distance between the rows of rivets, called the row pitch p_r , should not be less than $0.6p$ for zigzag riveting, and $0.8p$ for chain riveting.

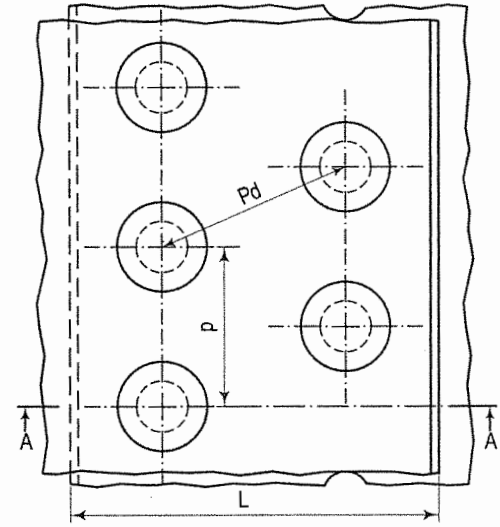
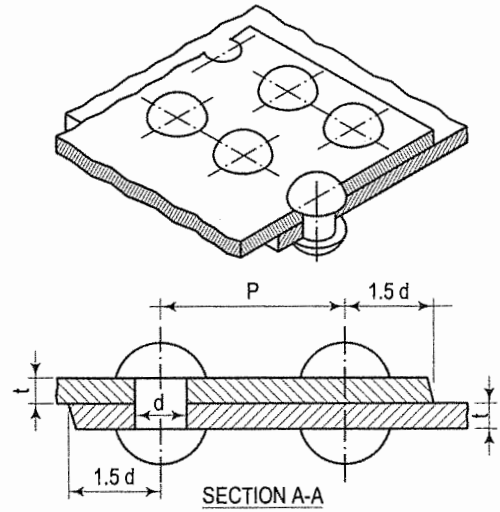
In terms of d , the approximate values of p_r are $2d$ and $2d + 6$ mm respectively. In zigzag arrangement, the distance between the centre of a rivet in one row and the centre of the nearest rivet in the adjoining row, is called the *diagonal pitch*.

This pitch is also often considered instead of the row pitch and its value is obtained by the formula:

$$p_d = \frac{2p + d}{3}$$



Double-riveted (chain) lap joint
FIG. 25-7



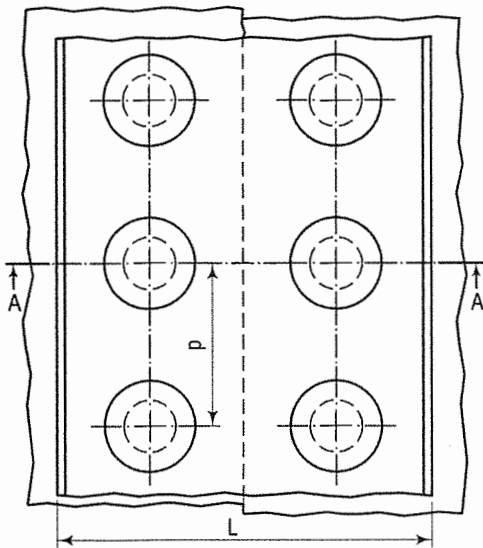
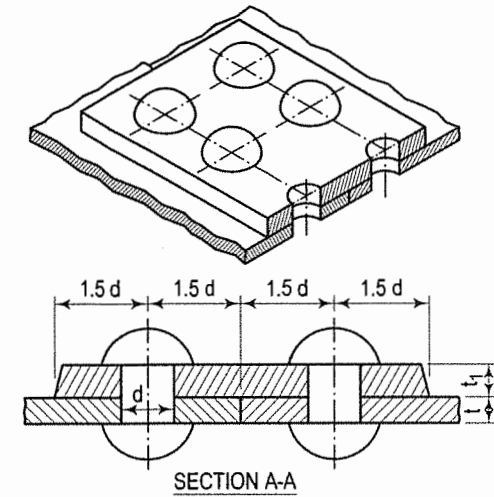
Double-riveted (zigzag) lap joint
FIG. 25-8

25-6-2. BUTT JOINT

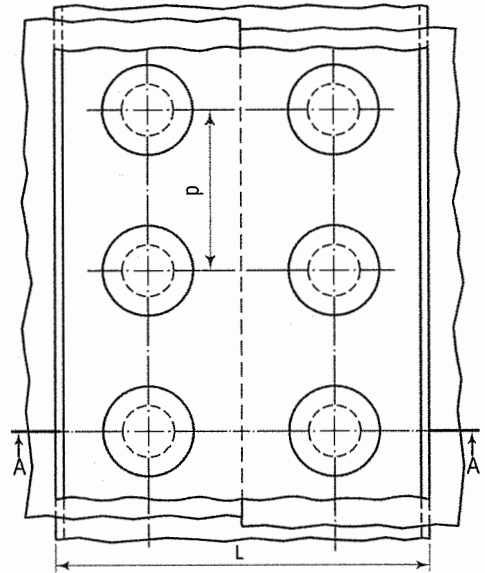
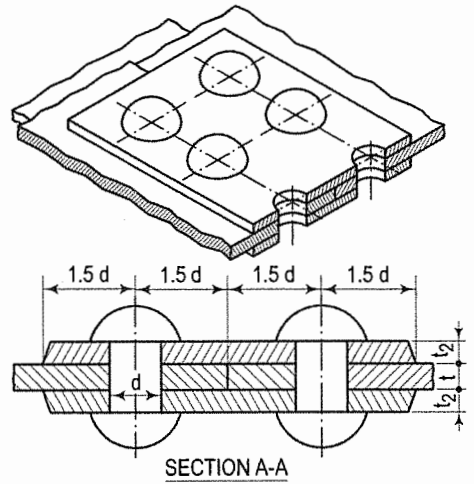
In a butt joint, edges of the plates to be connected butt against each other and the joint between them is covered by butt-plates or butt-straps (also called cover-plates or cover-straps) on one or both sides. At least two rows of rivets, one in each connected plate, are necessary to make the joint.

Thickness of butt-strap:

When only one strap is used, thickness $t_1 = t$ to $1.125t$;
when two straps are used, thickness $t_2 = 0.7t$ to $0.8t$,
where t is the thickness of the plates to be connected.



Single-riveted (single strap) butt joint
FIG. 25-9

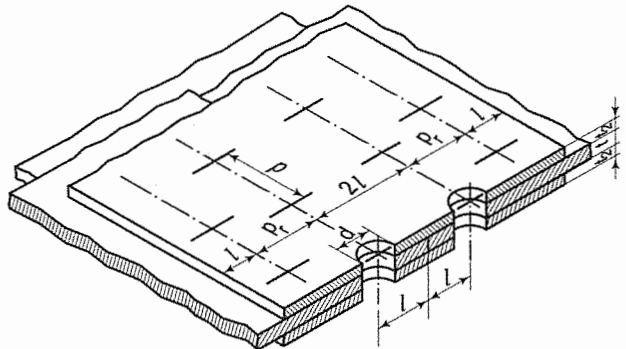


Single-riveted (double straps) butt joint
FIG. 25-10

Fig. 25-9 shows two views of a single-riveted butt joint with one cover-strap.

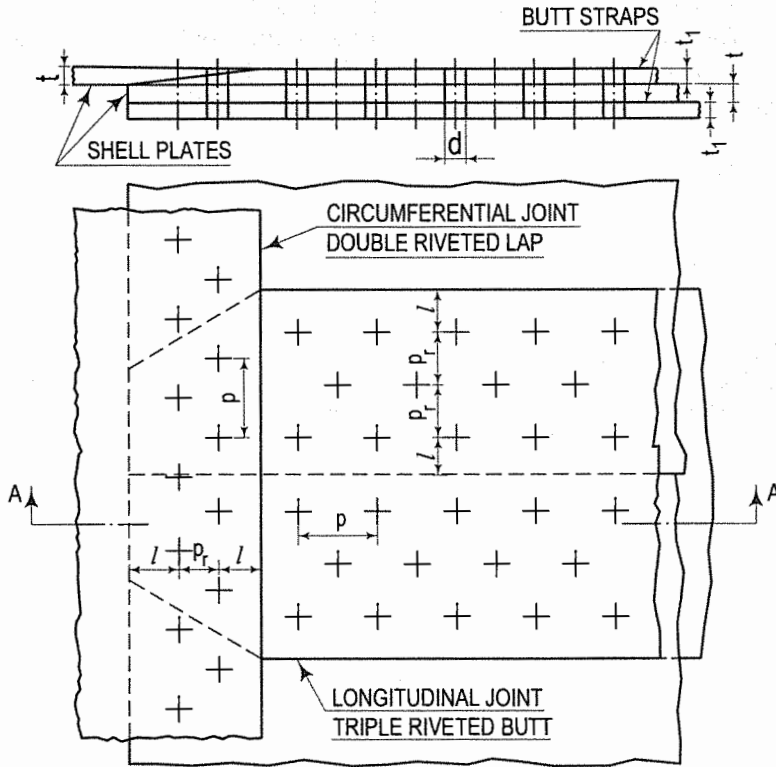
The same joint with two straps is shown in fig. 25-10.

A double-riveted butt joint with double straps is shown in pictorial view in fig. 25-11. Note that there are two rows of rivets in each of the main plates, and that the rivets are arranged in zigzag formation.



Double-riveted (zigzag) butt joint
FIG. 25-11

Two rings of a steam-boiler shell are connected as shown in fig. 25-12. The circumferential joint is a double-riveted lap joint. A triple-riveted or treble-riveted butt joint with two cover-strap is used for the longitudinal joint.

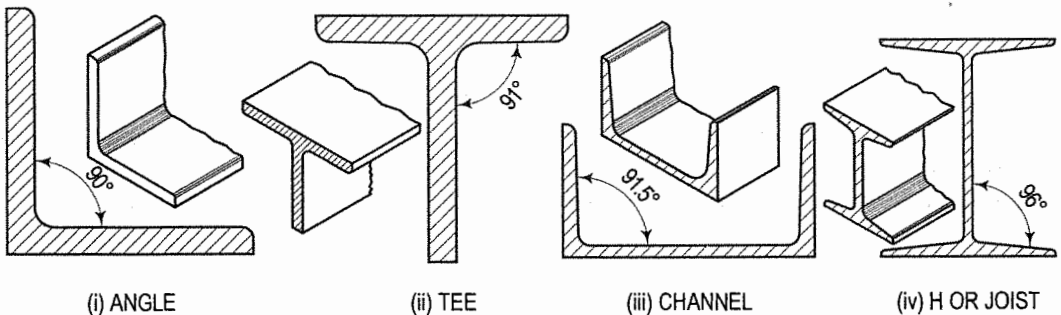


Boiler-shell plate joints

FIG. 25-12

25-7. ROLLED-STEEL SECTIONS (FIG. 25-13)

These are largely used in steel structures. The common shapes are: (i) angle, (ii) tee, (iii) channel and (iv) H or joist.

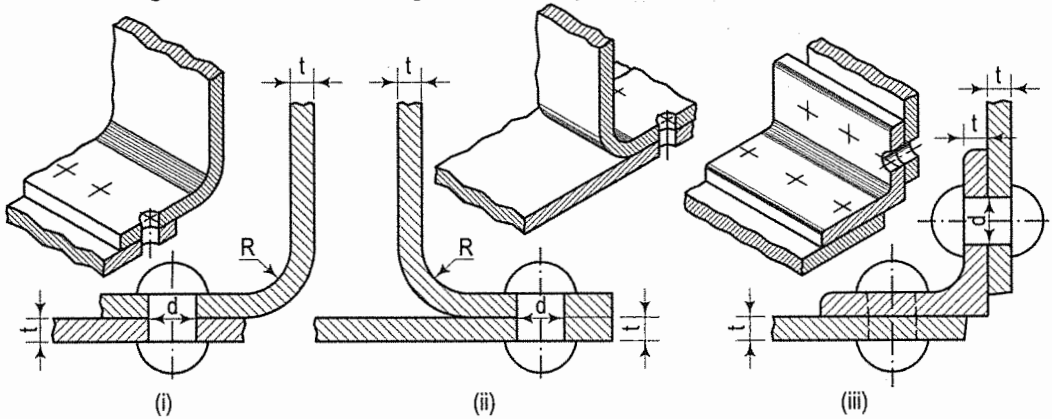


Rolled-steel sections

FIG. 25-13

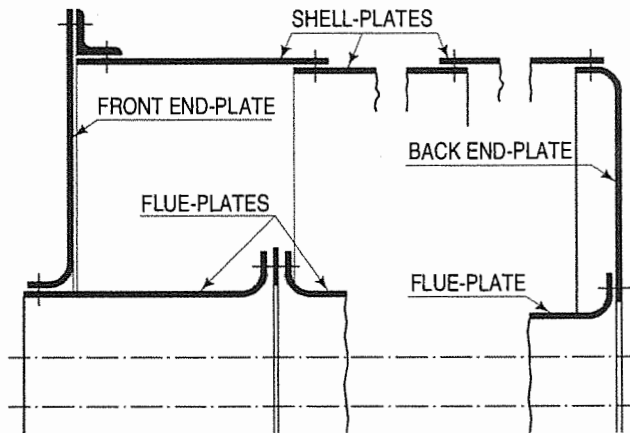
25-7-1. CONNECTION OF PLATES AT RIGHT ANGLES

Plates may be connected at right angles by flanging one of the plates. Fig. 25-14(i) shows a plate bent inside. The plate may also be bent outside as shown in fig. 25-14(ii). The radius R of the inside curve of the bent plate should not be less than twice its thickness. Another method, in which an angle-section is used, is shown in fig. 25-14(iii). The angle is often placed outside also.



Connection of plates at right angles

FIG. 25-14



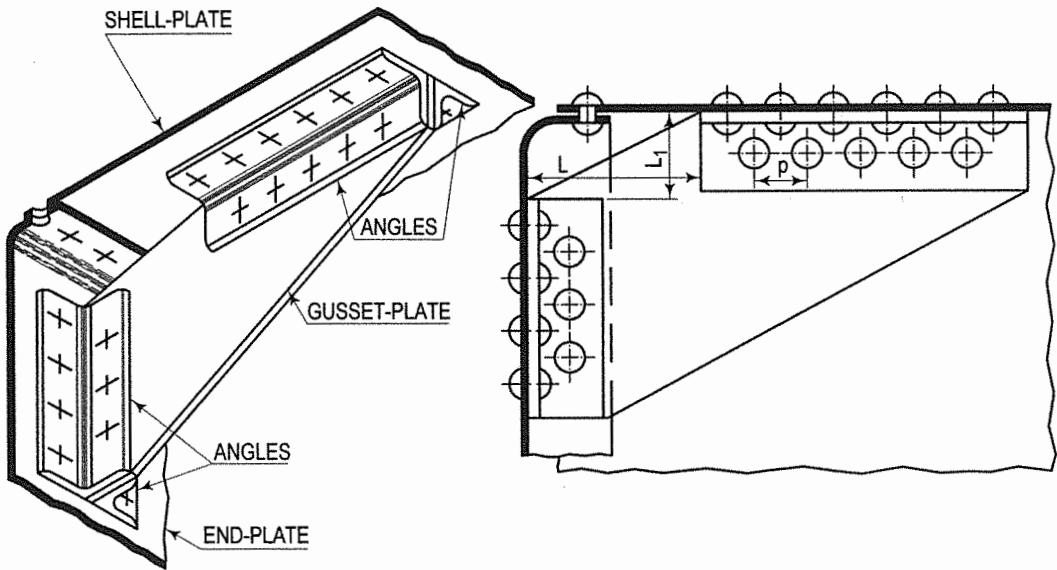
Boiler joints

FIG. 25-15

These methods of connection as employed in boiler construction are shown in fig. 25-15. The front end-plate is connected to the shell-plate by a ring of angle-section placed on the outside. The back end-plate is bent inside and flanged to the shell-plate. Note carefully how the flue-plates are jointed to each end-plate.

25-7-2. GUSSET STAY (FIG. 25-16)

It is a plate which connects the flat end and the cylindrical shell of a boiler. Lengths of angle-section are used to make the joints. The flat ends of a boiler are prevented from bulging out and are strengthened by means of these stays.



Gusset stay
FIG. 25-16

25-8. WELDED JOINTS

25-8-1. WELDING

Welding is a technique of making permanent joint. It is the process of joining two parts of metal by fusing them together. Until recently riveted joints were the main type of permanent joint extensively used in the construction of boilers, ships, bridges, steel structures, etc. During the last decade, however, the rapid development of welding methods has replaced the riveted joints. Since welding is used so widely, and for a large variety of purposes, it is essential to have an accurate method of showing on the working drawing of machines or structures, the types, sizes, and locations of weld desired by the machine designer.

25-8-2. TYPES OF WELDING PROCESS

The welding processes can be classified into the following *three* groups:

- (1) Pressure welding or forge welding
- (2) Fusion welding
- (3) Fusion and pressure welding.

(1) **Pressure welding or forge welding:** In this method of welding two metal parts are heated at the joint upto plastic condition and then joined together by applying external mechanical pressure.

(2) **Fusion welding:** It is a process of welding by local fusion with or without use of a filler metal. The most commonly used fusion welding processes are gas welding and arc welding. In case of gas welding, the welding puddle is produced by the effect of a flame generated by a fuel gas (generally acetylene and oxygen). The flame temperature varies from 2600° to 3200° depending upon fuel gases.

In case of arc fusion welding, the welding puddle is created by the effect of an arc. The arc produce between the electrode and the work-piece. The entrance of atmospheric gases into the arc and the welding puddle is prevented by the inert gases or slags produced by the electrode in case of consumable coated electrode.

(3) **Fusion and pressure welding:** In this welding heavy current (50,000A) is passed through the joint which gets melt. The welding is completely under external pressure. Spot welding, seam welding, flash butt welding and pressure butt are the examples of the pressure resistance welding.

25-8-3. TYPES OF WELDED AND WELDS JOINTS

(1) **Types of welded joints:** There are four important types of welded joints classified according to the positions of the workpiece being joined as shown in fig. 25-17.

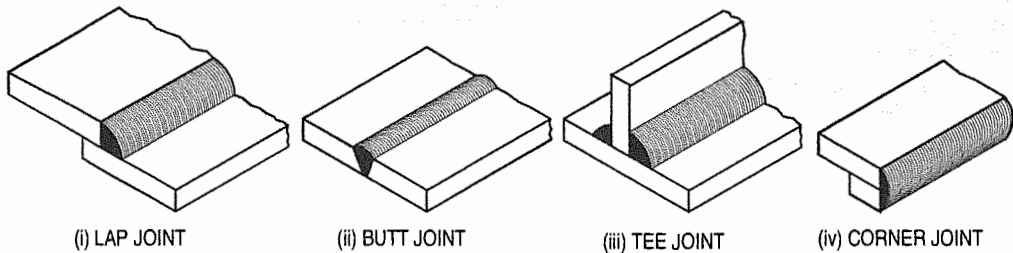


FIG. 25-17

(2) **Types of welds:** There are four types of basic welds of arc and gas welds as shown in fig. 25-18.

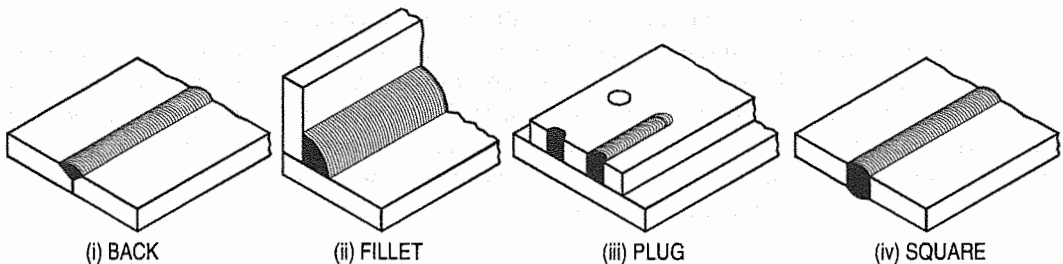


FIG. 25-18

Similarly, the resistance welds are also of four types

- (i) spot weld,
- (ii) projection weld,
- (iii) seam weld and
- (iv) flash and upset welds.

Symbolically they are represented as shown in fig. 25-19.

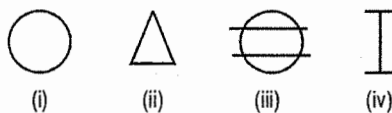


FIG. 25-19

25-8-4. REPRESENTATION OF WELDED JOINTS

A welded joint is shown on a drawing by means of (i) a symbol which specifies the form of weld and (ii) a bent arrow and a reference line indicating the location of the weld as shown in fig. 25-20.

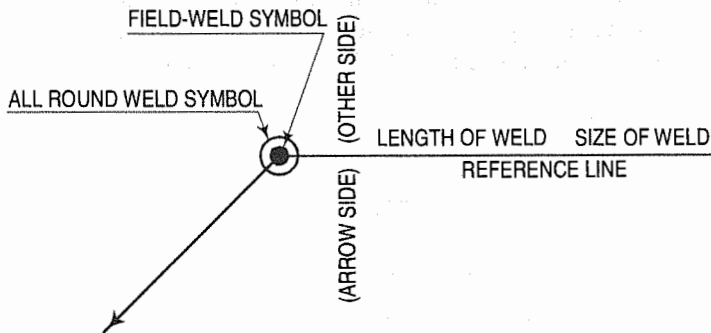


FIG. 25-20

Table 25-2 shows symbols for various forms of welded joints as recommended by the Bureau of Indian Standards.

TABLE 25-2

No.	FORM OF WELD	ILLUSTRATION	SYMBOL	No.	FORM OF WELD	ILLUSTRATION	SYMBOL
(i)	FILLET			(viii)	SINGLE-BEVEL BUTT		
(ii)	SQUARE BUTT			(ix)	SINGLE-BEVEL BUTT WELD WITH BROAD ROOT FACE		
(iii)	SINGLE-V BUTT			(x)	DOUBLE-BEVEL BUTT		
(iv)	DOUBLE-V BUTT			(xi)	DOUBLE-BEVEL BUTT WELD WITH BROAD ROOT FACE		
(v)	SINGLE-U BUTT			(xii)	SPOT		
(vi)	DOUBLE-U BUTT			(xiii)	SEAM		
(vii)	SINGLE-J BUTT			(xiv)	EDGE		

The method of showing the form and location of a fillet weld on a drawing is shown in fig. 25-21.

The symbol should be shown only in one of the views. When a weld is to be on the arrow side, the symbol should be inverted and placed below the reference line as shown in fig. 25-21(i). When a weld is to be on the other side, the symbol should be placed in its correct position, but over the reference line. For welds on both the sides, a symbol is placed above as well as below the reference line. See fig. 25-21(ii). Note that in each case, the vertical portion of the symbol is always kept on the left-hand side of the symbol. The size of weld is indicated on the left-side of the weld symbol.

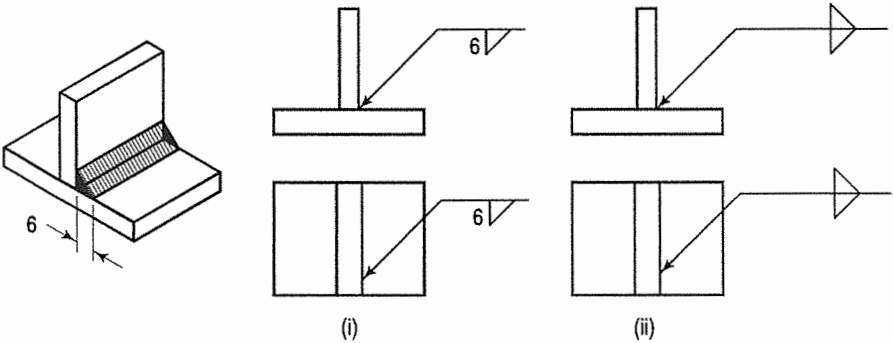


FIG. 25-21

The depth of a partially penetrated U-butt or V-butt weld is indicated by means of a dimension placed to the left of the symbol. Fig. 25-22(i) shows a V-butt weld partially penetrated on the top surface, while the dimensions of the same weld, partially penetrated on both the sides are shown in fig. 25-22(ii).

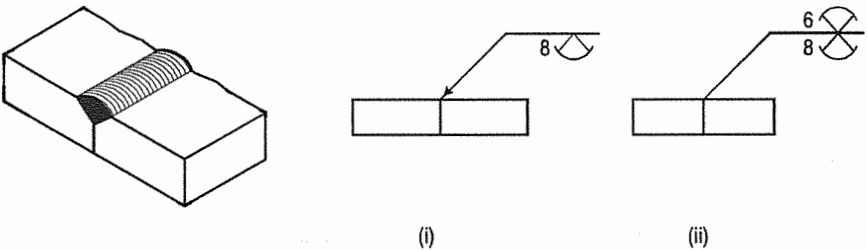


FIG. 25-22

A butt-weld, when flush with the surface of the plate, is indicated by placing a bar above the symbol as shown in fig. 25-23. An all-round weld is shown by means of a circle drawn around the point of intersection between the arrow and the reference lines as shown in fig. 25-24.

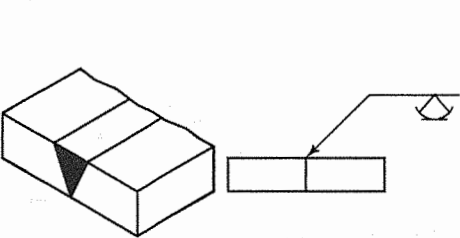


FIG. 25-23

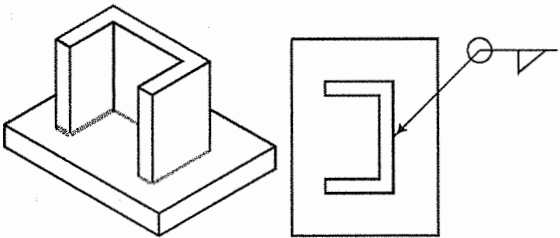


FIG. 25-24

Sometimes the welding is required to carry out at the site or in the field. This can be indicated on a drawing by placing a filled circle drawn around the point of intersection between the arrow and reference lines as shown in fig. 25-25.

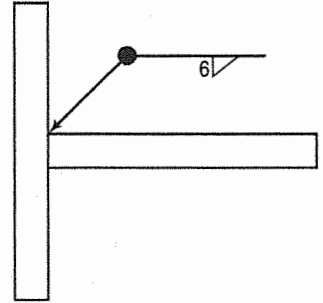


FIG. 25-25

EXERCISES 25

1. Show by means of neat, dimensioned sketches the shapes of the following rivets:
Cup head; pan head; conical head; countersunk head.
2. Explain with the aid of sketches the processes of (i) caulking and (ii) fullering, used in riveted joints.
3. Describe the ways in which a riveted joint may fail. What steps are taken to prevent failures? Illustrate your answer with necessary sketches.
4. Draw the sectional front view, the top view and a side view of a single riveted lap joint for 12 mm thick plates. Show the pitch, margin and width of overlap.
5. Show by means of sketches, the difference between chain riveting and zigzag riveting. What is the advantage of one over the other arrangement?
6. Sketch neatly, two views of a double-riveted lap joint using rivets in zigzag arrangement. State why this arrangement is used. Thickness of plates = 10 mm. Diameter of rivets = 20 mm. Give all other dimensions.
7. Draw three views of the triple-riveted lap joint shown in fig. 25-26, taking $t = 12$ mm, $d = 20$ mm, $p = 60$ mm and $p_r = 40$ mm.

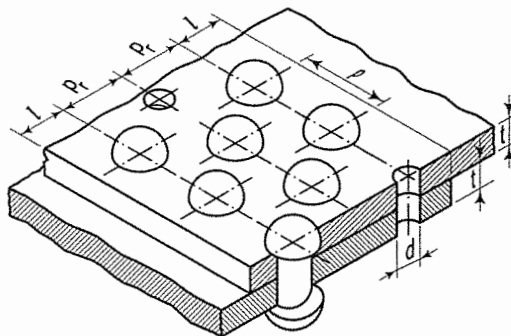


FIG. 25-26

8. Sketch neatly, a sectional front view and top view of a single-riveted butt joint for two 10 mm thick plates, using two butt-straps. Show all dimensions on your sketch.

9. Draw two views of the double-riveted butt joint shown in fig. 25-11. Take $t = 10$ mm and $d = 20$ mm. Insert all dimensions on your drawing.
10. Draw neat sketches to show the circumferential and longitudinal joints between two rings of shell plates in a boiler.
11. Show with the aid of sketches, three different ways in which two plates may be connected at right angles to each other by means of rivets.
12. Sketch neatly, two views to show how two plates, each 12 mm thick can be joined at right angles by using an angle-iron and rivets.
13. Draw neat sketches to show how the end-plates of a Cornish boiler are joined to (i) the shell-plates and (ii) the flue-plates.
14. Draw two views of the Gusset stay shown in fig. 25-16. Take $t = 16$ mm, $d = 24$ mm, $L = 200$ mm, $L_1 = 100$ mm, $p = 60$ mm.
15. Draw three dimensioned views of the riveted joint for a tie-bar shown in fig. 25-27. Take the thickness of the bar as 12 mm.

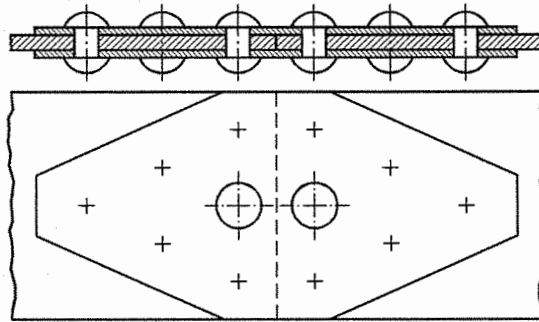


FIG. 25-27

16. Discuss in brief various methods of welding.
17. Why welding is more popular than riveted joints?
18. Show by means of sketches the method of showing location, symbol, size and depth of the following forms of weld:
 - (i) Single V-butt weld
 - (ii) Single bevel butt weld
 - (iii) Double J-butt weld
 - (iv) All-round fillet weld.

Chapter 26



COMPUTER AIDED DRAFTING (CADr)

26-1. INTRODUCTION



Manufacturing of a product is the main activity in engineering profession. The design of a product may start with trial designs in the form of sketches on paper. As the design improves and undergoes changes, the final form of design must be the scaled manufacturing drawings with finer details included. These drawings are two-dimensional representations of three-dimensional objects designed.

During the process of design, the designer may have to carry out a large amount of computations so that an optimum design is obtained. A computer with good graphic capabilities helps the designer to

- (i) realize his ideas.
- (ii) carry out complex computations.
- (iii) present the results of computations in a useful form for decision making and possible improvement.
- (iv) present the improved model for evaluation.

Interactive Computer Graphics (ICG) is the tool of the designer.

26-2. COMPUTER AIDED DRAFTING



A part to be manufactured is defined first in terms of its geometry which also includes dimensions, tolerances, surface finish, and in some cases the type of fit between two mating parts. The two-dimensional representation of a part, called an engineering drawing or a blueprint, shows three orthogonal views of the part. Sometimes, when three views are not enough to define the part, additional sectional views, auxiliary views may have to be added for conveying the right information.

Any design is finally represented in the form of orthographic views and auxiliary views so that production can be carried out. Hence, the computer aided drafting is an important tool for Computer Aided Design. The Computer Aided Drafting (CAD) system is the computerization of technical, production, electronics and architectural drawings.

CAD is the product of computer era. Its development originated from early computer graphics systems. CAD can find its roots to the development of Interactive Computer Graphics (ICG). A system called *Sketchpad* was developed at Massachusetts Institute of Technology, U.S.A., in 1963 by Ivan Sutherland.

In the beginning, CAD systems were no more than graphics editors with some built-in design symbols. The geometry available to the user was limited to lines, circular arcs, and a combination of the two. The development of free-form curves and surfaces such as Coon's patch, Ferguson's patch, Bezier's curve, and B-splines enabled a CAD system to be used for more sophisticated work. A 3D CAD system allows a user to do very sophisticated design and analysis work.

Computer aided drawing and drafting system uses the computer to assist in generation of blueprint data. CAD systems are essential in design and a large number of computer based systems are commercially available. 2D drawing systems correspond directly to traditional engineering drawings, and they are developed to substitute manual drafting.

The advantages offered by computerized drafting systems can be summarized as:

- (a) It increases the accuracy and productivity of designer.
- (b) It allows design alterations to be made easily.
- (c) It offers better drawing visualization through colours.
- (d) It improves the quality of drawings produced.
- (e) Drawings are easier to store and retrieve.
- (f) Storage space required is less.
- (g) Transfer of drawings is faster and cheaper.
- (h) It permits the use of library of standard symbols for more productive CAD work (refer to art. 26-6).

Although the capital investment required in setting up a computer aided drafting system is high, the greater capabilities offered by computers and software are making the systems more affordable.

26-3. COMPUTER (FIG. 26-1)

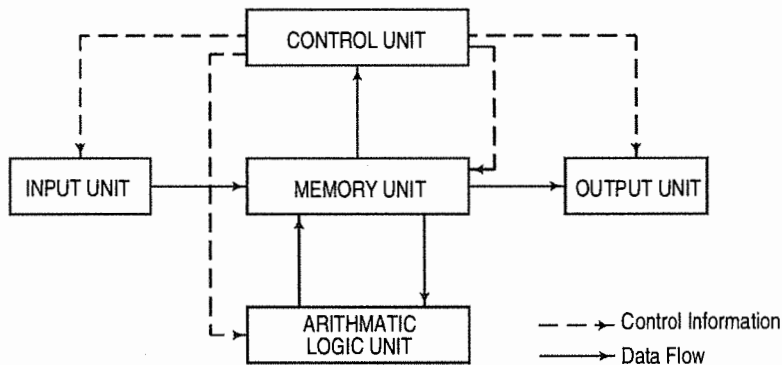


A computer system consists of

- (a) Central Processing Unit (CPU), also known as processor
- (b) Main memory
- (c) Input devices
- (d) Output devices
- (e) Secondary storage devices.

The CPU controls and supervises the entire computer system. The actual arithmetic and logical operations are performed by CPU with the help of main memory. The main memory stores program instructions and processing data. Secondary storage devices provide slow but high capacity memory for storing program and large amount of data that is not currently being processed. Typical secondary storage devices are hard disk, floppy diskettes, magnetic tape, C.D. (compact disk), D.V.D., pen drive, etc.

During operation, data flows between the CPU, the main memory, and the various input/output devices. Moreover, the control information flows between these units to tell what to do with the data. Control information is used to manage the overall operation of the system.



Organisation of a computer

FIG. 26-1

It must be remembered that the main memory is directly connected to the CPU. The main memory stores the program instructions and all the data being used for processing until the data are released as output under the instructions from the CPU. A control unit inside the CPU directs and coordinates all the operations of the system that are called for by the program. It involves control of the input and output devices, the storage and retrieval of information from memory, and the transfer of data between memory and ALU.

A CAD system consists of Hardware and Software. The system usually has the following major hardware elements:

26-3-1. PROCESSOR (CPU) [FIG. 26-2(vii)]



Every CAD system must have a CPU to process and store a very large amount of graphics data. The speed and processing power of a computer is mainly decided by the CPU speed and the amount of main memory (RAM) installed.

26-3-2. DISPLAY [FIG. 26-2(iv)]



The most common type of display device for CAD systems is a Cathode Ray Tube (CRT). The display device must be capable of displaying both graphical and alphanumeric data. Although the display looks like television monitors, they have special electronics for creating graphics images on the screen from computer processed data as per the instructions. But, they lack the television program receiving capability. Since both graphics and text are to be displayed simultaneously on the screen, the graphics monitor is usually divided into graphics area and a small text display area (see fig. 26-3).

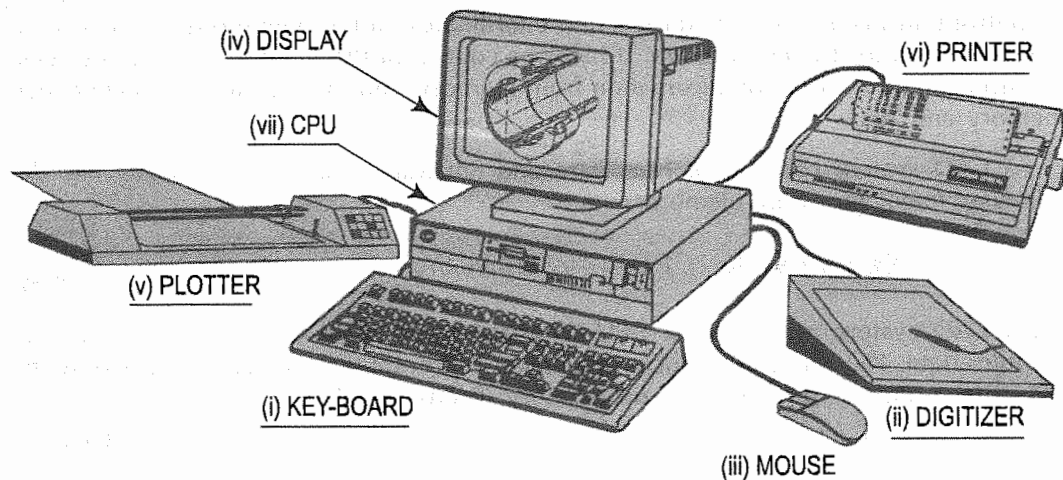
The capabilities of display devices are evaluated in terms of resolution, colours and picture refresh rate. Picture refresh rate is important to get a steady and flicker free display. For a general purpose graphics system, a display with 1024 (horizontal) by 768 (vertical) pixel resolution, 0.28 mm dot pitch and 256 shades of colours is the desirable configuration. The small dot pitch gives a high resolution and therefore good quality picture.

26-3-3. INPUT DEVICES (FIG. 26-2)

(1) **Keyboard** [fig. 26-2(i)]: This is the most common input device. It is preferred for entering commands, text or value such as coordinates of a point or radius of a circle. The 101-key keyboards have special function keys to support special graphic functions. Whenever a key is pressed, the key character is identified by the computer and a character is displayed on the screen. If function keys are pressed or a combination of keys is pressed, the software takes the appropriate action.

(2) **Mouse** [fig. 26-2(iii)]: The mouse is a pointing device which is moved across a flat surface (usually on a mouse pad) by hand to indicate X-Y movement. The rotation of rubber ball underneath the device is translated and the corresponding cursor movement on the screen provides a visual feedback. The mouse is used as a pointing and pick-up device.

Either two-button or three-button mice are available in the market. The mouse has become very popular with microcomputers in recent years, and it is an indispensable pointing device. It is inexpensive, small, and convenient to use. However, fine and very precise sketching is very difficult to obtain with the mouse.



CAD system hardware

FIG. 26-2

(3) **Tablet or Digitizer** [fig. 26-2(ii)]: This is also a very common and popular input device for CAD. It has a flat surface on which a stylus is placed and moved. The movement of stylus on the surface in terms of X and Y coordinates of surface is detected by sensors embedded in the tablet. The stylus incorporates a number of buttons for input. The parameters of a digitizer are resolution, accuracy and repeatability.

The digitizers are available in very large sizes to be used with full size engineering drawings. They have a very high resolution. The resolution is measured in terms of lines per inch (LPI). Digitizers with 2000 LPI resolution are commercially available.

The most common method of using digitizer is for tracing a drawing or a portion of the drawing. The drawing to be copied is placed on the digitizer and the stylus is used to trace the portions required. Thus, old drawings, maps, etc., can be entered into a CAD system.

26-3-4. GRAPHIC OUTPUT DEVICES



A hard copy i.e. copy on paper of what is displayed on a graphics terminal can be obtained by a variety of graphics printers and plotters. The computer graphic output device may be thought of as paper and a pen or a pencil.

(1) **Dot Matrix Printers (DMP) and Laser Printers:** Dot Matrix Printer is the most commonly used printer for text printing. The characters are formed by printing dots in a specific manner. The dot matrix printer [fig. 26-2(vi)] can also be used for printing of drawings, but the quality of output is poor.

(2) **Pen Plotters** [fig. 26-2(v)]: Pen plotters are the simplest output devices for CAD. A pen plotter consists of a device to hold the paper. Usually two orthogonal motorized carriages hold a pen and move it under computer control. There are *three* inputs to the pen plotter: (i) an *X* coordinate, (ii) a *Y* coordinate and (iii) a pen variable. The pen variable can specify the pen colour by pen number, the pen to be *up* (non-drawing position) or *down* (in contact with paper in drawing position).

The two varieties of plotters are flat-bed and drum plotters. The flat-bed plotter is limited by the paper size it can handle. The drum plotter utilizes a continuous roll of paper which rolls over the top of the drum. The capacity and capabilities of a plotter are evaluated by the size of paper it can handle, resolution, speed of plotting, number of pens it can handle, etc.

(3) **Ink-Jet printers/plotters:** These are dot matrix printers. The drawing which is made up of lines, arcs, characters and symbols is converted into dot form. Then the rows of dots are printed across the width of the paper by impelling a tiny jet of ink on the surface of paper. The jets are switched on and off at high frequency to create multicolour plots.

Typically, the resolution is 600 to 900 dots per inch, and each dot is arranged to overlap the adjacent ones. This provides a high quality photo-realistic picture. These plotters are quiet during operation. These plotters are used for colour plots of drawings, shaded images, contour plots and artistic work.

(4) **Laser printers:** A laser printer rapidly produces high quality text and graphics on plain paper. The fastest colour laser printers can print over 100 pages per minute. It can write with much greater speed than an ink jet and can draw more precisely, without spilling any excess ink. The toner powder of laser printer is cheap and lasts a long time compared to expensive ink cartridges.

26-4. CAD SOFTWARE



The CAD system creates an environment to prepare drawings interactively. Most CAD systems available commercially are menu driven. Commands can either be typed directly with the help of a keyboard or can be picked-up from the screen menu or from toolbar with the help of a mouse or can be selected from the digitizer menu. Some screen menus offer *pull-down* menus (also referred to as *pop-up* menus) and *dialogue boxes*.

For example, a variety of hatching patterns are displayed on the screen for better visualization and selection if a hatching command is chosen. The appropriate hatching pattern can be selected with the help of input device. The effect of every command is immediately displayed on the screen so that selection and corrections can be done interactively and immediately.

The major functions to be performed by a computer aided drafting system are:

- (a) Basic set-up of a drawing
- (b) Drawing the objects
- (c) Changing the object properties
- (d) Translating the objects
- (e) Scaling the objects
- (f) Clipping the objects to fit the image to the screen
- (g) Creating symbol libraries for frequently used objects
- (h) Text insertion
- (i) Dimensioning
- (j) Creates various layers (Transparent sheets)
- (k) Allows zoom-in and zoom-out of any components of drawing or complete drawing
- (l) Creates different numbers of print/plot layouts.

Some of the features of CAD systems are:

(1) Modelling and Drafting: The majority of systems provide 2D and 3D modelling capabilities. Some low cost CAD systems are dedicated to 2D drafting only.

(2) Ease of use: The users find CAD systems very easy to learn and use.

(3) Flexibility: Popular CAD systems provide greater flexibility when configuring the available hardware. Hundreds of computers, display devices, expansion boards, input and output devices are compatible and configurable with popular softwares.

(4) Modularity: Standard input and output devices are attached to standard connectors thereby making the system modular in nature.

(5) Low maintenance cost: Little maintenance is needed to keep the system functional.

Capabilities and versatilities of the drafting system vary depending on the system on which they are implemented. AutoCAD, VersaCAD, CADKey, DesignCAD, ZWCAD, etc. are few popular commercially available drafting systems in use. These systems provide a variety of features required for producing engineering drawings.

As an example of a popular, low-cost CAD software, we describe here some of the fundamental capabilities of AutoCAD. All these or similar facilities are very much desired in any CAD software for it to be useful.

26-5. AUTOCAD



AutoCAD (a product and registered trademark of Autodesk Inc., USA) is a low cost yet very effective computer aided design and drafting software. AutoCAD is accepted as the industry standard and it is preferred by a large community of CAD users in the world. Although AutoCAD is available for a variety of computer systems, majority of AutoCAD implementations are available on IBM or compatible personal computers with various operating systems. AutoCAD comes with a very large number of user-selectable options to support a great variety of commercially available display devices, digitizers, mouse, printers and plotters. AutoCAD supports 2D drafting and 3D modelling.

The basic drawing entities are lines, polylines [refer to art. 26-5-4 of any width, circles, arcs, ellipses and solids. There are many ways of defining a drawing entity, and the software always prompts the user for all options. Each drawing entity has an associated line-type, colour, layer [refer to art. 26-5-5(2)] and thickness. The thickness is a property associated with 3D entities.

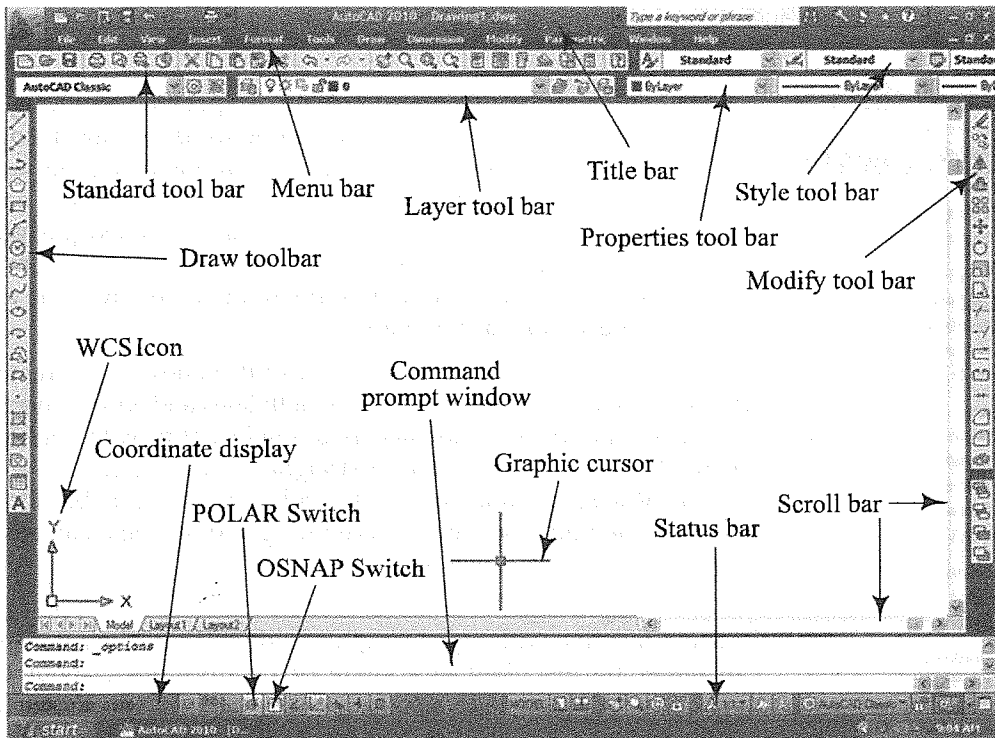
Before any drawing is started, the AutoCAD environment must be prepared for proper units of measurement, line-type, drawing size, layer, etc. In AutoCAD the drawings are always prepared at full scale, and the drawing size can be changed at any instant of time by using LIMITS command.

26-5-1. HARDWARE REQUIRED FOR AUTOCAD 2009/2010

- (1) P IV Intel processor and motherboard
- (2) 160/320 GB Hard disk
- (3) 1 GB/2 GB RAM
- (4) Microsoft Window XP Operating system software.

26-5-2. CLASSIC SCREEN LAYOUT OF AUTOCAD 2010

The default AutoCAD 2010 drawing screen can be divided into four areas such as drawing area, command area, menu bar area and tool bar area. The various component of Graphical User Interface (GUI) of AutoCAD 2010 is shown in fig. 26-3.



Components of AutoCAD Classic screen of AutoCAD 2010

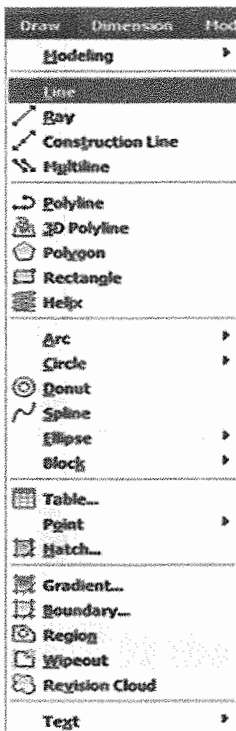
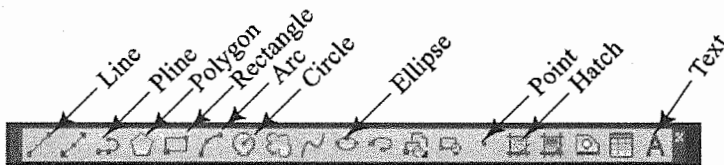
FIG. 26-3

26-5-3. FUNCTION KEYS



- | | | |
|-----|---|--------------|
| F1 | Online Help | |
| F2 | Toggles between Drawing screen to text screen | |
| F3 | Toggles between OSNAP | - On and Off |
| F4 | Toggles between Tablet | - On and Off |
| F5 | Switches among Isoplanes Top, Right and Left | |
| F6 | Toggles between Coordinates | - On and Off |
| F7 | Toggles between Grid | - On and Off |
| F8 | Toggles between Ortho Mode | - On and Off |
| F9 | Toggles between Snap Mode | - On and Off |
| F10 | Toggles between Polar Tracking | - On and Off |
| F11 | Toggles between Objects Snap Tracking | - On and Off |

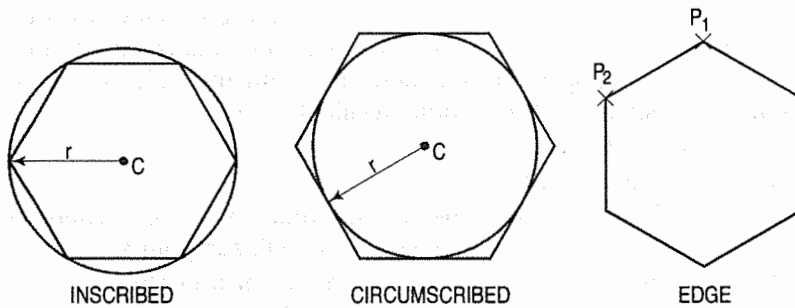
26-5-4. DRAWING ENTITIES (DRAW COMMANDS)



(1) **Line** (Refer module 26-1): A line is specified by giving its two endpoints. The LINE command can be used to draw a single line or a series of lines with the end-point of one being the start-point of the next. When a series of such lines is created, each line is treated as a separate entity. To create a closed polygon, the user has to type in C (close option) for the *To point:* prompt. This causes the last and the first points to be joined by a line and thus creating a closed boundary.

(2) **Pline** (Refer module 26-9, 26-25): Polylines are interesting drawing entities. Polylines can include both lines and arcs connected at end-points. Thus, a polyline is a single entity with multiple segments. The polylines can be straight or curved, can be wide (like a TRACE) or tapered. Fillets and chamfers can be added where needed on a polyline. Curve fitting and hatching can easily be performed on a polyline.

(3) **Polygon** (fig. 26-4) (Refer module 26-8, 26-12): A polygon is also a polyline with equal length of sides. The regular polygon can either be inscribed in a circle or circumscribed about the circle. The polygon may also be constructed by specifying the length of one side and the number of sides of polygon (called edges). In this method a polygon is constructed in anti-clockwise direction from the two edge end-points that have been specified.

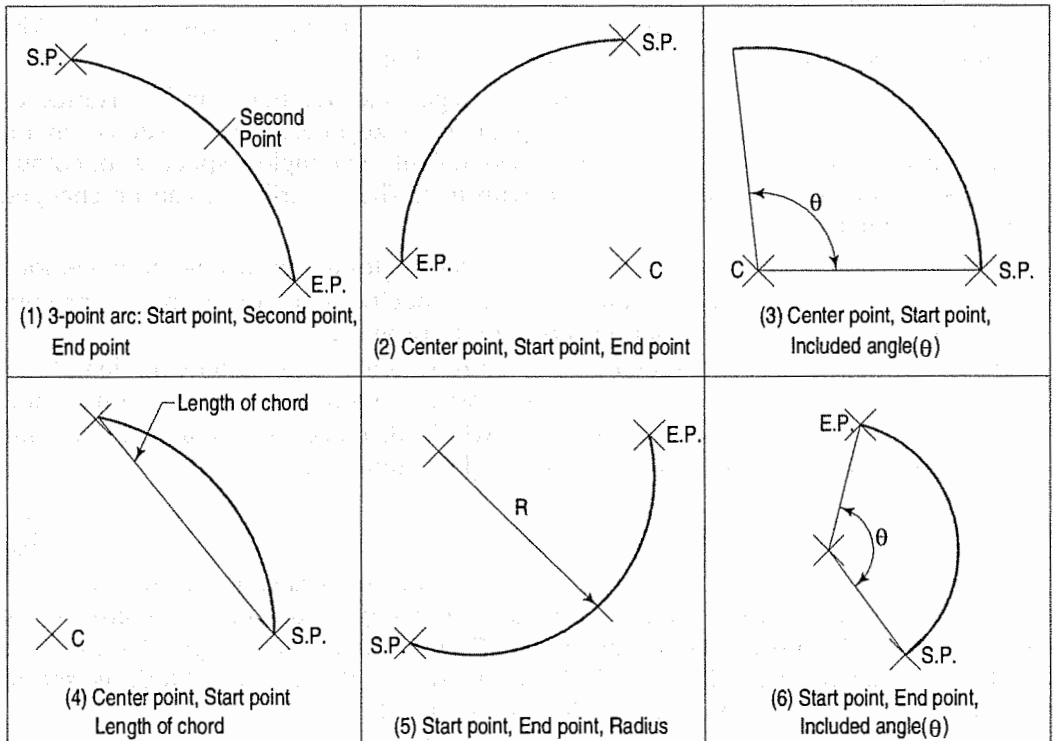


Polygon

FIG. 26-4

(5) **Arc** (fig. 26-5) (Refer module 26-8, 26-26, 26-27): This command is used to draw an arc accurately. Usually there are *three* parameters required for drawing an arc. Different ways of drawing circular arcs are:

- (i) **3 point arc:** The arc is drawn by specifying three points on the chord of arc. The first and third points define the start and end-points of an arc respectively.
- (ii) **Start, Center:** This option needs start point and center point of an arc. The third parameter may either be an end-point, included angle, or length of chord.
- (iii) **Start, End:** This option asks the user to enter the start and end-points of an arc. The arc is completed by either specifying radius or included angle or center point.



Arc definitions

FIG. 26-5

(6) **Circle** (Refer module 26-5, 26-11, 26-13): There are many ways of drawing a circle, the default being the centre point of circle and radius. Either on typing the command CIRCLE or selecting it from a menu bar with the help of mouse, all circle drawing options are displayed. The options available are:

- (a) Center point and Radius
- (b) Center point and Diameter
- (c) 3P: This specifies 3 points on the circumference of a circle. There is a unique circle passing through three given non-collinear points.
- (d) 2P: This specifies the end-points of diameter of a circle.
- (e) TTR (Tangent Tangent Radius): This command draws a circle of specified radius that is tangent to two lines, circles or arcs.
- (f) TTT (Tangent Tangent Tangent): This command draws a circle tangent to three entities.

(7) **Ellipse** (Refer module 26-11): An ellipse can be constructed by specifying the center point, radial length of major and minor axes. An ellipse can also be constructed by specifying end-points of one of its axis and the radial length of other axis.

(8) **Donut** (fig. 26-6) (Refer module 26-10): The DONUT is a special type of polyline which is made up of arc segments. A DONUT has two properties: it has width, and it is closed. The width of DONUT is set by specifying inside and outside diameters. The inside diameter may be zero thereby making it a filled circle.



Donut examples
FIG. 26-6

(9) **Hatch patterns** (Refer module 26-12): The HATCH command is used to fill up the area using a suitable pattern. The type of pattern and pattern variables can be chosen from a library of patterns available. The hatching will be carried out inside a closed defined area.

(10) **Text** (Refer module 26-6): Words, messages and numbers can be inserted as required on an engineering drawing. The alphanumeric keyboard is used extensively for non-graphical input such as text. The text style, height, text angle, aspect ratio, colour, etc. are some of the attributes associated with text. These attributes can be changed as per requirements.

(11) **Rectangle** (Refer module 26-6, 26-10): Rectangles are drawn by three methods

- (i) By specifying the co-ordinates at the "specify first corner point" prompt and at the "specify other corner point" prompt.
- (ii) By entering area of rectangle in current units and specifying length or width of rectangle. For this select "Area" option of rectangle command.
- (iii) By specifying length distance and width distance of rectangle. For this select "Dimensions" option of rectangle command.

26-5-5. DRAFTING AIDS

(1) **Limits** (Refer module 26-1 to 26-13): In AutoCAD, drawings are drawn in full scale therefore limits are needed to size up a drawing area. The limits are set by specifying X and Y co-ordinates of Lower Left Corner (L.L.C.) and Upper Right Corner (U.R.C.) of drawing area. By default L.L.C. is set to (0, 0) and U.R.C. is set to (12, 9) for imperial file and (420, 210) for metric file.

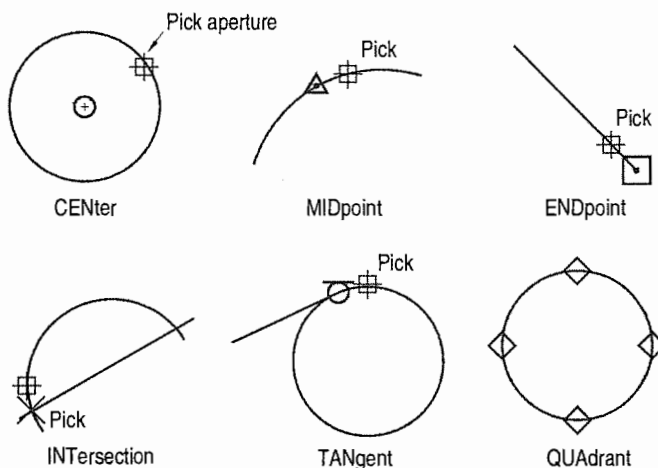
Note: By changing limits, the drawing display area does not change. It is required to use all option of the zoom command to display the limits inside the drawing area.

(2) **Layer** (Refer module 26-6, 26-13): A layer can be thought of as a transparent sheet on which drawings can be prepared. Drawings can be logically divided into different layers, and layers can be selectively displayed either individually or in any combination. Each layer is identified by a name. If the drawing becomes too dense or complicated, some layers can be turned off so that they do not interfere with the work. The drawing can be edited on any one layer at a time, called *current layer*. The colour, linetype and line weight are the properties associated with a layer. This controls an entity's colour, linetype and line weight drawn on that layer.

(3) **Dimensioning** (Refer module 26-7): The manufacturing drawing must be dimensioned for size and tolerances so that the right information can be conveyed. The appearance and size of dimension arrows, size and style of dimension text with or without tolerances, and the layer on which dimensions are placed can be controlled by setting dimension variables.

(4) **Object snap** (fig. 26-7) (Refer module 26-4): A very useful drawing aid, the OSNAP identifies the points on drawing entities that are visible on the screen. For example, the start point of an arc can be the endpoint of an existing line. This option allows the user to pick-up the points very accurately with respect to drawing displayed. Some of the OSNAP modes are:

- ⊗ NEarest : Closest point on an entity.
- CENter : Center of a circle or an arc.
- △ MIDpoint : Midpoint of a line or an arc.
- ENdpoint : Closest endpoint of a line or an arc.
- × INTersection : Intersection of two lines, two arcs, or a line with an arc or a circle.
- TANgent : Tangent to an arc or circle.
- ◇ QUadrant : Quadrant points will be located for circle and ellipses.

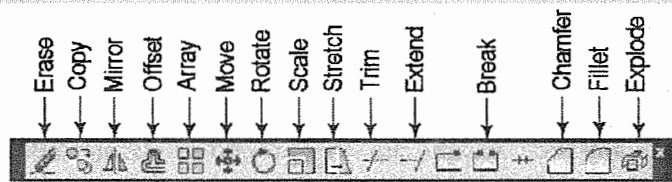


OSNAP options

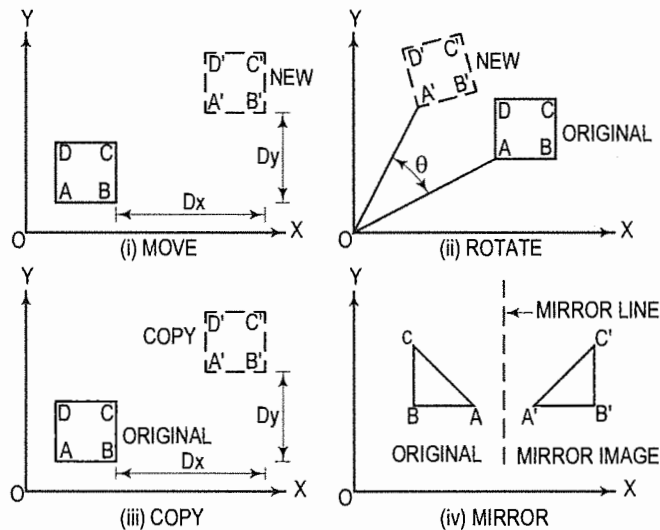
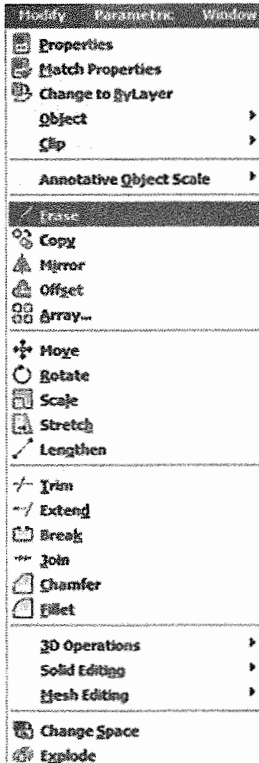
FIG. 26-7

(5) **Zoom** (Refer module 26-1): This is the most common method of magnifying a portion of current drawing on the screen. The portion of drawing to be zoomed is usually selected by a window. A window is identified by picking up the diagonal corners of a rectangle around the area of interest. Zoom also helps improving the accuracy of drawing by enlarging the details of drawing and providing better hand-eye coordination.

26-5-6. EDITING OF A DRAWING (MODIFY COMMANDS)



Modify Toolbar



Common drawing editing commands

FIG. 26-8

The fundamental commands to edit a drawing are:

(1) **Move** [fig. 26-8(i)]: Moves selected objects to another location about a base point.

(2) **Rotate** [fig. 26-8(ii)] (Refer module 26-12): Rotates selected objects through a specified angle about a base point.

(3) **Copy** [fig. 26-8(iii)] (Refer module 26-6): Creates one or more copies of selected objects at another location. The function of COPY command is similar to the MOVE command except that it preserves a copy of the objects selected at the original location.

(4) **Mirror** [fig. 26-8(iv)] (Refer module 26-8): Creates a mirror image of the selected objects about a specified line.

(5) **Array** (Refer module 26-10, 26-11): This command creates multiple copies of selected objects in rectangular or polar form. This is a form of COPY command.

(6) **Erase** (Refer module 26-8): This command deletes the selected entities. A record of entities erased is always maintained. The most recent entity can be *unerased* by OOPS command.

(7) **Oops** (Refer module 26-14): This command retrieves all objects erased by the last Erase and after executing Block or Wblock command.

(8) **Break**: This command erases a portion of line, arc, circle or a 2D polyline between two selected points.

(9) Fillet (Refer module 26-5, 26-13): This command is used to create a round corner between two lines. The lines are shortened or extended to fit a tangent arc of specified radius. FILLET works on any combination of two lines, arcs, circles, non-parallel lines, or a single polyline.

(10) Chamfer (Refer module 26-8): This command works on two lines or a single polyline to create a bevelled edge.

(11) Extend (Refer module 26-11): This command extends the lines, polylines and arcs to a boundary edge which can be a line, polyline, arc or circle. A closed polygon cannot be extended. When you invoke this command, you will be prompted to select the boundary edges. These edges can be lines, polylines, circles, arcs,, ellipse, xlines, rays, splines etc. After the boundary edges are selected, you must select each object to be extended. An object can be both a boundary edge and an object to be extended.

(12) Offset (Refer module 26-5): This command creates a parallel single copy of line, arc, circle, rectangle, polygon, or 2D polyline at a given offset distance. Each offset creates a new entity with the same linetype, color and layer settings.

(13) Stretch (Refer module 26-12): The STRETCH command can either lengthen entities or shorten them, and thus alter their shapes. The centre points of arcs or polyline arcs are adjusted accordingly.

(14) Trim (Refer module 26-5, 26-13): This command trims the objects that extend beyond a required point of intersection. When you invoke this command, you will be prompted to select the cutting edges. These edges can be lines, polylines, circles, arcs, ellipse, xlines, rays, splines, text, blocks or even viewports. After the cutting edges are selected, you must select each object to be trimmed. An object can be both a cutting edge and an object to be trimmed.

(15) Scale (Refer module 26-12): The SCALE command allows to shrink or enlarge the already existing drawing objects about a base point by specifying a scale factor.

(16) Pedit: A polyline is a single entity which is made up of a continuous series of line and arc segments. The PEDIT command is exclusively used for editing of polyline properties. The selected line, arc and polyline can be added to an existing polyline by a JOIN option. A smooth curve passing through all vertices of a polyline can be created by using FIT option. Similarly, a spline can also be constructed by using SPLINE option.

(17) Explode (Refer module 26-6): This command breaks a polyline into its individual segments. These segments can then individually be edited, and rejoined again to form an edited polyline.

(18) U (Refer module 26-8): The U command reverses the effects of a series of previously used commands and hence allows back-stepping. The REDO command wipes out the effect of U command.

The *advanced features* of AutoCAD are:

- (1) Built-in programming language AutoLISP provides programming environment so that AutoCAD commands can be called along with programs written for computations. This is very useful for parametric design and drawing.
- (2) Drawing Exchange Files (DXF) and script files can be used to interface AutoCAD with programs written in any other higher level language such as C. DXF and IGES file formats allow the exchange of drawing files among various drafting softwares.
- (3) AutoCAD provides techniques to define and extract attributes of entities. This feature is used in extracting information from a drawing for processing by other programs or to transfer it to a database.

Note: The commands not explained here above are covered up in the topics 26-6, 26-7, 26-8 and 26-9 as illustrative modules with on hand explanation.

26-6. SYMBOL LIBRARY (BLOCK, WBLOCK AND INSERT)



A complete drawing or a part of a drawing can be given a specific name and then stored as a block. A collection of such blocks form a library which is very useful for drawing purpose. The blocks, also referred to as symbols, can be scaled, rotated or mirrored if necessary and inserted into the drawing at the appropriate location e.g. bolt heads to be used at different locations in a drawing. Many suppliers of CAD add-on products offer symbol libraries of standard mechanical components, electronic components, architectural symbols, piping symbols, etc. The **block** and **wblock** command groups a number of selected entities together and treats them as a single object i.e. single block. Block commands saves the group of selected entities with block name and is inserted in the file in which it is created, but wblock command saves the group of selected entities as a drawing file and can be inserted in any drawing file. The blocks and wblocks can be scaled, rotated, stretched or mirrored. The explode command separates the entities of block and wblock.

INSERT (Refer module 26-14): INSERT command is used to insert blocks and wblocks in current drawing. Blocks will be inserted only in the drawing file in which they created and wblocks will be inserted in any drawing files.

26-7. TWO DIMENSIONAL DRAWINGS



The following modules (26-1 to 26-14) are self interactive and self learning practice modules to generate two dimensional drawings covering all possible 2-D commands. The configuration of computer will be as mentioned in art. 26-5-1.

(1) Absolute coordinate method: In the absolute coordinate method, the points are located with respect to the origin (0, 0). The syntax is **X, Y**.

Module 26-1. Draw a line diagram as shown in fig. 26-9. Use Absolute Coordinate Method.

To draw a line diagram using Absolute Coordinate Method, follow the steps mentioned below.

- (1) Command: **LIMITS** (↵)
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
Regenerating model.
- (3) Command: **LINE** (↵)
Specify first point: **20,20** (↵)
Specify next point or [Undo]: **100,20** (↵)
Specify next point or [Undo]: **100,50** (↵)
Specify next point or [Close/Undo]: **90,50** (↵)
Specify next point or [Close/Undo]: **85,65** (↵)
Specify next point or [Close/Undo]: **75,65** (↵)
Specify next point or [Close/Undo]: **70,50** (↵)
Specify next point or [Close/Undo]: **50,50** (↵)
Specify next point or [Close/Undo]: **50,65** (↵)

Specify next point or [Close/Undo]: **30,65** (↵)
Specify next point or [Close/Undo]: **30,50** (↵)
Specify next point or [Close/Undo]: **20,50** (↵)
Specify next point or [Close/Undo]: **C** (↵)

(4) Save This File As **Module 26-1.DWG**

Output of Module 26-1 (fig. 26-9):

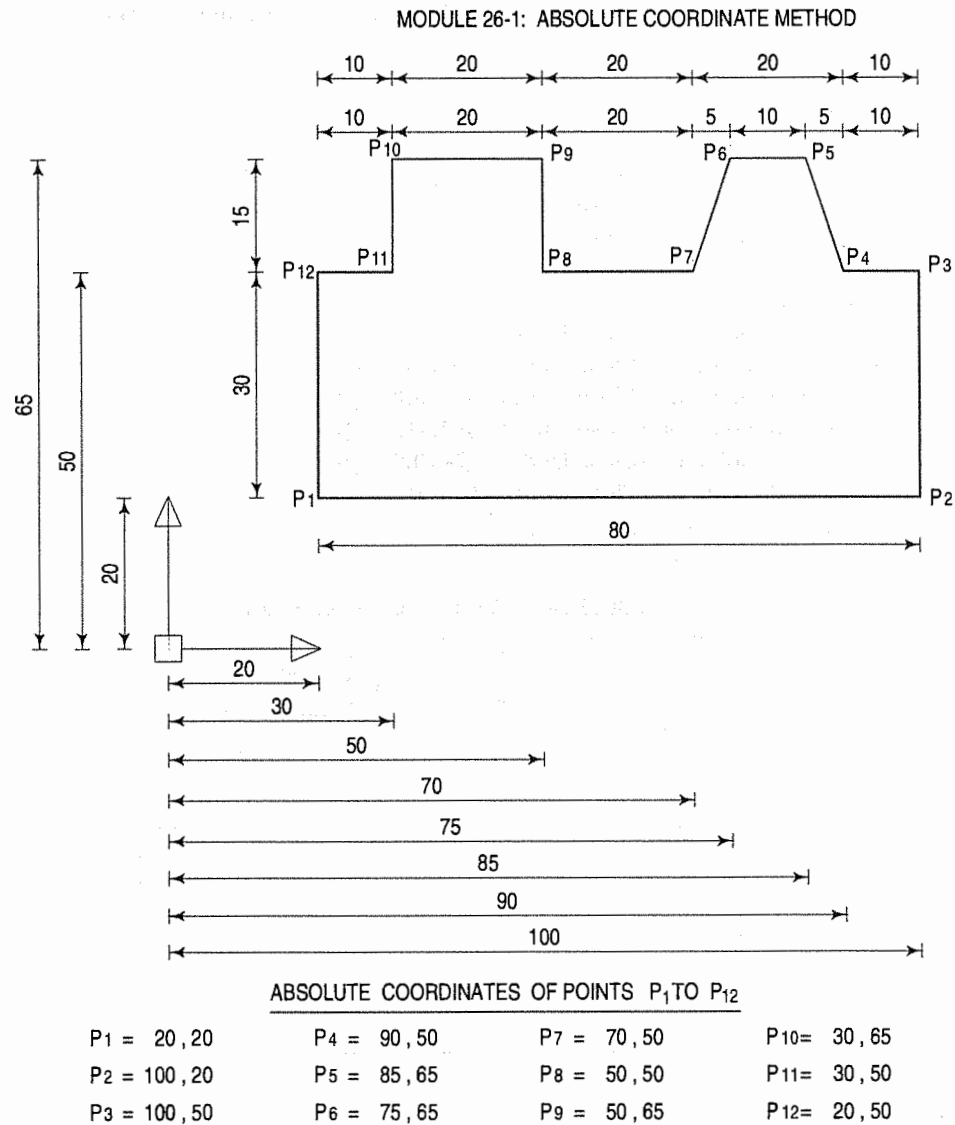


FIG. 26-9

(2) Relative coordinate method: In the relative coordinate method, the displacement along the X and Y axis are measured with reference to the previous point. The syntax is **@X, Y**.

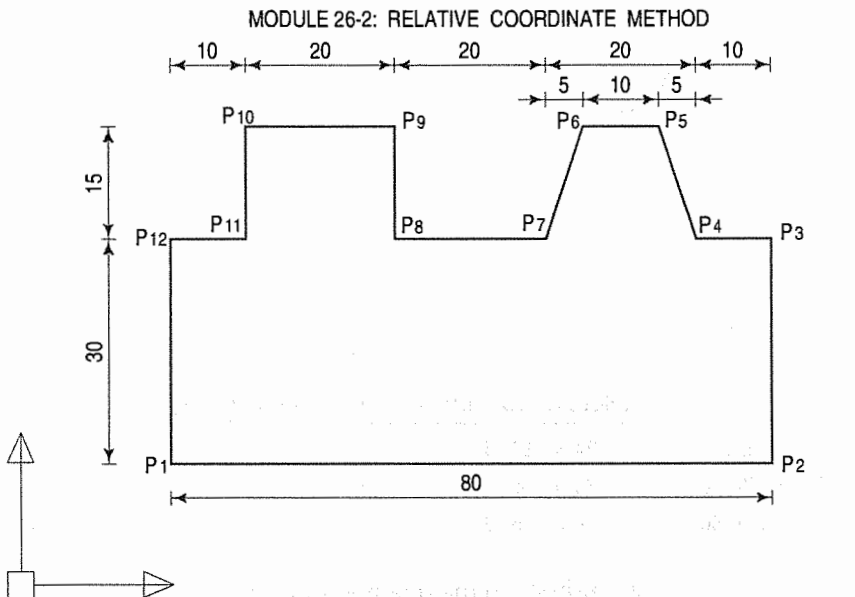
Module 26-2. Draw a line diagram as shown in fig. 26-10. Use Relative Coordinate Method.

To draw a line diagram using Relative Coordinate Method, follow the steps mentioned below.

- (1) Command: **LIMITS** (↵)
 Reset Model space limits:
 Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
 Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
 Specify corner of window, enter a scale factor (nX or nXP), or
 [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
 Regenerating model.
- (3) Command: **LINE** (↵)
 Specify first point: **20,20** (↵)
 Specify next point or [Undo]: **@80,0** (↵)
 Specify next point or [Undo]: **@0,30** (↵)
 Specify next point or [Close/Undo]: **@-10,0** (↵)
 Specify next point or [Close/Undo]: **@-5,15** (↵)
 Specify next point or [Close/Undo]: **@-10,0** (↵)
 Specify next point or [Close/Undo]: **@-5,-15** (↵)
 Specify next point or [Close/Undo]: **@-20,0** (↵)
 Specify next point or [Close/Undo]: **@0,15** (↵)
 Specify next point or [Close/Undo]: **@-20,0** (↵)
 Specify next point or [Close/Undo]: **@0,-15** (↵)
 Specify next point or [Close/Undo]: **@-10,0** (↵)
 Specify next point or [Close/Undo]: **C** (↵)

(4) Save This File As **Module 26-2.DWG**

Output of Module 26-2 (fig. 26-10):



RELATIVE COORDINATES OF POINTS P₁ TO P₁₂

P ₁ = 20,20	P ₄ = @-10,0	P ₇ = @-5,-15	P ₁₀ = @-20,0
P ₂ = @80,0	P ₅ = @-5,15	P ₈ = @-20,0	P ₁₁ = @0,-15
P ₃ = @0,30	P ₆ = @-10,0	P ₉ = @0,15	P ₁₂ = @-10,0

FIG. 26-10

(3) **Relative polar coordinate method:** In the relative polar coordinate method a point is located by defining the direct distance of the point from current point and the angle that the direct distance makes with the positive X-axis in counter clockwise direction. It's syntax is **@direct distance < angle**.

Module 26-3. Draw a line diagram as shown in fig. 26-11. Use Relative Polar Coordinate Method.

Output of Module 26-3 (fig. 26-11):

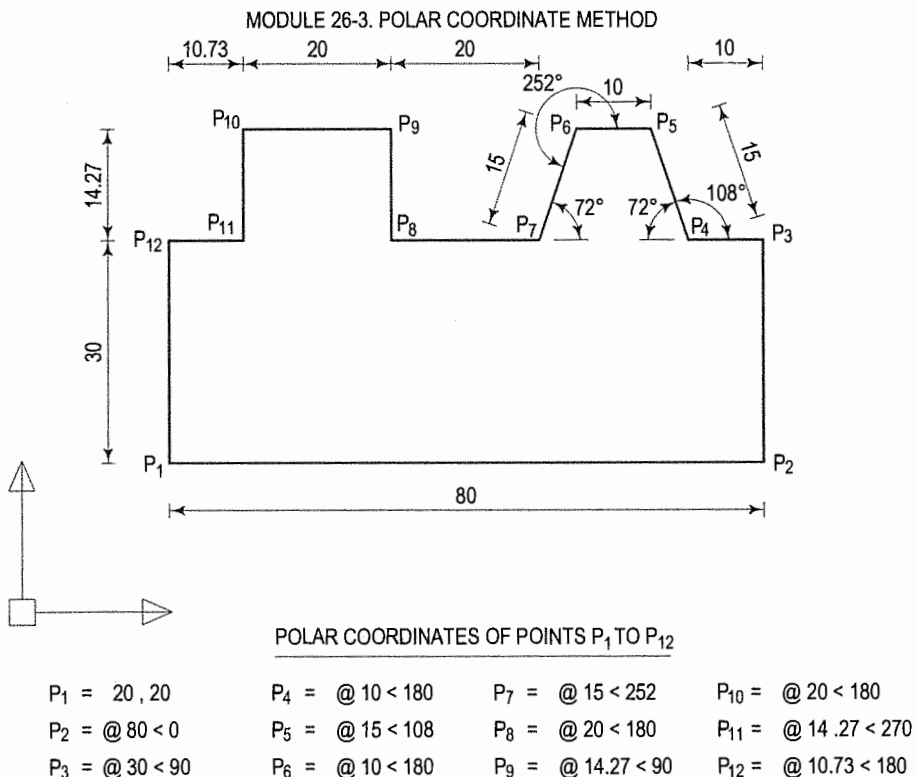


FIG. 26-11

To draw a line diagram using Polar Coordinate Method, follow the steps mentioned below.

- (1) Command: **LIMITS** (↵)
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
Regenerating model.
- (3) Command: **LINE** (↵)
Specify first point: **20,20** (↵)
Specify next point or [Undo]: **@80<0** (↵)
Specify next point or [Undo]: **@30<90** (↵)
Specify next point or [Close/Undo]: **@10<180** (↵)
Specify next point or [Close/Undo]: **@15<108** (↵)

Specify next point or [Close/Undo]: @10<180 (↵)
 Specify next point or [Close/Undo]: @15<252 (↵)
 Specify next point or [Close/Undo]: @20<180 (↵)
 Specify next point or [Close/Undo]: @14.27<90 (↵)
 Specify next point or [Close/Undo]: @20<180 (↵)
 Specify next point or [Close/Undo]: @14.27<270 (↵)
 Specify next point or [Close/Undo]: @10.73<180 (↵)
 Specify next point or [Close/Undo]: C (↵)

(4) Save This File As **Module 26-3.DWG**

Module 26-4. Draw a line diagram as shown in fig. 26-12. Use Object-Snap (Osnap) Method.

To draw a line diagram using Object-Snap (Osnap) Method, follow the steps mentioned below.

- (1) Command: **LIMITS** (↵)
 Reset Model space limits:
 Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
 Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
 Specify corner of window, enter a scale factor (nX or nXP), or [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
 Regenerating model.
- (3) Command: **LINE** (↵)
 Specify first point: **20,20** (↵)
 Specify next point or [Undo]: @80<0 (↵)
 Specify next point or [Undo]: @45<90 (↵)
 Specify next point or [Close/Undo]: @80<180 (↵)
 Specify next point or [Close/Undo]: C (↵)
- (4) Press **F3** key to Switch ON "OSNAP" if it is OFF
- (5) Command: **LINE** (↵)
 Specify first point: (Get Mouse Near To **P5** and CLICK When MIDPOINT magnet gets ON)
 Specify next point or [Undo]: (Get Mouse Near To **P6** and CLICK When MIDPOINT magnet gets ON)
 Specify next point or [Undo]: (↵)
- (6) Command: **LINE** (↵)
 Specify first point: (Get Mouse Near To **P7** and CLICK When MIDPOINT magnet gets ON)
 Specify next point or [Undo]: (Get Mouse Near To **P8** and CLICK When MIDPOINT magnet gets ON)
 Specify next point or [Undo]: (↵)
- (7) Command: **LINE** (↵)
 Specify first point: (Get Mouse Near To **P1** and CLICK When ENDPOINT magnet gets ON)
 Specify next point or [Undo]: (Get Mouse Near To **P3** and CLICK When ENDPOINT magnet gets ON)
 Specify next point or [Undo]: (↵)
- (8) Command: **LINE** (↵)
 Specify first point: (Get Mouse Near To **P2** and CLICK When ENDPOINT magnet gets ON)
 Specify next point or [Undo]: (Get Mouse Near To **P4** and CLICK When ENDPOINT magnet gets ON)
 Specify next point or [Undo]: (↵)
- (9) Save This File As **Module 26-4.DWG**

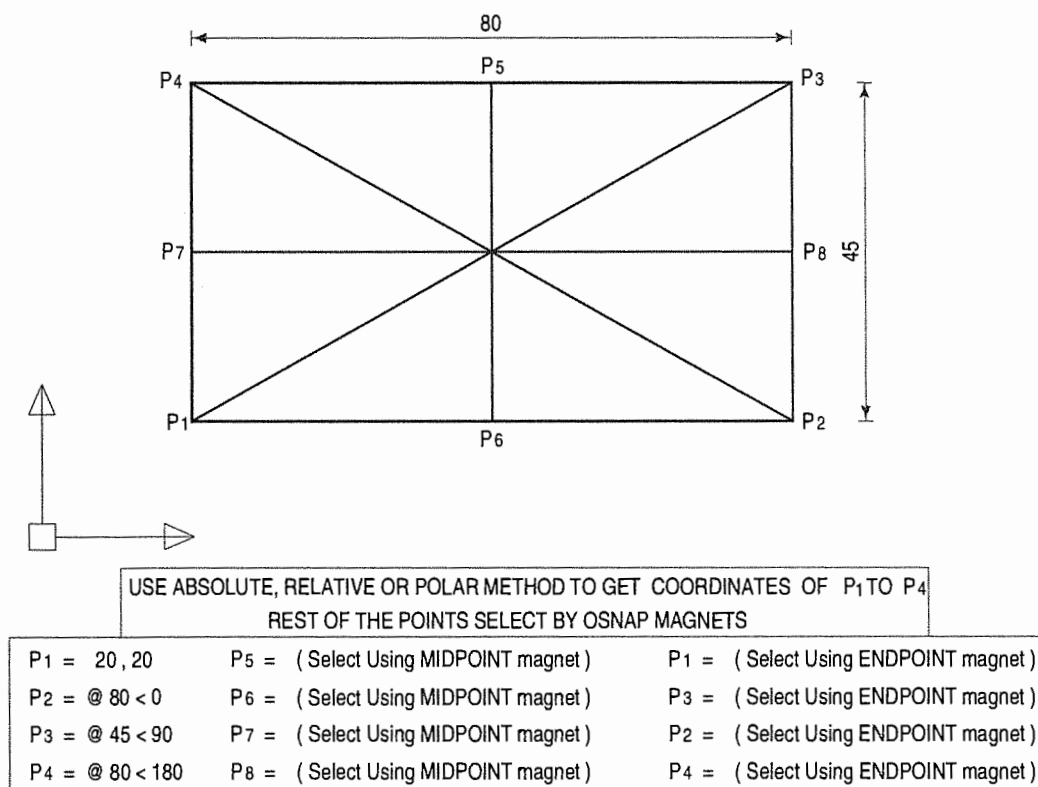
Output of Module 26-4 (fig. 26-12):**MODULE 26-4: USE OF OBJECT-SNAP (OSNAP)**

FIG. 26-12

Module 26-5. Draw a line diagram as shown in fig. 26-13. Use Circle, Offset, Trim and Fillet commands.

To draw a line diagram using circle, offset, trim and fillet commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-13(i) to fig. 26-13(x).

- (1) {Switch ON POLAR switch by Mouse Click or press **F10** key}
- (2) Command: **LIMITS** (↵)
 Reset Model space limits:
 Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
 Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (3) Command: **ZOOM** (↵)
 Specify corner of window, enter a scale factor (nX or nXP), or
 [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
- (4) Command: **LINE** (↵)
 Specify first point: **20,20** (↵)
 Specify next point or [Undo]: **90** (↵) (When 0° POLAR is ON)
 Specify next point or [Undo]: **20** (↵) (When 90° POLAR is ON)

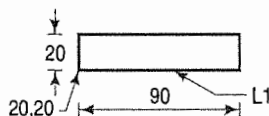


FIG. 26-13(i)

(13) Command: **TRIM** (↵)

Current settings: Projection=UCS, Edge=None

Select cutting edges ...

Select objects: (Select Line **L9**)Select objects: (Select Line **L10**)

Select objects: (↵)

Select object to trim or shift-select to extend or [Project/Edge/Undo]: (Select Line **L1** from Arrow Head)

Select object to trim or shift-select to extend or [Project/Edge/Undo]: (↵)

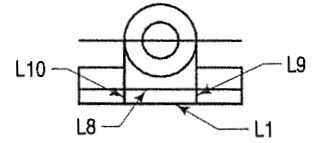


FIG. 26-13(vi)

(14) Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 0.0000

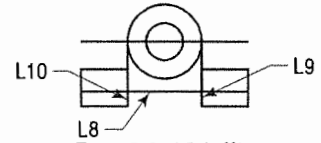
Select first object or [Polyline/Radius/Trim]: **R** (↵)Specify fillet radius <0.0000>: **2.5** (↵)Select first object or [Polyline/Radius/Trim]: (Select Line **L8** from Arrow Head)Select second object: (Select Line **L10** from Arrow Head)

FIG. 26-13(vii)

(15) Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 2.5000

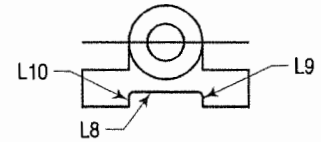
Select first object or [Polyline/Radius/Trim]: (Select Line **L8** from Arrow Head)Select second object: (Select Line **L9** from Arrow Head)

FIG. 26-13(viii)

(16) Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 2.5000

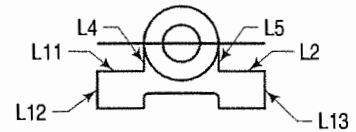
Select first object or [Polyline/Radius/Trim]: **R** (↵)Specify fillet radius <2.5000>: **5** (↵)Select first object or [Polyline/Radius/Trim]: (Select Line **L4** from Arrow Head)Select second object: (Select Line **L11** from Arrow Head)

FIG. 26-13(ix)

(17) Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 5.0000

Select first object or [Polyline/Radius/Trim]: (Select Line **L5** from Arrow Head)Select second object: (Select Line **L2** from Arrow Head)(18) Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 5.0000

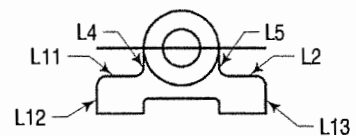
Select first object or [Polyline/Radius/Trim]: (Select Line **L11** from Arrow Head)Select second object: (Select Line **L12** from Arrow Head)

FIG. 26-13(x)

(19) Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 5.0000

Select first object or [Polyline/Radius/Trim]: (Select Line **L2** from Arrow Head)Select second object: (Select Line **L13** from Arrow Head)(20) Save This File As **Module 26-5.DWG****Output of Module 26-5** (fig. 26-13):

New color [Truecolor/Colorbook] : **MAGENTA** (↵)

Enter name list of layer(s) for color 6 (magenta) <0>: **TEXT** (↵)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/...]: (↵)

- (4) Command: **RECTANGLE** (↵)

Specify first corner point or [Chamfer/Elevation/Fillet/Thickness/Width]: **20,20** (↵)

Specify other corner point or [Dimensions]: **@100,45** (↵)

- (5) Command: **EXPLODE** (↵)

Select objects: (Select Rectangle **REC-1**)

Select objects: (↵)

- (6) Use **OFFSET** and **TRIM** Command to complete the Drawing as shown on Right Side

- (7) Command: **-LAYER** (↵)

Current layer: "0"

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight...]: **SET** (↵)

Enter layer name to make current <0>: **GRID** (↵)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight...]: (↵)

- (8) Command: **LINE** (↵)

Specify first point: (Select **B1** by **ENDPOINT** Osnap)

Specify next point or [Undo]: (Select **B6** by Osnap)

Specify next point or [Undo]: (↵)

- (9) Command: **COPY** (↵)

Select objects: (Select Line **L1**)

Select objects: (↵)

Specify base point or disp... : (Select **B1** by Osnap)

Specify second point of dis.. : (Select **B2** by Osnap)

Specify second point of dis.. : (Select **B3** by Osnap)

Specify second point of dis.. : (Select **B4** by Osnap)

Specify second point of dis.. : (Select **B5** by Osnap)

Specify second point of dis.. : (Select **B6** by Osnap)

Specify second point of dis.. : (Select **B7** by Osnap)

Specify second point of dis.. : (↵)

- (10) Command: **LINE** (↵)

Specify first point: (Select **B8** by Osnap)

Specify next point or [Undo]: (Select **B9** by Osnap)

Specify next point or [Undo]: (↵)

- (11) Command: **LINE** (↵)

Specify first point: (Select **B10** by Osnap)

Specify next point or [Undo]: (Select **B11** by Osnap)

Specify next point or [Undo]: (↵)

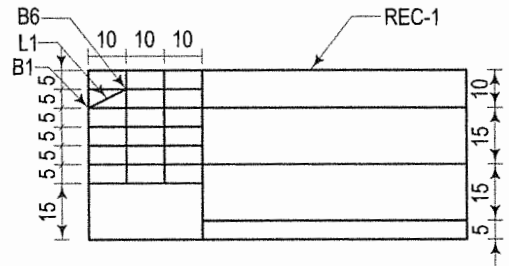


FIG. 26-14(ii)

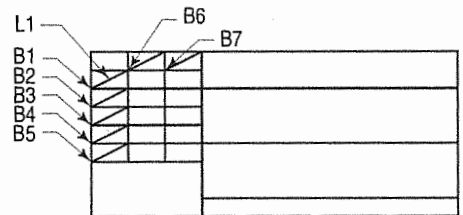


FIG. 26-14(iii)

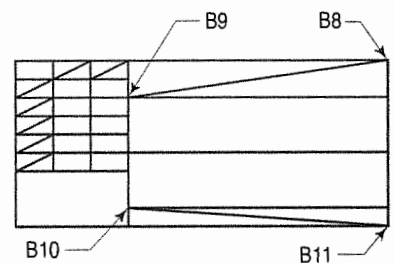


FIG. 26-14(iv)

(12) Command: **LINE** (↵)

Specify first point: (Select **B12** by Osnap)

Specify next point or [Undo]: **@70,7.5** (↵) (Point **B13**)

Specify next point or [Undo]: (Select **B9** by Osnap)

Specify next point or [Close/Undo]: (↵)

(13) Command: **LINE** (↵)

Specify first point: (Select **B15** by Osnap)

Specify next point or [Undo]: **@30,7.5** (↵) (Point **B16**)

Specify next point or [Undo]: (Select **B14** by Osnap)

Specify next point or [Close/Undo]: (↵)

(14) Command: **-LAYER** (↵)

Current layer: "0"

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight...]: **SET** (↵)

Enter layer name to make current <0>: **TEXT** (↵)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight...]: (↵)

(15) Command: **DTEXT** (↵)

Current text style: "Standard" Text height: 0.2000

Specify start point of text or [Justify/Style]: **J** (↵)

Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/ML/MC/...]: **MC** (↵)

Specify middle point of text: (Select **M1** by Osnap)

Specify height <0.2000>: **2** (↵)

Specify rotation angle of text <0>: **0** (↵)

Enter text: **NAME** (↵)

Enter text: (↵)

(16) Command: **DTEXT** (↵)

Current text style: "Standard" Text height: 2.0000

Specify start point of text or [Justify/Style]: **J** (↵)

Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/ML/MC/...]: **MC** (↵)

Specify middle point of text: (Select **M2** by Osnap)

Specify height <2.0000>: **3** (↵)

Specify rotation angle of text <0>: **0** (↵)

Enter text: **CHAROTAR PUBLISHING HOUSE** (↵)

Enter text: (↵)

(17) Command: **DTEXT** (↵)

Current text style: "Standard" Text height: 2.0000

Specify start point of text or [Justify/Style]: **J** (↵)

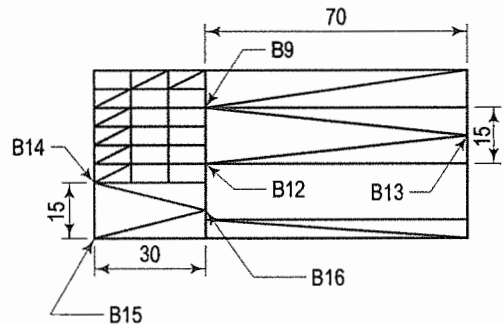


FIG. 26-14(v)

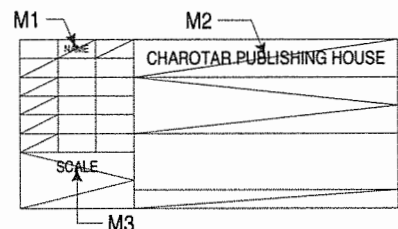


FIG. 26-14(vi)

Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/ML/MC/...]: **MC** (↵)

Specify middle point of text: (Select **M3** by Osnap)

Specify height <3.5000>: **3** (↵)

Specify rotation angle of text <0>: **0** (↵)

Enter text: **SCALE** (↵)

Enter text: (↵)

(18) Complete The Text Writing by DTEXT Command

(19) Command: **-LAYER** (↵)

Current layer: "0"

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/...]: **OFF** (↵)

Enter name list of layer(s) to turn off or <select objects>: **GRID** (↵)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/...]: (↵)

(20) Save This File As **Module 26-6.DWG**

Output of Module 26-6 (fig. 26-14):

MODULE 26-6: USE OF LAYER, RECTANGLE, EXPLODE, TEXT AND COPY COMMANDS

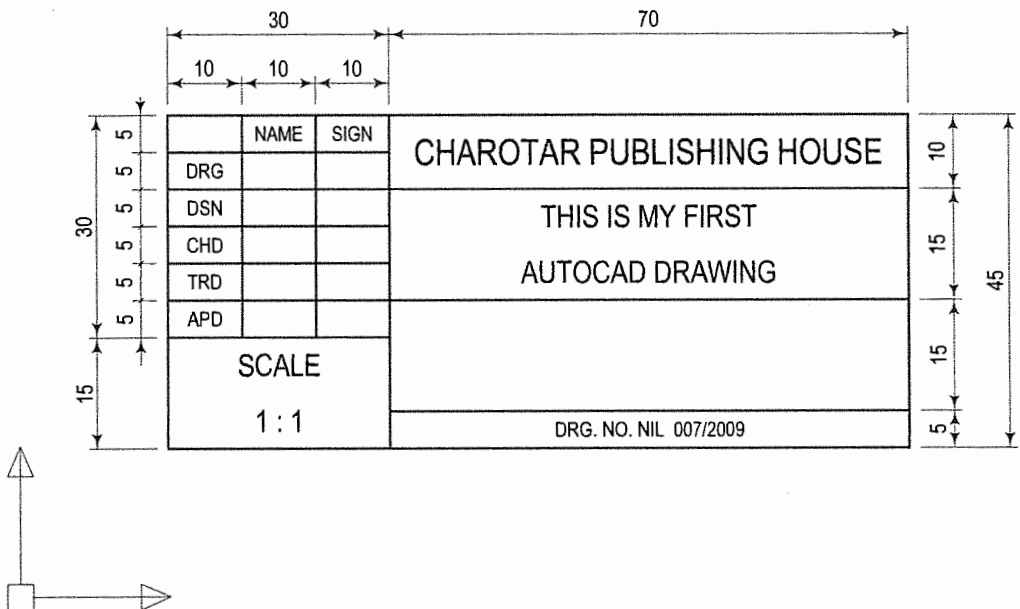


FIG. 26-14

Module 26-7. To generate Dimensions on the given diagram (fig. 26-15).

To generate Dimensions on the given diagram, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-15(i) to fig. 26-15(iv).

(1) Open drawing file **Module 26-5.DWG** (fig. 26-13)

(2) Command: **DIMSCALE** (↵)

Enter new value for DIMSCALE <1.0000>: **10** (↵)

- (3) Command: **DIMDEC** (↵)

Enter new value for DIMDEC <4>: **0** (↵)

- (4) Command: **DIMEXO** (↵)

Enter new value for DIMEXO <.0625>: **0.2** (↵)

- (5) Command: **DIMLINEAR** (↵)

Specify first extension line origin or <select object>:
(Select **B1** by OSNAP)

Specify second extension line origin: (Select **B2** by OSNAP)

Specify dimension line location or

[Mtext/Text/Angle/Horizontal/Vertical/Rotated]: (Select Point Below **B1** or **B2**)

Dimension text = **25**

- (6) Command: **DIMCONTINUE** (↵)

Specify a second extension line origin or [Undo/Select]
<Select>: (Select **B3** by OSNAP)

Dimension text = **40**

Specify a second extension line origin or [Undo/Select] <Select>: (Select **B4** by OSNAP)

Dimension text = **25**

Specify a second extension line origin or [Undo/Select] <Select>: (↵)

Select continued dimension: (↵)

- (7) Command: **DIMLINEAR** (↵)

Specify first extension line origin or <select object>: (Select **B4** by OSNAP)

Specify second extension line origin: (Select **B5** by OSNAP)

Specify dimension line location or

[Mtext/Text/Angle/Horizontal/Vertical/Rotated]: (Click on Right of **B4**)

Dimension text = **20**

- (8) Command: **DIMBASELINE** (↵)

Specify a second extension line origin or [Undo/Select] <Select>: (Select **B6** by OSNAP)

Dimension text = **35**

Specify a second extension line origin or [Undo/Select] <Select>: (↵)

Select base dimension: (↵)

- (9) Command: **DIMDIAMETER** (↵)

Select arc or circle: (Select Circle **C1**)

Dimension text = **40**

Specify dimension line location or [Mtext/Text/Angle]: (Click Outside **C1**)

- (10) Command: **DIMDIAMETER** (↵)

Select arc or circle: (Select Circle **C2**)

Dimension text = **20**

Specify dimension line location or [Mtext/Text/Angle]: (Click Outside **C2**)

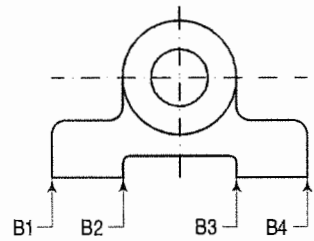


FIG. 26-15(i)

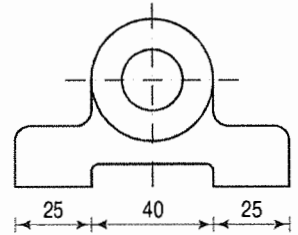


FIG. 26-15(ii)

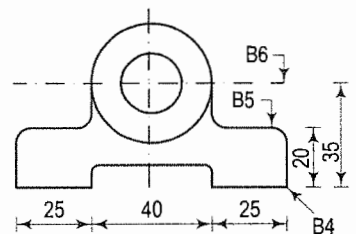
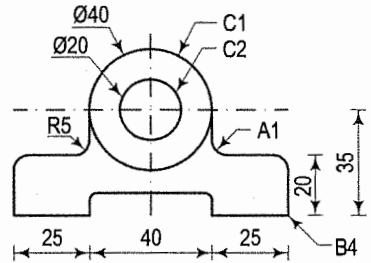


FIG. 26-15(iii)

- (11) Command: **DIMRADIUS** (↵)
 Select arc or circle: (Select Arc **A1**)
 Dimension text = **5**
 Specify dimension line location or [Mtext/Text/Angle]: (Click Outside **A1**)
- (12) Similarly complete the other dimensions.
- (13) Save This File As **Module 26-7.DWG**

Output of Module 26-7 (fig. 26-15):



MODULE 26-7: USE OF DIMENSIONS FIG. 26-15(iv)

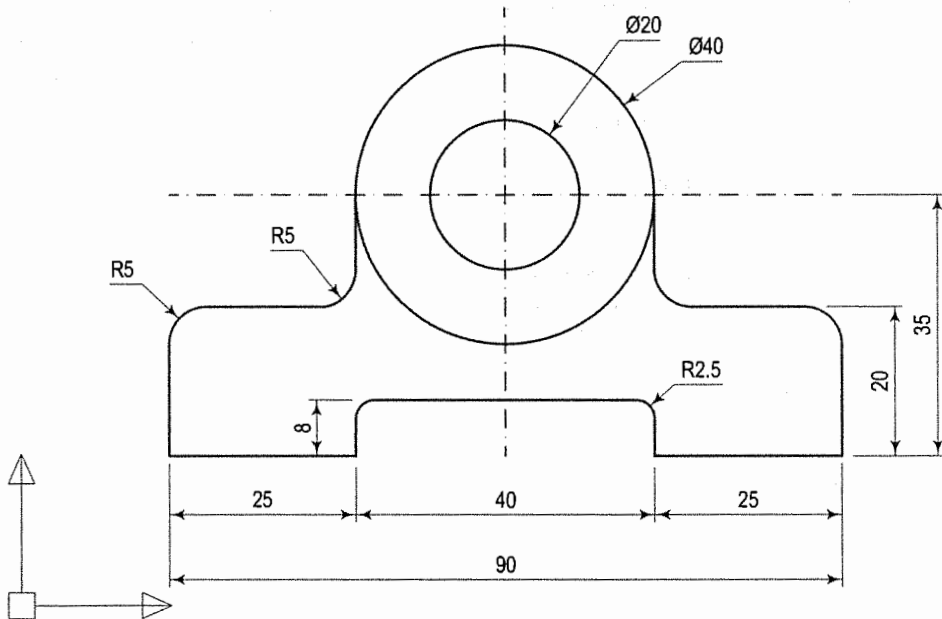


FIG. 26-15

Module 26-8. Draw a line diagram as shown in fig. 26-16. Use Polygon, Chamfer, Arc and Mirror commands.

To draw a line diagram using Polygon, Chamfer, Arc and Mirror commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-16(i) to fig. 26-16(v).

- (1) Command: **LIMITS** (↵)
 Reset Model space limits:
 Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
 Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
 Specify corner of window, enter a scale factor (nX or nXP), or
 [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
 Regenerating model.
- (3) Command: **LINE** (↵)
 Specify first point: **30,65** (↵)
 Specify next point or [Undo]: **20** (↵) (When 0° POLAR is ON)

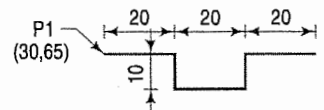


FIG. 26-16(i)

Specify first point of mirror line: (Click MIDPOINT of 13)

Specify second point of mirror line: (Click When 270° POLAR is ON)

Delete source objects? [Yes/No] <N>: N (↵)

(11) Command: **ERASE** (↵)

Select objects: **ALL** (↵)

Select objects: (↵)

(12) Command: **U** (↵)

(13) Save This File As **Module 26-8.DWG**

Output of Module 26-8 (fig. 26-16):

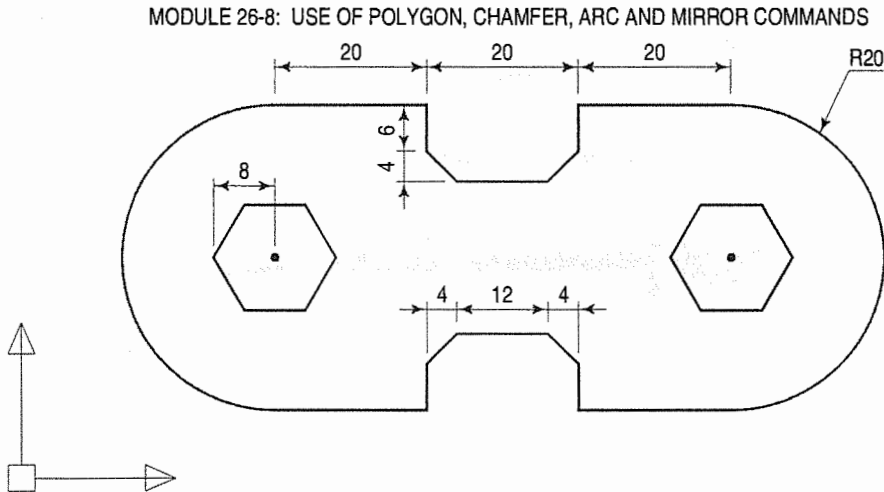


FIG. 26-16

Module 26-9. Draw a line diagram as shown in fig. 26-17. Use Polyline command.

To draw a line diagram using Polyline command, follow the steps mentioned below.

(1) Command: **LIMITS** (↵)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)

Specify upper right corner <12.0000,9.0000>: **120,90** (↵)

(2) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/

Window] <real time>: **ALL** (↵)

Regenerating model.

(3) Command: **PLINE** (↵)

Specify start point: **15,20** (↵)

Current line-width is 0.0000

Specify next point or [Arc/Halfwidth/Length/Undo/Width]: **WIDTH** (↵)

Specify starting width <0.0000>: **0** (↵)

Specify ending width <0.0000>: **10** (↵)

Specify next point or [Arc/Halfwidth/Length/...]: **30** (↵) (When 0° POLAR is ON)

Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: **WIDTH** (↵)

Specify starting width <10.0000>: **2.5** (↵)

Specify ending width <2.5000>: **2.5** (↵)

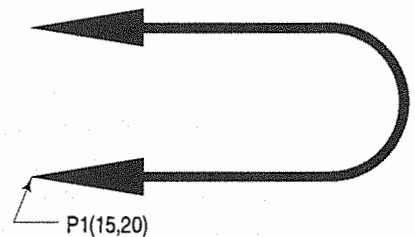


FIG. 26-17(i)

Specify next point or [Arc/Close/Halfwidth/...]: **50** (↵) (When 0° POLAR is ON)
 Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: **ARC** (↵)
 Specify endpoint of arc or
 [Angle/CEnter/CLose/Direction/Halfwidth/...]: **40** (↵) (When 90° POLAR is ON)
 Specify endpoint of arc or
 [Angle/CEnter/CLose/Direction/Halfwidth/Line/Radius/...]: **LINE** (↵)
 Specify next point or [Arc/Close/Halfwidth/...]: **50** (↵) (When 180° POLAR is ON)
 Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: **WIDTH** (↵)
 Specify starting width <2.5000>: **10** (↵)
 Specify ending width <10.0000>: **0** (↵)
 Specify next point or [Arc/Close/Halfwidth/...]: **30** (↵) (When 180° POLAR is ON)
 Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: (↵)

(4) Save this File As **Module 26-9.DWG**

Output of Module 26-9 (fig. 26-17):

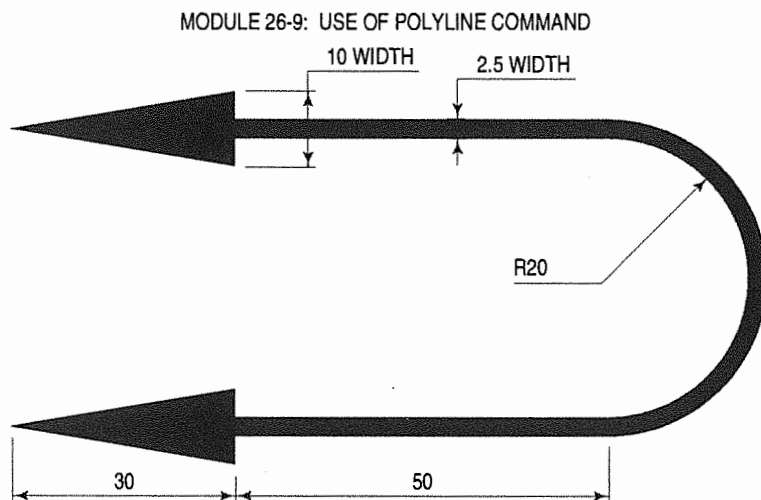


FIG. 26-17

Module 26-10. Draw a line diagram as shown in fig. 26-18. Use *Donut* and *Rectangular Array*, commands.

To draw a line diagram using *Donut* and *Rectangular Array* commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-18(i) to fig. 26-18(iii).

- (1) Command: **LIMITS** (↵)
 Reset Model space limits:
 Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
 Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
 Specify corner of window, enter a scale factor (nX or nXP), or
 [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
 Regenerating model.
- (3) Command: **RECTANGLE** (↵)
 Specify first corner point or [Chamfer/Elevation/Fillet/
 Thickness/Width]: **10,20** (↵)
 Specify other corner point or [Dimensions]:
@100,50 (↵)

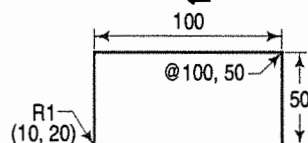


FIG. 26-18(i)

- (4) Command: **DONUT** (↵)
Specify inside diameter of donut <0.5000>: **4** (↵)
Specify outside diameter of donut <1.0000>: **6** (↵)
Specify center of donut or <exit>: **FROM** (↵)
Base point: (Select **R1** Point by ENDPOINT magnet) <Offset>:
@10,10 (↵)
Specify center of donut or <exit>: (↵)
- (5) Command: **ARRAY** (↵)
Select objects: (Select Donut **D1**)
Select objects: (↵)

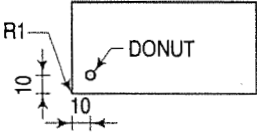


FIG. 26-18(ii)

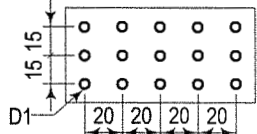
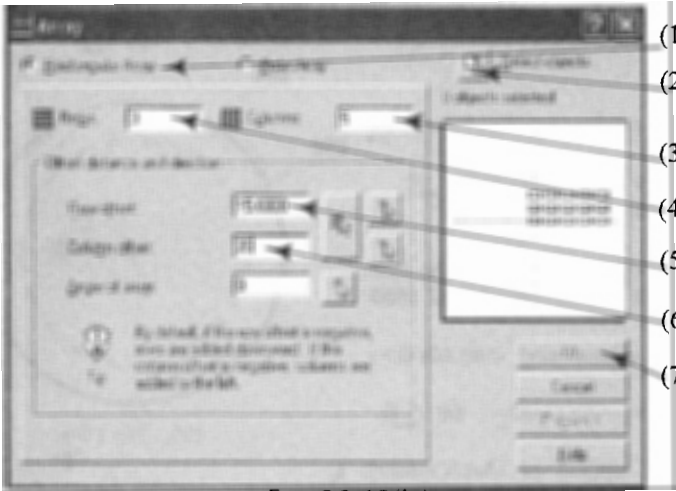


FIG. 26-18(iii)



- (1) Select Rectangular area
(2) Click here and select object
i.e. donut
(3) Enter No. of columns = 5
(4) Enter No. of rows = 3
(5) Enter row distance = 15
(6) Enter column distance = 20
(7) Finally click OK

FIG. 26-18(iv)

(6) Save this File As **Module 26-10.DWG**

Output of Module 26-10 (fig. 26-18):

MODULE 26-10: USE OF DONUT AND RECTANGULAR ARRAY COMMANDS

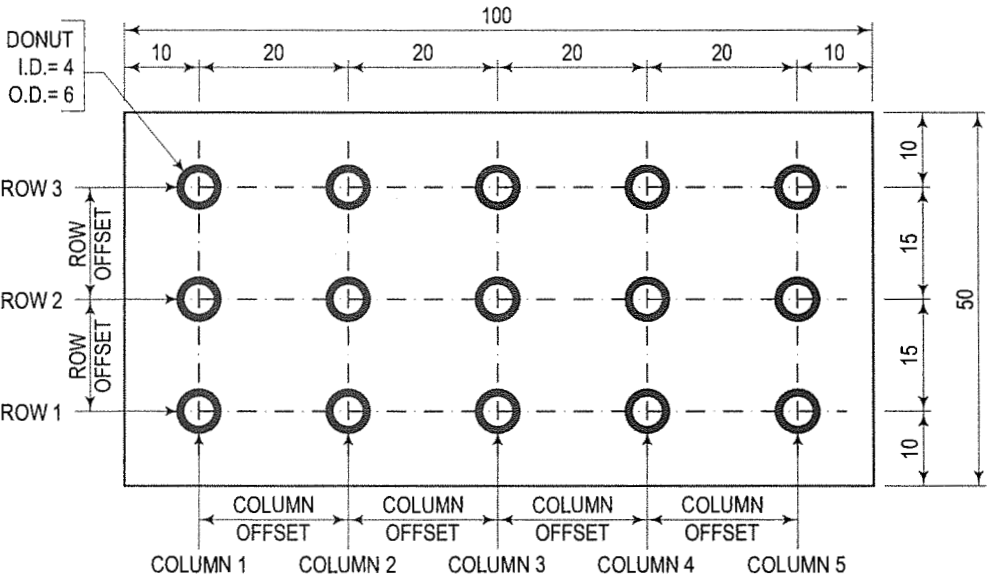


FIG. 26-18

Module 26-11. Draw a diagram as shown in fig. 26-19. Use *Ellipse* and *Polar Array* commands.

To draw a diagram using *Ellipse* and *Polar Array* commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-19(i) to fig. 26-19(iii).

- (1) Command: **LIMITS** (↵)
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
Regenerating model.
- (3) Command: **CIRCLE** (↵)
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: **60,45** (↵)
Specify radius of circle or [Diameter]: **D** (↵)
Specify diameter of circle: **80** (↵)
- (4) Command: **CIRCLE** (↵)
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: (Select CENTER magnet **C1** of **Ø80** Circle)
Specify radius of circle or [Diameter] <40.0000>:
D (↵)
Specify diameter of circle <80.0000>: **60** (↵)
- (5) Command: **ELLIPSE** (↵)
Specify axis endpoint of ellipse or [Arc/Center]: **C** (↵)
Specify center of ellipse: (Select **90° QUADRANT Q1** magnet of **Ø60** Circle)
Specify endpoint of axis: **6** (↵) (when **90° POLAR** is ON)
Specify distance to other axis or [Rotation]: **3** (↵)
- (6) Command: **ARRAY** (↵)
Specify center point of array: (Select CENTER magnet **C1**)
Select objects: (Select Ellipse **E1**)
Select objects: (↵)

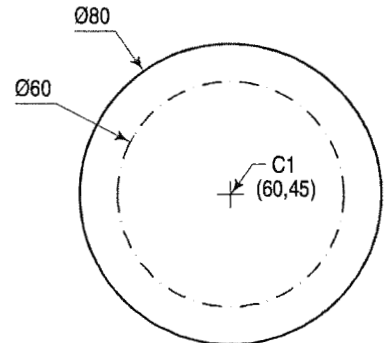


FIG. 26-19(i)

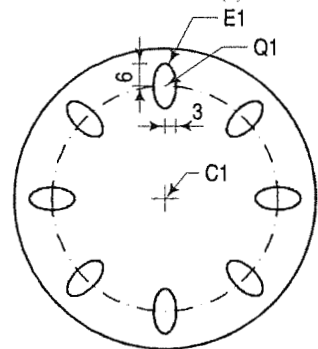


FIG. 26-19(ii)

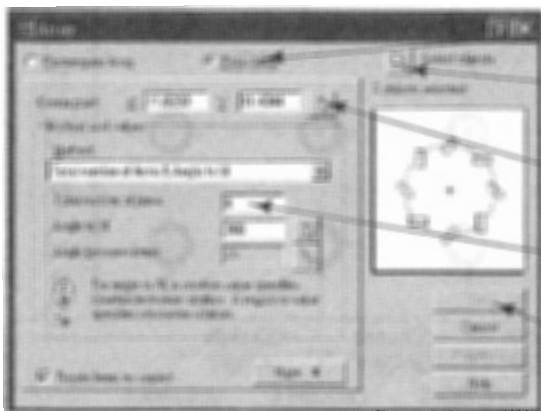


FIG. 26-19(iii)

- (1) Click Polar Array
- (2) Click here and select object as Ellipse
- (3) Click here and select Center C1
- (4) Enter No. of Items i.e. 8
- (5) Finally click OK

- (7) Command: **DIMSCALE** (↵)
Enter new value for DIMSCALE <1.0000>: **10** (↵)
- (8) Command: **DIMCENTER** (↵)
Select arc or circle:(Select Circle **CR1**)
- (9) Command: **EXTEND**
Current settings: Projection=UCS, Edge=None
Select boundary edges ...
Select objects or <select all>: (Select Circle **CR1**)
Select objects: (↵)
Select object to extend or shift-select to trim or
[Fence/Crossing/Project/Edge/Undo]:(Click Right End of
Line **L1**)
Select object to extend or shift-select to trim or
[Fence/Crossing/Project/Edge/Undo]: (Click Left End of
Line **L1**)
Select object to extend or shift-select to trim or
[Fence/Crossing/Project/Edge/Undo]: (Click Top End of
Line **L2**)
Select object to extend or shift-select to trim or
[Fence/Crossing/Project/Edge/Undo]: (Click Bottom End
of Line **L2**)
Select object to extend or shift-select to trim or
[Fence/Crossing/Project/Edge/Undo]: (↵)
- (10) Save This File As **Module 26-11.DWG**

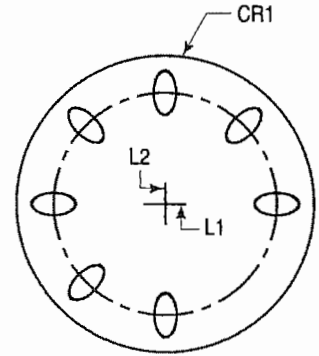


FIG. 26-19(iv)

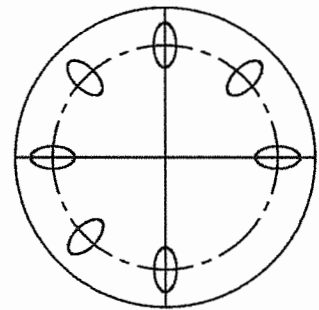


FIG. 26-19(v)

Output of Module 26-11 (fig. 26-19):

MODULE 26-11: USE OF ELLIPSE AND POLAR ARRAY COMMANDS

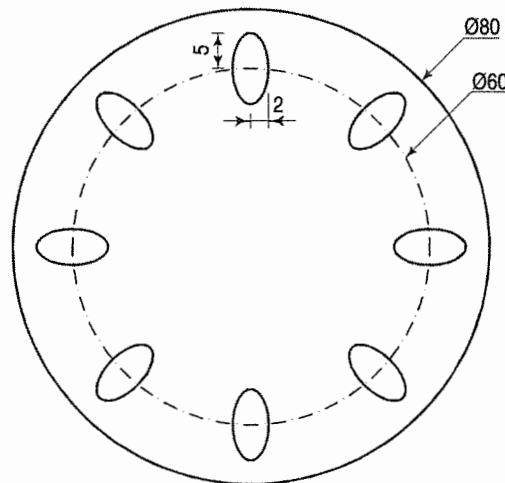


FIG. 26-19

Module 26-12. Draw a line diagram as shown in fig. 26-20. Use of Hatch, Rotate, Stretch and Scale commands.

To draw a line diagram using Hatch, Rotate, Stretch and Scale commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-20(i) to fig. 26-20(vi).

- (1) Command: **LIMITS** (↵)
 Reset Model space limits:
 Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
 Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
 Specify corner of window, enter a scale factor (nX or nXP), or
 [All/Center/Dynamic/Extents/Previous/Scale/Window] <real
 time>: **ALL** (↵)
- (3) Command: **POLYGON** (↵)
 Enter number of sides <4>: **5** (↵)
 Specify center of polygon or [Edge]: **E** (↵)
 Specify first endpoint of edge: **45,40** (↵)
 Specify second endpoint of edge: **30** (↵) (When 0°
 POLAR is ON)
- (4) Command: **BHATCH** (↵)

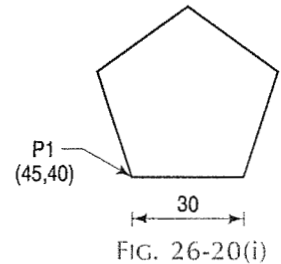
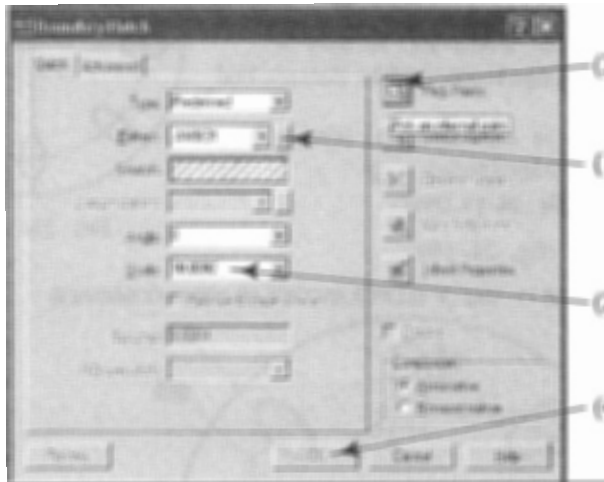


FIG. 26-20(i)



- Click here to select internal point
- Click here and select ANSI31 pattern
- Enter scale = 10
- Finally click OK

FIG. 26-20(ii)

Select internal point: (Click Inside Pentagon)
 Selecting everything visible...
 Analyzing the selected data...
 Analyzing internal islands...
 Select internal point: (↵)
 <Hit enter or right-click to return to the dialog> (↵)

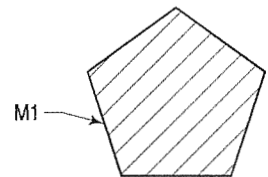


FIG. 26-20(iii)

- (5) Command: **ROTATE** (↵)
 Current positive angle in UCS: ANGDIR = counterclockwise
 ANGBASE=0
 Select objects: (Select Pentagon)
 Select objects: (Select Hatch Pattern)
 Select objects: (↵)
 Specify base point: (Select MIDPOINT magnet **M1**)
 Specify rotation angle or [Reference]: **-90** (↵)

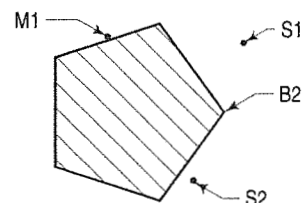


FIG. 26-20(iv)

(6) Command: **STRETCH** (↵)

Select objects to stretch by crossing-window or crossing-polygon...

Select objects: (Click near to **S1** and then click near to **S2**)

Select objects: (↵)

Specify base point or displacement: (Select ENDPOINT magnet **B2**)

Specify second point of displacement or <...>: **40** (↵) (When **0°** POLAR is ON)

(7) Command: **SCALE** (↵)

Select objects: (Select Pentagon)

Select objects: (Select Hatch Pattern)

Select objects: (↵)

Specify base point: (Select MIDPOINT magnet **M2**)

Specify scale factor or [Reference]: **0.5** (↵)

(8) Save This File As **Module 26-12.DWG**

Output of Module 26-12 (fig. 26-20):

MODULE 26-12: USE OF HATCH, ROTATE, STRETCH AND SCALE COMMANDS

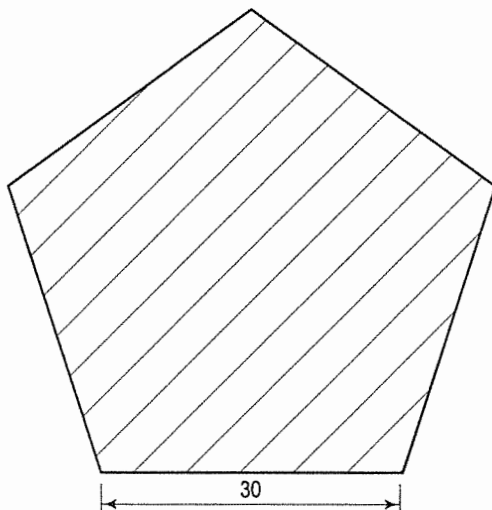


FIG. 26-20

Module 26-13. Draw a 2D diagram as shown in fig. 26-21.

To draw a 2D diagram using the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-21(i) to fig. 26-21(x).

(1) Command: **LIMITS** (↵)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)

Specify upper right corner <12.0000,9.0000>: **180,135** (↵)

(2) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/....] <real time>: **ALL** (↵)

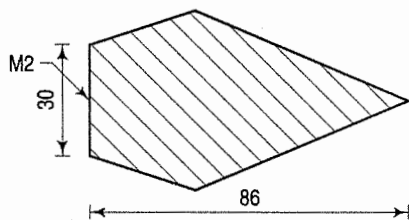


FIG. 26-20(v)

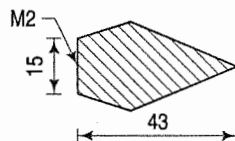


FIG. 26-20(vi)

- (3) Command: **LTSCALE** (↵)
Enter new linetype scale factor <1.0000>: **15** (↵)
- (4) Command: **-LAYER** (↵)
Enter an option
[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/Plot/Freeze/...]: **NEW** (↵)
Enter name list for new layer(s): **CENLIN** (↵)
Enter an option
[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/Plot/Freeze/...]: **COLOR** (↵)
New color [Truecolor/COLORbook] : **41** (↵)
Enter name list of layer(s) for color 41 <0>: **CENLIN** (↵)
Enter an option
[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/Plot/Freeze/...]: **LTTYPE** (↵)
Enter loaded linetype name or [?] <Continuous>: **DASHDOT** (↵)
Enter name list of layer(s) for linetype "DASHDOT" <0>: **CENLIN** (↵)
Enter an option
[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/Plot/Freeze/...]: (↵)

- (5) Command: **CIRCLE** (↵)
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:
90,100 (↵)
Specify radius of circle or [Diameter]: **d** (↵)
Specify diameter of circle: **68** (↵)

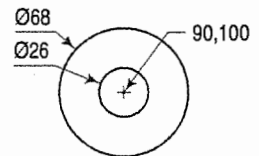


FIG. 26-21(i)

- (6) Command: **CIRCLE** (↵)
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:
90,100 (↵)
Specify radius of circle or [Diameter] <3.4000>: **d** (↵)
Specify diameter of circle <6.8000>: **26** (↵)

- (7) Command: **LINE** (↵)
Specify first point: (Select QUADRANT Magnet **Q2** of Circle **Ø68**)
Specify next point or [Undo]: (Type **121** and press enter when **270°** Polar is ON)
Specify next point or [Undo]: (Type **40** and press enter when **0°** Polar is ON)
Specify next point or [Close/Undo]: (Type **34** and press enter when **90°** Polar is ON)
Specify next point or [Close/Undo]: (Type **80** and press enter when **180°** Polar is ON)
Specify next point or [Close/Undo]: (Type **34** and press enter when **270°** Polar is ON)
Specify next point or [Close/Undo]: (Select ENDpoint Magnet **E1**)
Specify next point or [Close/Undo]: (↵)

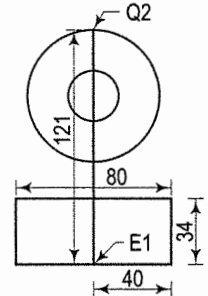


FIG. 26-21(ii)

- (8) Command: **FILLET** (↵)
Select first object or [Undo/Polyline/Radius/...]: (Select Line **L1** from Right end)
Select second object : (Select Line **L2** from Right end)
- (9) Command: **FILLET** (↵)
Select first object or [Undo/Polyline/Radius/...]: (Select Line **L1** from Left end)
Select second object : (Select Line **L2** from Left end)
- (10) Command: **CIRCLE** (↵)
Specify center point for circle or [3P/2P/...]: (Select MIDpoint Magnet **M1** of Line **L3**)

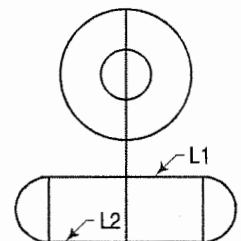


FIG. 26-21(iii)

Specify radius of circle or [Diameter] <1.7000>: **d**
(↵)

Specify diameter of circle <3.4000>: **16** (↵)

- (11) Command: **CIRCLE** (↵)

Specify center point for circle or [3P/2P/...]: (Select MIDpoint Magnet **M2** of Line **L4**)

Specify radius of circle or [Diameter] <1.7000>: **d**
(↵)

Specify diameter of circle <3.4000>: **16** (↵)

- (12) Command: **LINE** (↵)

Specify first point: (Select MIDpoint Magnet **M3** of Arc **A1**)

Specify next point or [Undo]: (Type **34** and press enter when **180°** Polar is ON)

Specify next point or [Undo]: (↵)

- (13) Command: **LINE** (↵)

Specify first point: (Select MIDpoint Magnet **M4** of Arc **A2**)

Specify next point or [Undo]: (Type **34** and press enter when **0°** Polar is ON)

Specify next point or [Undo]: (↵)

- (14) Command: **OFFSET** (↵)

Specify offset distance or [Through/Erase/Layer] <Through>: **15** (↵)

Select object to offset or [Exit/Undo] <Exit>: (Select Line **L5**)

Specify point on side to offset or [Exit/..] <Exit>: (Click on Right Side of Line **L5**)

Select object to offset or [Exit/Undo] <Exit>: (Select Line **L5**)

Specify point on side to offset or [Exit/..] <Exit>: (Click on Left Side of Line **L5**)

Select object to offset or [Exit/Undo] <Exit>: (↵)

- (15) Command: **FILLET** (↵)

Select first object or [Undo/Polyline/Radius/Trim/Multiple]: **r**
(↵)

Specify fillet radius <0.0000>: **10** (↵)

Select first object or [Undo/Polyline/Radius/...]: (Select Line **L6**)

Select second object or shift-select to apply corner: (select Circle **CR1**)

- (16) Command: **FILLET** (↵)

Select first object or [Undo/Polyline/Radius/Trim/Multiple]: **r**
(↵)

Specify fillet radius <0.0000>: **10** (↵)

Select first object or [Undo/Polyline/Radius/...]: (Select Line **L7**)

Select second object or shift-select to apply corner: (Select Circle **CR1**)

- (17) Command: **TRIM** (↵)

Current settings: Projection=UCS, Edge=Extend

Select cutting edges ...

Select objects or <select all>: (Select Line **L1**)

Select objects: (↵)

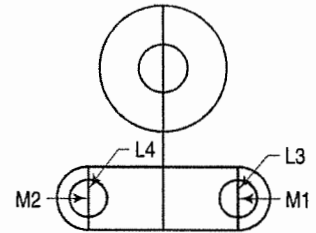


FIG. 26-21 (iv)

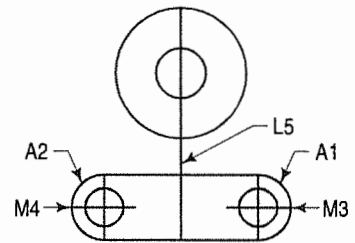


FIG. 26-21 (v)

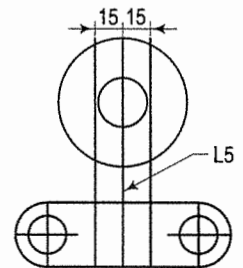


FIG. 26-21 (vi)

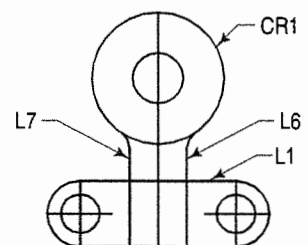


FIG. 26-21 (vii)

Select object to trim or shift-select to extend or
[Fence/...../Undo]: (Select Line **L6** from below Line **L1**)
Select object to trim or shift-select to extend or
[Fence/...../Undo]: (Select Line **L7** from below Line **L1**)
Select object to trim or shift-select to extend or
[Fence/Crossing/Project/Edge/eRase/Undo]: (↵)

(18) Command: **TRIM** (↵)

Current settings: Projection=UCS, Edge=Extend
Select cutting edges ...

Select objects or <select all>: (Select Line **L6**)

Select objects: (Select Line **L7**)

Select objects: (↵)

Select object to trim or shift-select to extend or

[Fence/..]: (Select Line **L1** from between Line **L6** and **L7**)

Select object to trim or shift-select to extend or
[Fence/Crossing/Project/Edge/eRase/Undo]: (↵)

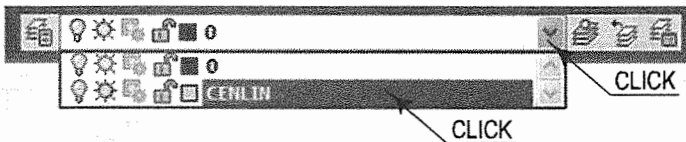
(19) Command: **LINE** (↵)

Specify first point: (Select QUADRANT Magnet **Q1** of Circle **Ø68**)

Specify next point or [Undo]: (Select QUADRANT Magnet **Q3** of Circle **Ø68**)

Specify next point or [Undo]: (↵)

(20) Select Line **L3, L4, L5, L8, L9** and **L10** by Mouse and Click in Layer tool box and select **CENLIN** Layer and Press Esc key.



(21) Save This File As **Module 26-13.DWG**

Output of Module 26-13 (fig. 26-21):

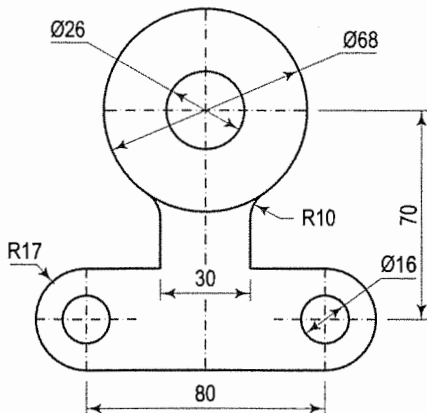


FIG. 26-21

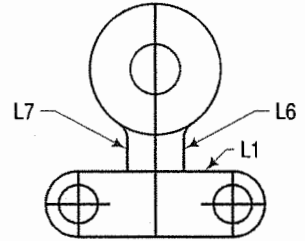


FIG. 26-21(viii)

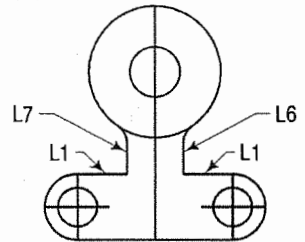


FIG. 26-21(ix)

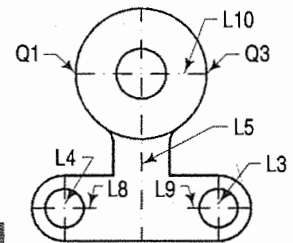


FIG. 26-21(x)

Module 26-14. Draw 2D diagram as shown in fig. 26-22(ii) and fig. 26-22(iii). Using **BLOCK** and **INSERT** command create assembly as shown in fig. 26-22(vii)

- (1) Use **LINE** command to draw the two given drawings as shown in fig. 26-22(ii) and fig. 26-22(iii). Use start point (1,1) and (8,1) for each respectively.

- (2) Command: **BLOCK** (↵)

(1) The Dialog Box Of **BLOCK** will Open as shown in fig. 26-22(i). Enter the name '**B**' in **NAME** box. Type **8.4** in **X** box and **2** in **Y** box. Select **Delete** radio button and finally click **Select Objects** button.

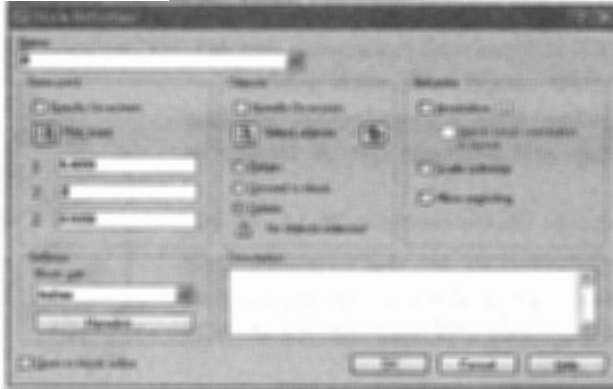


FIG. 26-22(i)

- (2) Select objects: (Select All Objects of PIN) Fig. 26-22(iii)

Select objects: (J)

- (3) Finally click **OK** button.

- (3) Command: **OOPS** (J)

- (4) Command: **INSERT** (J)

- (1) The Dialog Box Of INSERT will Open as shown below in fig. 26-22(iv).

- (2) Select 'B' block from **NAME** box. Enter **1** in **X** box. Enter **0** in **Rotation** box. Finally click **OK** button.

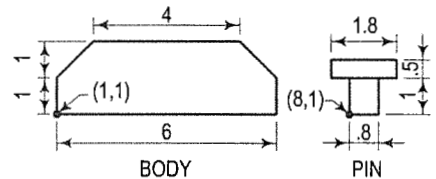


FIG. 26-22(ii)

FIG. 26-22(iii)

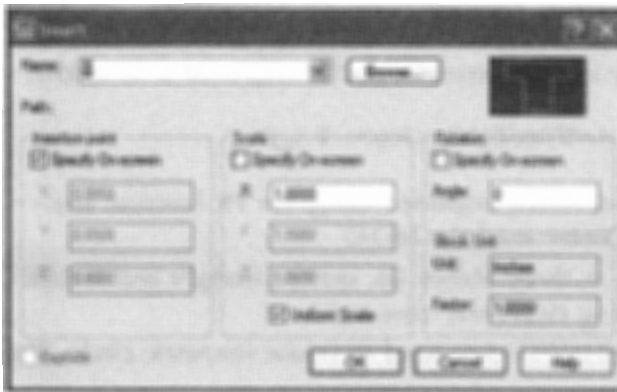


FIG. 26-22(iv)

- (3) Specify insertion point or [Basepoint/Scale/Rotate]: (Select MIDpoint **M1**)

- (5) Similarly Insert Block 'B' at MIDpoint **M2** and **M3** with **X** value of **0.5** and **Rotation** angle of **-45°** and **45°** respectively.

- (6) Save this drawing file as **Module 26-14.DWG**

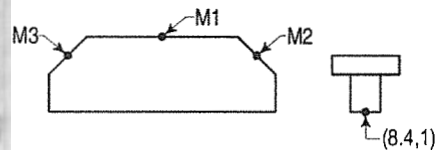


FIG. 26-22(v)

FIG. 26-22(vi)

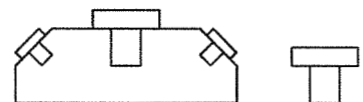


FIG. 26-22(vii)

26-8. ISOMETRIC DRAWINGS

- Isometric drawings are used to help visualize the shape of an object. It is just a two dimensional representation of a 3D drawing in a 2D plane.
- An isometric view is obtained by rotating the object 45° around the imaginary vertical axis and then tilting the object upward through a 35 degree and 16 minute angle.

- (iii) As shown in fig. 26-23, the three principal axes are AB (30° inclined with horizontal), AC (30° inclined with horizontal) and AD (90° inclined with horizontal). They also make 120° angle with each other.
- (iv) Isometric drawings have three principal planes, isoplane right, isoplane top and isoplane left as shown in fig. 26-23.

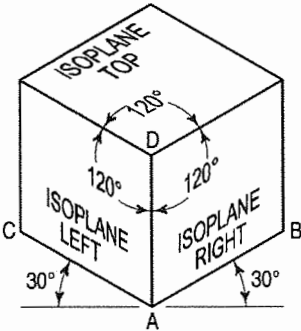


FIG. 26-23

(1) To set isometric mode (Refer module 26-15, 26-16)

- (i) Use the snap command to set the isometric grid and snap. The isometric grid lines are displayed at 30° of the horizontal axis and also set the vertical spacing between grid.
Command: **-GRID** (↵)
Specify grid spacing or [ON/OFF/...] <1.0000>: **1** (↵)
Command: **-GRID** (↵)
Specify grid spacing or [ON/OFF/...] <1.00000>: **ON** (↵)
Command: **-SNAP** (↵)
Specify snap spacing or [ON/OFF/.../style...] <0.5000>: **5** (↵)
Enter snap grid style [Standard/Isometric] <S>: **I** (↵)
Specify vertical spacing <0.5000>: **1** (↵)
- (ii) The command sequence is as below:

(iii) The cross hair now will be displayed at an isometric angle.

(2) To set Isoplane (Refer module 26-15, 26-16)

- (i) There are three orientation of crosshairs, isotop, isoright and isoleft.
- (ii) These orientation are set by isoplane command, the sequence is as below:
Command: **ISOPLANE** (↵)
Enter isometric plane setting [Left/Top/Right] <Top>: **T** (↵)
Current Isoplane: **Top**

(iii) We can also toggle among different isoplanes using the **F5** function key.

(3) Drawing isometric circles (Refer module 26-16)

- (i) Isometric circles are drawn in three planes i.e. isotop, isoright and isoleft.
- (ii) So first set the isoplane using ISOPLANE command.
- (iii) Now use isocircle option of ELLIPSE command to create isometric circle. The sequence is as below:
Command: **ELLIPSE** (↵)
Specify axis endpoint of ellipse or [Arc/Center/Isocircle]: **I** (↵)
Specify centre of isocircle: (Select a point)
Specify radius of isocircle or [Diameter]: (Enter radius value)

(4) Isometric text (Refer module 26-18)

- (i) Isometric text are created in three isoplanes.
- (ii) These text are obliqued at 30° or 330° angle with rotation angle of 30° or 330°.
- (iii) The oblique angle and rotation angle for isotext in three planes are given below:

Isoplane	Oblique angle	Rotation angle
Top	30	330
Right	30	30
Left	330	330

- (iv) For this use STYLE command to create two text styles (iso 30 and iso 330) having 30° and 330° oblique angle as shown in module 26-18.
- (v) Set the isoplane and use DT (Dynamic Text) command to write text in respective plane.
- (5) **Isometric dimension (Refer module 26-17)**
 - (i) Dimension the distance using DIMALIGNED command.
 - (ii) Change the oblique angle of dimension by using oblique option of DIMEDIT command. Note: The oblique angle is determined by the angle the extension line of the isometric dimension makes with the positive X-axis.

Module 26-15. Draw a isometric line diagram as shown in fig. 26-24(iii). Use Grid, Snap and Isoplane commands.

- (1) Command: **GRID** (↵)
Specify grid spacing(X) or [ON/OFF/Snap/Major/aDaptive/...] <0.5000>: **1** (↵)
- (2) Command: **GRID** (↵)
Specify grid spacing(X) or [ON/OFF/Snap/Major/aDaptive/...] <1.0000>: **ON** (↵)
- (3) Command: **SNAP** (↵)
Specify snap spacing or [ON/OFF/Aspect/Style/Type] <0.5000>: **STYLE** (↵)
Enter snap grid style [Standard/Isometric] <S>: **ISOMETRIC** (↵)
Specify vertical spacing <0.5000>: (↵)
- (4) Command: **SNAP** (↵)
Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: **OFF** (↵)
- (5) Command: **ISOPLANE** (↵)
Current isoplane: **Right**
Enter isometric plane setting [Left/Top/Right] <Left>: **RIGHT** (↵)
Current isoplane: **Right**
- (6) Command: **LINE** (↵)
Specify first point: **6,1** (↵)
Specify next point or [Undo]: **4** (↵) (When 30° POLAR is ON)
Specify next point or [Undo]: **4** (↵) (When 90° POLAR is ON)
Specify next point or [Close/Undo]: **4** (↵) (When 210° POLAR is ON)
Specify next point or [Close/Undo]: **C** (↵)
- (7) Command: **ISOPLANE** (↵)
Current isoplane: **Right**
Enter isometric plane setting [Left/Top/Right] <Left>: **LEFT** (↵)
Current isoplane: **Left**
- (8) Command: **LINE** (↵)
Specify first point: **6,1** (↵)
Specify next point or [Undo]: **4** (↵) (When 150° POLAR is ON)
Specify next point or [Undo]: **4** (↵) (When 90° POLAR is ON)
Specify next point or [Close/Undo]: **4** (↵) (When 330° POLAR is ON)
Specify next point or [Close/Undo]: (↵)
- (9) Command: **ISOPLANE** (↵)
Current isoplane: **Left**
Enter isometric plane setting [Left/Top/Right] <Top>: **TOP** (↵)
Current isoplane: **Top**

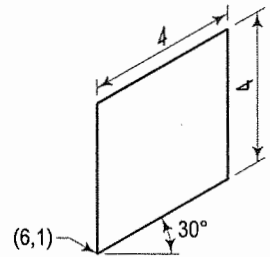


FIG. 26-24(i)

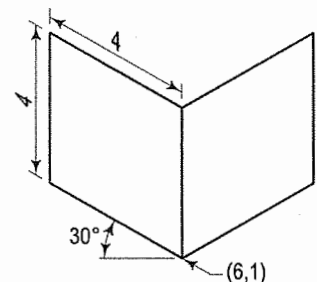


FIG. 26-24(ii)

- (10) Command: **LINE** (↵)
 Specify first point: (Select ENDPOINT magnet **E1**)
 Specify next point or [Undo]: **4** (↵) (When **150°** POLAR is ON)
 Specify next point or [Undo]: **4** (↵) (When **210°** POLAR is ON)
 Specify next point or [Close/Undo]: (↵)
- (11) Command: **SNAP** (↵)
 Specify snap spacing or [ON/OFF/Aspect/Style/Type] <0.5000>: **STYLE** (↵)
 Enter snap grid style [Standard/Isometric] <S>: **STANDARD** (↵)
 Specify vertical spacing <0.5000>: (↵)
- (12) Command: **SNAP** (↵)
 Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: **OFF** (↵)
- (13) Save this File As **Module 26-15.DWG**

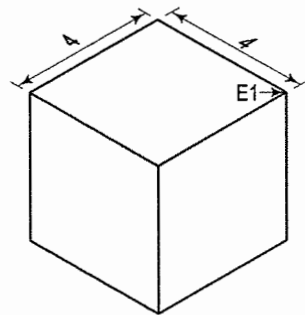


FIG. 26-24(iii)

Module 26-16. Open the file *Module 26-15.DWG* and create Isocircles in each isoplane using *Ellipse* command.

- (1) Open The File **Module 26-15.DWG**
- (2) Command: **GRID** (↵)
 Specify grid spacing(X) or [ON/OFF/Snap/Major/aDaptive/...] <0.5000>: **1** (↵)
- (3) Command: **GRID** (↵)
 Specify grid spacing(X) or [ON/OFF/Snap/Major/aDaptive/...] <1.0000>: **ON** (↵)
- (4) Command: **SNAP** (↵)
 Specify snap spacing or [ON/OFF/Aspect/Style/Type] <0.5000>: **STYLE** (↵)
 Enter snap grid style [Standard/Isometric] <S>: **ISOMETRIC** (↵)
 Specify vertical spacing <0.5000>: (↵)
- (5) Command: **SNAP** (↵)
 Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: **OFF** (↵)
- (6) Command: **LINE** (↵)
 Specify first point: (Select ENDPOINT magnet **E1**)
 Specify next point or [Close/Undo]: (Select ENDPOINT magnet **E2**)
 Specify next point or [Close/Undo]: (Select ENDPOINT magnet **E3**)
 Specify next point or [Close/Undo]: (↵)
- (7) Command: **ISOPLANE** (↵)
 Current isoplane: **Left**
 Enter isometric plane setting [Left/Top/Right] <Top>: **TOP** (↵)
 Current isoplane: **Top**
- (8) Command: **ELLIPSE** (↵)
 Specify axis endpoint of ellipse or [Arc/Center/Isocircle]: **I** (↵)
 Specify center of isocircle: (Select MIDPOINT magnet **M1** by mouse)
 Specify radius of isocircle or [Diameter]: **1.25** (↵)
- (9) Command: **ISOPLANE** (↵)
 Current isoplane: **Top**
 Enter isometric plane setting [Left/Top/Right] <Right>: **RIGHT** (↵)
 Current isoplane: **Right**

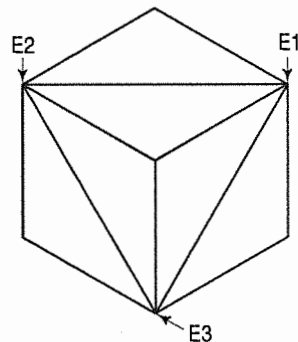


FIG. 26-25(i)

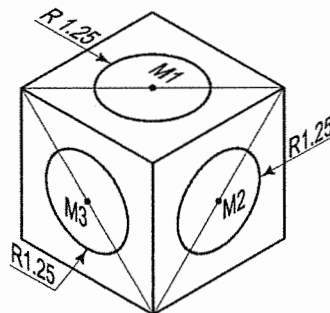


FIG. 26-25(ii)

- (10) Command: **ELLIPSE** (↵)
Specify axis endpoint of ellipse or [Arc/Center/ socircle]: **I** (↵)
Specify center of isocircle: (Select MIDPOINT magnet **M2** by mouse)
Specify radius of isocircle or [Diameter]: **1.25** (↵)
- (11) Command: **ISOPLANE** (↵)
Current isoplane: **Right**
Enter isometric plane setting [Left/Top/Right] <Left>: **LEFT** (↵)
Current isoplane: **Left**
- (12) Command: **ELLIPSE** (↵)
Specify axis endpoint of ellipse or [Arc/Center/ socircle]: **I** (↵)
Specify center of isocircle: (Select MIDPOINT magnet **M3** by mouse)
Specify radius of isocircle or [Diameter]: **1.25** (↵)
- (13) Command: **SNAP** (↵)
Specify snap spacing or [ON/OFF/Aspect/Style/Type] <0.5000>: **STYLE** (↵)
Enter snap grid style [Standard/ sometric] <S>: **STANDARD** (↵)
Specify vertical spacing <0.5000>: (↵)
- (14) Command: **SNAP** (↵)
Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: **OFF** (↵)
- (15) Save this File As **Module 26-16.DWG**

Module 26-17. Open the file *Module 26-15.DWG* and create Aligned dimensions.

- (1) Open the file **Module 26-15.DWG**
- (2) Command: **DIMDEC** (↵)
Enter new value for DIMDEC <4>: **1** (↵)
- (3) Command: **DIMALIGNED** (↵)
Specify first extension line origin or <select object>: (Select ENDpoint **E1**)
Specify second extension line origin: (Select ENDpoint **E2**)
Specify dimension line location or [Mtext/Text/Angle]: (Click below line **L1**)
Dimension text = 4.0000
- (4) Command: **DIMEDIT** (↵)
Enter type of dimension editing [Home/New/Rotate/Oblique] <Home>: **O** (↵)
Select objects: (Select dimension **D1**)
Select objects: (↵)
Enter obliquing angle (press ENTER for none): **90** (↵)
- (5) Command: **DIMALIGNED** (↵)
Specify first extension line origin or <select object>: (Select ENDpoint **E3**)
Specify second extension line origin: (Select ENDpoint **E4**)
Specify dimension line location or [Mtext/Text/Angle]: (Click Right of line **L2**)
Dimension text = 4.0000
- (6) Command: **DIMEDIT** (↵)
Enter type of dimension editing [Home/New/Rotate/Oblique] <Home>: **O** (↵)
Select objects: (Select dimension **D2**)
Select objects: (↵)
Enter obliquing angle (press ENTER for none): **30** (↵)
- (7) Command: **DIMALIGNED** (↵)
Specify first extension line origin or <select object>: (Select **E5**)
Specify second extension line origin: (Select ENDpoint **E6**)

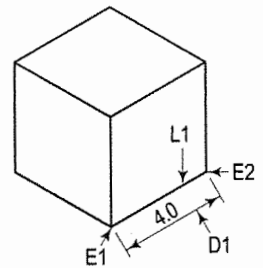


FIG. 26-26(i)

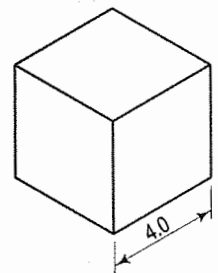


FIG. 26-26(ii)

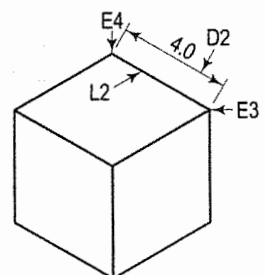


FIG. 26-26(iii)

Specify dimension line location or
[Mtext/Text/Angle]: (Click Left of line L3)
Dimension text = 4.0000

- (8) Enter type of dimension editing [Home/New/Rotate/Oblique]
<Home>: **O** (↵)
Select objects: (Select dimension D3)
Select objects: (↵)
Enter obliquing angle (press ENTER for none): **150** (↵)
- (9) Save this File As **Module 26-17.DWG**

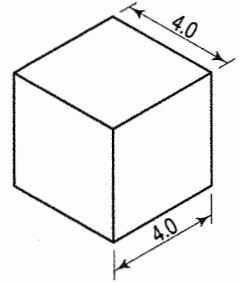


FIG. 26-26(iv)

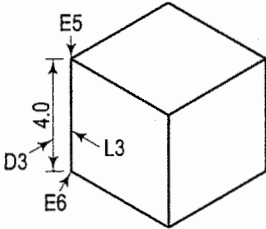


FIG. 26-26(v)

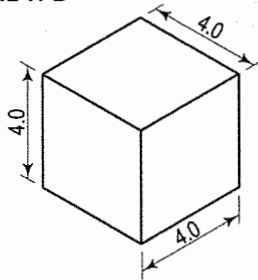


FIG. 26-26(vi)

Module 26-18. Open the file *Module 26-15.DWG* and create *Isotext* in each plane as shown in fig. 26-27(iii).

- (1) Open the File **Module 26-15.DWG**
- (2) Command: **SNAP** (↵)
Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: **S** (↵)
Enter snap grid style [Standard/Isometric] <I>: **I** (↵)
Specify vertical spacing <0.5000>: (↵)
- (3) Command: **SNAP** (↵)
Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: **OFF** (↵)
- (4) Command: **ISOPLANE** (↵)
Current isoplane: **Left**
Enter isometric plane setting [Left/Top/Right] <Top>: **TOP** (↵)
Current isoplane: **Top**
- (5) Command: **-STYLE** (↵)
Enter name of text style or [?] <ISO330>: **ISO30** (↵)
Specify full font name or font filename (TTF or SHX) <ROMANS.shx>: **ROMANTIC** (↵)
Specify height of text or [Annotative] <0.2000>: **0.18** (↵)
Specify width factor <1.0000>: (↵)
Specify obliquing angle <0>: **30** (↵)
Display text backwards? [Yes/No] <No>: (↵)
Display text upside-down? [Yes/No] <No>: (↵)
Vertical? [Yes/No] <No>: (↵)
"ISO30" is now the current text style.
- (6) Command: **DT** (↵)
Current text style: "ISO30" Text height: 0.1800 Annotative: No
Specify start point of text or [Justify/Style]: **J** (↵)
Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/ML/MC/MR/BL/BC/BR]: **BC** (↵)
Specify bottom-center point of text: (Select MIDpoint M1)
Specify rotation angle of text <0>: **330** (↵)
Enter text: **CHAROTAR** (↵)
Enter text: (↵)

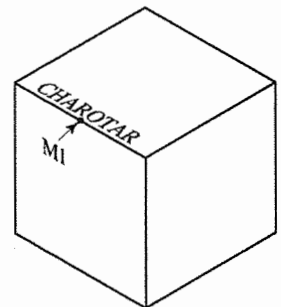


FIG. 26-27(i)

- (7) Command: **ISOPLANE** (↵)
 Current isoplane: **Top**
 Enter isometric plane setting [Left/Top/Right] <Right>: **RIGHT** (↵)
 Current isoplane: **Right**
- (8) Command: **DT** (↵)
 Current text style: "ISO30" Text height: 0.1800 Annotative: No
 Specify start point of text or [Justify/Style]: **J** (↵)
 Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/ML/MC/MR/BL/BC/BR]: **BC** (↵)
 Specify bottom-center point of text: (Select MIDpoint **M2**)
 Specify rotation angle of text <330>: **30** (↵)
 Enter text: **HOUSE** (↵)
 Enter text: (↵)
- (9) Command: **ISOPLANE** (↵)
 Current isoplane: **Right**
 Enter isometric plane setting [Left/Top/Right] <Top>: **FRONT** (↵)
 Current isoplane: **Front**
- (10) Command: **-STYLE** (↵)
 Enter name of text style or [?] <ISO30>: **ISO330** (↵)
 Specify full font name or font filename (TTF or SHX) <ROMANS.shx>: **ROMANTIC** (↵)
 Specify height of text or [Annotative] <0.2000>: **0.18** (↵)
 Specify width factor <1.0000>: (↵)
 Specify obliquing angle <30>: **330** (↵)
 Display text backwards? [Yes/No] <No>: (↵)
 Display text upside-down? [Yes/No] <No>: (↵)
 Vertical? [Yes/No] <No>: (↵)
 "ISO330" is now the current text style.
- (11) Command: **DT** (↵)
 Current text style: "ISO330" Text height: 0.1800 Annotative: No
 Specify start point of text or [Justify/Style]: **J** (↵)
 Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/ML/MC/MR/BL/BC/BR]: **BC** (↵)
 Specify bottom-center point of text: (Select MIDpoint **M3**)
 Specify rotation angle of text <30>: **330** (↵)
 Enter text: **PUBLISHING** (↵)
 Enter text: (↵)
- (12) Command: **SNAP** (↵)
 Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: **S** (↵)
 Enter snap grid style [Standard/Isometric] <I>: **S** (↵)
 Specify snap spacing or [Aspect] <0.5000>: (↵)
- (13) Command: **SNAP** (↵)
 Specify snap spacing or [ON/OFF/Aspect/Style/Type] <0.5000>: **OFF** (↵)
- (14) Save this File As **Module 26-18.DWG**

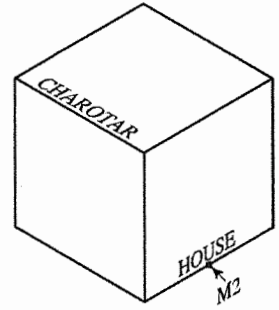


FIG. 26-27(ii)

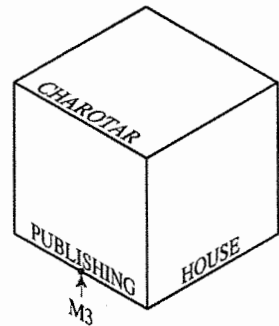


FIG. 26-27(iii)

26-9. 3D GEOMETRICAL MODELING

It is a process of generating 3D models for analysis, design, drafting, assembly, drawing views, realistic look, animation of the assemblies, mass properties, manufacturing and simulation. There are three types of 3D modeling.

- (1) 3D Wireframe modelling; (2) 3D Surface modelling (3) 3D Solid modelling.

26-9-1. 3D WIREFRAME MODELLING



It represents a model by its edge only. Nothing is between the edges and so cannot hide objects that are behind them. Fig. 26-28 shows a simple 3D Wireframe model. It is made of 18 lines with four circles representing the edges of a round hole. It gives the look of 3D but it is not real 3D. You cannot apply Boolean operations on them and cannot calculate the mass properties. It can be used as skeleton for surface and solid modeling.

Commonly used 3D command to generate 3D wire frame model are discussed below:

(1) VPOINT command: The VPOINT command is used to set the viewpoint in 3D space for viewing the 3D models. There are basic 4 isometric views to set view point. They are given below with their coordinates value to set. Enter any of this coordinate value against the command prompt.

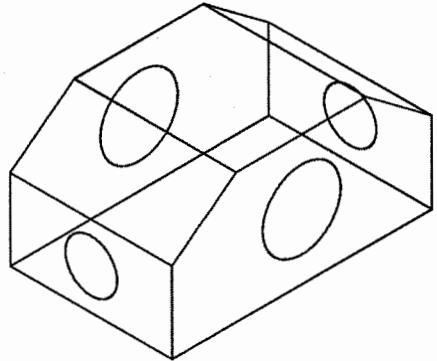


FIG. 26-28

- (i) South West Isometric view -1, -1, 1
- (ii) South East Isometric view 1, -1, 1
- (iii) North East Isometric view 1, 1, 1
- (iv) North West Isometric view -1, 1, 1

The command sequence is given below.

Command : **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1, -1, 1** (↵)

Regenerating model.

(2) UCS Command: The UCS command is used to set a new coordinate system by shifting the working XY plane to the desired location. "New" option will create new UCS by "3point" or "Zaxis" method and "Move" option will shift the UCS at desired location by entering coordinates (X, Y, Z) or using Osnap magnet. The command sequence is given below.

Command: **UCS** (↵)

Current ucs name: *WORLD*

Enter an option [New/Move/orthoGraphic/Prev/Restore/.../World] <World>:

Methods to Locate Points in Space to generate 3D Wireframe Model.

(1) 3D Absolute Coordinate Method (Refer module 26-19): In this all the points in space are specified by distance in X, Y and Z direction from fixed origin (0,0,0) i.e. WCS (World Coordinate System)

Format: X, Y, Z

(2) 3D Rectangular Coordinate Method (Refer module 26-19): This is relative coordinate method. In this the last point in space is treated as temporary origin for the immediate next point.

Format: @ X, Y, Z

(3) 3D Cylindrical Coordinate Method (fig. 26-29) (Refer module 26-20): The points for some 3D models such as helix and spirals are entered by using this method.

Format: $D < A, Z$

where D = Point's distance in XY plane from fixed origin

A = Point's angle on the XY plane from X axis

Z = Point's distance from XY plane

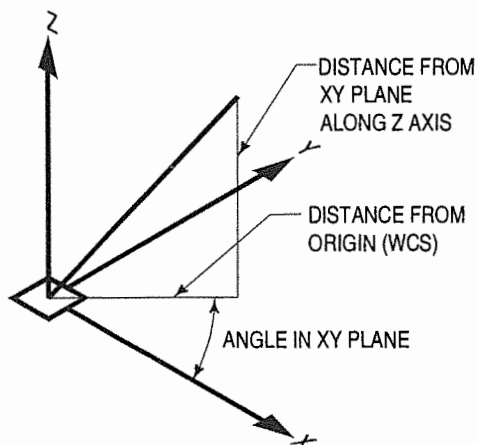


FIG. 26-29

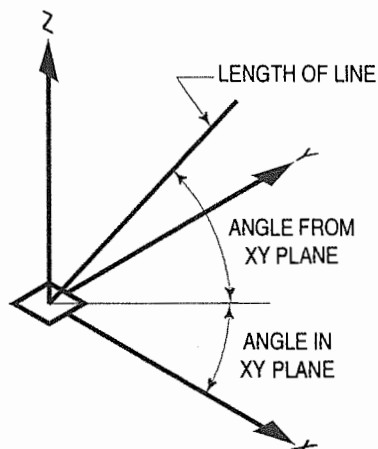


FIG. 26-30

(4) 3D Spherical Coordinate Method (fig. 26-30) (Refer module 26-20): It uses a distance and two angles to specify points in space.

Format: $D < HA < VA$

where D = Point's straight line distance from fixed origin

HA = Point's angle on XY plane from X axis

VA = Point's vertical angle from XY plane.

Methods (1) to (4) are explained below vide self interactive module 26-19 and module 26-20.

Module 26-19. To generate 3D Wireframe model as shown in fig. 26-31 using 3D Absolute Coordinate Method and 3D Rectangular Coordinate Method.

(1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

(2) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/...] <real time>: **ALL** (↵)

(3) Command: **LINE** (↵)

Specify first point: **0,0,0** (↵)

Specify next point or [Undo]: **5,0,0** (↵)

Specify next point or [Undo]: **5,8,0** (↵)

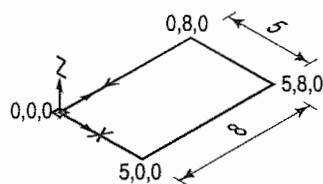


FIG. 26-31(i)

Specify next point or [Close/Undo]: **0,8,0** (↵)

Specify next point or [Close/Undo]: **C** (↵)

(4) Command: **LINE** (↵)

Specify first point: **5,0,0** (↵)

Specify next point or [Undo]: **@0,0,4** (↵)

Specify next point or [Undo]: **@0,2,0** (↵)

Specify next point or [Close/Undo]: **@0,2,-1.5** (↵)

Specify next point or [Close/Undo]: **@0,2,1.5** (↵)

Specify next point or [Close/Undo]: **@0,2,0** (↵)

Specify next point or [Close/Undo]: **@0,0,-4** (↵)

Specify next point or [Close/Undo]: (↵)

(5) Command: **LINE** (↵)

Specify first point: **0,0,0** (↵)

Specify next point or [Undo]: **@0,0,4** (↵)

Specify next point or [Undo]: **@0,2,0** (↵)

Specify next point or [Close/Undo]: **@0,2,-1.5** (↵)

Specify next point or [Close/Undo]: **@0,2,1.5** (↵)

Specify next point or [Close/Undo]: **@0,2,0** (↵)

Specify next point or [Close/Undo]: **@0,0,-4** (↵)

Specify next point or [Close/Undo]: (↵)

(6) Similarly using LINE Command and Osnap Magnet create remaining Lines and complete the diagram.

(7) Save This File As **3D-WF-1.DWG**

Output of Module 26-19 (fig. 26-31):

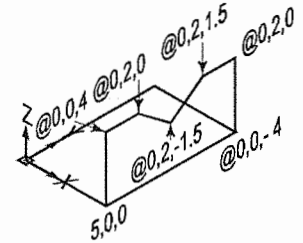


FIG. 26-31(ii)

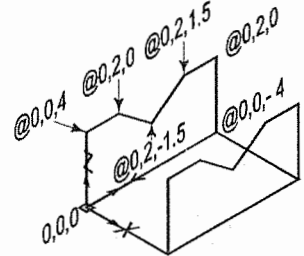


FIG. 26-31(iii)

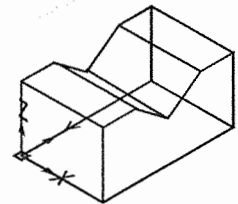


FIG. 26-31(iv)

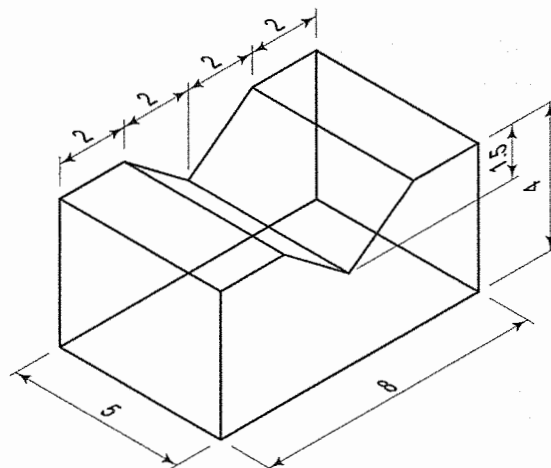


FIG. 26-31

Module 26-20. To generate 3D Wireframe model as shown in fig. 26-32. Using 3D Cylindrical Coordinate Method and 3D Spherical Coordinate Method.

(1) Command: **LINE** (↵)

Specify first point: **0,0,0** (↵)

Specify next point or [Undo]: **8<90,0** (↵)

Specify next point or [Undo]: **6<90,3.461** (↵)

Specify next point or [Close/Undo]: **2<90,3.4641** (↵)

Specify next point or [Close/Undo]: **C** (↵)

(2) Command: **UCS** (↵)

Current ucs name: ***TOP***

Enter an option [New/Move/orthoGraphic/Prev/...] <World>:

M (↵)

Specify new origin point or [Zdepth]<0,0,0>:

5,0,0 (↵)

(3) Command: **LINE** (↵)

Specify first point: **0,0,0** (↵)

Specify next point or [Undo]: **8<90<0** (↵)

Specify next point or [Undo]: **@4<90<120** (↵)

Specify next point or [Close/Undo]: **@4<90<180** (↵)

Specify next point or [Close/Undo]: **C** (↵)

(4) Using Line command and Osnap magnet complete the diagram as shown in fig. 26-32

(5) Save This File As **3D-WF-2.DWG**

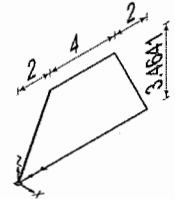


FIG. 26-32(i)

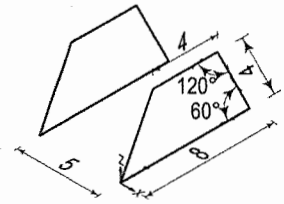


FIG. 26-32(ii)

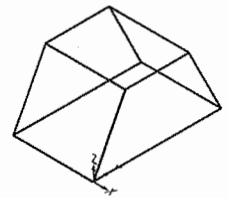


FIG. 26-32

26-9-2. 3D SURFACE MODELLING

In this modeling the objects are created by use of surfaces attached to 3D Wireframe models. The attached surfaces have zero thickness so cannot calculate the mass properties. It can hide objects that are behind them. These models are just empty shells. The surfaces are flat, with curved and rounded surfaces approximated by small rectangular or triangular faces.

Commands used to create 3D Surface Models:

(1) **Vpoint**: Refer detail on page no. 666

(2) **UCS**: Refer detail on page no. 666

(3) **SHADEMODE** command: The SHADEMODE command generates a simple shaded picture of the solids and surface models displayed in the current viewport. It will hide all back portion by "Hide" option. "Realistic" and "Conceptual" option will shade the model with current color. "3Dwireframe" option will get back the shaded model to wireframe model. The command sequence is given below.

Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]:
(Select option)

(4) **ELEV** command: This command sets elevation and thickness properties for new 2D wireframe objects such as line, point, circle, polygon, arc etc. Elevation sets the starting height of surface in Z direction from XY plane and thickness will give height to the surface in Z direction. The command sequence is explained below vide self interactive module 26-21.

Module 26-21. To generate 3D Surface using *ELEV* command model as shown in fig. 26-33.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass...>: **1,-1,1** (↵)
Regenerating model.
- (2) Command: **ELEV** (↵)
Specify new default elevation <0.0000>: (↵)
Specify new default thickness <0.0000>: **4** (↵)
- (3) Command: **CIRCLE** (↵)
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: **0,0** (↵)
Specify radius of circle or [Diameter]: **4** (↵)
- (4) Command: **ELEV** (↵)
Specify new default elevation <0.0000>: **4** (↵)
Specify new default thickness <4.0000>: (↵)
- (5) Command: **POLYGON** (↵)
Enter number of sides <4>: **6** (↵)
Specify center of polygon or [Edge]: **0,0** (↵)
Enter an option [Inscribed in circle/Circumscribed about circle] <I>: (↵)
Specify radius of circle: **2** (↵)
- (6) Command: **ELEV** (↵)
Specify new default elevation <4.0000>: **8** (↵)
Specify new default thickness <4.0000>: (↵)
- (7) Command: **POINT** (↵)
Current point modes: PDMODE=0 PDSIZE=0.0000
Specify a point: **0,0** (↵)
- (8) Save This File As **3D-SURFACE-ELEV-1.DWG**

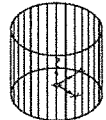


FIG. 26-33(i)

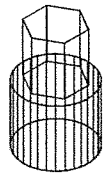


FIG. 26-33(ii)

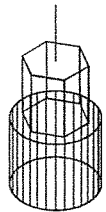


FIG. 26-33

(5) **3DFACE** command: It is used for making planar unmeshed surfaces that have three or four sides. They can hide objects, and can be coloured during rendering and shading. The command sequence is explained below vide self interactive module 26-22. The edges between the faces are made invisible by invisible option given in the command.

Module 26-22. To generate surface using *3DFACE* command on 3D wireframe model as shown in fig. 26-34.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass...>: **1,-1,1** (↵)
Regenerating model.
- (2) Create 3D Wireframe as shown in fig. 26-34
- (3) Command: **3DFACE** (↵)
Specify first point or [Invisible]: (Select Point **P1** by Osnap magnet)

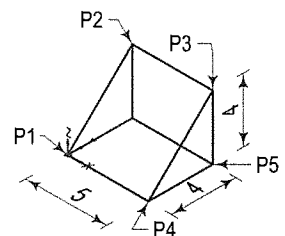


FIG. 26-34

- Specify second point or [Invisible]: (Select Point **P2** by Osnap magnet)
 Specify third point or [Invisible] <exit>: (Select Point **P3** by Osnap magnet)
 Specify fourth point or [Invisible] <create...>: (Select Point **P4** by Osnap magnet)
 Specify third point or [Invisible] <exit>: (Select Point **P5** by Osnap magnet)
 Specify fourth point or [Invisible] <create...>: (Select Point **P4** by Osnap magnet)
 Specify third point or [Invisible] <exit>: (↵)
- (4) Command: **SHADEMODE** (↵)
 Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]
 <Conceptual>: **R** (↵)
- (5) Save This File As **3DFACE-1.DWG**

(6) **PFACE command**: It creates planar surfaces on more than 3 vertices of 3D wireframe model. The faces made are tied together as a single object and edges between the faces are invisible. The surfaces generated are polyline variation called a polyline mesh. The command sequence is explained below vide self interactive module 26-23.

Module 26-23. To generate surface using **PFACE** command on 3D wireframe model as shown in fig. 26-35.

- (1) Command: **VPOINT** (↵)
 Current view direction: VIEWDIR=0.0000,0.0000,1.0000
 Specify a view point or [Rotate] <display compass...>: **1,-1,1** (↵)
 Regenerating model.
- (2) Create 3D Wireframe as shown in fig. 26-35
- (3) Command: **PFACE** (↵)
 Specify location for vertex 1: (Select **P1** by Osnap magnet)
 Specify location for vertex 2 or <define faces>: (Select **P2** by Osnap magnet)
 Specify location for vertex 3 or <define faces>: (Select **P3** by Osnap magnet)
 Specify location for vertex 4 or <define faces>: (Select **P4** by Osnap magnet)
 Specify location for vertex 5 or <define faces>: (Select **P5** by Osnap magnet)
 Specify location for vertex 6 or <define faces>: (↵)
 Face 1, vertex 1:
 Enter a vertex number or [Color/Layer]: **1** (↵)
 Face 1, vertex 2:
 Enter a vertex number or [Color/Layer] <next face>: **2** (↵)
 Face 1, vertex 3:
 Enter a vertex number or [Color/Layer] <next face>: **3** (↵)
 Face 1, vertex 4:
 Enter a vertex number or [Color/Layer] <next face>: **4** (↵)
 Face 1, vertex 5:
 Enter a vertex number or [Color/Layer] <next face>: (↵)
 Face 2, vertex 1:
 Enter a vertex number or [Color/Layer]: **1** (↵)
 Face 2, vertex 2:
 Enter a vertex number or [Color/Layer] <next face>: **4** (↵)
 Face 2, vertex 3:
 Enter a vertex number or [Color/Layer] <next face>: **5** (↵)
 Face 2, vertex 4:
 Enter a vertex number or [Color/Layer] <next face>: (↵)
 Face 3, vertex 1:
 Enter a vertex number or [Color/Layer]: (↵)

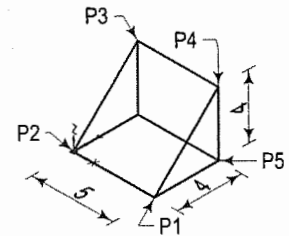


FIG. 26-35

- (4) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/.../Conceptual/Other] <Conceptual>: **C** (↵)
- (5) Save This File As **PFACE-1.DWG**

(7) **Revolve surface command:** REVSURF command creates a polygon mesh surface by revolving 2D profile object about an axis. The profile can be open or closed and axis will be a line, an open 2D polyline or an open 3D polyline. The command sequence is explained below vide self interactive module 26-24. The surfaces are generated by the matrix of number of surfaces in longitudinal and latitudinal direction. Their default values are 6×6 and are controlled by the command **Surftab1** and **Surftab2**.

Module 26-24. Create revolved surface using REVSURF command as per fig. 26-36.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.
- (2) Command: **UCS** (↵)
Current ucs name: ***WORLD***
Enter an option [New/Move/orthoGraphic/Prev/Restore/Save/...]<World>: **G** (↵)
Enter an option [Top/Bottom/Front/Back/Left/Right]<Top>:
FRONT (↵)
- (3) Command: **PLINE** (↵)
Specify start point: **4,0** (↵)
Current line-width is 0.0000
Specify next point or [Arc/Halfwidth/Length/Undo/Width]:
@-2,4 (↵)
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]:
@0,2 (↵)
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]:
@-2,0 (↵)
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: (↵)
- (4) Command: **LINE** (↵)
Specify first point: **0,0** (↵)
Specify next point or [Undo]: **0,6** (↵)
Specify next point or [Undo]: (↵)
- (5) Command: **SURFTAB1** (↵)
Enter new value for SURFTAB1 <6>: **24** (↵)
- (6) Command: **REVSURF** (↵)
Current wire frame density: SURFTAB1=24
SURFTAB2=6
Select object to revolve: (Select Polyline **PL1**)
Select object that defines the axis of revolution:
(Select Axis Line **AL1**)
Specify start angle <0>: (↵)
Specify included angle (+ = ccw, - = cw) <360>:
(↵)
- (7) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]
<Conceptual>: **H** (↵)
- (8) Save This File As **REVOLVESURFACE-1.DWG**

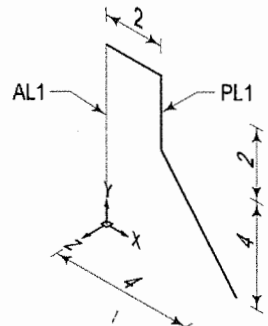


FIG. 26-36(i)

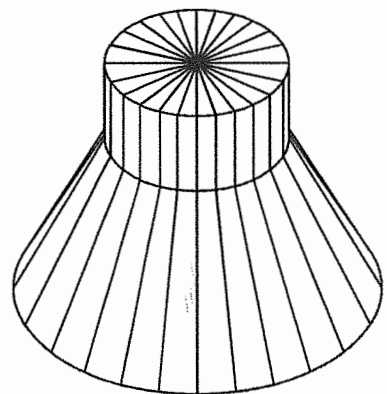


FIG. 26-36

(8) **Tabulated surfaces:** TABSURF command creates a polygon mesh surface by extruding a defining curve in a direction specified by an existing object. The first object to be selected is a path curve (open or closed) and second to be selected is a direction vector, must be a line, an open polyline or an open 3D polyline. The path curve can be a line, circle, arc, 2Dpolyline, 3Dpolyline, ellipse or spline. It uses SURFTAB1 command to generate number of surfaces. The command sequence is explained below vide self interactive module 26-25.

Module 26-25. To generate Surface using TABSURF command as per fig. 26-37.

- (1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

- (2) Command: **PLINE** (↵)

Specify start point: **0,0** (↵)

Current line-width is 0.0000

Specify next point or [Arc/Halfwidth/Length...]:

ARC (↵)

Specify endpoint of arc or

[Angle/Center/Direction/Halfwidth/Line/Radius/...]:

0,4 (↵)

Specify endpoint of arc or

[Angle/Center/Close/Direction/Halfwidth/Line/Radius/...]: **@0,4** (↵)

Specify endpoint of arc or

[Angle/Center/Close/Direction/Halfwidth/Line/Radius/...]: (↵)

- (3) Command: **LINE** (↵)

Specify first point: **0,0** (↵)

Specify next point or [Undo]: **@0,0,4** (↵)

Specify next point or [Undo]: (↵)

- (4) Command: **SURFTAB1** (↵)

Enter new value for SURFTAB1 <6>: **24** (↵)

- (5) Command: **TABSURF** (↵)

Current wire frame density: SURFTAB1=24

Select object for path curve: (Select Polyline **PL1**)

Select object for direction vector: (Select Line **L1**)

- (6) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **H** (↵)

- (7) Save This File As **TABULATEDSURFACE-1.DWG**

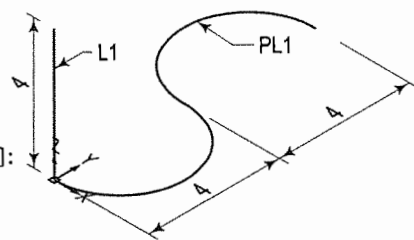


FIG. 26-37(i)

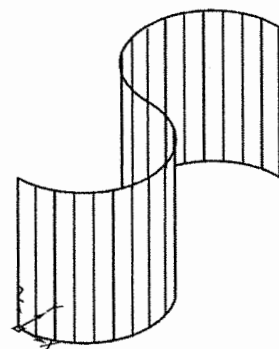


FIG. 26-37

(9) **Ruled surfaces:** RULESURF command creates a surface between two existing 2D Curve. Both curve should be open or both should be closed. Curves can be a line, 2D or 3D polyline, spline, arc, polygon, ellipse, circle, donuts or even a point. It uses SURFTAB1 command to generate number of surfaces. The command sequence is explained below vide self interactive module 26-26.

Module 26-26. To generate Surface using **RULESURF** command as per fig. 26-38.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.
- (2) Command: **LINE** (↵)
Specify first point: **0,0** (↵)
Specify next point or [Undo]: **0,4** (↵)
Specify next point or [Undo]: (↵)
- (3) Command: **ARC** (↵)
Specify start point of arc or [Center]: **0,0,4** (↵)
Specify second point of arc or [Center/End]: **E** (↵)
Specify end point of arc: **@0,4** (↵)
Specify center point of arc or [Angle/Direction/Radius]: **A** (↵)
Specify included angle: **-180** (↵)
- (4) Command: **SURFTAB1** (↵)
Enter new value for SURFTAB1 <6>: **24** (↵)
- (5) Command: **RULESURF** (↵)
Current wire frame density: SURFTAB1=24
Select first defining curve: (Select Line L1)
Select second defining curve: (Select Arc A1)
- (6) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **H** (↵)
- (7) Save This File As **RULESURFACE-1.DWG**

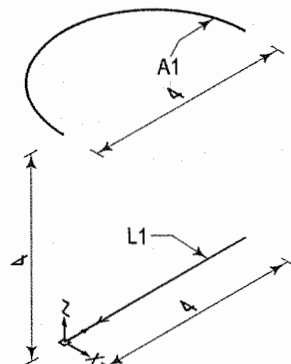


FIG. 26-38(i)

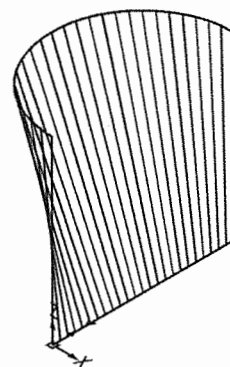


FIG. 26-38

(10) **Edge surfaces:** **EDGESURF** command creates a polygon mesh surface between four boundary 2D curves. The curves can be a lines, arcs, open 2D or 3D polylines or splines. Their ends must touch. It uses **SURFTAB1** and **SURFTAB2** commands to generate number of surfaces. The command sequence is explained below vide self interactive module 26-27 and fig. 26-39.

Module 26-27. To generate Surface using **EDGESURF** command as per fig. 26-39.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.
- (2) Command: **LINE** (↵)
Specify first point: **0,0** (↵)
Specify next point or [Undo]: **4,0** (↵)
Specify next point or [Undo]: (↵)
- (3) Command: **LINE** (↵)
Specify first point: **0,4** (↵)
Specify next point or [Undo]: **4,4** (↵)
Specify next point or [Undo]: (↵)
- (4) Command: **ARC** (↵)
Specify start point of arc or [Center]: **0,4** (↵)

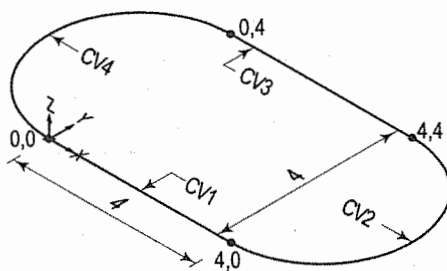


FIG. 26-39(i)

Specify second point of arc or [Center/End]: E (↵)

Specify end point of arc: 0,0 (↵)

Specify center point of arc or [Angle/Direction/Radius]: A (↵)

Specify included angle: 180 (↵)

(5) Command: ARC (↵)

Specify start point of arc or [Center]: 4,0 (↵)

Specify second point of arc or

[Center/End]: E (↵)

Specify end point of arc: 4,4 (↵)

Specify center point of arc or [Angle/

Direction/Radius]: A (↵)

Specify included angle: 180 (↵)

(6) Command: SURFTAB1 (↵)

Enter new value for SURFTAB1 <6>:

24 (↵)

(7) Command: SURFTAB2 (↵)

Enter new value for SURFTAB1 <6>:

24 (↵)

(8) Command: EDGESURF (↵)

Current wire frame density: SURFTAB1=24 SURFTAB2=24

Select object 1 for surface edge: (Select curve CV1)

Select object 2 for surface edge: (Select curve CV2)

Select object 3 for surface edge: (Select curve CV3)

Select object 4 for surface edge: (Select curve CV4)

(9) Command: SHADEMODE (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]

<Conceptual>: H (↵)

(10) Save This File As EDGESURFACE-1.DWG

(11) **3DMESH**: 3DMESH command creates a polygon mesh surface of M and N direction. If we consider M as number of vertices in X direction then N becomes number of vertices in Y direction. The command sequence is explained below vide self interactive module 26-28.

Module 26-28. To generate Surface using 3DMESH command as per fig. 26-40.

(1) Command: VPOINT (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: 1,-1,1 (↵)

Regenerating model.

(2) Create 2D wireframe as shown fig. 26-40(i)

(3) Command: 3DMESH (↵)

Enter size of mesh in M direction: 3 (↵)

Enter size of mesh in N direction: 3 (↵)

Specify location for vertex (0,0): (Select P1)

Specify location for vertex (0,1): (Select P2)

Specify location for vertex (0,2): (Select P3)

Specify location for vertex (1,0): (Select P4)

Specify location for vertex (1,1): (Select P5)

Specify location for vertex (1,2): (Select P6)

Specify location for vertex (2,0): (Select P7)

Specify location for vertex (2,1): (Select P8)

Specify location for vertex (2,2): (Select P9)

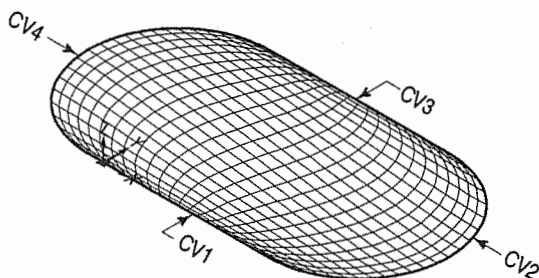


FIG. 26-39

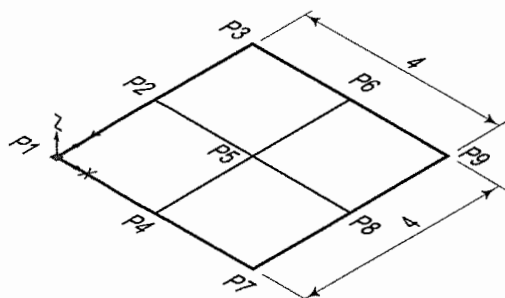


FIG. 26-40(i)

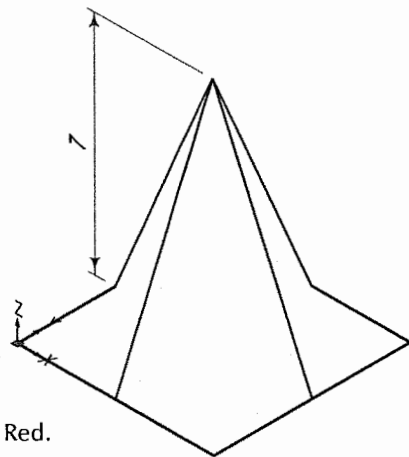


FIG. 26-40

- (4) 1. Select any line
2. Select middle blue box so it will become Red.
3. Type @0,0,7 and press Enter key.
4. Press Esc key.
- (5) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **H** (↵)
- (6) Save This File As **3DMESH-1.DWG**

(12) **3D Standard shaped surface:** Eight numbers of standard shapes are available in CAD to create 3D basic shaped surfaces. They are Box, Pyramid, Wedge, Dome, Dish, Sphere, Cone and Torus. The command sequence for each shape is explained below vide self interactive examples. (module 26-29 to module 26-36 and fig. 26-41 to fig. 26-48).

Module 26-29. To generate Surface using **BOX** option of 3D command as per fig. 26-41.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.
- (2) Command: **3D** (↵)
[Box/Cone/Dish/Dome/Mesh/Pyramid/Sphere/Torus/Wedge]: **BOX** (↵)
Specify corner point of box: **0,0** (↵)
Specify length of box: **8** (↵)
Specify width of box or [Cube]: **8** (↵)
Specify height of box: **4** (↵)
Specify rotation angle of box about the Z axis or [Reference]: **0** (↵)
- (3) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **H** (↵)
- (4) Save This File As **3DBOXSURFACE-1.DWG**

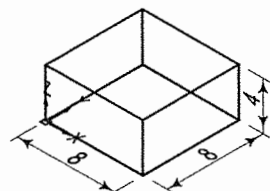


FIG. 26-41

Module 26-30. To generate Surface using *PYRAMID* option of 3D command as per fig. 26-42.

- (1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

- (2) Command: **3D** (↵)

Initializing... 3D Objects loaded.

Enter an option

[Box/Cone/DIsh/DOme/Mesh/Pyramid/Sphere/Torus/Wedge]: **PYRAMID** (↵)

Specify first corner point for base of pyramid: **0,0** (↵)

Specify second corner point for base of pyramid: **8,0** (↵)

Specify third corner point for base of pyramid: **8,8** (↵)

Specify fourth corner point for base of pyramid or [Tetrahedron]: **0,8** (↵)

Specify apex point of pyramid or [Ridge/Top]: **TOP** (↵)

Specify first corner point for top of pyramid: **@2,2,4** (↵)

Specify second corner point for top of pyramid: **@-2,2,4** (↵)

Specify third corner point for top of pyramid: **@-2,-2,4** (↵)

Specify fourth corner point for top of pyramid: **@2,-2,4** (↵)

- (3) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]
<Conceptual>: **H** (↵)

- (4) Save This File As **3DPYRAMIDSURFACE-1.DWG**

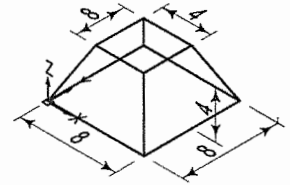


FIG. 26-42

Module 26-31. To generate Surface using *WEDGE* option of 3D command as per fig. 26-43.

- (1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

- (2) Command: **3D** (↵)

[Box/Cone/DIsh/DOme/Mesh/Pyramid/Sphere/Torus/Wedge]:
WEDGE (↵)

Specify corner point of box: **0,0** (↵)

Specify length of box: **8** (↵)

Specify width of box or [Cube]: **8** (↵)

Specify height of box: **4** (↵)

Specify rotation angle of box about the Z axis or [Reference]: **0** (↵)

- (3) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]
<Conceptual>: **H** (↵)

- (4) Save This File As **3DWEDGESURFACE-1.DWG**

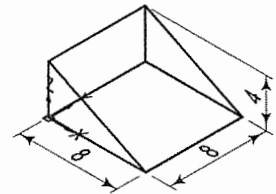


FIG. 26-43

Module 26-32. To generate Surface using *SPHERE* option of 3D command as per fig. 26-44.

- (1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

- (2) Command: **3D** (↵)

Initializing... 3D Objects loaded.

Enter an option

[Box/Cone/Dish/Dome/Mesh/Pyramid/Sphere/Torus/Wedge]: **SPHERE**

(↵)

Specify center point of sphere: **0,0** (↵)

Specify radius of sphere or [Diameter]: **8** (↵)

Enter number of longitudinal segments for surface of sphere <16>: (↵)

Enter number of latitudinal segments for surface of sphere <16>: (↵)

- (3) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]

<Conceptual>: **H** (↵)

- (4) Save This File As **3DSPHERESURFACE-1.DWG**



FIG. 26-44

Module 26-33. To generate Surface using *CONE* option of 3D command as per fig. 26-45.

- (1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

- (2) Command: **3D** (↵)

Initializing... 3D Objects loaded.

Enter an option

[Box/Cone/Dish/Dome/Mesh/Pyramid/Sphere/Torus/Wedge]:

CONE (↵)

Specify center point for base of cone: **0,0** (↵)

Specify radius for base of cone or [Diameter]: **4** (↵)

Specify radius for top of cone or [Diameter] <0>: (↵)

Specify height of cone: **8** (↵)

Enter number of segments for surface of cone <16>: (↵)

- (3) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]

<Conceptual>: **H** (↵)

- (4) Save This File As **3DCONESURFACE-1.DWG**

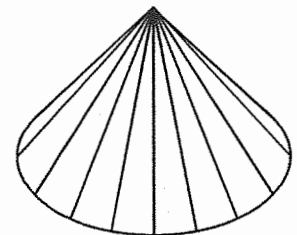


FIG. 26-45

Module 26-34. To generate Surface using *DOME* or *DISH* option of 3D command as per fig. 26-46.

- (1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

(2) Command: **3D** (↵)

Initializing... 3D Objects loaded.

Enter an option

[Box/Cone/Dish/DOME/Mesh/Pyramid/Sphere/Torus/Wedge]: **DOME**
or **DISH** (↵)

Specify center point of sphere: **0,0** (↵)

Specify radius of sphere or [Diameter]: **8** (↵)

Enter number of longitudinal segments for surface of sphere
<16>: (↵)

Enter number of latitudinal segments for surface of sphere <16>:
(↵)

**3D DOME****3D DISH**

FIG. 26-46

(3) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/
Conceptual/Other] <Conceptual>: **H** (↵)

(4) Save This File As **3DDOMESURFACE-1.DWG**

Module 26-35. To generate Surface using **TORUS** option of **3D** command as per fig. 26-47.

(1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

(2) Command: **3D** (↵)

Initializing... 3D Objects loaded.

Enter an option

[Box/Cone/Dish/DOME/Mesh/Pyramid/Sphere/Torus/
Wedge]: **TORUS** (↵)

Specify center point of torus: **0,0** (↵)

Specify radius of torus or [Diameter]: **8** (↵)

Specify radius of tube or [Diameter]: **2** (↵)

Enter number of segments around tube circumference <16>: (↵)

Enter number of segments around torus circumference <16>: (↵)

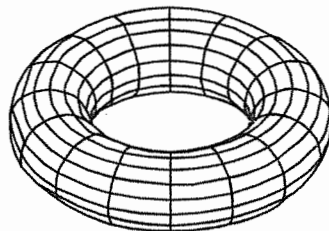


FIG. 26-47

(3) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]
<Conceptual>: **H** (↵)

(4) Save This File As **3DTORUSSURFACE-1.DWG**

Module 26-36. To generate Surface using **MESH** option of **3D** command as per fig. 26-48.

(1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

(2) Command: **3D** (↵)

Initializing... 3D Objects loaded.

Enter an option

[Box/Cone/Dish/DOME/Mesh/Pyramid/Sphere/Torus/Wedge]: **MESH** (↵)

Specify first corner point of mesh: **0,0** (↵)
 Specify second corner point of mesh: **8,0** (↵)
 Specify third corner point of mesh: **8,8** (↵)
 Specify fourth corner point of mesh: **0,8** (↵)
 Enter mesh size in the M direction: **4** (↵)
 Enter mesh size in the N direction: **8** (↵)

(3) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other]
 <Conceptual>: **C** (↵)

(4) Save This File As **MESHSURFACE-1.DWG**

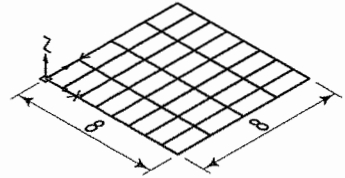


FIG. 26-48

26-9-3. 3D SOLID MODELLING

Solid modeling is the process of building objects that have all the attributes of an actual solid object. It has mass, weight, volume, center of gravity, moment of inertia etc, in addition to surfaces and edges. We can cut, join and create holes and cavities. It is a single object and are used as a part to create assemblies and interference check. It eliminates the need for building expensive prototypes and makes the product development cycle shorter. The solid model created on the cad software are based on a program called the ACIS Geometrical Modeler. So they are also called ACIS solids. There are two methods to create solids.

Method 1: Standard 3D shapes based solids:

In this method six types of standard shapes available in cad are used to generate solids of different shapes. These shapes are Box, Wedge, Cylinder, Cone, Sphere and Torus. They are also called primitives. The command sequence for each shape is explained below vide self interactive examples (Module 26-37 to Module 26-42 and fig. 26-49 to fig. 26-54).

Module 26-37. To generate 3D Solid box using **BOX** command as per fig. 26-42(i).

(1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

(2) Command: **ISOLINES** (↵)

Enter new value for ISOLINES <4>: **16** (↵)

(3) Command: **BOX** (↵)

Specify first corner or [CEnter]: **0,0,0** (↵)

Specify other corner or [Cube/Length]: **L** (↵)

Specify length: **8** (↵)

Specify width: **8** (↵)

Specify height or [2Point]: **4** (↵)

(4) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/
 Realistic/Conceptual/Other] <Conceptual>: **H** (↵)

(5) Save This File As **3DBOXSOLID-1.DWG**

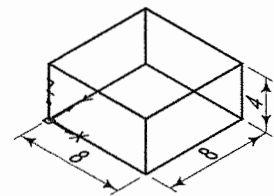


FIG. 26-49

Module 26-38. To generate 3D Solid wedge using **WEDGE** command as per fig. 26-50.

(1) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

- (2) Command: **ISOLINES** (↵)
Enter new value for ISOLINES <4>: **16** (↵)
- (3) Command: **WEDGE** (↵)
Specify first corner or [CEnter]: **0,0,0** (↵)
Specify other corner or [Cube/Length]: **1** (↵)
Specify length: **8** (↵)
Specify width: **8** (↵)
Specify height or [2Point]: **4** (↵)
- (4) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **H** (↵)
- (5) Save This File As **3DWEDGESOLID-1.DWG**

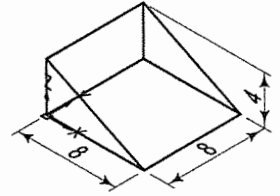


FIG. 26-50

Module 26-39. To generate 3D Solid cylinder using **CYLINDER** command as per fig. 26-51.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.
- (2) Command: **ISOLINES** (↵)
Enter new value for ISOLINES <4>: **16** (↵)
- (3) Command: **CYLINDER** (↵)
Specify center point of base or [3P/2P/Ttr/Elliptical]: **0,0,0** (↵)
Specify base radius or [Diameter]: **4** (↵)
Specify height or [2Point/Axis endpoint]: **8** (↵)
- (4) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **H** (↵)
- (5) Save This File As **3DCYLINDERSOLID-1.DWG**

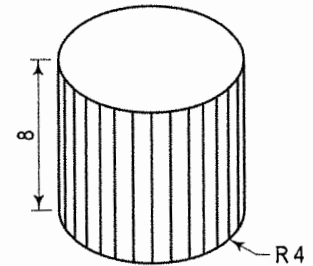


FIG. 26-51

Module 26-40. To generate 3D Solid cone using **CONE** command as per fig. 26-52.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.
- (2) Command: **ISOLINES** (↵)
Enter new value for ISOLINES <4>: **16** (↵)
- (3) Command: **CONE** (↵)
Specify center point of base or [3P/2P/Ttr/Elliptical]: **0,0,0** (↵)
Specify base radius or [Diameter]: **6** (↵)
Specify height or [2Point/Axis endpoint/Top radius]: **8** (↵)
(When cursor is above **XY** plane) (↵)
- (4) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **H** (↵)
- (5) Save This File As **3DCONESOLID-1.DWG**

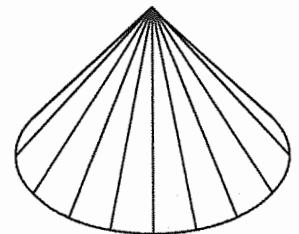


FIG. 26-52

Module 26-41. To generate 3D Solid sphere using *SPHERE* command as per fig. 26-53.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.
- (2) Command: **ISOLINES** (↵)
Enter new value for ISOLINES <4>: **16** (↵)
- (3) Command: **SPHERE** (↵)
Specify center point or [3P/2P/Ttr]: **3** (↵)
Specify radius or [Diameter]: **4** (↵)
- (4) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/
Conceptual/Other] <Conceptual>: **H** (↵)
- (5) Save This File As **3DSPHERESOLID-1.DWG**



FIG. 26-53

Module 26-42. To generate 3D Solid torus using *TORUS* command as per fig. 26-54.

- (1) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.
- (2) Command: **ISOLINES** (↵)
Enter new value for ISOLINES <4>: **16** (↵)
- (3) Command: **TORUS** (↵)
Specify center point or [3P/2P/Ttr]: **0,0,0** (↵)
Specify radius or [Diameter] <1.5000>: **8** (↵)
Specify tube radius or [2Point/Diameter] <2.0000>: **2** (↵)
- (4) Command: **SHADEMODE** (↵)
Enter an option [2dwireframe/3dwireframe/3dHidden/
Realistic/Conceptual/Other] <Conceptual>: **H** (↵)
- (5) Save This File As **3DTORUSSOLID-1.DWG**

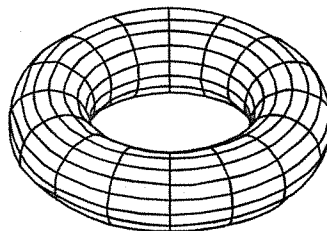


FIG. 26-54

Method II: Profile based solids:

In this method the following steps are performed.

- (1) Create 2D closed profile in XY plane using 2D commands.
- (2) Generate surface on this profile by using **REGION** command.
- (3) Use **EXTRUDE** command to provide height to this profile in Z direction OR use **REVOLVE** command to revolve this profile about X or Y axis of UCS and thus solid will be created.
- (4) Finally use **SHADEMODE** command to get real look of solid.

26-9-4. COMMANDS TO GENERATE PROFILE BASED 3D SOLIDS

Commonly used commands to generate 3D solids are discussed below and their use is shown in self interactive module 26-43 to module 26-49.

- (1) **VPOINT** command: Refer details on page no. 666
- (2) **UCS** Command: Refer details on page no. 666
- (3) **Shademode**: Refer details on page no. 669

(4) **REGION command:** Creates regions from the selected 2D closed objects. They are the 2D objects with properties of 3D solids. All 3D commands are used on regions to generate 3D solids. The command sequence is given below.

Command: **REGION** (↵)
 Select objects: (Select 2D closed objects)
 Select objects: (↵)
 1 loops extracted.
 1 Region created.

(5) **EXTRUDE command:** The EXTRUDE command creates solids by extruding 2D closed single object or region in Z direction of UCS or about a specified path. The command sequence is given below.

Command : **EXTRUDE** (↵)
 Current wire frame density: ISOLINES=4
 Select objects to extrude:
 Select objects to extrude: (↵)
 Specify height of extrusion or [Direction/Path/Taper angle]: (Enter height value) (↵)

(6) **REVOLVE command:** The REVOLVE command creates solids by revolving 2D closed single object or region about an object or X or Y axis of UCS. It can revolve only one object at a time. The command sequence is given below.

Command: **REVOLVE** (↵)
 Current wire frame density: ISOLINES=4
 Select objects to revolve:
 Select objects to revolve: (↵)
 Specify axis start point or define axis by [Object/X/Y/Z] <Object>: (Select any option) (↵)
 Specify angle of revolution or [Start angle] <360>: 360 (↵)

(7) **BOOLEAN OPERATIONS:**

- (i) **UNION:** The UNION command combines selected regions or solids by addition. The command sequence is given below.

Command: **UNION** (↵)
 Select objects: (Select more than one solid)
 Select objects: (↵)

- (ii) **SUBTRACT:** The SUBTRACT command subtracts one set of solids from another set of solids to create holes, cavities etc. The command sequence is given below.

Command: **SUBTRACT** (↵)
 Select solids, surfaces and regions to subtract from ..
 Select objects: (Select solids to subtract from)
 Select objects: (↵)
 Select solids, surfaces and regions to subtract ..
 Select objects: (Select solids to subtract)
 Select objects: (↵)

- (iii) **INTERSECT:** The INTERSECT command creates composite solids or regions from the intersection of two or more solids or regions and removes the areas outside of the intersection. The command sequence is given below.

Command: **INTERSECT** (↵)

Select objects: (Select more than one solid)

Select objects: (↵)

(8) **3DARRAY command:** 3DARRAY can create a Rectangular array or a Polar array of objects in 3D. In Rectangular array we have to specify number of columns (X direction), number of rows (Y direction), and number of levels (Z direction). And in Polar array we have to select solids and axis about which solids will be arranged. The command sequence is given below.

- (i) *Command sequence for 3D Rectangular Array*

Command: **3DARRAY** (↵)

Initializing... 3DARRAY loaded.

Select objects: (Select solids)

Select objects: (↵)

Enter the type of array [Rectangular/Polar] <R>: **R** (↵)

Enter the number of rows (—) <1>: (Enter Integer value)

Enter the number of columns (|||) <1>: (Enter Integer value)

Enter the number of levels (...) <1>: (Enter Integer value)

Specify the distance between rows (—): (Enter Integer or Real value)

Specify the distance between columns (|||): (Enter Integer or Real value)

Specify the distance between levels (...): (Enter Integer or Real value)

- (ii) *Command sequence for 3D Polar Array*

Command: **3DARRAY** (↵)

Initializing... 3DARRAY loaded.

Select objects: (Select solids)

Select objects: (↵)

Enter the type of array [Rectangular/Polar] <R>: **P** (↵)

Enter the number of items in the array: (Enter Integer value)

Specify the angle to fill (+ = ccw, - = cw) <360>: (↵)

Rotate arrayed objects? [Yes/No] <Y>: (↵)

Specify center point of array: (Select a point on axis)

Specify second point on axis of rotation: (Select another point on same axis)

(9) **FILLET on 3D solids:** FILLET command is used to round the edges or corners of the 3D solid models. This command is used both in 2D and 3D. In 3D first we have to select edge of solid to get options of 3D. The command sequence is given below.

Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 0.0000

Select first object or [Undo/Polyline/Radius/Trim/Multiple]: (Select edge of solid)

Enter fillet radius: (Enter radius value)

Select an edge or [Chain/Radius]: (Select another edge to fillet)

Select an edge or [Chain/Radius]: (↵)

(10) CHAMFER on 3D solids: CHAMFER command is used to create beveled edges on edges of solids. This command is used both in 2D and 3D. In 3D first we have to select edge of solid to get options of 3D. The command sequence is given below.

Command: **CHAMFER** (↵)

(TRIM mode) Current chamfer Dist 1 = 0.0000, Dist 2 = 0.0000

Select first line or [Undo/Polyline/Distance/Angle/Trim/...]: (Select edge of solid)

Base surface selection...

Enter surface selection option [Next/OK (current)] <OK>: (↵)

Specify base surface chamfer distance: (Enter distance value from base surface)

Specify other surface chamfer distance <2.0000>: (Enter distance value from other surface)

Select an edge or [Loop]: (Select edges on base surface)

Select an edge or [Loop]: (↵)

(11) ROTATE3D command: The ROTATE3D command is used to rotate the solids about X, Y or Z axis of UCS or about an Object to required angle. Positive angle will rotate in counter clock wise direction. The command sequence is given below.

Command: **ROTATE3D** (↵)

Current positive angle: ANGDIR=counterclockwise ANGBASE=0

Select objects: (Select solids)

Select objects: (↵)

Specify first point on axis or define axis by

[Object/Last/View/Xaxis/Yaxis/Zaxis/2points]: (Select required option for axis say Z)

Specify a point on the Z axis <0,0,0>: (Specify base point)

Specify rotation angle or [Reference]: (Enter angle value)

(12) MIRROR3D command: The MIRROR3D command is used to mirror the solids about a specified plane in the space. The plane can be the 2D object or three planes of UCS i.e. XY, YZ, or ZX. The command sequence is given below.

Command: **MIRROR3D** (↵)

Select objects: (Select solids)

Select objects: (↵)

Specify first point of mirror plane (3 points) or

[Object/Last/Zaxis/View/XY/YZ/ZX/3points] <3points>: (Select required option as mirror plane say ZX)

Specify point on ZX plane <0,0,0>: (Select base point)

Delete source objects? [Yes/No] <N>: (↵)

(13) SLICE command: The SLICE command is used to slice the selected solid with the help of specified plane. Generally 2D object or three planes of UCS i.e. XY, YZ or ZX are selected as slicing plane from options. After slicing, the option will be given to retain either or both solids. The command sequence is given below.

Command: **SLICE** (↵)

Select objects to slice: (Select solid)

Select objects to slice: (↵)

Specify start point of slicing plane or [planar Object/Surface/Zaxis/View/XY/YZ/ZX/3points] <3points>: (Select required option as slicing plane say ZX)

Specify a point on the YZ-plane <0,0,0>: (Select the point to slice from)

Specify a point on desired side or [keep Both sides] <Both>: (Click either side to retain one solid or type B to retain both solids)

26-10. THREE DIMENSIONAL DRAWINGS

Module 26-43. Draw a three dimensional diagram as shown in fig. 26-55. Use 3D commands such as Region, Extrude and Subtract.

To draw a three dimensional diagram using the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-55(i) to fig. 26-55(vii).

- (1) Command: **LIMITS** (↵)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)

Specify upper right corner <12.0000,9.0000>: **120,90** (↵)

- (2) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/.....] <real time>: **ALL** (↵)

- (3) Create 2D diagram (fig. 26-55(i)) Using **Pline** command, take Coordinate of **P1** as **0,0**

- (4) Command: **VPOINT** (↵)

Current view direction: VIEWDIR = 0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

- (5) Command: **REGION** (↵)

Select objects: (Select any Line)

Select objects: (↵)

1 loop extracted.

1 Region created.

- (6) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **H** (↵)

- (7) Command: **EXTRUDE** (↵)

Current wire frame density: ISOLINES=4

Select objects: (Select any line)

Select objects: (↵)

Specify height of extrusion or [Path]: **40** (↵)

- (8) Command: **ROTATE3D** (↵)

Current positive angle: ANGDIR = counterclockwise
ANGBASE=0

Select objects: (Select Solid **S1**)

Select objects: (↵)

Specify first point on axis or define axis by

[Object/Last/View/Xaxis/Yaxis/Zaxis/2points]: **X** (↵)

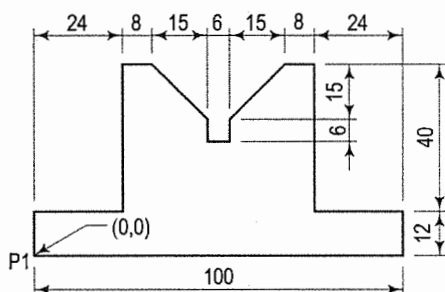


FIG. 26-55(i)

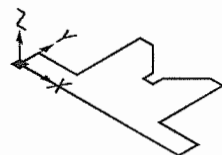


FIG. 26-55(ii)

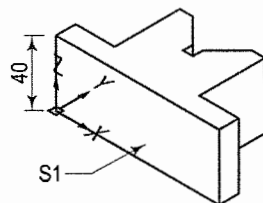


FIG. 26-55(iii)

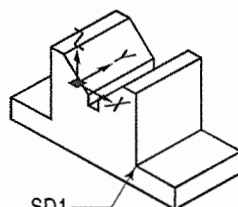


FIG. 26-55(iv)

Specify a point on the X axis <0,0,0>: (↵)

Specify rotation angle or [Reference]: 90 (↵)

(9) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/.....] <real time>: **ALL** (↵)

(10) Command: **UCS** (↵)

Current ucs name: *TOP*

Enter an option [New/Move/orthoGraphic/...] <World>: **M** (↵)

Specify new origin point or [Zdepth]<0,0,0>: (Select Point **SD1**)

(11) Command: **CIRCLE** (↵)

Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: **12,20** (↵)

Specify radius of circle or [Diameter]: **D** (↵)

Specify diameter of circle: **13** (↵)

(12) Command: **CIRCLE** (↵)

Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: **-64,20** (↵)

Specify radius of circle or [Diameter] <0.6500>: **D** (↵)

Specify diameter of circle: **13** (↵)

(13) Command: **EXTRUDE** (↵)

Current wire frame density: ISOLINES=4

Select objects: (Select Circle **C1**)

Select objects: (Select Circle **C2**)

Select objects: (↵)

Specify height of extrusion or [Path]: **-40** (↵)

(14) Command: **SUBTRACT** (↵)

Select solids and regions to subtract from ..

Select objects: (Select Solid **SL1**)

Select objects: (↵)

Select solids and regions to subtract ..

Select objects: (Select Cylinder **CL1**)

Select objects: (Select Cylinder **CL2**)

Select objects: (↵)

(15) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **C** (↵)

(16) Save This File As **Module 26-43.DWG**

Output of Module 26-43 (fig. 26-55):

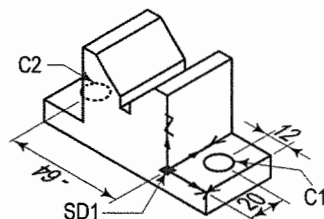


FIG. 26-55(v)

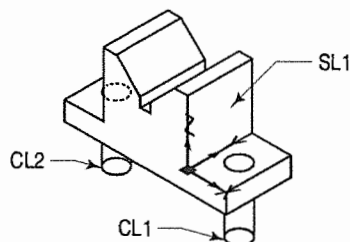


FIG. 26-55(vi)

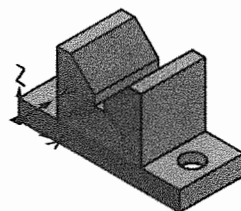


FIG. 26-55(vii)

MODULE 26-43: USE OF 3D COMMANDS REGION, EXTRUDE AND SUBTRACT

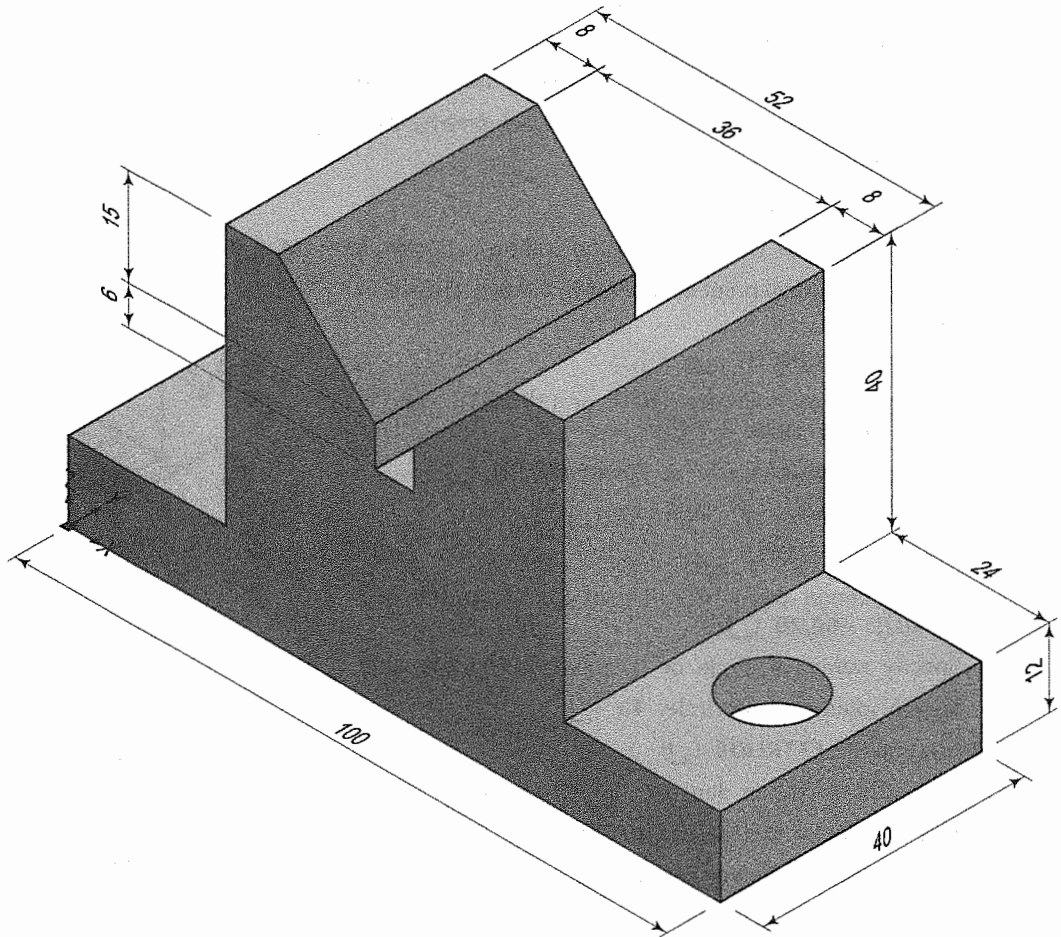


FIG. 26-55

Module 26-44. Draw a three dimensional diagram as shown in fig. 26-56. Use 3D commands such as *Region*, *Revolve* and *Fillet*.

To draw a three dimensional diagram using the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-56(i) to fig. 26-56(iv).

- (1) Command: **LIMITS** (↵)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)

Specify upper right corner <12.0000,9.0000>: **360,270** (↵)

- (2) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)

Regenerating model.

- (3) Using **PLINE** Command Create 2D diagram (fig. 26-56(i)), take co-ordinate of **P1 0,0**

- (4) Command: **VPOINT** (↵)

Current view direction: VIEWDIR = 0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

- (5) Command: **REGION** (↵)

Select objects: (Select any Line)

Select objects: (↵)

1 loop extracted.

1 Region created.

- (6) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/...] <Conceptual>: **H** (↵)

- (7) Command: **REVOLVE** (↵)

Current wire frame density: ISOLINES=4

Select objects to revolve: (Select any Line)

Select objects to revolve: (↵)

Specify axis start point or define axis by [Object/X/Y/Z] <Object>: **X** (↵)

Specify angle of revolution or [SStart angle] <360>: **360** (↵)

- (8) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)

- (9) Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 0.0000

Select first object or [Polyline/Radius/Trim]: (Select Edge **E1**)

Enter fillet radius: **6** (↵)

Select an edge or [Chain/Radius]: (↵)

1 edge(s) selected for fillet.

- (10) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/...] <Conceptual>: **C** (↵)

- (11) Save this File as **Module 26-44.DWG**

Output of Module 26-44 (fig. 26-56):

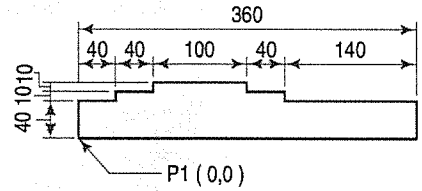


FIG. 26-56(i)

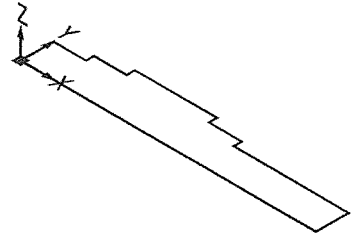


FIG. 26-56(ii)

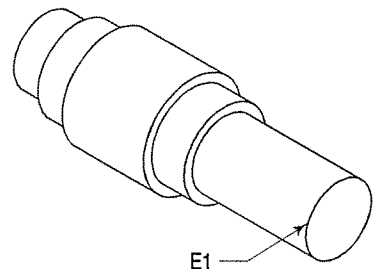


FIG. 26-56(iii)

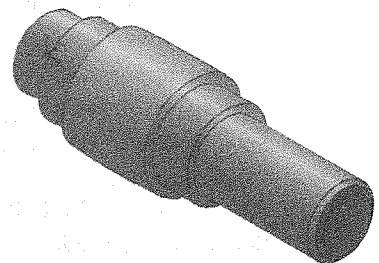


FIG. 26-56(iv)

MODULE 26-44: USE OF 3D COMMANDS REGION, REVOLVE AND FILLET

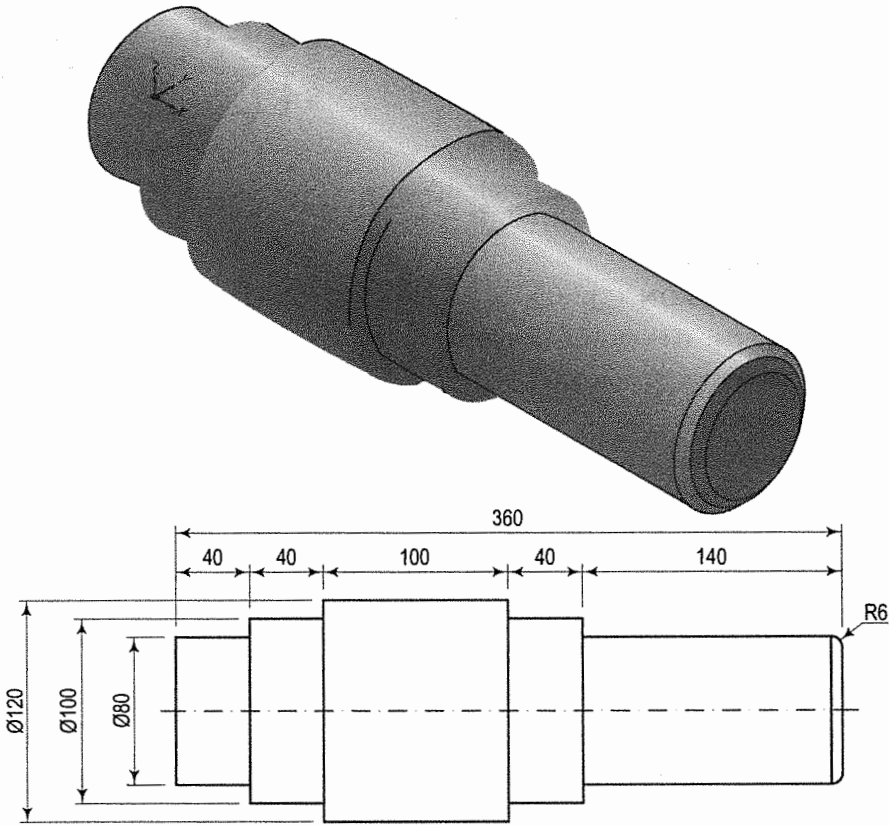


FIG. 26-56

Module 26-45. Draw a three dimensional diagram as shown in fig. 26-57. Use 3D commands such as Region, Extrude and Union.

To draw a three dimensional diagram using the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-57(i) to fig. 26-57(ix).

- (1) Command: **LIMITS** (↵)
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
Specify upper right corner <12.0000,9.0000>: **600,450** (↵)
- (2) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
Regenerating model.
- (3) Command: **RECTANGLE** (↵)
Specify first corner point or [Chamfer/Elevation/Fillet/Thickness/Width]: **0,0** (↵)
Specify other corner point or [Dimensions]: **@100,50** (↵)

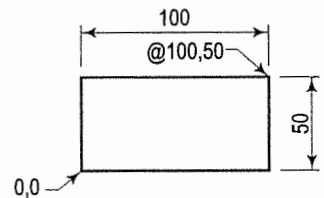


FIG. 26-57(i)

(4) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

(5) Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 0.0000

Select first object or [Polyline/Radius/Trim]: **R** (↵)

Specify fillet radius <0.0000>: **25** (↵)

Select first object or [Polyline/Radius/Trim]: (Select **L1**)

Select second object: (Select **L2**)

(6) Command: **FILLET** (↵)

Current settings: Mode = TRIM, Radius = 25.0000

Select first object or [Polyline/Radius/Trim]: (Select **L2**)

Select second object: (Select **L3**)

(7) Command: **CIRCLE** (↵)

Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:
(Select CENTER magnet **C1**)

Specify radius of circle or [Diameter]: **D** (↵)

Specify diameter of circle: **24** (↵)

(8) Command: **EXTRUDE** (↵)

Current wire frame density: ISOLINES=4

Select objects to extrude: (Select Rectangle **R1**)

Select objects to extrude: (↵)

Specify height of extrusion or [Direction/Path/
Taper angle] <40.0000>: **40** (↵)

(9) Command: **EXTRUDE** (↵)

Current wire frame density: ISOLINES=4

Select objects to extrude: (Select Circle **CR1**)

Select objects to extrude: (↵)

Specify height of extrusion or [Direction/Path/
Taper angle] <40.0000>: **70** (↵)

(10) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX
or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window]
<real time>: **ALL** (↵)

Regenerating model.

(11) Command: **SUBTRACT** (↵)

Select solids, surfaces and regions to subtract from ..

Select objects: (Select Solid 3D Box **S1**)

Select objects: (↵)

Select solids, surfaces and regions to subtract ..

Select objects: (Select Solid 3D Cylinder **S2**)

Select objects: (↵)

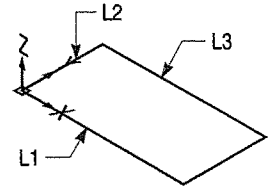


FIG. 26-57(ii)

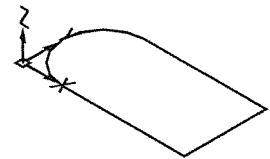


FIG. 26-57(iii)

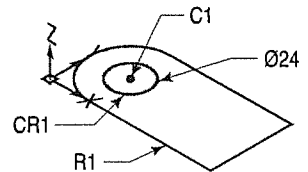


FIG. 26-57(iv)

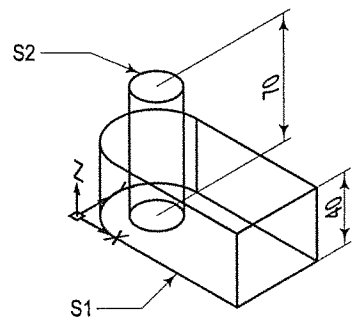


FIG. 26-57(v)

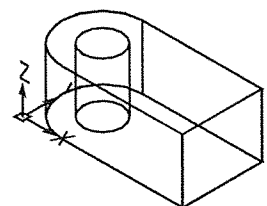


FIG. 26-57(vi)

(12) Command: **UCS** (↵)

Current ucs name: *WORLD*

Enter an option [New/Move/orthoGraphic/Prev/.../World]<World> : **G** (↵)

Enter an option [Top/Bottom/Front/BAck/Left/Right]<Top>: **RIGHT** (↵)

(13) Command: **UCS** (↵)

Current ucs name: *RIGHT*

Enter an option [New/Move/orthoGraphic/Prev/.../World] <World>: **M** (↵)

Specify new origin point or [Zdepth]<0,0,0>: (Select
ENDPOINT magnet **E1**)

(14) Command: **CIRCLE** (↵)

Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:
2P (↵)

Specify first end point of circle's diameter: (Select MIDPOINT
magnet **M2**)

Specify second end point of circle's diameter: (Select
MIDPOINT magnet **M3**)

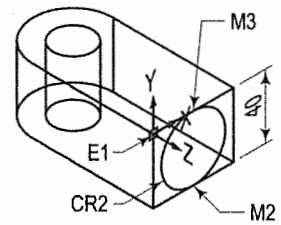


FIG. 26-57(vii)

(15) Command: **EXTRUDE** (↵)

Current wire frame density:
ISOLINES=4

Select objects to extrude: (Select Circle
CR2)

Select objects to extrude: (↵)

Specify height of extrusion or [Direction/
Path/Taper angle]<40.0000>: **150** (↵)

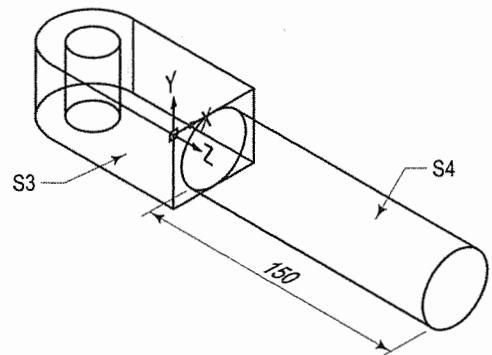


FIG. 26-57(viii)

(16) Command: **ZOOM** (↵)

Specify corner of window, enter a scale
factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/
Scale/Window] <real time>: **ALL** (↵)

(17) Command: **UNION** (↵)

Select objects: (Select Solid **S3**)

Select objects: (Select Solid **S4**)

Select objects: (↵)

(18) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/
3dHidden/Realistic/...] <Conceptual>:
C (↵)

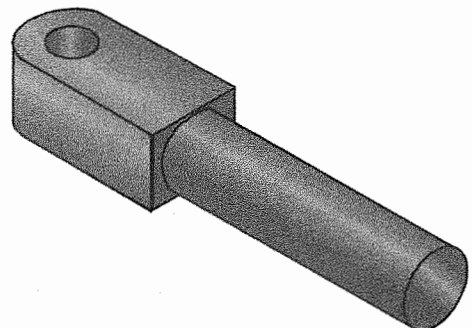


FIG. 26-57(ix)

(19) Save this File As **Module 26-45.DWG**

Output of Module 26-45 (fig. 26-57):

MODULE 26-45: USE OF 3D COMMANDS REGION, EXTRUDE, SUBTRACT AND UNION

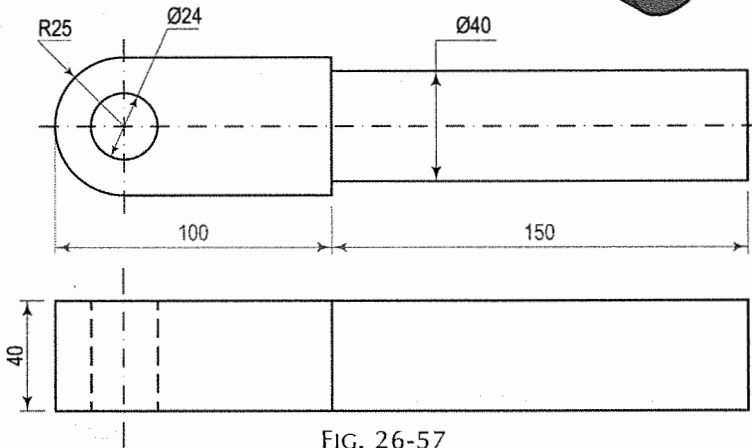
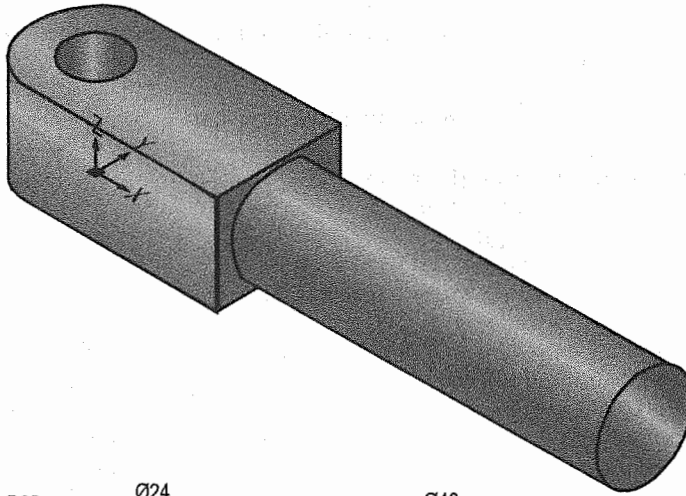


FIG. 26-57

Module 26-46. Draw a three dimensional diagram as shown in fig. 26-58. Use 3D Polar Array.

To draw a three dimensional diagram using the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-58(i) to fig. 26-58(vi).

- (1) Command: **LIMITS** (↵)
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
- (3) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.

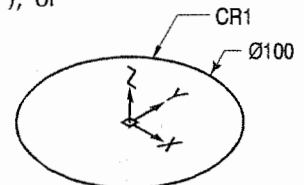


FIG. 26-58(i)

- (4) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL**
Regenerating model.

- (5) Command: **CIRCLE** (↵)
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: **0,0** (↵)
Specify radius of circle or [Diameter]: **D** (↵)
Specify diameter of circle: **100** (↵)

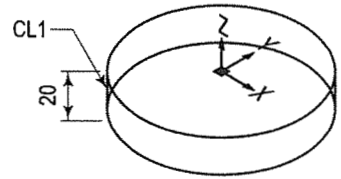


FIG. 26-58(ii)

- (6) Command: **EXTRUDE** (↵)
Current wire frame density: ISOLINES=4
Select objects to extrude: (Select Circle **CR1**)
Select objects to extrude: (↵)
Specify height of extrusion or [Direction/Path/Taper angle]<40.0000>: **-20** (↵)

- (7) Command: **CIRCLE** (↵)
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: **0,0** (↵)
Specify radius of circle or [Diameter] <50.0000>: **D** (↵)
Specify diameter of circle <100.0000>: **70** (↵)

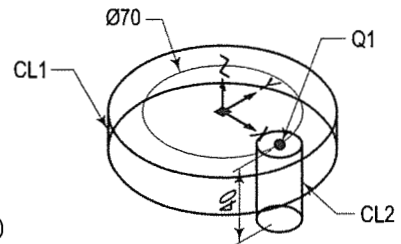


FIG. 26-58(iii)

- (8) Command: **CYLINDER** (↵)
Specify center point of base or [3P/2P/Ttr/Elliptical]: (Select QUADRANT **Q1** of **Ø70** circle)
Specify base radius or [Diameter]: **D** (↵)
Specify diameter: **20** (↵)
Specify height or [2Point/Axis endpoint] <-20.0000>: **-40** (↵)

- (9) Command: **3DARRAY** (↵)
Initializing... 3DARRAY loaded.
Select objects: (Select Cylinder **CL2**)
Select objects: (↵)
Enter the type of array [Rectangular/Polar] <R>: **P** (↵)

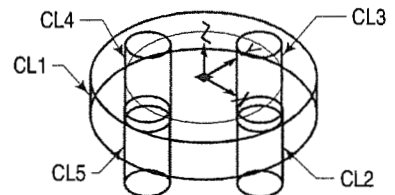


FIG. 26-58(iv)

- Enter the number of items in the array: **4** (↵)
Specify the angle to fill (+ = ccw, - = cw) <360>: (↵)
Rotate arrayed objects? [Yes/No] <Y>: (↵)
Specify center point of array: **0,0,0** (↵)
Specify second point on axis of rotation: **0,0,40** (↵)

- (10) Command: **SUBTRACT** (↵)
Select solids, surfaces and regions to subtract from ..
Select objects: (Select Cylinder **CL1**)
Select objects: (↵)
Select solids, surfaces and regions to subtract ..
Select objects: (Select Cylinder **CL2**)
Select objects: (Select Cylinder **CL3**)
Select objects: (Select Cylinder **CL4**)
Select objects: (Select Cylinder **CL5**)
Select objects: (↵)

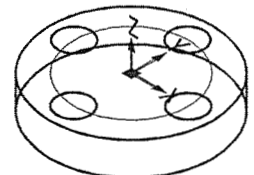


FIG. 26-58(v)

- (11) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or [All/Center/Dynamic/Extents/...] <real time>: **ALL** (↵)



FIG. 26-58(vi)

(12) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/.../Conceptual/...] <Conceptual>: **C** (↵)

(13) Save this File As **Module 26-46.DWG**

Output of Module 26-46 (fig. 26-58):

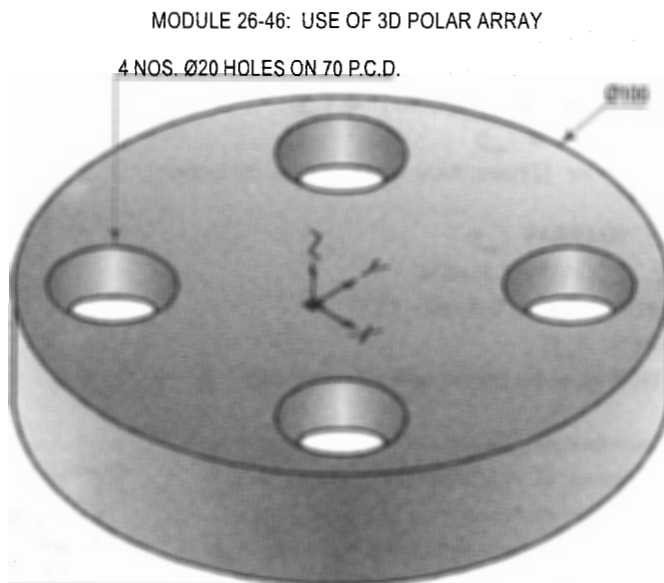


FIG. 26-58

Module 26-47. Draw a three dimensional diagram as shown in fig. 26-59. Use 3D Rectangular Array.

To draw a three dimensional diagram using the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-59(i) to fig. 26-59(vi).

(1) Command: **LIMITS** (↵)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)

Specify upper right corner <12.0000,9.0000>: **120,90** (↵)

(2) Command: **VPOINT** (↵)

Current view direction: VIEWDIR=0.0000,0.0000,1.0000

Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)

Regenerating model.

(3) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX or nXP), or [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)

(4) Command: **RECTANGLE** (↵)

Specify first corner point or [Chamfer/Elevation/Fillet/Thickness/Width]: **0,0** (↵)

Specify other corner point or [Dimensions]: **80,60** (↵)

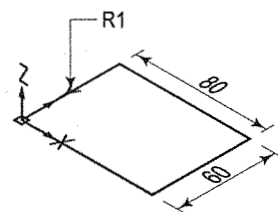


FIG. 26-59(i)

- (5) Command: **EXTRUDE** (↵)

Current wire frame density: ISOLINES=4

Select objects to extrude: (Select rectangle **R1**)

Select objects to extrude: (↵)

Specify height of extrusion or [Direction/Path/Taper angle]<40.0000>: **-10** (↵)

- (6) Command: **CYLINDER** (↵)

Specify center point of base or [3P/2P/Ttr/Elliptical]: **15,15** (↵)

Specify base radius or [Diameter]: **d** (↵)

Specify diameter: **10** (↵)

Specify height or [2Point/Axis endpoint]<-20.0000>: **-30** (↵)

- (7) Command: **3DARRAY** (↵)

Initializing... 3DARRAY loaded.

Select objects: (Select Cylinder **CL1**)

Select objects: (↵)

Enter the type of array [Rectangular/Polar] <R>: **R** (↵)

Enter the number of rows (—) <1>: **2** (↵)

Enter the number of columns (|||) <1>: **3** (↵)

Enter the number of levels (...) <1>: **1** (↵)

Specify the distance between rows (—): **30** (↵)

Specify the distance between columns (|||): **25** (↵)

- (8) Command: **SUBTRACT** (↵)

Select solids, surfaces and regions to subtract from ..

Select objects: (Select solid **S1**)

Select objects: (↵)

Select solids, surfaces and regions to subtract ..

Select objects: (Select Cylinder **CL1**)

Select objects: (Select Cylinder **CL2**)

Select objects: (Select Cylinder **CL3**)

Select objects: (Select Cylinder **CL4**)

Select objects: (Select Cylinder **CL5**)

Select objects: (Select Cylinder **CL6**)

Select objects: (↵)

- (9) Command: **ZOOM** (↵)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window]

<real time>: **ALL** (↵)

- (10) Command: **SHADEMODE** (↵)

Enter an option [2dwireframe/3dwireframe/.../Conceptual/...] <Conceptual>: **C** (↵)

- (11) Save This File As **Module 26-47.DWG**

Output of Module 26-47 (fig. 26-59):

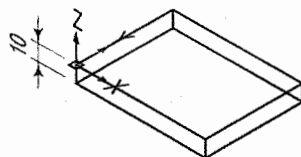


FIG. 26-59(ii)

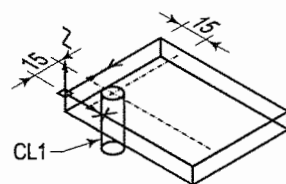


FIG. 26-59(iii)

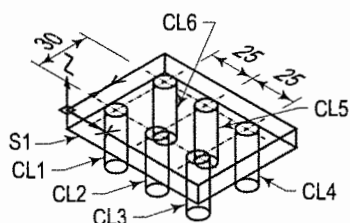


FIG. 26-59(iv)

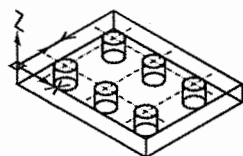


FIG. 26-59(v)

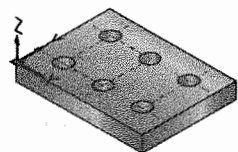


FIG. 26-59(vi)

MODULE 26-47: USE OF 3D RECTANGULAR ARRAY

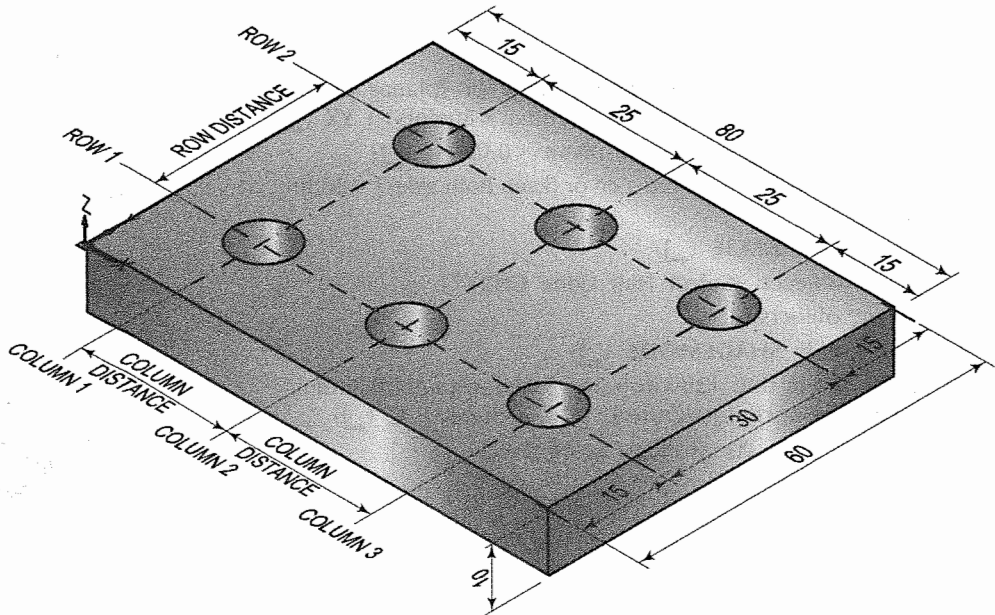


FIG. 26-59

Module 26-48. Draw a three dimensional diagram as shown in fig. 26-60. Use 3D Slice command.

To draw a three dimensional diagram using the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-60(i) to fig. 26-60(iii).

- (1) Command: **LIMITS** (↵)
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
Regenerating model.
- (3) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or [All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: **ALL** (↵)
Regenerating model.
- (4) Command: **CONE** (↵)
Specify center point of base or [3P/2P/Ttr/Elliptical]: **0,0** (↵)
Specify base radius or [Diameter] <5.0000>: **d** (↵)
Specify diameter <10.0000>: **80** (↵)
Specify height or [2Point/Axis endpoint/Top radius] <-30.0000>: **90** (↵)

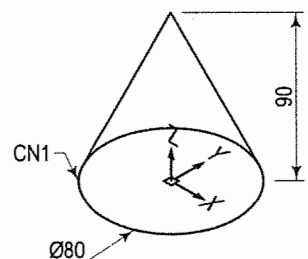


FIG. 26-60(i)

(5) Command: **SLICE** (↵)Select objects to slice: (Select Cone **CN1**)

Select objects to slice: (↵)

Specify start point of slicing plane or [planar Object/
Surface/Zaxis/View/XY/YZ/ZX/3points] <3points>: **YZ** (↵)

Specify a point on the YZ-plane <0,0,0>: (↵)

Specify a point on desired side or [keep Both sides] <Both>:
B (↵)(6) Command: **ERASE** (↵)Select objects: (Select Half Cone **CN2**)

Select objects: (↵)

(7) Command: **SHADEMODE** (↵)Enter an option [2dwireframe/3dwireframe/3dHidden/
Realistic/Conceptual/Other] <Conceptual>: **C** (↵)(8) Save This File As **Module 26-48.DWG**

Output of Module 26-48 (fig. 26-60):

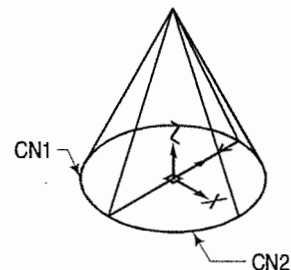


FIG. 26-60(ii)

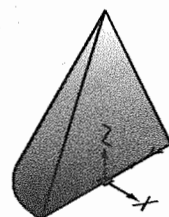


FIG. 26-60(iii)

MODULE 26-48: USE OF 3D SLICE COMMAND

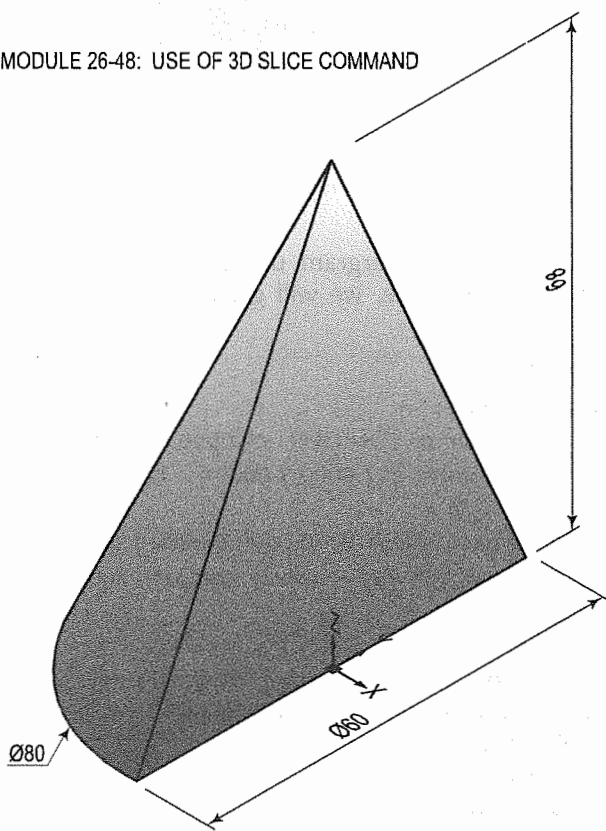


FIG. 26-60

Module 26-49. Draw a three dimensional diagram as shown in fig. 26-61. Use 3D commands such as *Mirror3D* and *Chamfer*.

To Draw a three dimensional diagram using steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-61(i) to fig. 26-61(iii).

- (1) Command: **LIMITS** (↵)
 Reset Model space limits:
 Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (↵)
 Specify upper right corner <12.0000,9.0000>: **120,90** (↵)
- (2) Command: **ZOOM** (↵)
 Specify corner of window, enter a scale factor (nX or nXP), or
 [All/Center/Dynamic/Extents/Previous/Scale/...] <real time>: **ALL** (↵)
 Regenerating model.
- (3) Command: **VPOINT** (↵)
 Current view direction: VIEWDIR=0.0000,0.0000,1.0000
 Specify a view point or [Rotate] <display compass and tripod>: **1,-1,1** (↵)
 Regenerating model.
- (4) Command: **BOX** (↵)
 Specify first corner or [Center]: **0,0,0** (↵)
 Specify other corner or [Cube/Length]: **L** (↵)
 Specify length: **50** (↵) (When 0° polar is ON)
 Specify width: **40** (↵) (When 90° polar is ON)
 Specify height or [2Point]: **30** (↵)
- (5) Command: **CHAMFER** (↵)
 (TRIM mode) Current chamfer Dist1 = 0.0000, Dist2 = 0.0000
 Select first line or [Undo/Polyline/Distance/Angle/Trim/mEthod/Multiple]: (Select Edge **E1**)
 Base surface selection...
 Enter surface selection option [Next/OK (current)] <OK>: (↵)
 Specify base surface chamfer distance: **20** (↵)
 Specify other surface chamfer distance <20.0000>: **40** (↵)
 Select an edge or [Loop]: (Select Edge **E1**)
 Select an edge or [Loop]: (↵)
- (6) Command: **MIRROR3D** (↵)
 Select objects: (Select solid)
 Select objects: (↵)
 Specify first point of mirror plane (3 points) or [Object/Last/Zaxis/View/XY/YZ/ZX/3points] <3points>: **YZ** (↵)
 Specify point on YZ plane <0,0,0>: (Select Osnap magnet **M1**)
 Delete source objects? [Yes/No] <N>: (↵)
- (7) Command: **SHADEMODE** (↵)
 Enter an option [2dwireframe/3dwireframe/3dHidden/Realistic/Conceptual/Other] <Conceptual>: **C** (↵)
- (8) Save This File as **Module 26-49.DWG**

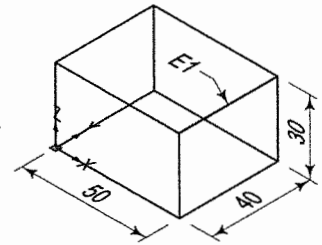


FIG. 26-61(i)

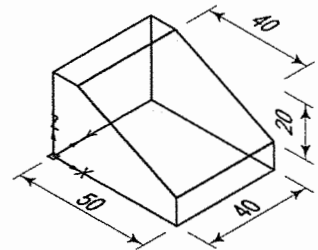


FIG. 26-61(ii)

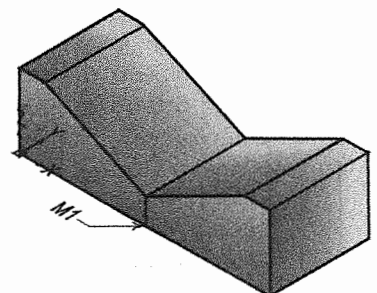


FIG. 26-61(iii)

Output of Module 26-49 (fig. 26-61):

MODULE 26-49: USE OF MIRROR3D AND CHAMFER COMMANDS

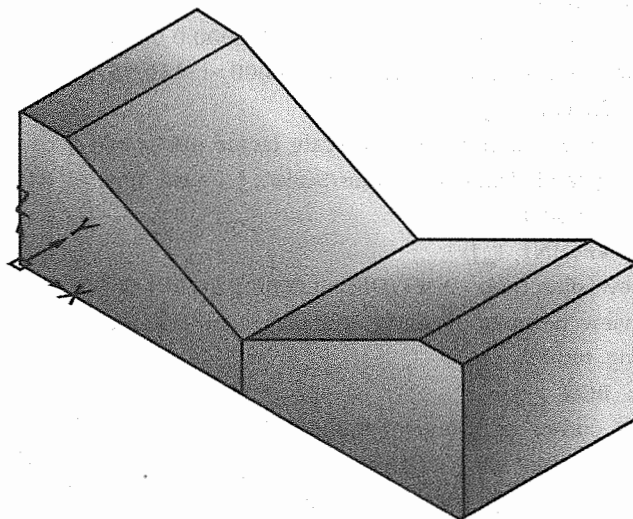


FIG. 26-61

26-11. PERSPECTIVE VIEW IN AUTOCAD

Refer module 26-50

- (i) Create 3D solid model or open file.
- (ii) Set the isometric view using VPOINT command.
- (iii) To set perspective view, use DVIEW command and select DISTANCE option.
- (iv) To turnoff perspective view. Use OFF option of DVIEW command.

Module 26-50. Open file Module 26-43.DWG and set perspective view.

- (1) Open File **Module 26-43.DWG**
- (2) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or
[All/Extents/Window/Previous/Object] <real time>: **ALL** (↵)
- (3) Command: **VPOINT** (↵)
Current view direction: VIEWDIR=0.0000,0.0000,1.0000
Specify a view point or [Rotate] <display compass and tripod>:
1,-1,1 (↵)
- (4) Command: **DVIEW** (↵)
Select objects or <use DVIEWBLOCK>: (Select 3DSolid S1)
Select objects or <use DVIEWBLOCK>: (↵)
Enter option
[CAmera/TARget/Distance/POints/PAn/Zoom/TWist/CLip/Hide/Off/
Undo]: **D** (↵)
Specify new camera-target distance <29.3885>: (↵)
Enter option
[CAmera/TARget/Distance/POints/PAn/Zoom/TWist/CLip/Hide/Off/
Undo]: (↵)

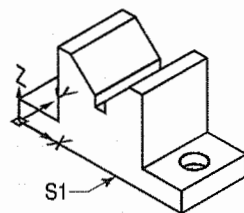


FIG. 26-62(i)

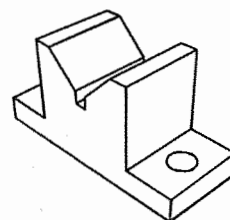


FIG. 26-62(ii)

- (5) Command: **ZOOM** (↵)
Specify corner of window, enter a scale factor (nX or nXP), or
[All/Extents/Window/Previous/Object] <real time>: **ALL** (↵)
- (6) Click Save from File menu

EXERCISES 26

1. Prepare a report on the computer hardware, peripherals and CAD software available in your laboratory.
2. What are the reasons for implementing a CAD system? What are the disadvantages?
3. List the main functions of a CAD system.
4. Explain the following 2D drawing entities with the help of sketch:
Line, Arc, Polyline, Ellipse, Donut, Polygon.
5. With suitable examples explain the following transformations:
 - (i) Moving an object
 - (ii) Scaling an object
 - (iii) Rotation of an object
 - (iv) Copying an object
 - (v) Mirroring an object.
6. What is the difference between COPY and MOVE?
7. Make sketches of 3D solid modelling primitives and name them.
8. Explain in brief:
 - (i) Wireframe modelling; (ii) Surface modelling; (iii) Solid modelling.
9. With the help of suitable examples, explain the boolean operations that can be performed on solid primitives.
10. Using the CAD software available in your laboratory, create the drawings of fig. 2-13 to fig. 2-26 shown on pages 31 and 32.
11. Study modules 26-1 to 26-14 carefully. Draw Hexagonal-headed bolt with hexagonal nut and washer as shown in fig. 26-63.

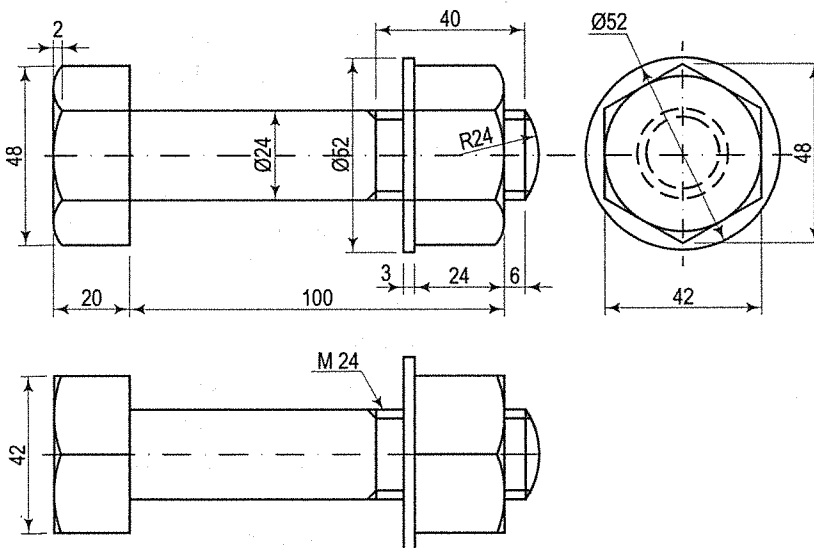


FIG. 26-63

12. Draw fig 26-64 using line, circle, rectangle, fillet, rectangular array, polar array, dimension, hatch, layer, hidden line, center line etc., commands.

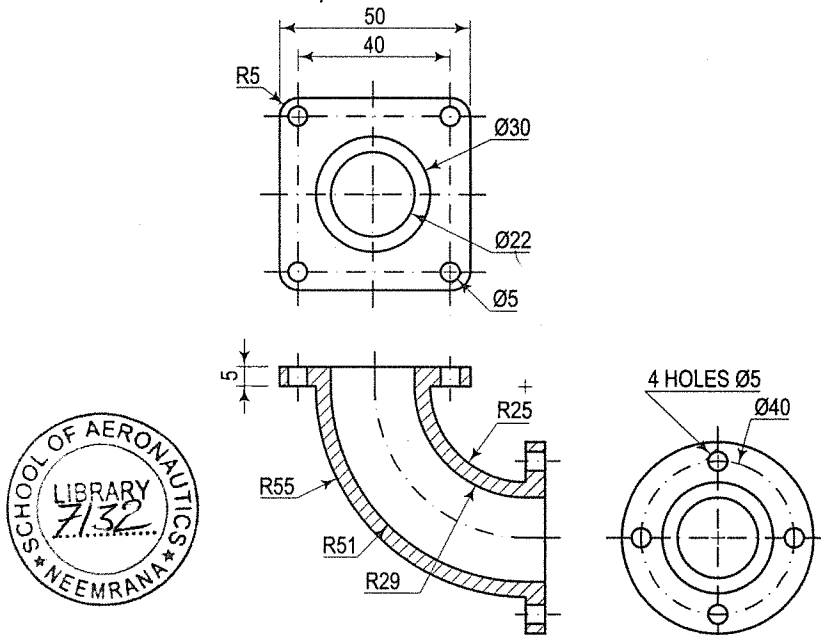


FIG. 26-64

13. Name four UCS options that only rotates the UCS about the origin but do not move the origin.
14. Write differences between Array and 3DArray.
15. How does the TABSURF command differ from the RULESURF command?
16. What are the use of Surftab1 and Surftab2 command?
17. Explain the use of Elev command.
18. How does shademode's flat shading option differs from 3D wireframe option.
19. List basic differences between surface models and solid models.
20. What are the requirements for profile objects of extruded and removed solids.
21. Write the name and sequence of command to create hole in solid.
22. What are the differences between the Union command and Slice command.
23. Write the uses of blocks and wblocks and write differences between them.
24. Mention the use of Insert command and explain it's scale option.
25. How layer helps in organizing the drawing objects.
26. What are the ways to assign properties to drawing objects?
27. How can we fillet edges of solids. Explain the sequence.
28. Give difference between Subtract and Intersect command.
29. List the hardware configuration required for AutoCAD 2010.
30. What is the use of VPOINT command.
31. Write the difference between Line and Polyline command.
32. Why do we use Polyline command to create solid filled arrow.

INDEX

Acme thread	578	3D Spherical co-ordinate method	667
Adhesive tapes	14	UCS command	666
Aligned system of dimensions	42	VPOINT command	666
Angle of vision	479	AutoCAD–Drafting aids	630
Angular perspective	487	Dimensioning (DIM)	631
Apparent section	314	Layer	631
Approximate method for surface		Limits	630
development of spheres	376	Object snap (OSNAP)	631
Archimedian Spirial	134	Zoom	631
Arrowhead	41	AutoCAD–Draw commands	628
AutoCAD	626	Arc	629
AutoCAD–3D Geometrical modelling	665	Circle	630
AutoCAD–3D Solid modelling	680	Donuts	630
3DARRAY command	684	Ellipse	630
BOOLEAN operations	683	Hatch pattern	630
CHAMFER on 3D solids	685	Line	628
EXTRUDE command	683	Pline	628
FILLET on 3D solids	684	Polygon	628
INTERSECT command	684	Rectangle	630
MIRROR3D comman	685	Text	630
REGION command	683	AutoCAD–Isometric drawing	659
REVOLVE command	683	Isometric circles	660
ROTATE3D command	685	Isometric dimension	661, 663
SLICE command	685	Isometric mode	660
SUBTRACT command	683	Isometric text	660
UNION command	683	Isoplane	660
3D Surface modelling	669	AutoCAD–Modify commands (Editing)	632
3D standard shaped surface ..	676	Array	632
3DMESH command	675	Break	632
EDGESURF command for		Chamfer	633
edge surfaces	674	Copy	632
ELEV command	669	Erase	632
PFACE command	671	Explode	633
REVSURF command for		Extend	633
revolve surface method ...	672	Fillet	633
RULESURF command for		Mirror	632
ruled surfaces	673	Move	632
SHADEMODE command	669	Offset	633
TABSURF command for		Oops	632
tabulated surfaces	673	Pedit	633
AutoCAD–3D Solid modelling	680	REDO	633
3D Wireframe modelling	666	Rotate	632
3D Absolute co-ordinate method	666	Scale	633
3D Cylindrical co-ordinate		Stretch	633
method	667	Trim	633
3D Rectangular co-ordinate		U (UNDO)	633
method	666	AutoCAD–Perspective view	700

AutoCAD-Symbol library	634	Computer	622
Block	634	Computer aided drafting (CAD) ...	621
Insert block	634	Cone	273
Wblock	634	Conic sections	101
AutoCAD-Three dimensional drawings	686	Conjugate axes	103
AutoCAD-Two dimensional drawing	634	Connection of plates at right angles	614
Absolute co-ordinate method .	634	Construction of a hyperbola	112
Relative co-ordinate method ..	635	Construction of a parabola	110
Relative polar co-ordinate method	637	Construction of ellipse	102
Auxiliary front view	241	Construction of tangents to conics	115
Auxiliary ground plane (A.G.P.)	478	Conventional representation	
Auxiliary inclined plane (A.I.P.)	241	of threads	579
Auxiliary top view	241	Conversion of pictorial views	
Auxiliary vertical plane (A.V.P.)	241	into orthographic views	517
B.I.S. code of practice for		Co-ordinate or offset method	427
first-angle projection	177	Copper bolt	603
Bar stay	604	Counter-sunk headed bolt	594
Bent bolt	603	Crest	573
Blue-print reading	511	C-spanner	604
Bolts	591	Cube	271
Border lines	17	Cup-headed bolt	593
Box method	426	Curved bolt	603
British Association (B.A.) thread ...	578	Curves	101
British standard fine (B.S.F.) thread	576	Cutting-plane method	382
British standard pipe (B.S.P.) thread	576	Cycloid	116
British standard whitworth (B.S.W.)		Cycloidal curves	116
thread	576	Cylinder	273
Butt joint	611	Cylindrical bolt	593
Buttress thread	578	Cylindrical nut	589
Cabinet projection	467	D evelopment of surfaces	351
Cad software	625	Development of transition pieces ..	372
Cam	143	Developments of lateral surfaces	
Cap nut	589	of right solids	352
Cap-screw	595	Diagonal scales	55
Capstan nut	589	Difference between oblique and	
Castle nut	600	isometric projection	466
Caulking	608	Digitizer	624
Cavalier projection	467	Dimension line	41
Central plane (C.P.)	478	Dimensioning	40
Central processing unit (CPU)	622	Dimensions of a riveted joint	609
Centre of gravity	539	Display	623
Centre of vision	478	Distance points	489
Check-nut	598	Dodecahedron	272
Cheese-headed bolt	593	Dome nut	589
Circle method	332	Dot matrix printer (DMP)	625
Circular vernier	64	Double-riveted butt joint	612
Clips	14	Drafting machine	15
Comparative scales	59	Drawing a helical curve	138

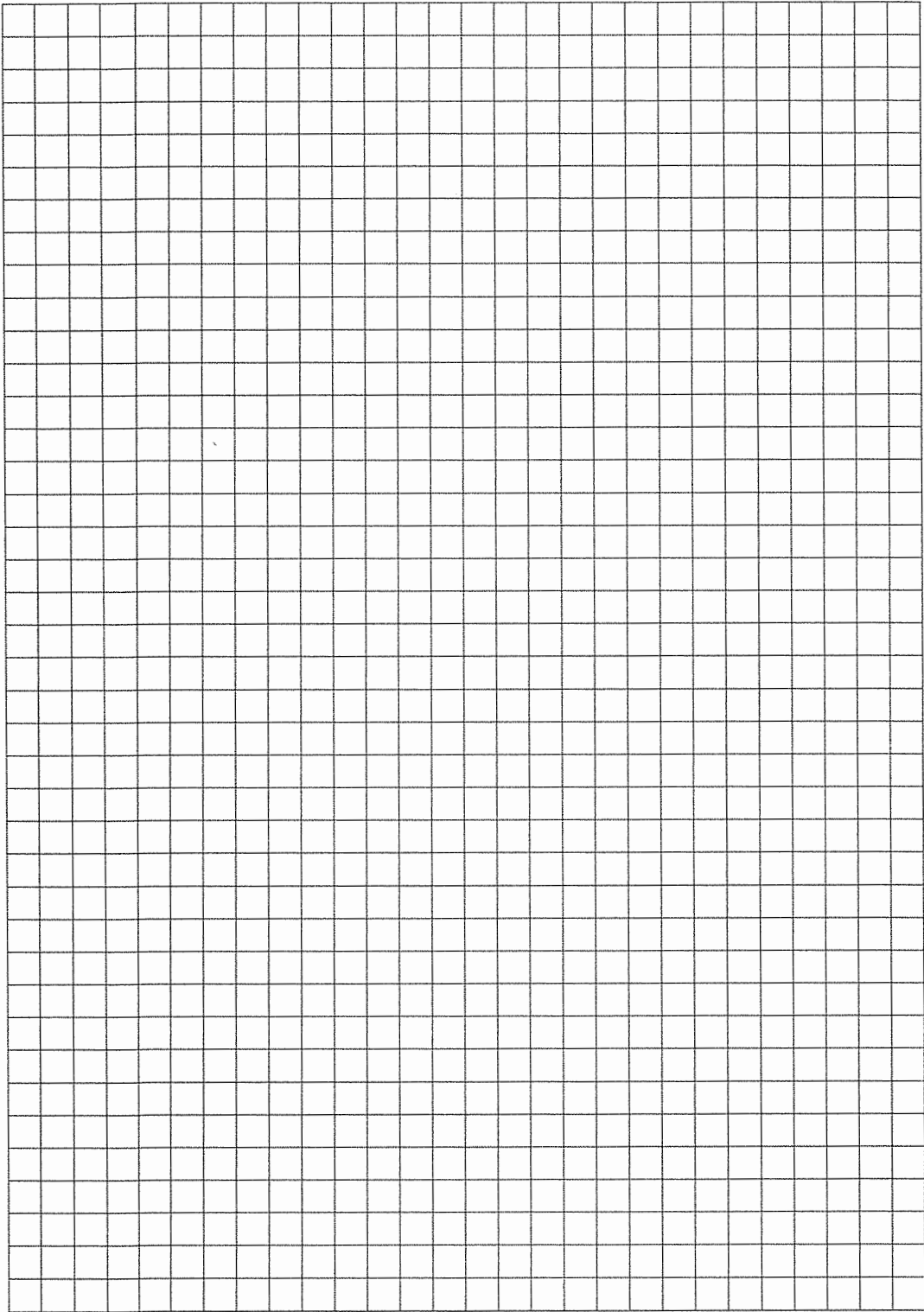
- Drawing board 2
 Drawing instrument box 7
 Drawing instruments 1
 Drawing papers 13
 Drawing pencils 13
 Drawing pins 14
 Duster 15
E
 Ellipse 102
 Engineer's scale 51
 Epicycloid 120
 Epitrochoid 122
 Equiangular spiral 135
 Eraser 14
 Evolutes 130
 Extension line 41
 External square thread 580
 External threads 579
 External V threads 580
 Eye-bolt 595, 602
F
 Failure of riveted joint 609
 First-angle projection 171
 Flanged nut 589
 Folding marks 24
 Follower 143
 Forge welding 615
 Forms of bolts 591
 Forms of rivet-heads 608
 Forms of screw threads 574
 Foundation bolts 602
 Four quadrants 171
 Four-bar mechanism 156
 Four-centre method 423
 Free-hand sketching 21
 French curves 12
 Frustum 273
 Fullering 608
 Functional modulus 556
 Fusion and pressure welding 616
 Fusion welding 615
G
 General rules for dimensioning 43
 Generator method 331
 Geometrical construction 69
 Bisecting a line 69
 Special methods of drawing
 regular polygons 84
 To bisect an angle 74
 To construct an ogee or
 reverse curve 79
 To construct equilateral triangles 80
 To construct regular polygons 82
 To construct squares 81
 To determine lengths of arcs 91
 To divide a circle 74
 To divide a line 73
 To draw circles and lines
 in contact 92
 To draw inscribed Circles 94
 To draw parallel lines 72
 To draw perpendiculars 70
 To draw regular figures using
 T-square and Set-squares 88
 To draw regular polygons
 inscribed in circles 86
 To draw tangents 89
 To find the centre of an arc 76
 To trisect an angle 75
 Gothic letters 40
 Graphic output devices 625
 Graphical scale 52
 Grid reference system 22
 Ground line (G.L.) 478
 Ground plane (G.P.) 478
 Gusset stay 614
H
 Hardware requirement for Autocad 627
 Headless tapered bolt 594
 Helical springs 139
 Helix 138
 Helix upon a cone 142
 Hexagonal nut 586
 Hexagonal-headed bolt 591
 Hexahedron 271
 Hook bolt 594
 Hoop bolt 602
 Horizon line (H.L.) 478
 Horizontal plane (H.P.) 170, 478
 Hyperbola 112
 Hypocycloid 121
 Hypotrochoid 123
I
 Icosahedron 272
 Identification of planes 515
 Inferior trochoid 118
 Inked drawings 33
 Inking pen 10

Ink-jet plotters	625	Logarithmic spiral	135
Ink-jet printers	625	Longitudinal stay	604
Input devices	624	Lune method for surface development of spheres	376
Internal threads	579	Lune of a sphere	372
Internal V threads	580	M asuring line	491
Intersection of cone and cone	411	Methods of drawing mon-isometric lines	426
Intersection of cone and cylinder ..	401	Method of points	422
Intersection of cone and prism	409	Method of projection	169
Intersection of cylinder and cylinder	390	Methods of development of surfaces	352
Intersection of sphere and cylinder	412	Methods of drawing perspective view	480
Intersection of sphere and prism ..	412	Metric thread	576
Intersection of surfaces	381	Missing lines	512
Involute	124	Missing views	513
Isometric axes, lines and planes ...	418	Moments of inertia of areas	547
Isometric drawing	420	Mouse	624
Isometric drawing of cones	429	Multiple-start threads	581
Isometric drawing of cylinders	429	N omographs of three variables ...	557
Isometric drawing of planes	421	Nomography	555
Isometric drawing of prisms	425	Normal and tangent to a cycloid curve	117
Isometric drawing of pyramids	425	Normal and tangent to a epicycloid and hypocycloid	121
Isometric drawing of sphere	430	Normal and tangent to an archemedian curve	134
Isometric graph	420	Normal and tangent to an involute	125
Isometric projection	417	Nuts	585
Isometric scale	418	O blique drawing of circle	470
Isometric view	420	Oblique drawing of cylinder	471
K eyboard	624	Oblique drawing of prism	472
Knuckle thread	578	Oblique drawing of pyramid	470
L ap joint	610	Oblique perspective	488
Large size divider	9	Oblique planes	257
Large-size compass	8	Oblique projection	465
Laser printer	625	Octahedron	272
Layout of nomographs	563	Offset slider crank mechanism	154
Lead	574	One point perspective	486
Leader	41	Orthographic projection	169, 517
Least count of a vernier	62	Orthographic reading	511
Left-hand threads	582	P arabola	109
Lengthening bar	8	Parallel perspective	486
Lettering	37	Parallel scale nomographs	559
Lewis bolt	602	Pen plotters	625
Lifting eye-bolt	595	Pencil drawings	33
Line method	382		
Line of heights	491		
Lines	33		
Lines of intersection	381		
Loci of points	151		
Locking arrangements for nuts	598		
Locking-plate	601		
Lock-nut	598		

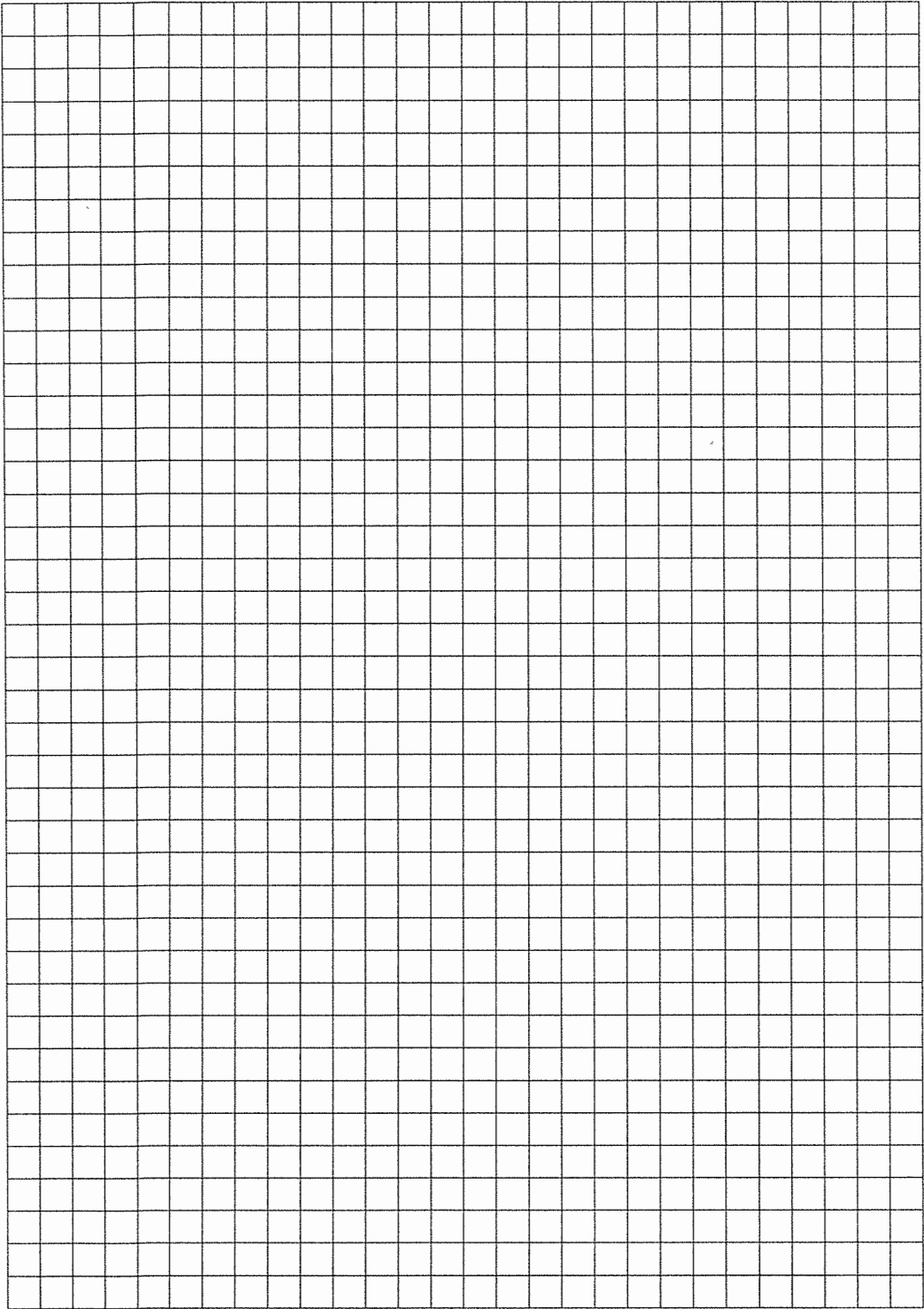
- Penn, ring and grooved nut 600
 Perpendicular axis (P.A.) 478
 Perpendicular planes 255
 Perspective elements 477
 Perspective of circles 492
 Perspective of hut 504
 Perspective of lamp-post 507
 Perspective of sign-post 505
 Perspective of solids 492
 Perspective of steps 506
 Perspective of steps 508
 Perspective of window-frame 502
 Perspective projection 477
 Picture plane (P.P.) 478, 480
 Pin spanner 604
 Pitch 574
 Placing of dimensions 42
 Plain scales 52
 Planes of projection 171
 Polyhedra 271
 Positions of traces of a line 212
 Practical hints on dimensioning 43
 Pressure welding 615
 Principle of oblique projection 465
 Principle of perspective projection 477
 Principle of projection 169
 Prism 272
 Procedure for preparing
 a scale-drawing 522
 Processor 623
 Projections of oblique planes 261
 Projections of planes 255
 Projections of points 189
 Projections of solids 271
 Projections of solids
 in simple positions 274
 Projections of spheres 300
 Projections of straight lines 195
 Projections on auxiliary planes 241
 Proportions of rivet-heads 608
 Protractor 12
 Pyramid 272
Rag bolt 602
 Reading of orthographic views 511
 Receding angles 467
 Receding lines 467
 Rectangular hyperbola 112
 Reference line 173
 Representation of welded joints 617
 Representative fraction (R.F.) 52
 Right circular cone 273
 Right circular cylinder 273
 Right-hand threads 582
 Ring nut 590
 Riveted joints 607
 Riveting 607
 Rivets 607
 Rolled-steel sections 613
 Roll-n-draw 16
 Root 573
 Round-headed bolt 593
 Rules for object position
 in oblique projection 468
Sand-paper block 15
 Sawn nut 600
 Scale modulus 556
 Scale of chords 65
 Scales 10
 Scales 51
 Scales and scale drawing 25
 Scales on Drawings 52
 Screen layout of AutoCAD 2010 ... 627
 Screw threads 141, 573
 Screwed fastenings 585
 Section planes 313
 Sections of cones 329
 Sections of cylinders 326
 Sections of prisms 314
 Sections of pyramids 320
 Sections of solids 313
 Sections of spheres 338
 Sellers thread 577
 Set-screws 597
 Set-squares 4
 Sheet layout 21
 Simmond's lock-nut 600
 Simple mechanisms 153
 Simple slider crank mechanism ... 154
 Single-stroke letters 37
 Sketching 26
 Slider crank mechanism 153
 Slope of a thread 574
 Slotted nut 599
 Small bow compass 8
 Small bow divider 9

Small bow ink-pen	9	Types of auxiliary planes	241
Solids of revolution	273	Types of lines	35
Spanner	604	Types of Machine drawings	22
Sphere	273	Types of nomographs	555
Spheres in contact with each other	302	Types of nuts	585
Spirals	133	Types of oblique projection	467
Split-pin	599	Types of perspective	486
Spring-washer	601	Types of planes	255
Square nut	588	Types of riveted joints	610
Square thread	578	Types of scales	52
Square-headed bolt	592, 604	Types of solids	271
Station point (S.P.)	478, 479	Types of welded joints	616
Stays	604	Types of welding process	615
Steps for drawing		Types of welds	616
the oblique projection	469	Unequal spheres	303
Stop-plate	601	Unidirectional system of dimensions	42
Stud	595	Unified thread	575
Stud-bolt	595	Unit of dimensioning	43
Superior trochoid	119	V threads	575
Symbols for nuts and bolts	605	Vanishing points	485
Tablet	624	Vanishing-point method	485
Tangents and normals to conics	115	Vernier scales	61
Tap-bolt	595	Vetical plane (V.P.)	170
Tetrahedron	271	Views of an object	175
T-headed bolt	594	Visual-ray method	481
Third-angle projection	172	Washers	590
Three point perspective	488	Welded joints	615
Title block	22	Welding	615
Traces of a line	209	Whitworth thread	576
Traces of planes	257	Wiles nut	600
Transition pieces	372	Wing nut	590
Triangular threads	575	Zone method for surface	
Trochoid	118	development of spheres	376
True shape of a section	314	Zone of a sphere	372
Truncated solid	273	Z-type nomographs	568
T-square	2		
Two point perspective	487		

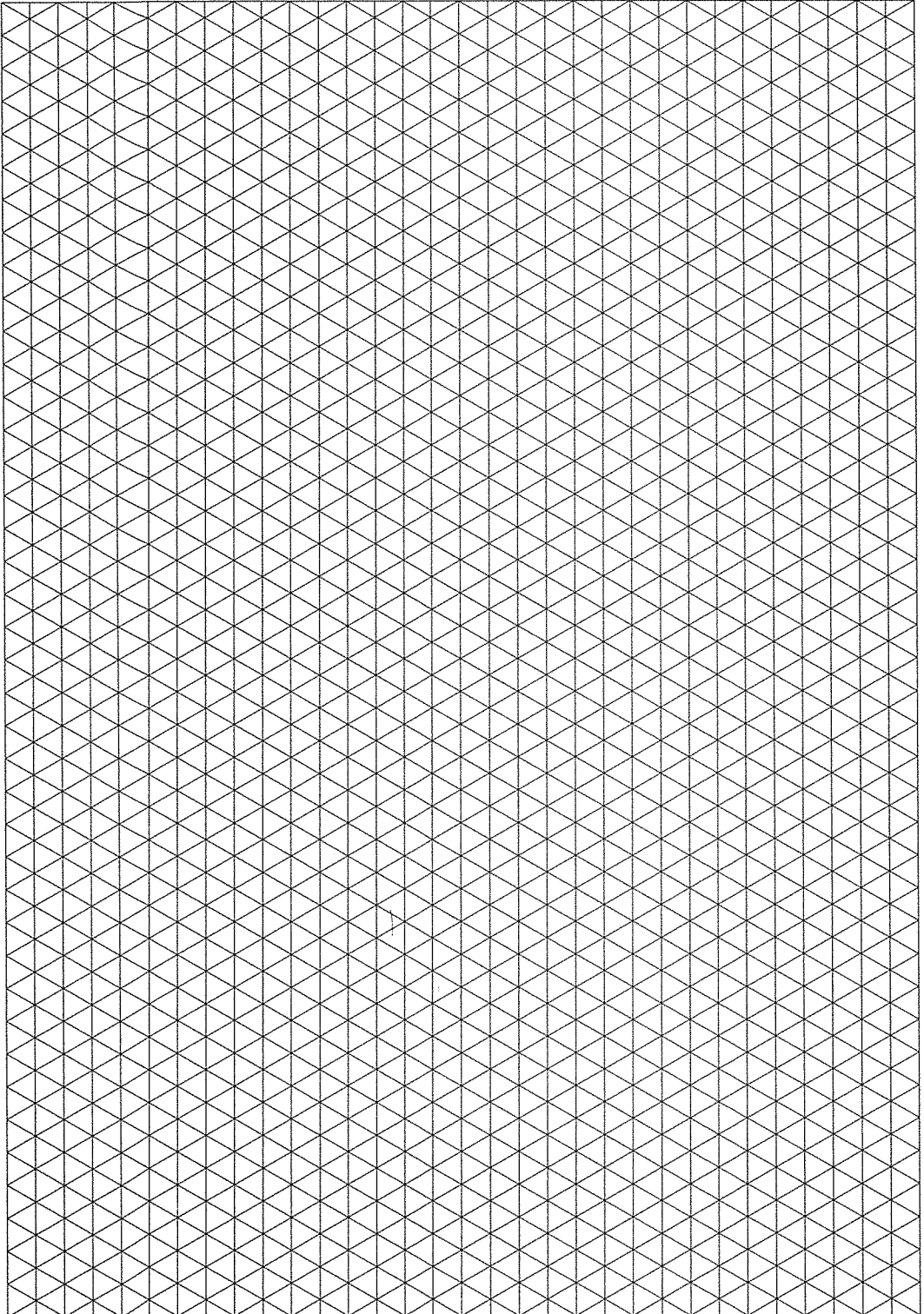




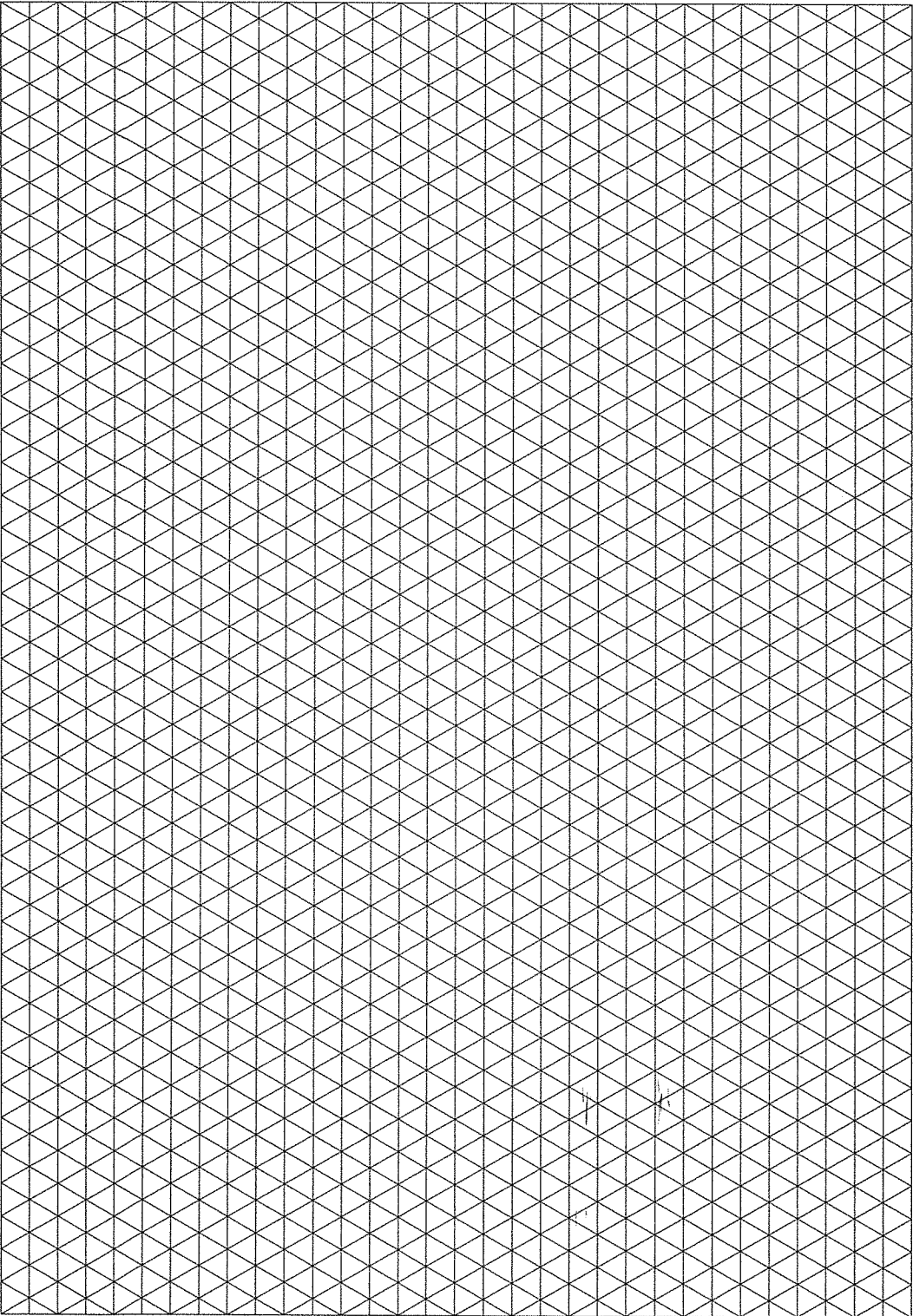
PARALLEL GRID



PARALLEL GRID



ISOMETRIC GRID



ISOMETRIC GRID

- 1 : INTRODUCTION
- 2 : VECTORS
- 3 : COPLANAR CONCURRENT FORCES
- 4 : MOMENTS
- 5 : PARALLEL FORCES AND COUPLES
- 6 : RESULTANT OF COPLANAR FORCE SYSTEMS
- 7 : SPATIAL FORCES
- 8 : EQUILIBRIUM OF COPLANAR FORCE SYSTEMS
- 9 : REACTIONS
- 10 : EQUILIBRIUM OF SPATIAL FORCE SYSTEMS
- 11 : PROPERTIES OF LINES, AREAS AND SOLIDS
- 12 : TRUSSES
- 13 : FRAMES AND MECHANISMS
- 14 : GRAPHIC STATICS
- 15 : MOMENTS OF INERTIA
- 16 : FRICTION
- 17 : RECTILINEAR MOTION OF A PARTICLE
- 18 : MOTION UNDER VARIABLE ACCELERATION
- 19 : RELATIVE MOTION OF A PARTICLE

- 20 : CURVILINEAR MOTION OF A PARTICLE
- 21 : KINEMATICS OF RIGID BODIES
- 22 : KINEMATICS OF PARTICLES LAWS OF MOTION
- 23 : KINETICS OF PARTICLES : WORK AND ENERGY
- 24 : KINETICS OF PARTICLES : IMPULSE AND MOMENTUM
- 25 : CENTRAL FORCE MOTION
- 26 : COLLISIONS
- 27 : KINETICS OF SYSTEMS OF PARTICLES
- 28 : KINETICS OF RIGID BODIES
- 29 : MOTION OF VEHICLES
- 30 : BALANCING OF ROTATING MASSES
- 31 : VIRTUAL WORK
- 32 : THE CATENARY
- 33 : BELT AND ROPE DRIVE
- 34 : TOOTHED GEARING
- 35 : LIFTING MACHINES
- 36 : MECHANICAL VIBRATIONS
- 37 : HYDROSTATICS
- 38 : IMPACT OF JETS
- APPENDIX 1 SELECTED MATHEMATICS
- APPENDIX 2 OBJECTIVE QUESTIONS

APPLIED MECHANICS

INCLUDING VECTOR APPROACH

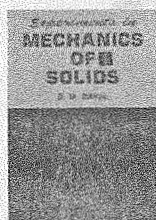
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EXPERIMENTS IN MECHANICS OF SOLIDS

B. M. RAVAL

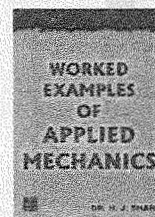


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- 2 : LABORATORY MACHINES
- 3 : TENSION TEST
- 4 : COMPRESSION TEST
- 5 : TORSION TEST
- 6 : TRANSVERSE TEST
- 7 : IMPACT TEST
- 8 : HARDNESS TEST
- 9 : FATIGUE TEST
- 10 : BEND TEST
- 11 : EXPERIMENTS USING E. S. A. TECHNIQUE
- 12 : MECHANICAL EXTENSOMETER
- 13 : PRESENTATION OF REPORT
- 14 : PLANNING A STRENGTH OF MATERIALS LABORATORY
- 15 : WORKING MODELS IN EXPERIMENTAL MECHANICS OF SOLIDS
- 16 : OBJECTIVE EXPERIMENTS

WORKED EXAMPLES OF APPLIED MECHANICS

DR. H. J. SHAH

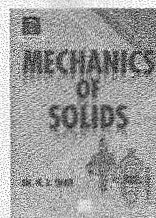


651 SOLUTIONS
OF THE EXAMPLES
GIVEN AT THE END OF
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MECHANICS OF SOLIDS

DR. H. J. SHAH

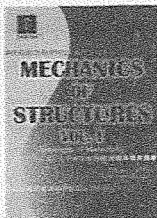


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- 6 : RESULTANT OF COPLANAR FORCE SYSTEMS
- 7 : EQUILIBRIUM OF COPLANAR FORCE SYSTEMS
- 8 : REACTIONS
- 9 : TRUSSES
- 10 : GRAPHIC STATICS
- 11 : PROPERTIES OF LINES, AREAS AND SOLIDS
- 12 : AREA MOMENTS OF INERTIA
- 13 : FRICTION
- 14 : BELT AND ROPE DRIVES
- 15 : LIFTING MACHINES
- 16 : SIMPLE STRESS
- 17 : SIMPLE STRAIN
- 18 : STATICALLY INDETERMINATE MEMBERS
- 19 : THERMAL STRESSES AND STRAINS
- 20 : STRESSES ON INCLINED PLANES
- 21 : COMBINED STRESSES : PRINCIPAL STRESSES
- 22 : MOHR'S CIRCLE METHOD
- 23 : TESTING OF MATERIALS - I
- 24 : SHEAR FORCES AND BENDING MOMENTS - I
- 25 : SHEAR FORCES AND BENDING MOMENTS - II
- 26 : BENDING STRESSES IN BEAMS
- 27 : SHEAR STRESSES IN BEAMS
- 28 : TESTING OF MATERIALS - II

MECHANICS OF STRUCTURES VOL. I**STRENGTH OF MATERIALS**

DR. H. J. SHAH
S. B. JUNNARKAR



ISBN : 978-93-80358-05-5
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| 2 : SIMPLE STRAIN | 15 : SHEAR STRESSES IN BEAMS |
| 3 : STATICALLY INDETERMINATE MEMBERS | 16 : DEFLECTIONS I |
| 4 : THERMAL STRESSES AND STRAINS | 17 : DEFLECTIONS II |
| 5 : STRESSES ON INCLINED PLANES | 18 : FIXED BEAMS |
| 6 : COMBINED STRESSES | 19 : CONTINUOUS BEAMS |
| 7 : MOHR'S CIRCLE METHOD | 20 : COMPOSITE BEAMS |
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| 9 : TESTING OF MATERIALS – I | 22 : DIRECT AND BENDING STRESSES |
| 10 : SHEAR FORCES AND BENDING MOMENTS – I | 23 : SHAFTS AND SPRINGS IN TORSION |
| 11 : SHEAR FORCES AND BENDING MOMENTS – II | 24 : TESTING OF MATERIALS – II 780-798 |
| 12 : CENTROIDS OF LINES AND AREAS | 25 : COLUMNS AND STRUTS OF UNIFORM SECTION |
| 13 : AREA MOMENTS OF INERTIA | 26 : RADIAL PRESSURE - CYLINDRICAL AND SPHERICAL SHELLS |
| | 27 : RIVETED AND BOLTED JOINTS |
| | 28 : WELDED JOINTS |
| | 29 : SHEAR CENTRE |
| | 30 : UNSYMMETRICAL BENDING |
| | 31 : BENDING STRESSES IN CURVED BARS |

MECHANICS OF STRUCTURES VOL. II**THEORY AND ANALYSIS OF STRUCTURES**

DR. H. J. SHAH
S. B. JUNNARKAR



ISBN : 978-93-80358-16-1
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PAGES : 970 + 60

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| 1 : INFLUENCE LINES FOR BEAMS | 12 : CONSISTENT DEFORMATION II |
| 2 : ROLLING LOADS | 13 : LEAST WORK AND THEOREM OF THREE MOMENT |
| 3 : INFLUENCE LINES FOR TRUSSES | 14 : INDETERMINATE ARCHES |
| 4 : MASONRY DAMS AND RETAINING WALLS | 15 : SLOPE DEFLECTION METHOD |
| 5 : CABLES AND SUSPENSION BRIDGES | 16 : MOMENT DISTRIBUTION METHOD |
| 6 : THREE HINGED ARCHES | 17 : COLUMN ANALOGY METHOD |
| 7 : FORCES IN FRAMED STRUCTURES | 18 : THE ELASTIC CENTRE |
| 8 : INDETERMINATE STRUCTURES | 19 : THE RECIPROCAL THEOREM |
| 9 : DEFLECTIONS I | 20 : PLASTIC THEORY |
| 10 : DEFLECTIONS II | 21 : INTRODUCTION TO MATRIX METHODS OF ANALYSIS |
| 11 : CONSISTENT DEFORMATION I | |

KEY TO MECHANICS OF STRUCTURES VOL. I**WORKED EXAMPLES IN STRENGTH OF MATERIALS AND THEORY OF STRUCTURES**

S. B. JUNNARKAR



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| 1 : SI UNITS | 11 : RADIAL PRESSURE – CYLINDRICAL AND SPHERICAL SHELLS |
| 2 : SIMPLE STRESSES AND STRAINS | 12 : RIVETED JOINTS |
| 3 : PRINCIPAL PLANES AND PRINCIPAL STRESSES | 13 : SHAFTS AND SPRINGS IN TORSION |
| 4 : IMPACT OR SHOCK LOADING : STRAIN ENERGY | 14 : ELEMENTS OF REINFORCED CONCRETE |
| 5 : BEAMS AND BENDING – I | 15 : STRESSES IN FRAMES : STATIC LOADING |
| 6 : BEAMS AND BENDING – II | 16 : METHOD OF TENSION COEFFICIENTS |
| 7 : BEAMS AND BENDING – III | 17 : UNSYMMETRICAL BENDING |
| 8 : BEAMS AND BENDING – IV | 18 : CONJUGATE BEAM METHOD |
| 9 : DIRECT AND BENDING STRESSES | 19 : WELDED JOINTS |
| 10 : COLUMNS AND STRUTS OF UNIFORM SECTION | 20 : SHEAR CENTRE |
| | 21 : BENDING STRESSES IN CURVED BARS |

REINFORCED CONCRETE VOL. I

ELEMENTRY REINFORCED CONCRETE

DR. H. J. SHAH



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- 2 : PROPERTIES OF MATERIALS
- 3 : STRUCTURAL CONCRETE
- 4 : DESIGN FOR FLEXURE : FUNDAMENTALS
- 5 : DESIGN FOR FLEXURE :
WORKING STRESS METHOD
- 6 : LIMIT STATE METHOD
- 7 : SHEAR AND DEVELOPMENT LENGTH
- 8 : DEFLECTION AND CRACKING
- 9 : SIMPLY SUPPORTED AND CANTILEVER
BEAMS
- 10 : SIMPLE SUPPORTED AND CANTILEVER
SLABS
- 11 : CONTINUOUS BEAMS AND SLABS
- 12 : TORSION
- 13 : STAIRS

- 14 : LOAD CALCULATIONS - 1
- 15 : SIMPLE DESIGNS
- 16 : FRAMED BEAMS
- 17 : COLUMNS
- 18 : DESIGN OF FOUNDATIONS :
FUNDAMENTALS
- 19 : ISOLATED FOOTINGS
- 20 : COMBINED FOOTINGS
- 21 : PILE FOUNDATIONS
- 22 : RETAINING WALLS
- 23 : FORM WORK
- 24 : DETAILING OF REINFORCEMENT

APPENDICES

APPENDIX A SHORT QUESTIONS WITH ANSWERS

APPENDIX B USEFUL TABLES

REINFORCED CONCRETE VOL. II

ADVANCED REINFORCED CONCRETE

DR. H. J. SHAH



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- 1 : CIRCULAR SLABS
- 2 : RIBBED SLABS AND WAFFLE SLABS
- 3 : FLAT SLABS
- 4 : DOMES
- 5 : DEEP BEAMS AND CORBELS
- 6 : BEAMS CURVED IN PLAN
- 7 : GRID OR COFFERED FLOORS
- 8 : CIRCULAR RAFT FOUNDATIONS
- 9 : MULTI-STORYED BUILDINGS :
FUNDAMENTALS
- 10 : ANALYSIS AND DESIGN FOR
GRAVITY LOADS : AN OVERVIEW
- 11 : LATERAL LOADS

- 12 : ANALYSIS FOR LATERAL LOADS
- 13 : DUCTILITY CONSIDERATIONS
- 14 : UNBRACED BUILDING : DESIGN
EXAMPLE
- 15 : SHEAR WALLS
- 16 : INTRODUCTION TO TALL BUILDINGS
- 17 : WATER TANKS : FUNDAMENTALS
- 18 : CIRCULAR TANKS
- 19 : RECTANGULAR TANKS
- 20 : ELEVATED TANKS
- 21 : INTZ TANKS
- 22 : ELEMENTS OF PRESTRESSED CONCRETE

UNDERSTANDING CONCEPT OF STRUCTURAL ANALYSIS AND DESIGN

JANAK PARIKH

PART I ANALYSIS

- 1 : STRUCTURES, FORCES, MOMENT -
A COUPLE OF FORCES
- 2 : EQUILIBRIUM
- 3 : RESISTANCE TO FORCE BY
MEMBER - STRESS, STRAIN AND FAILURE
- 4 : STRUCTURAL MEMBERS AND THEIR
ANALYSIS
- 5 : APPROXIMATION OF STRUCTURES AND
APPROXIMATE METHODS OF ANALYSIS

PART II DESIGN

- 6 : DESIGN CRITERIA
- 7 : FORM, STIFFNESS,
STRENGTH AND DEFORMATION

- 8 : STEEL STRUCTURES
- 9 : RCC STRUCTURES
- 10 : STRUCTURAL FRAMING

**PART III CONCEPTUAL ANALYSIS AND DESIGN OF
FOUR STRUCTURES**

- 11 : A LOAD BEARING SINGLE STOREY
STRUCTURE
- 12 : RCC BUILDING FRAME
- 13 : BRACING FOR WIND FORCES IN A STEEL
STRUCTURE
- 14 : PRELIMINARY COLUMN DESIGN OF 30
STOREYED STEEL TOWER

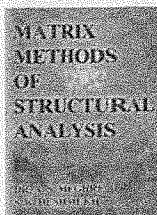


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MATRIX METHODS OF STRUCTURAL ANALYSIS

THEORY, EXAMPLES AND PROGRAMS

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S. K. DESHMUKH



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- 3 : STIFFNESS METHOD
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- 5 : SPACE TRUSS
- 6 : PLANE FRAME
- 7 : GRID
- 8 : SPACE FRAME
- 9 : ADDITIONAL TOPICS – I
- 10 : ADDITIONAL TOPICS – II
- 11 : ADDITIONAL TOPICS – III
- 12 : NON-LINEAR ANALYSIS

APPENDICES

APPENDIX A To APPENDIX C

LOGARITHMS, STEAM AND OTHER TABLES

ENTIRELY IN SI UNIT REFRIGERANTS :
PROPERTIES, TABLES AND CHARTS

DR. N. C. PANDYA
DR. C. S. SHAH



LOGARITHMS, STEAM AND
OTHER TABLES
[INCLUDING MATHEMATICAL TABLES,
FORMULAS, STEEL TABLES,
REFRIGERANT TABLES AND LIST OF
INDIAN STANDARDS]

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HEAT ENGINES VOL. I

THERMODYNAMICS AND THERMAL ENGINEERING

DR. N. C. PANDYA, DR. C. S. SHAH, S. S. KHANDARE



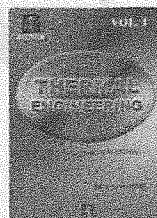
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|---|--|
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BOILER |
| 2 : PROPERTIES OF GASES | 9 : DRAUGHT |
| 3 : AIR STANDARD CYCLES | 10 : FUELS AND COMBUSTION |
| 4 : FLOW PROCESSES AND
THERMODYNAMIC RELATIONS | 11 : STEAM ENGINE |
| 5 : PROPERTIES OF STEAM AND STEAM
CYCLE | 12 : TESTING AND PERFORMANCE OF
STEAM ENGINES |
| 6 : STEAM BOILERS | 13 : STEAM NOZZLE |
| 7 : BOILER MOUNTINGS AND
ACCESSORIES | 14 : STEAM TURBINE |
| | 15 : CONDENSER |

THERMAL ENGINEERING VOL. I

THERMODYNAMICS AND HEAT ENGINES

DR. S. S. KHANDARE



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|--|---|
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| 2 : THERMODYNAMIC PROCESS AND
CYCLES | 12 : CONDENSER |
| 3 : PROPERTIES OF STEAM AND STEAM
CYCLE | 13 : INTERNAL COMBUSTION ENGINE |
| 4 : STEAM BOILERS | 14 : TESTING OF INTERNAL COMBUSTION
ENGINE |
| 5 : BOILER MOUNTINGS AND ACCESSORIES | 15 : GAS TURBINE |
| 6 : TESTING AND PERFORMANCE OF BOILER | 16 : AIR POLLUTION AND CONTROL |
| 7 : DRAUGHT | APPENDIX A |
| 8 : FUELS AND COMBUSTION | SHORT QUESTIONS FOR VIVA-OCÉ |
| 9 : STEAM ENGINE AND TESTING | APPENDIX B |
| 10 : STEAM NOZZLE | STEAM TABLES |

LOGARITHMS AND OTHER TABLES

ABRIDGED FROM LOGARITHMS, STEAM AND OTHER TABLES

PREFIXES USED WITH UNITS

LIST OF SYMBOLS AND GREEK ALPHABETS

LOGARITHMS

ANTILOGARITHMS

NATURAL SINES

NATURAL COSINES

NATURAL TANGENTS

LOGARITHMIC SINES

LOGARITHMIC COSINES

LOGARITHMIC TANGENTS

NATURAL LOGARITHMS

BESSEL'S FUNCTIONS (CYLINDRICAL FUNCTIONS)

IMPORTANT FORMULA FOR BESSEL'S FUNCTIONS

VALUES OF GAMMA FUNCTIONS AND

LEGENDRE'S POLYNOMIALS

HYPERBOLIC FUNCTIONS (TABLES)

DEGREES TO RADIANS

MINUTES TO RADIANS

SECONDS TO RADIANS

DECIMAL EQUIVALENTS OF COMMON

FRACTIONS

BINOMIAL COEFFICIENTS

AREA OF THE CIRCLE WHEN DIAMETER

IS GIVEN

CIRCUMFERENCE OF THE CIRCLE WHEN

DIAMETER IS GIVEN

DR. N. C. PANDYA

DR. C. S. SHAH



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STEEL TABLES

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SYMBOLS : GREEK ALPHABETS

PREFIXES USED WITH UNITS

TABLE - I : DIMENSIONS AND PROPERTIES OF ROLLED STEEL BEAMS

TABLE - II : PLATES
STANDARD NOMINAL THICKNESS
'T' OF PLATES IN MM

TABLE - III : DIMENSIONS AND PROPERTIES OF EQUAL ANGLES :
(STRUCTURAL STEEL)

TABLE - IV : DIMENSIONS AND PROPERTIES OF UNEQUAL ANGLES :
(STRUCTURAL STEEL)

TABLE - V : DIMENSIONS AND PROPERTIES OF ROLLED STEEL CHANNELS

TABLE - VI : DIMENSIONS AND PROPERTIES OF ROLLED STEEL TEE BARS

TABLE - VII : MAXIMUM STANDARD NOMINAL THICKNESS AND SIZE OF PLATES ROLLED

TABLE - VIII : FLATS

STANDARD NOMINAL DIMENSIONS
AND WEIGHT OF FLATS IN KG. PER
METRE

TABLE - IX : STRIPS

STANDARD NOMINAL DIMENSIONS
AND WEIGHT OF STRIPS IN KG. PER
METRE

TABLE - X : STANDARD DIMENSIONS AND WEIGHT
OF ROUND AND SQUARE BARS

TABLE - XI : ALLOWABLE STRESSES IN STEEL
CONSTRUCTION

TABLE - XII : ALLOWABLE STRESSES IN TENSION
RODS, BOLTS AND RIVETS

TABLE - XIII : EFFECTIVE LENGTH OF
COMPRESSION MEMBERS

TABLE - XIV : MAXIMUM SLENDERNESS RATIO OF
COMPRESSION MEMBERS

TABLE - XV : PROPERTIES OF STEEL TUBES

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DR. C. S. SHAH



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STEAM TABLES IN SI UNITS

INCLUDING MOLLIER CHART

DR. N. C. PANDYA

DR. C. S. SHAH

TABLE - I : PROPERTIES OF SATURATED STEAM AND
SATURATED WATER (TEMPERATURE BASE)

TABLE - II : PROPERTIES OF SATURATED STEAM AND
SATURATED WATER (PRESSURE BASE)

TABLE - III : PROPERTIES OF SUPERHEATED STEAM

TABLE - IV : PROPERTIES OF SATURATED MERCURY VAPOUR
MOLLIER DIAGRAM ENTHALPY-ENTROPY CHART
FOR STEAM IN SI UNITS.



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| 4 : SCALES | 17 : ISOMETRIC PROJECTION |
| 5 : GEOMETRICAL CONSTRUCTION | 18 : OBLIQUE PROJECTION |
| 6 : CURVES USED IN ENGINEERING PRACTICE | 19 : PERSPECTIVE PROJECTION |
| 7 : LOCI OF POINTS | 20 : ORTHOGRAPHIC READING AND CONVERSION OF VIEWS |
| 8 : ORTHOGRAPHIC PROJECTION | 21 : CENTRES OF GRAVITY AND MOMENTS OF INERTIA OF AREAS |
| 9 : PROJECTIONS OF POINTS | 22 : NOMOGRAPHY |
| 10 : PROJECTIONS OF STRAIGHT LINES | 23 : SCREW THREADS |
| 11 : PROJECTIONS ON AUXILIARY PLANES | 24 : SCREWED FASTENINGS |
| 12 : PROJECTIONS OF PLANES | 25 : RIVETED JOINTS AND WELDED JOINTS |
| 13 : PROJECTIONS OF SOLIDS | 26 : COMPUTER AIDED DRAFTING (CADR) |

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| 2 : SHEET LAYOUT AND SKETCHING | 11 : VALVES |
| 3 : LINES, LETTERING AND DIMENSIONING | 12 : RIVETED JOINTS AND WELDED JOINTS |
| 4 : CONVERSION OF PICTORIAL VIEWS INTO OR THOGRAPHIC VIEWS | 13 : SHAFT BEARINGS, BRACKETS AND HANGERS |
| 5 : SECTIONAL VIEWS | 14 : SHAFT COUPLINGS, CLUTCHES AND BRAKES |
| 6 : ORTHOGRAPHIC READING OR INTERPRETATION OF VIEWS | 15 : PULLEYS |
| 7 : SCREW THREADS | 16 : SPUR GEARING |
| 8 : SCREWED FASTENINGS | 17 : ENGINE PARTS |
| 9 : KEYS, COTTER-JOINTS, PIN-JOINTS | 18 : ELEMENTS OF PRODUCTION DRAWING |
| | 19 : ASSEMBLY DRAWINGS |
| | 20 : ELEMENTS OF COMPUTER-AIDED DRAFTING |

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| 2 : SHEET LAYOUT AND SKETCHING | 13 : ORTHOGRAPHIC READING OR INTERPRETATION OF VIEWS |
| 3 : LINES, LETTERING AND DIMENSIONING | 14 : SCREW THREADS |
| 4 : SCALES | 15 : SCREWED FASTENINGS |
| 5 : GEOMETRICAL CONSTRUCTION | 16 : KEYS, COTTER-JOINTS, PIN-JOINTS |
| 6 : CURVES USED IN ENGINEERING PRACTICE | 17 : SHAFT COUPLINGS, CLUTCHES AND BRAKES |
| 7 : LOCI OF POINTS | 18 : PIPE JOINTS |
| 8 : PRINCIPLES OF PROJECTION | 19 : RIVETED JOINTS AND WELDED JOINTS |
| 9 : ISOMETRIC PROJECTION | 20 : SHAFT BEARINGS, BRACKETS AND HANGERS |
| 10 : OBLIQUE PROJECTION | 21 : PULLEYS |
| 11 : CONVERSION OF PICTORIAL VIEWS INTO ORTHOGRAPHIC VIEWS | 22 : ASSEMBLY DRAWINGS |

BUILDING PLANNING AND DRAWING

**DR. N. KUMARA SWAMY
A. KAMESHWARA RAO**

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- 2 : BRIEF HISTORY OF BUILDING DRAWING
- 3 : DIFFERENT TYPES OF BUILDINGS
- 4 : DIFFERENT TYPES OF RESIDENTIAL BUILDINGS
- 5 : SITE SELECTION FOR RESIDENTIAL BUILDINGS
- 6 : CLIMATE AND ITS INFLUENCE ON BUILDING PLANNING
- 7 : ORIENTATON OF BUILDINGS
- 8 : PRINCIPLES OF PLANNING OF BUILDINGS
- 9 : DIFFERENT METHODS OF CONSTRUCTION
- 10 : PREFABRICATED CONSTRUCTION
- 11 : ECONOMY MEASURES IN BUILDING CONSTRUCTION
- 12 : BUILDING BYE-LAWS
- 13 : PLANNING OF RESIDENTIAL BUILDINGS
- 14 : PLANNING OF PUBLIC BUILDINGS

- 15 : CONSTRUCTION MANAGEMENT TECHNIQUES
- 16 : BASIC CONCEPTS OF THE BUILDING ELEMENTS
- 17 : NOMENCLATURE OF BUILDING ELEMENTS
- 18 : STANDARD GUIDELINES FOR BUILDING DRAWING
- 19 : GUIDELINES FOR PLANNING AND DRAWING OF RESIDENTIAL BUILDING
- 20 : DRAFTING MATERIALS AND THEIR UTILIZATION
- 21 : CONVENTIONAL SIGNS AND B.I.S. CODE COLOURS
- 22 : A FEW FACTS OF THE VASTU SASTRA
- 23 : PERSPECTIVE DRAWINGS
- 24 : COMPUTER AIDED BUILDING DRAWING
- 25 : TYPICAL BUILDING DRAWINGS
- 26 : QUESTION BANK

APPENDIX A UNIVERSITY EXAMINATION
QUESTIONS



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ELEMENT OF CIVIL ENGINEERING

DR. ANURAG KANDYA

PART I ELEMENTS OF ENGINEERING MATERIALS

- 2 : ENGINEERING MATERIALS
- 3 : ORGANIC MATERIALS
- 4 : TIMBER
- 5 : CERAMICS
- 6 : ROCKS AND STONES
- 7 : BRICKS
- 8 : TILES
- 9 : REFRACTORIES
- 10 : CEMENT
- 11 : CONCRETE
- 12 : GLASS
- 13 : PAINTS, VARNISHES AND DISTEMPERS

PART II ELEMENTS OF BUILDING CONSTRUCTION

- 14 : BUILDINGS AND THEIR CLASSIFICATIONS
- 15 : FUNCTIONAL REQUIREMENTS OF A BUILDING
- 16 : PRINCIPLES OF PLANNING OF A BUILDING
- 17 : BRICK MASONRY
- 18 : FLOORS AND FLOORINGS
- 19 : DOORS AND WINDOWS
- 20 : STAIRS
- 21 : ROOFS
- 22 : LOADS ON A STRUCTURE
- 23 : FOUNDATION SA
- 24 : EARTHQUAKE AND EARTH QUAKE RESISTANT STRUCTURES

PART III ELEMENTS OF SURVEYING AND LEVELLING

- 25 : BASICS OF SURVEYING
- 26 : TOOLS OF SURVEYING
- 27 : SCALES
- 28 : CHAINING AND RANGING
- 29 : LINEAR MEASUREMENTS
- 30 : COMPASS SURVEYING
- 31 : LEVELLING
- 32 : CONTOURING
- 33 : AREA AND VOLUME
- 34 : THEODOLITE
- 35 : RECENT ADVANCEMENTS IN SURVEYING

PART IV ELEMENTS OF TRANSPORTATION ENGINEERING

- 36 : TRANSPORTATION ENGINEERING
- 37 : DEVELOPMENT OF ROADS IN INDIA
- 38 : HIGHWAY CONSTRUCTION
- 39 : TRAFFIC ENGINEERING
- 40 : RAILWAY ENGINEERING
- 41 : BRIDGES
- 42 : TUNNELS

PART V ELEMENTS OF ENVIRONMENTAL ENGINEERING

- 43 : BASICS OF ENVIRONMENT
- 44 : OUR RESOURCES
- 45 : GLOBEL ENVIROMENTAL PROBLEMS

APPENDICES

APPENDIX A TO APPENDIX C



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CIVIL ENGINEERING DRAWING

S. C. RANGWALA

**CIVIL ENGINEERING
DRAWING**

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- 3 : METHODS OF DRAWING
- 4 : SUBMISSION AND WORKING DRAWINGS
- 5 : PERSPECTIVE DRAWINGS
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- 7 : ARCHITECTURAL COMPOSITION
- 8 : BUILDING BYE - LAWS
- 9 : PLANNING OF RESIDENTIAL BUILDINGS
- 10 : PLANNING OF INDUSTRIAL STRUCTURES
- 11 : PLANNING OF PUBLIC BUILDINGS
- 12 : MISCELLANEOUS TOPICS

SIZE : 270 mm x 210 mm
PAGES : 148 + 16



SURVEYING AND LEVELLING**S. C. RANGWALA**

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- | | |
|--------------------------------------|---------------------------------------|
| 1 : INTRODUCTION | 13 : THEODOLITE |
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| 3 : LINEAR MEASUREMENTS OF DISTANCES | 15 : TACHEOMETRIC SURVEYING |
| 4 : CROSS-STAFF SURVEY | 16 : CIRCULAR CURVES |
| 5 : CHAIN SURVEYING | 17 : TRANSITION CURVES |
| 6 : COMPUTATION OF AREAS | 18 : VERTICAL CURVES |
| 7 : COMPASS SURVEY | 19 : FIELD ASTRONOMY |
| 8 : PLANE TABLE SURVEY | 20 : THEORY OF ERRORS |
| 9 : LEVELLING | 21 : SETTING OUT WORKS |
| 10 : CONTOURING | 22 : HYDROGRAPHIC SURVEYING |
| 11 : MINOR INSTRUMENTS | 23 : PHOTOGRAPHIC SURVEYING |
| 12 : COMPUTATION OF VOLUMES | 24 : ELECTRONIC DISTANCE MEASUREMENTS |

ENGINEERING MATERIALS**MATERIAL SCIENCE****S. C. RANGWALA**

ISBN : 978-93-80358-13-0
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- | | |
|---|---|
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| 2 : STONES | 13 : CORROSION |
| 3 : CLAY PRODUCTS AND REFRACTORIES | 14 : GLASS |
| 4 : BRICKS | 15 : PAINTS, VARNISHES AND DISTEMBERS |
| 5 : LIME | 16 : PLASTICS |
| 6 : CEMENT | 17 : MISCELLANEOUS MATERIALS |
| 7 : MORTAR | 18 : MATERIAL SCIENCE OF METALS |
| 8 : CEMENT CONCRETE | APPENDIX I |
| 9 : TIMBER | B.I.S C ZODES FOR IMPORTANT ENGINEERING MATERIALS |
| 10 : FERROUS METALS | APPENDIX II |
| 11 : STEEL | ABBREVIATED TERMS USED IN THIS BOOK |

BUILDING CONSTRUCTION**S. C. RANGWALA**

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| 2 : FUNCTIONAL PLANNING OF BUILDINGS | 16 : CEMENT CONCRETE CONSTRUCTION |
| 3 : IMPORTANT BUILDING COMPONENTS | 17 : ARCHES |
| 4 : SITE INVESTIGATION AND GROUND TECHNIQUES | 18 : LINTELS |
| 5 : FOUNDATIONS | 19 : STAIRS |
| 6 : DEEP EXCAVATIONS | 20 : FORM WORK |
| 7 : DEEP FOUNDATIONS (PILE FOUNDATIONS) | 21 : DOORS, WINDOWS AND VENTILATORS |
| 8 : COFFERDAMS | 22 : CARPENTRY AND JOINERY |
| 9 : CAISSONS | 23 : FLOORS AND FLOORING |
| 10 : STONE MASONRY | 24 : ROOFS |
| 11 : BRICK MASONRY | 25 : POINTING AND PLASTERING |
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| 13 : PARTITIONS | 27 : STRUCTURAL STEEL WORK |
| 14 : SCAFFOLDING, SHORING AND UNDERPINNING | 28 : ACOUSTICS |
| | 29 : FIRE PROTECTION IN BUILDINGS |
| | 30 : VENTILATION AND AIR-CONDITIONING |
| | 31 : CONSTRUCTION EQUIPMENTS |
| | 32 : CPM AND PERT |

ESTIMATING COSTING AND VALUATION

PROFESSIONAL PRACTICE AND QUANTITY SURVEYING

PART I ELEMENTS OF ESTIMATING AND COSTING

- 1 : APPROXIMATE METHODS OF COSTING
- 2 : SPECIFICATIONS
- 3 : CONDITIONS OF CONTRACT
- 4 : CONTRACTS AND TENDERS
- 5 : ARBITRATION
- 6 : ACCOUNTS
- 7 : CONSTRUCTION MANAGEMENT AND PLANNING

PART II ESTIMATING AND COSTING IN PRACTICE

- 8 : RATE ANALYSIS
- 9 : TAKING OUT QUANTITIES
- 10 : ESTIMATES OF VARIOUS TYPES OF BUILDINGS
- 11 : ESTIMATES OF DIFFERENT R.C.C. STRUCTURES AND THEIR FORM WORK
- 12 : ESTIMATES OF DIFFERENT TYPES OF ROOFS AND STEEL STRUCTURES

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- 14 : ESTIMATES OF BRIDGES, CULVERTS AND PIERS
- 15 : ESTIMATES OF IRRIGATION WORKS
- 16 : ESTIMATES OF ROAD WORKS

PART III VALUATION

- 17 : VALUE
- 18 : INVESTMENT IN REAL PROPERTY
- 19 : INTERESTS IN REAL PROPERTY
- 20 : METHODS OF VALUATION
- 21 : OUTGOINGS
- 22 : DEPRECIATION
- 23 : VALUATION TABLES
- 24 : MISCELLANEOUS TOPICS
- 25 : EASEMENTS

APPENDICES

S. C. RANGWALA



ISBN : 978-93-80358-19-2
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VALUATION OF REAL PROPERTIES

S. C. RANGWALA

- 1 : VALUE
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- 3 : INTERESTS IN REAL PROPERTY
- 4 : METHODS OF VALUATION
- 5 : OUTGOINGS
- 6 : DEPRECIATION
- 7 : VALUATION OF LICENSED PREMISES
- 8 : VALUATION OF LIFE INTERESTS
- 9 : VALUATION FOR RATING
- 10 : VALUATION OF AGRICULTURAL LANDS
- 11 : FINANCE FOR INVESTMENT IN REAL PROPERTIES

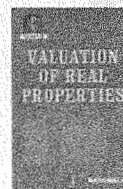
- 12 : VALUATION TABLES
- 13 : MISCELLANEOUS TOPICS
- 14 : EASEMENTS
- 15 : STANDARD RENT
- 16 : COMPULSORY ACQUISITION OF LAND
- 17 : THE TRANSFER OF PROPERTY ACT
- 18 : URBAN LAND CEILING ACT
- 19 : TYPICAL PROBLEMS

APPENDIX I

IMPORTANT JUDGEMENTS AND DECISIONS

APPENDIX II

VALUATION TABLES I TO VIII



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TOWN PLANING

S. C. RANGWALA

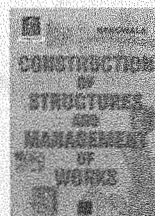


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- 2 : SURVEYS
- 3 : ZONING
- 4 : HOUSING
- 5 : SLUMS
- 6 : PARKS AND PLAYGROUNDS
- 7 : INDUSTRIES
- 8 : PUBLIC BUILDINGS
- 9 : RE-PLANNING EXISTING TOWNS
- 10 : BUILDING BYE - LAWS
- 11 : DEVELOPMENT PLAN OR MASTER PLAN
- 12 : URBAN ROADS
- 13 : TRAFFIC MANAGEMENT
- 14 : MISCELLANEOUS TOPICS

CONSTRUCTION OF STRUCTURES AND MANAGEMENT OF WORKS

S. C. RANGWALA



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- 2 : CPM AND PERT
- 3 : CONSTRUCTION OF DAMS
- 4 : CONSTRUCTION OF MULTI-STOREYED BUILDINGS
- 5 : ERECTION OF STEEL STRUCTURES
- 6 : TUNNELLING
- 7 : COMPRESSED AIR
- 8 : CONSTRUCTION EQUIPMENTS
- 9 : CONSTRUCTION ACCIDENTS
- 10 : MANAGEMENT AND ORGANIZATION
- 11 : CONSTRUCTION LABOUR
- 12 : CONSTRUCTION MATERIALS
- 13 : ECONOMIC ANALYSIS
- 14 : MISCELLANEOUS TOPICS

WATER SUPPLY AND SANITARY ENGINEERING**ENVIRONMENTAL ENGINEERING****S. C. RANGWALA**

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- 1 : INTRODUCTION
- 2 : QUANTITY OF WATER
- 3 : SOURCES OF WATER SUPPLY
- 4 : PUMPS FOR WATER SUPPLY PROJECT
- 5 : QUALITY OF WATER
- 6 : SEDIMENTATION TANKS
- 7 : COAGULATION OF WATER
- 8 : FILTRATION OF WATER
- 9 : DISINFECTION OF WATER
- 10 : WATER SOFTENING
- 11 : MISCELLANEOUS METHODS OF WATER TREATMENT
- 12 : CONVEYANCE OF WATER
- 13 : DISTRIBUTION SYSTEM OF WATER
- 14 : PIPE APPURTENANCES
- 15 : WATER POLLUTION CONTROL AND WATER MANAGEMENT
- 16 : RADIOACTIVITY AND WATER SUPPLIES

PART II SANITARY ENGINEERING

- 17 : SANITARY ENGINEERING – AND INTRODUCTION
- 18 : COLLECTION AND CONVEYANCE OF REFUSE
- 19 : QUANTITY OF SEWAGE

- 20 : CONSTRUCTION OF SEWERS
- 21 : DESIGN OF SEWER
- 22 : SEWER APPURTENANCES
- 23 : SANITARY ENGINEERING – PUMPS
- 24 : HOUSE DRAINAGE
- 25 : QUALITY OF SEWAGE
- 26 : NATURAL METHODS OF SEWAGE DISPOSAL
- 27 : PRIMARY TREATMENT OF SEWAGE
- 28 : FILTRATION OF SEWAGE
- 29 : ACTIVATED SLUDGE PROCESS
- 30 : SLUDGE DISPOSAL
- 31 : MISCELLANEOUS METHODS OF SEWAGE TREATMENT
- 32 : MISCELLANEOUS TOPICS
- 33 : WASTE WATER

PART II ENVIRONMENTAL ENGINEERING

- 34 : ENVIRONMENTAL ENGINEERING – AN INTRODUCTION
- 35 : AIR POLLUTION
- 36 : NOISE POLLUTION
- 37 : MISCELLANEOUS TOPICS

APPENDIX A TYPICAL DESIGN OF SEWAGE TREATMENT PLAN**APPENDIX B TERMINOLOGY****FLUID MECHANICS****S. C. RANGWALA**

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- 5 : FUNDAMENTALS OF FLUID FLOW
- 6 : ENERGY EQUATIONS
- 7 : ORIFICES, MOUTHPIECES
- 8 : NOTCHES AND WEIRS
- 9 : FLOW THROUGH PIPES
- 10 : IMPACT OF JETS
- 11 : FLOW IN OPEN CHANNELS
- 12 : HYDRAULIC TURBINES
- 13 : RECIPROCATING PUMPS
- 14 : CENTRIFUGAL PUMPS
- 15 : HYDRAULIC SYSTEMS AND COMPONENTS
- 16 : DIMENSIONAL ANALYSIS AND HYDRAULIC SIMILITUDE

RAILWAY ENGINEERING

S. C. RANGWALA

- 1 : INTRODUCTION
- 2 : RAILWAY SURVEYS
- 3 : TRAIN RESISTANCES AND ROLLING STOCK
- 4 : RAIL GAUGES
- 5 : RAILS
- 6 : RAIL FASTENINGS
- 7 : SLEEPERS
- 8 : BALLAST
- 9 : PLATE-LAYING
- 10 : MAINTENANCE
- 11 : CREEP
- 12 : CURVATURE OF TRACK
- 13 : STATIONS AND YARDS
- 14 : STATION MACHINERY
- 15 : POINTS AND CROSSINGS
- 16 : COMBINATIONS OF POINTS AND CROSSINGS
- 17 : SIGNALLING

- 18 : INTERLOCKING
 - 19 : RAILWAY TRACTION
 - 20 : EARTHWORK AND DRAINAGE
 - 21 : TUNNELLING
 - 22 : RAPID TRANSIT SYSTEM
 - 23 : MATERIALS MANAGEMENT
- APPENDIX I**
UNITS OF THE INDIAN RAILWAYS
- APPENDIX II**
TRAINING INSTITUTIONS OF THE INDIAN RAILWAYS
- APPENDIX III**
FAMOUS INDIAN TRAINS
- APPENDIX IV**
ABBREVIATED TERMS
- APPENDIX V**
MULTIPLE CHOICE



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HIGHWAY ENGINEERING

S. C. RANGWALA

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- 2 : HIGHWAY LANNING AND ALIGNMENT
- 3 : GEOMETRICAL DESIGN OF HIGHWAYS
- 4 : LOW COST ROADS
- 5 : BITUMINOUS ROADS
- 6 : CEMENT CONCRETE ROADS
- 7 : TESTS FOR HIGHWAY MATERIALS
- 8 : DESIGN OF HIGHWAY PAVEMENTS
- 9 : OTHER TYPES OF PAVEMENTS
- 10 : HILL ROADS
- 11 : HIGHWAY DRAINAGE
- 12 : HIGHWAY MAINTENANCE

- 13 : HIGHWAY MAKING MACHINERIES
 - 14 : TRAFFIC ENGINEERING
 - 15 : HIGHWAY ARBORICULTURE AND LIGHTING
 - 16 : HIGHWAY ECONOMICS
- APPENDIX 1** TYPICAL TRAFFIC SIGNS
- APPENDIX 2** ABBREVIATED TERMS



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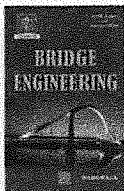
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S. C. RANGWALA

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 - 2 : AIRPORT SURVEYS
 - 3 : AIRPORT PLANNING
 - 4 : RUNWAY DESIGN
 - 5 : TAXIWAY DESIGN
 - 6 : PLANNING AND DESIGN OF TERMINAL AREA
 - 7 : AIRPORT PAVEMENT DESIGN
 - 8 : AIRPORT GRADING AND DRAINAGE
 - 9 : VISUAL AIDS
 - 10 : AIR TRAFFIC CONTROL
 - 11 : HELIPORTS AND STOLPORTS
- APPENDIX I** ABBREVIATED TERMS



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BRIDGE ENGINEERING**S. C. RANGWALA**

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- 1 : INTRODUCTION
 - 2 : FOUNDATIONS-I : SPREAD AND PILE FOUNDATION
 - 3 : FOUNDATIONS - II
CAISSONS AND COFFERDAMS
 - 4 : SUBSTRUCTURES
 - 5 : FIXED-SPAN SUPER STRUCTURES
 - 6 : LOW COST BRIDGES :
MOVABLE-SPAN SUPER STRUCTURES
 - 7 : LOW COST BRIDGES : CULVERTS AND CAUSEWAYS
 - 8 : LOW COST BRIDGES
TIMBER OR TEMPORARY BRIDGES
 - 9 : FLOORING
 - 10 : LOADINGS ON BRIDGES
 - 11 : BEARINGS
 - 12 : MISCELLANEOUS TOPICS
- APPENDIX**
SOME MARVELLOUS BRIDGES

BRIDGE ENGINEERING**J. S. ALAGIA**

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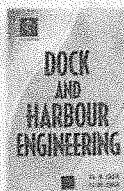
PART I DESCRIPTION OF BRIDGES

- 1 : GENERAL INTRODUCTION AND TYPES OF BRIDGES
- 2 : PERMANENT BIG BRIDGES
[SUPER STRUCTURES]
- 3 : PERMANENT BIG BRIDGES
[SUB STRUCTURES]
- 4 : PERMANENT SMALL BRIDGES
- 5 : TEMPORARY BRIDGES
[TEMPORARY FIXED BRIDGES]

PART II DESIGN, CONSTRUCTION AND MAINTENANCE OF BRIDGES

- 1 : CHOICE OF TYPE OF A BRIDGE
- 2 : GENERAL LAYOUT
- 3 : PERMANENT BIG BRIDGES
[SUPER STRUCTURES]
- 4 : PERMANENT BIG BRIDGES
[SUB STRUCTURES]
- 5 : PERMANENT SMALL BRIDGES AND TEMPORARY BRIDGES
- 6 : CONSTRUCTION AND MAINTENANCE

APPENDIX
BRIDGE ARCHITECTURE

DOCK AND HARBOUR ENGINEERING**H. P. OZA****G. H. OZA**

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 - 2 : HYDROGRAPHIC SURVEYS AND CHARTS
 - 3 : WINDS, WAVES AND CYCLONES
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 - 5 : INVESTIGATIONS AND MODEL TESTS
 - 6 : SHIP FEATURES RELATED TO PORT PLANNING
 - 7 : TRAFFIC FORECASTING AND HINTERLAND
 - 8 : HARBOUR LAYOUT
 - 9 : CHANNEL, BASIN AND BERTHS
 - 10 : BREAKWATERS
 - 11 : WHARVES
 - 12 : JETTIES, DOLPHINS AND MOORINGS
 - 13 : BERTHS FOR CRUDE OIL AND PETROLEUM PRODUCTS
 - 14 : RO-RO FERRY SERVICE
 - 15 : LOCKS
 - 16 : SHORE PROTECTION WORKS
 - 17 : DRY DOCKS AND SLIPWAYS
 - 18 : CARGO HANDLING EQUIPMENT
 - 19 : APRONS, TRANSIT SHEDS AND WAREHOUSES
 - 20 : SUPPORTING FACILITIES AND ANCILLARIES
 - 21 : NAVIGATIONAL AIDS
 - 22 : DREDGING AND DREDGERS
 - 23 : CONSTRUCTION MATERIALS AND METHODS
- APPENDIX I TO APPENDIX XII**

HARBOUR DOCK AND TUNNEL ENGINEERING

SECTION I

HARBOUR AND DOCK ENGINEERING

- 1 : HARBOURS AND PORTS
- 2 : NATURAL PHENOMENA : TIDES, WIND AND WAVES
- 3 : PROTECTION FACILITIES : MOUND BREAKWATER
- 4 : PROTECTION FACILITIES : WALL TYPE AND SPECIAL BREAKWATERS
- 5 : PLANNING AND LAYOUT OF PORTS
- 6 : DOCKING FACILITIES
- 7 : REPAIRING FACILITIES
- 8 : APPROACH FACILITIES
- 9 : LOADING UNLOADING FACILITIES
- 10 : STORING FACILITIES
- 11 : DREDGING FACILITIES
- 12 : GUIDING FACILITIES

SECTION II

TUNNEL ENGINEERING

- 13 : GENERAL ASPECTS
- 14 : STAGES IN TUNNEL CONSTRUCTION
- 15 : SOIL CLASSIFICATION AND TUNNELLING METHODS
- 16 : OTHER METHODS OF TUNNELLING IN SOFT SOILS
- 17 : TUNNELLING IN WATER BEARING SOILS
- 18 : TUNNELLING IN ROCK
- 19 : THE NEW AUSTRIAN TUNNELLING METHOD (NATM)
- 20 : SHAFTS
- 21 : TUNNEL LINING
- 22 : DRAINAGE OF TUNNELS
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- 24 : HEALTH PROTECTION IN TUNNELS

R.SRINIVASAN



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**A. ALAVUDEEN, KHALID HUSSAIN SYED,
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- 7 : TROUBLESHOOTING IN HYDRAULIC
- 8 : HYDRAULIC ELECTRICAL DEVICES TROUBLESHOOTING AND SAFETY
- 9 : PNEUMATICS
- 10 : COMPONENT OF PNEUMATICS

- 11 : BASIC PNEUMATIC CIRCUITS
- 12 : APPLICATION FOR PNEUMATICS
- 13 : SERVO SYSTEMS, PROPORTIONAL VALVES AND PNEUMATIC SAFETY
- 14 : FLUIDICS
- 15 : DESIGN OF PNEUMATIC CIRCUITS

APPENDIX A TO APPENDIX C



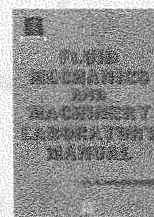
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FLUID MECHANICS AND MACHINERY LABORATORY MANUAL

DR. N. KUMARA SWAMY

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- 2 : PRACTICAL APPLICATION OF FLUID MECHANICS IN VARIOUS FIELDS
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- 4 : MEASUREMENT OF HYDRAULIC PARAMETERS
- 5 : PERFORMANCE OF HYDRAULIC MACHINES
- 6 : EXPERIMENTS ON FLUID MECHANICS
- 7 : EXPERIMENTS ON FLUID MACHINERY
- 8 : VIVA QUESTIONS ON FLUID MECHANICS
- 9 : VIVA QUESTIONS ON FLUID MACHINERY
- 10 : PLANNING, OPERATION AND MAINTENANCE OF LABORATORY

APPENDIX A TO APPENDIX I



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PROGRAMMING IN C**JIGNESH SHAH**

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- 3 : STATEMENT, BRANCHING AND LOOPING
- 4 : 'C' FUNCTIONS – USER DEFINED FUNCTION
- 5 : STORAGE CLASSES
- 6 : ARRAY
- 7 : POINTERS
- 8 : STRUCTURES AND UNIONS
- 9 : WORKING WITH FILES
- 10 : INTRODUCTION TO OBJECT ORIENTED PROGRAMMING

- 11 : SPECIAL PROGRAMS
- APPENDICES**
- APPENDIX A**
FLOWCHART AND ALGORITHM
- APPENDIX B**
BIT OPERATORS
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CNC FUNDAMENTALS AND PROGRAMMING

**P. M. AGRAWAL
V. J. PATEL**



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- 1 : INTRODUCTION TO NC/CNC MACHINE TOOLS
- 2 : BASICS OF NC/CNC MACHINE TOOLS
- 3 : CNC MACHINE TOOLS - STRUCTURE AND ELEMENTS
- 4 : BASICS OF CNC PROGRAMMING
- 5 : BASICS OF TURNING CENTER PROGRAMMING
- 6 : SINGLE PASS CANNED CYCLES FOR TURNING CENTERS
- 7 : MULTIPASS CANNED CYCLES IN TURNING
- 8 : BASICS OF MACHINING CENTER PROGRAMMING
- 9 : DRILLING CANNED CYCLES IN MILLING
- 10 : SUBPROGRAMMING
- 11 : MACRO PROGRAMMING
- 12 : DNC AND DATA COMMUNICATION
- 13 : APT PROGRAMMING
- 14 : CAD/CAM INTEGRATION

APPENDIX I

MULTIPLE CHOICE QUESTIONS

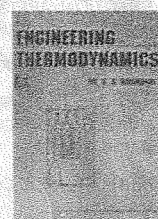
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PROGRAMMING CODES**FUNDAMENTALS OF PROGRAMMING
WITH FORTAN 77****DR. N. D. JOTWANI**

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 PAGES : 216 + 12

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 - 2 : TYPES OF DATA
 - 3 : BASIC OPERATIONS ON DATA
 - 4 : FLOW OF CONTROL IN A PROGRAM
 - 5 : CONDITIONAL EXECUTION OF STATEMENTS
 - 6 : ITERATIVE EXECUTION OF STATEMENTS
 - 7 : FORMATTED INPUT AND OUTPUT
 - 8 : ARRAYS
 - 9 : SUBPROGRAMS
 - 10 : DATA STATEMENT AND SPECIFICATION STATEMENTS
 - 11 : FILE OPERATIONS
 - 12 : MISCELLANEOUS STATEMENTS
- APPENDICES A TO APPENDICES F**

ENGINEERING THERMODYNAMICS**DR. S. S. KHANDARE**

- 1 : INTRODUCTION
- 2 : GAS LAWS AND PROPERTIES
- 3 : THERMODYNAMIC PROCESSES
- 4 : AVAILABILITY
- 5 : AIR STANDARD CYCLES
- 6 : PROPERTIES OF STEAM AND STEAM CYCLE
- 7 : FLOW PROCESSES
- 8 : FUELS AND COMBUSTION



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